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Artificial Intelligence and Personality Traits: Evidence from Online Job Vacancies

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GEP

Artificial Intelligence and Personality Traits: Evidence from Online Job Vacancies *

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Abstract

This paper examines how demand for artificial intelligence (AI)-related skills is associated with firms' demand for worker personality traits. Using text analysis of 11.7 million UK online job postings from 2017 to 2022, drawn from a near-universe dataset of online vacancies, we construct measures of Big Five personality traits and AI-related skill requirements from job descriptions. At the vacancy level, positions requiring AI-related skills are significantly more likely to demand openness to experience and less likely to demand the remaining four traits, after controlling for firm, occupation, location, and time fixed effects. This pattern is distinct from the profile of postings requiring general computer skills, suggesting that the association with openness is specific to AI knowledge rather than broader digital requirements. At the firm level, staggered difference-in-differences estimates show that when firms expand AI-related hiring, their demand for openness rises across all postings, including non-AI vacancies within occupation groups. The results are consistent with firms that seek to integrate AI knowledge valuing creative workers who can adapt to new tools and approaches, and extending this preference beyond the roles that directly involve AI.

Keywords: Artificial intelligence, labour demand, personality, online job postings
JEL Codes: J23, J24, O33, C55

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1 Introduction

The diffusion of artificial intelligence (AI) has created concerns about labour displacement and reorganisation of tasks (Acemoglu et al., 2022, 2026; Brynjolfsson, Chandar, & Chen, 2025; Webb, 2019). Existing research examines how AI affects tasks and the demand for cognitive, technical skills, as well as experience and expertise levels (Alekseeva et al., 2026, 2021; Autor & Thompson, 2025; Brynjolfsson, Li, & Raymond, 2025; Squicciarini & Nachtigall, 2021). Beyond altering task content, AI can modify how tasks are performed, problems approached, and solutions developed. It can also help to generate new ideas and processes (Acemoglu et al., 2023; Brynjolfsson, Li, & Raymond, 2025). As a result, firms that seek to integrate AI-related capabilities into their workforce may perceive certain personality traits as more relevant in the new work environment. In particular, personality traits such as openness, curiosity, or cooperation may influence workers' ability to effectively interact with and adapt to AI-based processes. These traits are fundamental components of human capital that are not fully captured by conventional measures of education, experience, or technical expertise (D. Deming & Silliman, 2025; Heckman et al., 2021). This raises a fundamental yet unexplored question: what personality traits do firms seek in workers as they demand AI-related skills?

Using text analysis of UK online job postings from 2017 to 2022, drawn from a near-universe dataset of online vacancies, we construct measures of AI-related skill requirements and employer demand for the Big Five personality traits (openness, conscientiousness, extraversion, agreeableness, and emotional stability). We focus on personality traits because they are relatively stable traits that affect labour market outcomes (Alderotti et al., 2023; Bowles et al., 2001; Fitzenberger et al., 2022; Flinn et al., 2025; Heckman et al., 2006; Nyhus & Pons, 2005; Woessmann, 2024), yet remain largely fixed after early adulthood and cannot be easily developed through training or work experience (Cobb-Clark & Schurer, 2012; Heckman & Kautz, 2012; McCrae & John, 1992). This means that when firms demand particular personality attributes, they may do so at the recruitment stage rather than developing them internally (Grugulis & Vincent, 2009;

Matteson et al., 2016; McGee & McGee, 2020). Because firms cannot emphasize every desirable characteristic with equal prominence in a job advertisement, the traits that appear in vacancy text are likely to reflect attributes that employers consider especially salient for the position.[1] Consequently, job postings are a particularly informative source for studying which personality traits firms demand as they seek to integrate AI knowledge into their production processes.

AI may influence the nature of jobs (Babina, 2026) and modify how firms organise production, process information, and make decisions (Barua et al., 2025). In industrial settings, AI knowledge is used to support prediction, pattern recognition, anomaly detection or optimisation.[2] These capabilities are central to firms' use of AI, including more recent generative AI systems. Therefore, AI may influence the personality traits firms value in workers, and potentially the types of workers they seek to hire. On the one hand, AI may increase coordination and interaction requirements within firms. Consequently, it can raise the value of interpersonal traits such as agreeableness and extraversion. Task-specific AI systems are often imperfect substitutes for human interaction in activities involving communication, teamwork, and organisational integration. As Johnson and Acemoglu (2023) observe: "It is therefore not surprising that despite the spread of AI technologies, many companies are increasingly seeking workers with social, rather than mathematical, or technical skills..."

On the other hand, task-based models consider that AI can complement non-routine cognitive tasks while substituting for routine cognitive activities (Acemoglu & Restrepo, 2019; Autor, 2014). Therefore, AI tools may be associated with greater relative demand for traits linked to creativity and complex problem-solving, such as openness, while reducing the importance of traits associated with routine cognitive task execution, such as

[1] Online job advertisements are not strictly constrained by space the way print ads are, but employers still face meaningful incentives to be selective. Employers face attention constraints on both sides of the advertisement: they must decide which requirements to feature prominently, and applicants selectively read and respond to vacancy text (Abraham et al., 2024; Flory et al., 2015). Traits that appear in a posting are therefore more likely to reflect attributes employers consider especially salient for the position.

[2] For example, Siemens and GE use predictive analytics to monitor machinery and anticipate maintenance needs; BMW applies AI-based quality-control systems that analyse sensor and image data on production lines; Nike analyses customer behaviour to personalise products; and Bosch uses AI in factories to detect deviations in production processes and improve efficiency.

conscientiousness. More broadly, as firms seek to integrate AI knowledge into production processes, they may value workers who can adapt to new tools and are open to integrating these technologies.

We analyse 11.7 million job advertisements posted in the UK by 128,559 companies, drawn from Adzuna, a job vacancy aggregator providing near-universe coverage of online job postings in the UK. A key advantage of this dataset is that it includes full job descriptions, which we use to identify personality-trait requirements using dictionaries grounded in the psychology literature (Costa Jr & McCrae, 1992; Goldberg, 1992; John, 1990). To ensure that dictionary matches capture genuine trait demand rather than unrelated usage, we manually inspected random samples of advertisements separately for AI and non-AI vacancies, and systematically excluded from the entire sample word combinations where the surrounding context indicated the term described a product, task, or organisational characteristic rather than a required worker trait.

We identify AI-related postings using an expanded dictionary based on Acemoglu et al. (2022) and Babina et al. (2024), whose sample periods overlap with ours. This measure captures firms seeking to bring internal AI-related capabilities or knowledge (such as machine learning, neural networks, or natural language processing) rather than procuring AI-powered services externally. We use the terms "AI-related skills" and "AI-related knowledge" interchangeably to refer to the technical competencies identified by our dictionary, following the convention in prior work (Acemoglu et al., 2022; Babina et al., 2024). Importantly, we distinguish AI-related demand from broader computer-skill demand. Our AI measure is based on terms related to artificial intelligence and machine learning, while general computer skills, such as software development and data analysis requirements, are measured separately and used as a comparison in the empirical analysis. To ensure the relevance and accuracy of the AI dictionary in our context, we manually validate AI-related terms against random samples of job advertisements. The expanded dictionary covers more than 100 terms related to artificial intelligence and machine learning. We combine the vacancy data with firm-level data from FAME, a financial database within the Bureau van Dijk suite of business databases. Our analysis is conducted at both the

vacancy and firm levels.

First, at the vacancy level, we show systematic differences in personality requirements between AI and non-AI vacancies. Positions requiring AI-related skills are significantly more likely to demand openness, while showing lower demand for the other personality traits. These patterns remain robust after controlling for detailed job characteristics, as well as firm, occupation, location, and time fixed effects. Importantly, the pattern for AI-related vacancies differs from the pattern for postings requiring general computer skills. This comparison shows that the association between AI and personality-trait demand is not simply capturing broader digital, software, or data-analysis requirements. Instead, the results suggest that vacancies seeking AI-related knowledge place a distinctive emphasis on openness.

Second, we examine whether these vacancy-level differences translate into changes in firms' overall hiring strategies using an event-study design. We define the event date as the first observed vacancy requiring AI-related skills posted by a firm. This timing captures the point at which a firm first appears in our data as demanding AI-related capabilities. Following prior work that uses demand for AI-related skills in job postings as a proxy for firms' engagement with AI technologies (Acemoglu et al., 2022; Alekseeva et al., 2026, 2021; Babina et al., 2024), we refer to firms that post at least one AI-related vacancy as AI-demanding firms.

We then estimate changes in personality-trait demand around the first observed AI-related vacancy using a weighted staggered difference-in-differences estimator (Callaway & Sant'Anna, 2021), where weights are derived from entropy balancing (Hainmueller, 2012). Because AI-related hiring is virtually absent before 2016 and only begins to expand meaningfully thereafter (Acemoglu et al., 2022), our analysis focuses on the period from 2017 onward.[3] The estimator exploits within-firm variation over time, implicitly absorbing

[3]Firms already observed demanding AI-related skills in the first sample period are treated as left-censored and are excluded from the identifying comparison. The estimates are therefore identified from firms whose first observed AI-related vacancy occurs after the initial period and from firms in the control group, comprising those that have not yet posted any AI-related vacancies at each point of comparison, including firms that never do so during the sample.

time-invariant unobserved firm heterogeneity, while the entropy balancing weights ensure that AI-demanding and non-AI-demanding firms are comparable on observed pre-event characteristics, including baseline personality requirements, firm attributes, occupational composition, and skill demands. Our estimates, therefore, condition on both time-invariant firm characteristics and balanced pre-event observables, and should be interpreted as net of pre-existing differences in firms' personality orientation. These estimates document associations between firms' first observed demand for AI-related skills and subsequent changes in their hiring patterns, conditional on observable pre-event characteristics and time-invariant firm heterogeneity.

The firm-level results reveal a pattern that differs from the vacancy-level substitution. When comparing AI-demanding firms with non-AI-demanding firms, we observe higher demand for openness among AI-demanding firms across both AI-related and non-AI-related jobs. This pattern is visible in the year of first observed AI-related demand and remains apparent for at least two subsequent years. By contrast, we do not observe significant differences in the demand for other personality traits. The analysis of non-AI vacancies suggests that this pattern is not confined to AI-specific hiring. A shift-share decomposition reveals two offsetting forces: AI-demanding firms shift their non-AI hiring toward occupation groups with lower baseline openness requirements, while within most occupation groups their non-AI vacancies become more likely to mention openness. The net firm-level effect on non-AI vacancies is positive, indicating that the within-occupation increase dominates. Taken together, these findings are consistent with firms seeking AI-related capabilities placing greater emphasis on openness not only within narrowly defined AI positions, but also more broadly across the organisation.

We contribute to the literature on technological change and labour demand by showing that AI is associated with the firms' demand for personality characteristics. Existing work documents how AI influences task content and the demand for cognitive, technical skills, expertise or managerial skills (Acemoglu et al., 2022; Alekseeva et al., 2026, 2021; Autor & Thompson, 2025; Webb, 2019), while a separate literature highlights the growing importance of social and interpersonal skills in labour markets (Deming, 2017; Deming &

Kahn, 2018; Izadi & Tuhkuri, in press). In this paper, we connect these literatures and provide, to our knowledge, the first evidence on the relationship between firms' demand for AI-related skills and their demand for personality traits. Prior studies using vacancy data show that personality traits are widely demanded and vary across occupations (Brenčič & McGee, 2023a, 2023b), but they do not link trait demands to technological change. In contrast, we examine how AI-related job postings differ from otherwise comparable vacancies within firms in the personality traits demanded. Our work is related to Alekseeva et al. (2021) and Squicciarini and Nachtigall (2021), who show that AI shifts labour demand toward non-routine analytical and interpersonal skills. We extend this literature by showing that firms' demand for AI-related skills is associated not only with changes in skill requirements, but also with changes in demand for workers' stable behavioural characteristics. In doing so, we highlight personality traits as an additional and previously underexplored margin through which AI affects the composition of labour demand.

We also contribute to the literature on how firms adjust their recruitment following AI demand. Most existing work focuses on the characteristics of AI-related jobs or on aggregate labour market effects and productivity (Acemoglu et al., 2022; Aghion et al., 2025; Autor & Thompson, 2025; Brynjolfsson, Chandar, & Chen, 2025). We analyse both vacancy-level and firm-level responses. We show that AI is linked to substitution across personality traits within AI-intensive roles, while being associated with a persistent increase in demand for openness across firms' entire hiring portfolios. This contrast suggests that AI demand is accompanied by organisational adjustments beyond the requirements of specific jobs, consistent with evidence that AI-investing firms experience complementary changes in employment and organisational structures (Babina et al., 2023).

We contribute methodologically to the measurement of employer demand for personality traits. We construct text-based indicators of Big Five trait requirements using dictionaries grounded in the psychology literature (Costa Jr & McCrae, 1992; Goldberg, 1992; John, 1990) and apply them to UK job postings. Our approach follows the methodology of Brenčič and McGee (2026), who develop personality demand measures from US

job ads using the same lexical tradition. We extend their approach in three respects: we apply it to a substantially larger dataset of UK vacancies; we examine how personality demand varies with firms' AI-related hiring rather than characterising occupational demand in the cross-section; and we exploit within-firm variation over time to study how firms' overall personality requirements shift as they begin recruiting AI-capable workers. Our approach complements individual-level studies that measure workers' personality endowments (Collischon, 2020; Gensowski, 2018) by directly capturing firms' stated hiring requirements.

The rest of the paper proceeds as follows. Section 2 describes the data sources and construction of the variables. Section 3 presents the empirical methodology for both vacancy-level and firm-level analyses. Section 4 reports the results. Section 5 concludes.

2 Data

This section describes the data sources and construction of key variables. Comprehensive information on the datasets, data cleaning procedures, and variable definitions is provided in the Online Appendix.

2.1 Data Sources

The primary data source comes from Adzuna, a job vacancy aggregator providing near-universe coverage of online job postings in the UK. Adzuna's algorithms collect advertisements from multiple job boards and company websites, offering the most comprehensive collection of online vacancy information available. The UK Office for National Statistics uses this database to produce its weekly vacancy indicator. Our dataset contains weekly snapshots of open vacancies from January 2017 to March 2022, with each snapshot in our final database capturing between 2,500 and 93,000 listings. Each record includes the full text of the job advertisement, the job title, and the employer's name.

One advantage of this dataset is that it provides the raw description text of each advertisement, allowing flexible construction of measures for personality traits and AI-

related skills. From job descriptions, we extract information on job schedule and contract type. We recode job locations to the two-digit Nomenclature of Territorial Units for Statistics (NUTS) level. We also create indicators for soft and hard skills requirements, as well as education, experience, and seniority level.

To determine Standard Occupational Classification (SOC) codes, we implement the machine learning algorithm by Turrell et al. (2019), which classifies advertisements into 3-digit SOCs based on title, description, and industry. To obtain industry classifications, we match companies listed in Adzuna's job postings with data from Companies House, the UK government's official register of companies. Through this process, we retrieve Standard Industrial Classification (SIC) codes for the relevant companies. We drop advertisements from recruitment agencies because they lack information on the user firm, preventing industry code identification. Additional details of the different databases used and the matching process are provided in Online Appendix A and Online Appendix B, respectively.

We complement the job vacancy data with information at both the occupational and firm level. For occupations, we obtain data from the UK Household Longitudinal Study (UKHLS), a household panel survey provided by the Institute for Social and Economic Research. The UKHLS asks individuals to self-assess their personality using the Big Five framework, allowing us to validate our personality trait measures by comparing demands in job vacancies with workers' self-reported traits in different occupations. At the firm level, we match Adzuna data with the FAME database to identify publicly listed companies and obtain balance sheet information. To link vacancies to firm characteristics, we consider non-agency postings with posted salary information that can be uniquely matched to Companies House and to firms with the necessary firm identifiers and industry information.[4] After combining the datasets, our final database contains 11.7 million vacancies from 128,559 companies.

A possible concern is the representativeness of the dataset. Not all UK job advertise-

[4]Wage structures are used as a control variable in the main specifications and to verify that personality-trait demand does not differ systematically between agency and non-agency postings, providing reassurance that our restriction to direct employer postings does not drive the results.

ments are posted online, and certain types of work, such as casual, freelance, or informal jobs, may be advertised through alternative channels. Table A.1 in the Appendix compares the occupational composition of Adzuna vacancies in 2021 with that of the 2021 UK Census at the 1-digit SOC level. The Adzuna data differ from the Census distribution, with Caring and Leisure occupations over-represented and Administrative and Managers and Directors occupations under-represented. Since part of this difference may reflect the distinction between vacancy flows and employment stocks, as well as online-posting coverage, we include occupational fixed effects in the analysis to mitigate concerns related to broad occupational composition.

2.2 Measuring Personality Traits

Similar to Brenčič and McGee (2026), we measure personality trait demands using the Big Five framework, the most widely used taxonomy for studying personality in psychology and economics. The five dimensions are openness, conscientiousness, extraversion, agreeableness, and emotional stability. Each trait encompasses specific behavioural characteristics (Costa Jr & McCrae, 1992; Goldberg, 1992; Stanek & Ones, 2023): *Conscientiousness* relates to being organised, responsible, and diligent; *openness* refers to interest in new aesthetic, cultural, or intellectual experiences and creativity; *extraversion* involves an outward orientation towards social interactions and activities; *agreeableness* appears in cooperative and unselfish behaviour; and *emotional stability* refers to consistency in emotional responses. It is considered the opposite of *neuroticism*. This framework originated from the lexical hypothesis: important personality characteristics become encoded in everyday language. Its structure has been extensively validated through factor analyses of natural language adjectives and psychometric questionnaires, and across different languages and cultures (Borghans et al., 2008; King et al., 2005). In job advertisements, employers likely use words associated with known traits, and job seekers recognise and respond to these terms when applying (Abraham et al., 2024; Flory et al., 2015).

To identify personality traits in job advertisements, we construct a binary indicator equal to one if a job posting contains at least one keyword from a Big Five trait dictionary,

either in the job title or job requirements section. We build these dictionaries using trait-descriptive adjectives from authoritative sources in psychology literature (Costa Jr & McCrae, 1992; Goldberg, 1992; Heckman & Kautz, 2012; John, 1990; McCrae & John, 1992; Saucier, 1994). Since vacancies often mention many traits simultaneously, we also construct a binary indicator for whether the advertisement requires *any* keyword from the Big Five trait dictionary.

As personality-trait retrieval from online vacancies can produce false positives (Brenčič & McGee, 2026), we mitigate this risk through a two-step approach grounded in both the source vocabulary and systematic manual validation. First, the starting vocabulary is drawn exclusively from established Big Five personality measures and lexical taxonomies in psychology (Costa Jr & McCrae, 1992; Goldberg, 1992; Heckman & Kautz, 2012; John, 1990; McCrae & John, 1992; Saucier, 1994). Because these sources catalogue adjectives that describe stable individual characteristics, the initial word list already excludes generic task-language or product-related language by construction.

Second, we refine the dictionaries through iterative manual inspection. For each word, we examine up to eight rounds of 100 randomly selected sentences from job advertisements containing that word, reading the ten surrounding words to verify that the term refers to a worker or team characteristic in context. This process identifies terms whose meaning in vacancy text systematically differs from their psychological sense. For instance, "flexible" in vacancy descriptions predominantly refers to work schedules, not to a personality trait. Rather than removing the word entirely, we replace its schedule-related variants with the modified string "flx" before matching, so the dictionary captures only the personality-relevant usage. Words whose vacancy-context meaning cannot be disambiguated are removed from the dictionary entirely. We repeat the inspection cycle until a fresh sample of 100 advertisements yields no spurious matches. In total, 1,702 text replacements were made before the final dictionary matching. The complete list is reported in Online Appendix C. This procedure substantially reduces the risk that the trait indicators capture product or firm descriptive language rather than demand for a personality characteristic.[5]

[5]We also compare our trait terms with the dictionaries used by Brenčič and McGee (2026). For

An example of a job vacancy and the classification using this approach is shown in Table 1. The complete dictionaries are presented in Table A.2 in the Appendix. Further details on the dictionary construction process are provided in Online Appendix C.

Table 1: **Example of a job vacancy with AI skills, personality trait requirements and the dummy construction.**

Vacancy: Data Scientist

As a data scientist you will use your expertise in ***machine learning****,* ***neural networks****, and* ***deep learning*** *to develop products. You must have experience with* ***TensorFlow*** *and* ***PyTorch****. You should be* ***creative*** *and* ***imaginative*** *with the ability to develop* ***innovative*** *solutions. We are looking for someone who is* ***persistent*** *and can work independently. Strong analytical skills are required.*

Categorization

Personality Trait	
Agreeableness	0
Conscientiousness	1
Extraversion	0
Openness	1
Emotional Stability	0
AI Skills	1

To examine the validity of our text-based measures, we assess whether personality traits identified in vacancy texts align with self-reported personality characteristics of workers in the same occupations using data from the UKHLS. If our measures capture personality demands in job advertisements, occupations with higher advertised demand for a given trait should tend to employ workers who score higher on that trait. This validation exercise helps assess whether personality-related language in job postings reflects job requirements rather than only differences in advertising style.

Figure 1 shows the relationship at the occupational level, using 2-digit SOC groups,

agreeableness, extraversion, conscientiousness, and openness, our dictionaries largely correspond to a narrower subset of their positive-pole Big Five terms, or to close variants adapted to vacancy language. This comparison is reassuring because it shows that our measures are grounded in the same lexical tradition, while adopting a more parsimonious approach. This is important given that Brenčič and McGee (2026) show that broad dictionary-based searches in job advertisements can generate a large number of false positives, with many trait-descriptive terms used in contexts unrelated to worker personality. Our narrower dictionaries therefore reflect a conservative measurement choice: rather than maximizing the number of potentially relevant terms, we prioritize terms that are more likely to capture genuine employer demand for worker attributes in vacancy text.

between personality requirements in job advertisements and workers' self-reported traits in the UKHLS. Labels are shown for the three occupations with the highest advertised demand for each trait in Adzuna. Consistent with Brenčič and McGee (2026), the correlations are positive for all traits except emotional stability, for which the relationship is negative.[6] This pattern suggests that the emotional-stability measure should be interpreted more cautiously than the other trait measures. Importantly, the main conclusions of the paper do not rely on the emotional-stability results. Overall, the evidence suggests that personality-related language in job postings generally aligns with the traits observed among workers in the same occupations, although emotional stability appears to capture a more context-specific dimension of job requirements. While this validation exercise exploits cross-occupation variation, our main specifications include occupational fixed effects to account for systematic differences across occupations.

Table 2 presents descriptive statistics at the advertisement level. In the data, 72% of advertisements contain at least one personality trait-related word. Extraversion appears most frequently (44%), followed by conscientiousness (34%), openness (30%), emotional stability (14%) and agreeableness (13%).[7] Personality traits appear more often in job postings than traditional human capital measures such as soft skills (62%), hard skills (55%), or education requirements (11%), as shown in Figure A.1 in the Appendix.

[6] One possible explanation is that advertised emotional-stability language captures situational job demands rather than the average personality profile of workers in the occupation. The occupations with the highest advertised demand for emotional stability include Elementary Administration, Caring Personal Services, and Protective Services, where postings often refer to working under pressure, handling difficult situations, or dealing with stressful interactions. These requirements may be especially salient in high-stress roles, even if workers in those occupations do not report the highest levels of emotional stability. This interpretation is related to Brenčič and McGee (2026), who suggest that employers may emphasize emotional stability precisely in occupations where this trait is scarce or difficult to find. In such occupations, firms may advertise for emotional regulation, resilience, or the ability to work under pressure because these attributes are particularly important for performance, even if workers currently employed in those occupations do not report high levels of emotional stability on average. Measurement differences may also contribute, since UKHLS emotional stability is derived from inverted neuroticism items, whereas our dictionary captures language related to job requirements in vacancies.

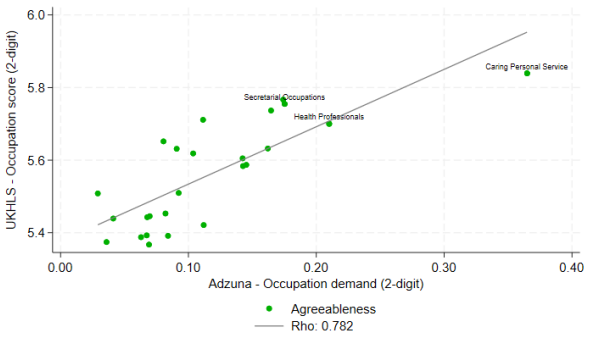
[7] The most common terms within each trait dictionary are shown in Figure OD1 in the Online Appendix D.

Figure 1: Scatter-plot between mean standardized trait scores in UKHLS and share of vacancies that mention Personality traits in Adzuna. 2-digit SOC groups

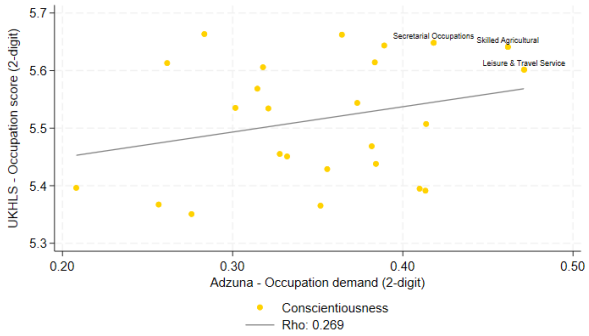
(a) Any Personality Trait



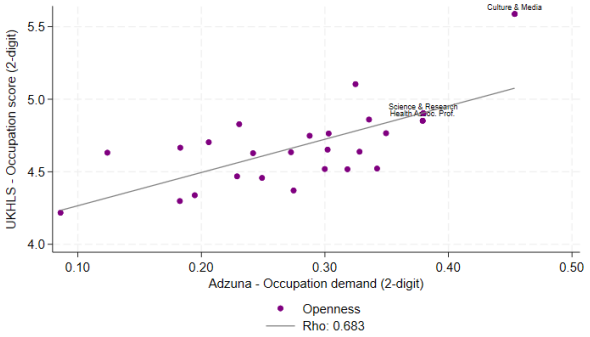
(b) Agreeableness



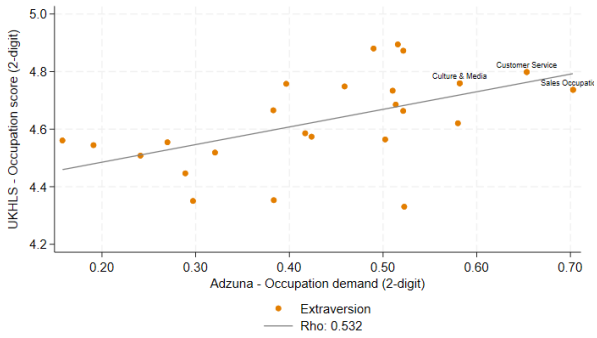
(c) Conscientiousness



(d) Openness



(e) Extraversion



(f) Emotional Stability

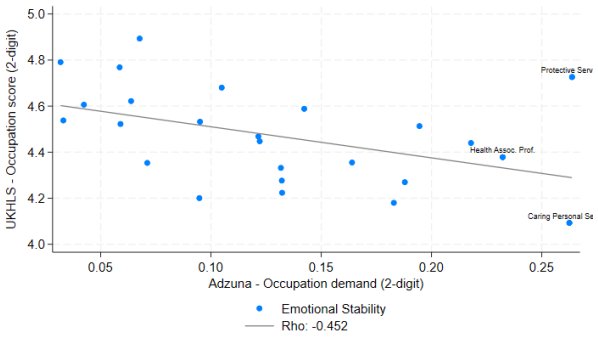


Table 2: Summary statistics

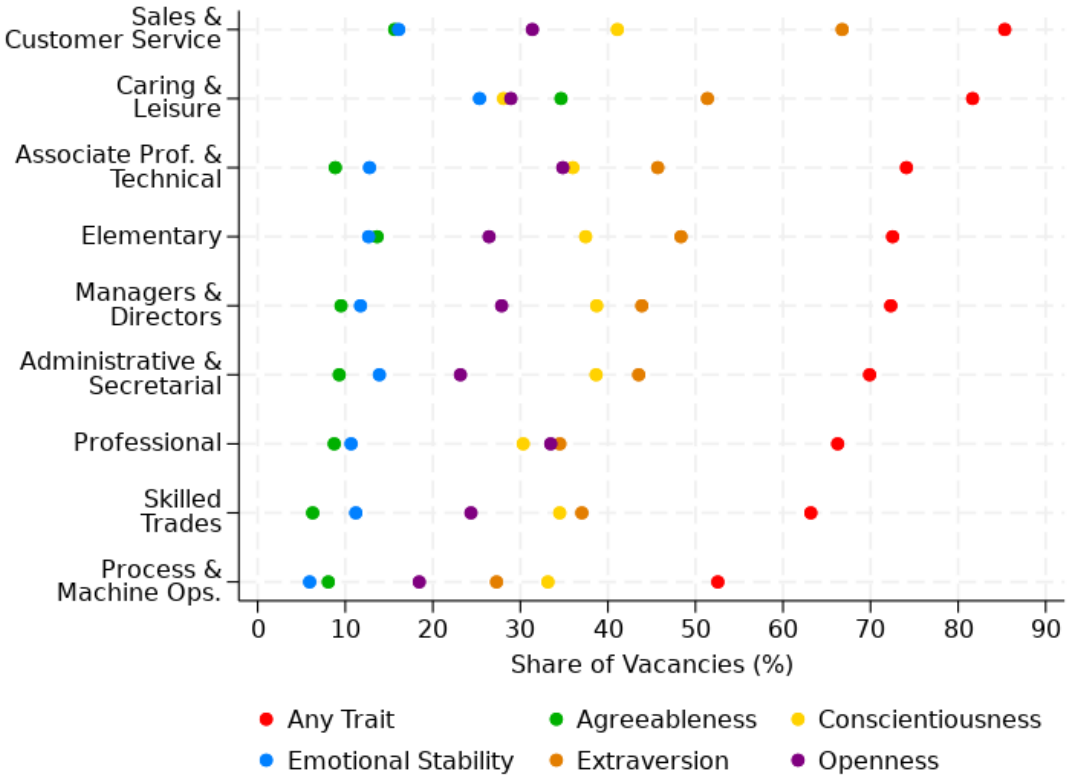
	Category	Mean	SD
Basic Statistics	Number of Ads	11,662,124	
	Mean length characters	2,297	1,236
	Number of firms	128,559	
Baseline	No trait requirements	0.28	0.45
	No traits nor skills requirements	0.09	0.29
	Obs. with no requirements	89,620	
Traits	Any Personality Trait	0.72	0.45
	Conscientiousness	0.34	0.47
	Openness	0.30	0.46
	Emotional Stability	0.14	0.35
	Extraversion	0.44	0.50
	Agreeableness	0.13	0.34
Skills	Soft skills	0.62	0.48
	Hard skills	0.55	0.50
	AI skills	0.01	0.11
Seniority	High Seniority	0.01	0.08
	Middle Seniority	0.19	0.39
	Low Seniority	0.51	0.50
Education	No education specified	0.89	0.31
	GCSE	0.03	0.16
	A-level	0.04	0.19
	Bachelor's degree	0.03	0.17
	Master or Doctoral degree	0.02	0.12
Job schedule	Full time job	0.30	0.46
	Part time job	0.08	0.28
Job duration	Temporary job	0.13	0.34
	Permanent job	0.35	0.48
Experience	Experience requested	0.69	0.46
	No experience requested	0.01	0.09
Other job info	Remote Work	0.03	0.18
	Average Salary	31,940	18,690
	Negotiable salary	0.49	0.50
	Bonus offered	0.23	0.42

Notes: No trait requirements denotes the share of vacancies that do not contain any words from the personality-trait dictionaries. *No traits nor skills* denotes the share of vacancies that do not mention personality traits or either soft or hard skills. *Obs. with no requirements* refers to observations in which no job requirements were matched in the vacancy text; these observations constitute the baseline sample used in the regression analysis.

Personality trait demands vary systematically across occupations. Figure 2 plots trait demands by 1-digit SOC groups. Jobs involving complex social interactions or non-routine cognitive activities show higher demands for personality traits, supporting the idea that these characteristics are increasingly important in the service-oriented economy (Borghans et al., 2014; Deming, 2017). Sales and Customer Service occupations show the highest overall trait demands (85.3%), with particular emphasis on extraversion (66.8%) and conscientiousness (41.1%). Caring and Leisure Services follow with 81.7% overall trait

demand and show the highest requirements for agreeableness (34.7%) and emotional stability (25.3%). In contrast, Process and Machine Operatives show the lowest overall trait demands (52.6%). More detailed patterns at the 3-digit SOC level are presented in Table OD1 in the Online Appendix D. The demand for personality traits remains relatively stable during our sample period. As shown in Online Appendix E., after accounting for location, occupation, and firm, most personality traits exhibit limited systematic changes over time.[8]

Figure 2: Personality Trait demand by 1-digit SOC group



2.3 Measuring AI Demand

The period covered by our data, 2017–2022, captures demand for established AI, such as machine learning technologies used systematically in business functions for customer analytics, forecasting, optimisation, fraud detection, and decision support. These forms

[8]The exception is Conscientiousness, which declines gradually and becomes significantly negative by 2022. Extraversion shows some positive and statistically significant increases between 2019 and 2021 as compared to 2017. Overall, this suggests that changes in the demand for personality traits over time are modest and that the observed patterns are largely stable across the sample period.

of AI are widely used in firms today. In our measurement, AI demand is defined as demand for AI-related methods, applications, and tools, rather than as general demand for digital or computer skills. We therefore distinguish AI-related skill requirements from broader computer-skill requirements, such as software development, programming, data analysis, and other general IT skills, which are measured separately and used as controls and comparison variables in the empirical analysis.

To identify AI-related vacancies, we build on the AI skill dictionaries developed by Acemoglu et al. (2022) and Babina et al. (2024), which we expand to improve coverage and consistency in our data.[9] The expanded dictionary is intended to capture terms directly related to artificial intelligence and machine learning, including AI methods, AI applications, and AI-specific software tools.

We manually review 100 job advertisements for each AI-related term from the original dictionaries, examining the surrounding text for related tools, skills, or technologies. This process serves a similar function to the co-occurrence methods in Babina et al. (2024) and Choi and Leigh (2024), identifying contextually related AI terms. Additionally, we review over 2,000 AI-related vacancies to capture new AI tools and software packages that might be absent from structured databases. Throughout this process, we distinguish AI-specific requirements from broader IT, programming, software-development, and data-analysis requirements. Terms that refer to general digital proficiency are not used to classify vacancies as AI-related, while ambiguous terms are assessed in context and retained only when they refer to AI methods, applications, or tools. Based on this dictionary, we construct a binary indicator for AI demand: a job posting is coded as AI-related if it contains at least one keyword from the extended AI dictionary, either in the job title or in the job requirements. We report the complete list of AI skills, sources, and text replacements in Online Appendix C.

In our dataset, 1.3% of job advertisements mention AI-related skills. As shown in Figure A.2 in the Appendix, this demand has increased over time, rising from 0.75% in

[9]For example, we group closely related concepts such as *cross-domain learning*, *federated learning*, and *reinforcement learning* under broader categories such as *machine learning* when searching job postings.

2017 to a peak of 2.3% in early 2021, before stabilising around 1.5%. These numbers mirror findings by Schmidt et al. (2024), who show that AI hiring intensity in the UK peaked in late 2020 and then returned to pre-pandemic levels.

Of the 128,559 firms in the dataset, 6,436 (5%) posted at least one vacancy requiring AI-related skills over the sample period, collectively accounting for 153,079 AI-related job advertisements, equivalent to 1.3% of all postings. First-time AI-related postings were spread across the sample window: 1,396 firms posted their first AI-related vacancy in 2017, 1,129 in 2018, 1,173 in 2019, 990 in 2020, 1,385 in 2021, and 363 in the first quarter of 2022. The variation in the demand timing across firms forms the basis of our staggered difference-in-differences analysis at the firm level, described in Section 3.

A natural concern for the identification strategy is whether a firm's first AI-related vacancy reflects a durable shift in its demand for AI-related capabilities or a one-off change in advertisement language. The evidence suggests the former: 69.6% of the 6,436 firms that ever posted AI-related vacancies did so on more than one occasion, with a median of 3 such vacancies per firm (75th percentile: 8). Among firms whose first AI-related vacancy appeared at least eighteen months before the end of the sample, allowing three consecutive post-demand half-year periods to be observed, 46.9% posted an AI-related vacancy in the first six months after first observed AI-related demand, 40.7% in the following six months, and 38.2% in the subsequent six months. This indicates that firms' demand for AI-related skills was sustained rather than concentrated at the initial demand event. This pattern holds across AI-demand cohorts, with positive persistence rates recorded for all cohorts for which at least twelve post-demand months of data are available.

AI vacancies are strongly associated with technical skill requirements: 91.3% list at least one hard skill, compared to 54.2% of non-AI postings. The most frequently co-demanded technical skills are software proficiency (59.2%), engineering knowledge (45.3%), and data analysis (35.1%). These roles also call for interpersonal abilities such as communication (29.3%), teamwork (16.4%), and problem-solving (13.8%). The patterns of hard and soft skill co-demands with AI are illustrated in Figure A.3 in the Appendix.

Despite these overlaps, AI demand remains relatively specialised. Among vacancies requiring software skills, only 8.5% also mention AI; for engineering and data analysis roles, the figures are 7.3% and 5.5% respectively. This suggests that AI complements broader technical profiles, but constitutes a distinct requirement. To ensure our results are not confounded by general programming demand, we separately examine the effect of computer science skills in the robustness analysis.

AI demand is unevenly distributed across occupations. AI-related terms appear in all 90 3-digit SOC occupational categories, but 78% of AI-demanding advertisements are concentrated in just five: IT and Telecommunications Professionals (49.1%), Business, Research and Administrative Professionals (10.3%), Engineering Professionals (7.2%), Legal Associate Professionals (6.2%), and Electrical and Electronic Trades (5.1%). This concentration aligns with Schmidt et al. (2024), who find that AI hiring is most intense in the Information and Communication, Finance and Insurance, and Professional, Scientific and Technical sectors.

We find distinct differences in personality trait demand between AI and non-AI positions, which we show in Table 3. Openness is more common in AI-related postings (52.1% versus 29.6%), suggesting that employers seek adaptability and creativity alongside technical skills. This is related to Alekseeva et al. (2021), who document that AI-related jobs are associated with higher demand for complementary cognitive and managerial skills, which may reflect a need for broader capabilities beyond purely technical expertise. Conversely, AI postings mention lower levels of all the other personality traits, especially agreeableness and emotional stability. However, the overall incidence of soft skills and personality traits is similar across both groups (62.8% versus 62.4% for soft skills; 73.1% versus 72.1% for personality traits). This suggests that AI demand is related to specific types of personality traits rather than reducing the overall importance of non-technical attributes in hiring.

Table 4 shows descriptive statistics aggregated at the firm-year level. On average, firms in our sample are large in terms of number of employees, posting 46 vacancies per

Table 3: Demand for Personality Traits and Skills in vacancies by their AI demand (percentages)

Demands AI	Hard	Soft	P.T.	Agree	Consc	Emot	Extra	Open
Yes	91.3	62.8	73.1	6.2	29.9	7.8	35.2	52.1
No	54.2	62.4	72.1	13.6	34.2	14.1	44.4	29.6

Notes: *Hard* stands for *Hard-Skill*, *Soft* for *Soft-Skill*, *P.T.* for *Any Personality Traits*, *Agree* for *Agreeableness*, *Consc* for *Conscientiousness*, *Emot* for *Emotional Stability*, *Extra* for *Extraversion*, and *Open* for *Openness*

year. In the sample, 5% of companies advertised AI skills. Approximately half of the companies have advertisements mentioning personality traits, hard skills, and soft skills.

Table 4: Summary statistics - Year Aggregation

	Category	Mean	SD
Basic Statistics	Number of observations	254,451	
	Number of vacancies per period	46	
	Number of firms	128,559	
Personality Traits	Any Personality Trait	0.56	0.50
	Conscientiousness	0.38	0.42
	Openness	0.31	0.41
	Emotional Stability	0.13	0.29
	Extraversion	0.46	0.44
	Agreeableness	0.10	0.26
	No trait requirements	0.18	0.38
	No traits nor skills requirements	0.08	0.27
Skills	Soft skills	0.64	0.42
	Hard skills	0.59	0.44
	AI skills	0.05	0.21
Firm Characteristics	Firm size	632.13	8497.23
	Profit	35,422	354,118

Notes: Data aggregated at firm-year level. *No trait requirements* refers to firm-periods with no personality traits. *No traits nor skills* refers to firm-periods with no traits or skills.

3 Methodology

We employ two complementary empirical approaches to examine how AI is associated with personality trait demands. First, we analyse vacancy-level patterns using high-dimensional fixed effects models to compare AI and non-AI positions, controlling for firm, occupation, location and time fixed effects. This analysis provides information on how personality requirements differ between AI-intensive roles and other positions within firm, occupation, location, and time periods. Second, we examine firm-level dynamics using staggered difference-in-differences estimation to study how firms' efforts to demand AI knowledge are associated with changes in their overall hiring strategies. This approach captures

whether firms adjust their demand for personality traits across all positions as they begin seeking AI knowledge, rather than only within AI-specific roles.

The advantage of combining these two approaches is that they provide complementary perspectives on the relationship between AI and personality demands. The vacancy-level analysis captures differences between AI and non-AI positions, isolating how personality requirements vary across roles, holding firm, occupation, location, job characteristics, and time constant. The firm-level analysis examines how firms' overall hiring patterns evolve as they begin seeking AI knowledge, capturing changes in the composition of personality demands across all vacancies. Together, these approaches distinguish between role-specific differences and broader changes in firms' hiring behaviour.

3.1 Vacancy-Level Analysis

Our vacancy-level specification examines the relationship between AI skill requirements and personality trait demands using a linear probability model with high-dimensional fixed effects:

$$P(T_i = 1) = \alpha + \beta AI_i + \theta X_i + \delta_l + \phi_j + \gamma_k + \lambda_t + \epsilon_i \tag{1}$$

where T_i is a binary indicator equal to one if personality trait T is mentioned in vacancy i, AI_i is a binary indicator equal to one if the vacancy requires AI skills, X_i is a vector of job posting characteristics, δ_l are location fixed effects (2-digit NUTS), ϕ_j are firm fixed effects, γ_k are occupation fixed effects (3-digit SOC), λ_t are month-year fixed effects (capturing each specific month-year combination such as September 2019, October 2019, etc.), and ϵ_i is the error term. Standard errors are clustered at both the occupation and firm level to account for potential correlation within these groups. We estimate Equation 1 separately for each personality trait (openness, conscientiousness, extraversion, agreeableness, emotional stability) and for an overall indicator equal to one if the vacancy mentions any personality trait.

The vector X_i includes observable job characteristics that might correlate with both AI requirements and personality demands. In particular, we consider: other personality

traits (when the dependent variable is a specific trait), soft skills, hard skills (including computer science skills), remote work, contract type (temporary/permanent), work schedule (full-time/part-time), seniority level, education requirements, experience requirements, performance-based pay, wage negotiability, and advertisement length. Computer science skills are defined as requirements related to software development and data analysis capabilities within the broader hard skills category.[10] Including these controls addresses concerns that the AI-personality relationship reflects correlations with other job requirements rather than AI-specific demands. The coefficient β identifies how personality trait demands differ between AI and non-AI positions after controlling for occupation, location, firm, time, and other job characteristics.[11]

Several threats could bias our estimates. First, personality trait measures may be endogenous if they correlate with unobserved job requirements. We address this by including extensive observable controls in X_i, as previously explained. Second, personality demands vary systematically across occupations. We include occupation fixed effects to absorb time-invariant, occupation-specific heterogeneity and focus on within-occupation variation. Third, firm-specific HR practices might jointly influence job advertisement phrasing and underlying requirements. We address this with firm fixed effects. Fourth, local labour market conditions may vary across regions and influence both personality and AI demands. We include location fixed effects at the 2-digit NUTS level. Finally, month-year fixed effects account for temporal variation, controlling for any aggregate trends or seasonal patterns in personality or AI demands. This identification strategy isolates within-occupation, within-location, within-firm, and within-time-period variation. The high-dimensional fixed effects absorb systematic differences across these dimensions, allowing us to compare similar positions that differ primarily in AI requirements. While we cannot rule out all sources of endogeneity, the extensive controls and fixed effects address the most plausible confounders. As a robustness check, we examine whether the

[10] This appears in 1,911,086 vacancies, approximately 12 times more than AI-demanding positions.

[11] For example, consider two software developer roles posted by Google in London in September 2019, one requiring AI skills and the other not. Despite having the same firm, occupation, location, and month-year, these positions may differ in other respects such as required experience or contract terms. By controlling for these observable characteristics and including fixed effects, we isolate the association between AI requirements and personality trait demands.

AI-personality relationship differs from that of general computer skills. By comparing AI and computer skill effects, we assess whether our results reflect AI-specific demands or broader technological requirements. The comparison also helps rule out that we are simply capturing general programming or technical skill requirements.

3.2 Firm-Level Analysis

The vacancy-level analysis described above examines how personality requirements differ between AI and non-AI positions within the same firm. When firm fixed effects are included, identification comes from variation across vacancies posted by the same firm, so the estimates can be interpreted as within-firm differences in advertised personality requirements between AI and non-AI vacancies. In addition to these role-specific differences, the emergence of demand for AI-related skills may be associated with broader organisational adjustments that affect hiring strategies beyond AI-specific roles. To study this possibility, we use a difference-in-differences approach in which the treatment is defined by a firm's first observed demand for AI-related skills. Specifically, we define the event date as the date of the firm's first job advertisement requiring AI-related skills. Given that our dataset begins in 2017, when relatively few firms were demanding AI-related skills, and that job advertisements typically reflect emerging skill needs and strategic shifts in technology use, the first AI-related job posting provides a meaningful indicator of when a firm begins seeking AI-related capabilities and may adjust its workforce accordingly.

Following Callaway and Sant'Anna (2021), we employ a staggered difference-in-differences approach that addresses potential bias from treatment effect heterogeneity across cohorts and time periods. The main building block is the group-time average treatment effect (ATT), defined as:

$$ATT(g,t) = \mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1] \tag{2}$$

where $Y_t(g)$ denotes the outcome at time t for firms whose first observed AI-related demand occurs in period g, and $Y_t(0)$ denotes the counterfactual outcome that would

be observed in the absence of AI-related demand. G indicates the period in which a firm first appears in the data as demanding AI-related skills. The group-time average treatment effects are aggregated using non-negative weights that sum to one to obtain overall treatment effects.[12] The analysis is conducted at the firm-period level, where the outcome variable measures average personality-trait demand across all vacancies posted by a firm within a given time period. Additionally, we separately examine the association with non-AI vacancies to assess whether changes in personality-trait demand extend beyond positions explicitly requiring AI-related skills. We consider two levels of temporal aggregation: yearly and half-yearly. These aggregations improve the precision of adoption timing and increase the number of observations per firm-period.

A concern is that AI-demanding firms may differ systematically from non-AI-demanding firms before their first observed AI-related vacancy along dimensions correlated with personality-trait demand. For example, more innovative or forward-looking firms may be both more likely to demand AI-related skills and more inclined to value traits such as openness, independently of the emergence of AI-related demand itself (Acemoglu et al., 2023). In this case, estimated treatment effects could reflect pre-existing differences between firms rather than changes associated with first observed AI-related demand. To address this potential selection issue, we employ an entropy balancing approach (Hainmueller, 2012), which reweights non-AI-demanding firms so that their pre-event characteristics match those of AI-demanding firms along key covariates. This procedure reduces bias from observable selection into AI-related demand.

The balancing is implemented separately for each personality trait and includes: initial demands for other personality traits, firm characteristics (public status, industry at 2-digit SIC), average skill requirements, location distribution indicators, and occupational composition indicators. The weights are estimated targeting the first and second moments (mean and variance) of each covariate, computed as averages over each firm's pre-treatment

[12] Weights on each $ATT(g,t)$ vary depending on the aggregation scheme chosen (by group, by calendar period, by event time, or overall). The weights are non-negative and sum to one (Callaway & Sant'Anna, 2021). The estimation excludes left-censored observations, that is, firms already advertising AI-related skills in the first observed period.

periods. Standard errors are clustered at the firm level throughout.

Our methodology uses a weighted staggered difference-in-differences estimator. The estimator implicitly accounts for time-invariant unobserved firm heterogeneity by exploiting within-firm variation over time (analogous to the role of firm fixed effects in a two-way fixed effects regression). The balancing weights ensure that treated and control firms are comparable on observable pre-treatment characteristics.

4 Results

4.1 AI and Personality Traits within Vacancies

Table 5 presents the relationship between AI demand and personality traits across different specifications. Each row in the table corresponds to a different personality trait used as the dependent variable. In panel A, the dependent variable is a dummy variable that takes the value of one if the advertisement presents at least one personality trait, while in the rest of the panels, the dependent variable is each of the Big Five. Column (1) reports the baseline specification without any controls or fixed effects. Column (2) adds controls for job requirements. Column (3) further includes time fixed effects. Column (4) builds on this by incorporating location fixed effects. Column (5) adds firm fixed effects to the specification with job controls, time, and location fixed effects. Column (6) replaces firm fixed effects with occupation fixed effects. Finally, column (7) includes all controls and fixed effects: time, location, firm, and occupation.

In Panel A, where the dependent variable indicates the presence of any personality trait requirement, AI is positively associated with personality requirements in most specifications. When we examine each personality trait separately, however, the aggregate relationship masks substantial heterogeneity across traits. Panels B to F show that, across specifications, AI is positively and statistically significantly associated with openness, but negatively associated with the other personality traits. Openness captures attributes such as adaptability, creativity, and intellectual curiosity, which are particularly relevant in tasks involving novel technologies and problem-solving. Accordingly, AI-related vacancies

are more likely to value these individual characteristics. This might reflect the need for workers who can adapt to evolving tools and engage with complex, non-routine tasks. In contrast, the negative associations with the remaining traits suggest that AI-related roles place relatively less emphasis on the other personality traits. The inclusion of firm fixed effects is associated with the largest reduction in estimated coefficients, highlighting that firm-specific characteristics account for a substantial part of the relationship between AI and personality trait demands at the vacancy level. In contrast, occupation fixed effects explain less of the variation, suggesting that firm-specific differences in how AI-related capabilities are integrated matter more than occupation-specific technological needs. This is in line with Deming and Kahn (2018), who highlight the importance of firm heterogeneity in skill demands.

In the most comprehensive specifications (columns 6 and 7), the relationship with openness remains substantially high, while the coefficients for the other traits are smaller and negative. Overall, the net association on the aggregate measure of *any* trait is small but positive. For example, in column (7), the estimated coefficient of openness is 6.7 percentage points and the estimated coefficients of the other traits range from 1.2 to 2.5 percentage points. This pattern is consistent with AI being associated with a reallocation across traits, and it is in line with Johnson and Acemoglu (2023), who argue that AI tends to automate routine cognitive tasks but increases demand for human attributes such as adaptability and creativity. This pattern is also consistent with Autor (2014)'s observations on how technological change tends to amplify the comparative advantage of workers in supplying problem-solving, adaptability, and creativity (p. 2), while substituting for more routine activities.

Table 5: AI and Personality Traits within vacancies

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. Any personality trait							
AI_{it}	0.010***	-0.012** *	-0.012***	-0.011***	0.011	0.021***	0.016**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.007)	(0.005)	(0.007)
Panel B: Dependent variable Agreeableness							
AI_{it}	-0.074***	-0.028***	-0.031***	-0.033***	-0.015***	-0.011***	-0.012***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.004)	(0.003)	(0.004)
Panel C: Dependent variable Conscientiousness							
AI_{it}	-0.043***	-0.065***	-0.064***	-0.064***	-0.023***	-0.038***	-0.017**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.007)	(0.009)	(0.009)
Panel D: Dependent variable: Emotional Stability							
AI_{it}	-0.063***	-0.044***	-0.046***	-0.047***	-0.027***	-0.026***	-0.022***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.004)	(0.010)	(0.007)
Panel E: Dependent variable: Extraversion							
AI_{it}	-0.092***	-0.082***	-0.082***	-0.083***	-0.034***	-0.038***	-0.025***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.008)	(0.009)	(0.007)
Panel F: Dependent variable: Openness							
AI_{it}	0.225***	0.141***	0.143***	0.143***	0.073***	0.111***	0.067***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.008)	(0.014)	(0.013)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes
R-squared range	[0.001, 0.012]	[0.080, 0.121]	[0.082, 0.122]	[0.084, 0.123]	[0.273, 0.351]	[0.105, 0.169]	[0.281, 0.364]
N observations	11662124	11662124	11662124	9923701	9888315	9923701	9888315
N firms	128559	128559	128559	117229	81843	117229	81843

Notes: AI_{it} is AI at the vacancy level. The R-squared range shows the minimum and maximum values across the six trait regressions for each specification. Controls include: computer skills, traits and skills demand, remote work requirements, length of advertisement, contract type, work schedule, bonus offerings, education, and negotiable salary option. Standard errors are clustered at the occupation and firm level in specifications (5)-(7). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Distinguishing AI demand from general computer-skill demand: A potential concern is that the estimated association between AI-related skill demand and personality-trait requirements may reflect hiring for general software-development or data-analysis roles rather than demand for AI-specific capabilities. We address this concern by distinguishing AI-related skill demand from general computer-skill demand. We estimate three complementary specifications. First, we re-estimate our baseline model excluding computer skills as a control variable, allowing us to capture the overall association between AI-related skill demand and personality-trait requirements without conditioning on general computer-skill requirements. Second, we include both AI-related skill demand and computer-skill demand jointly to assess whether the association with AI is distinct from this general computer-skill measure. Third, we replace AI with computer skills as the main explanatory variable. This allows us to compare the personality-trait profile of AI-related vacancies with that of vacancies requiring software-development or data-analysis skills.

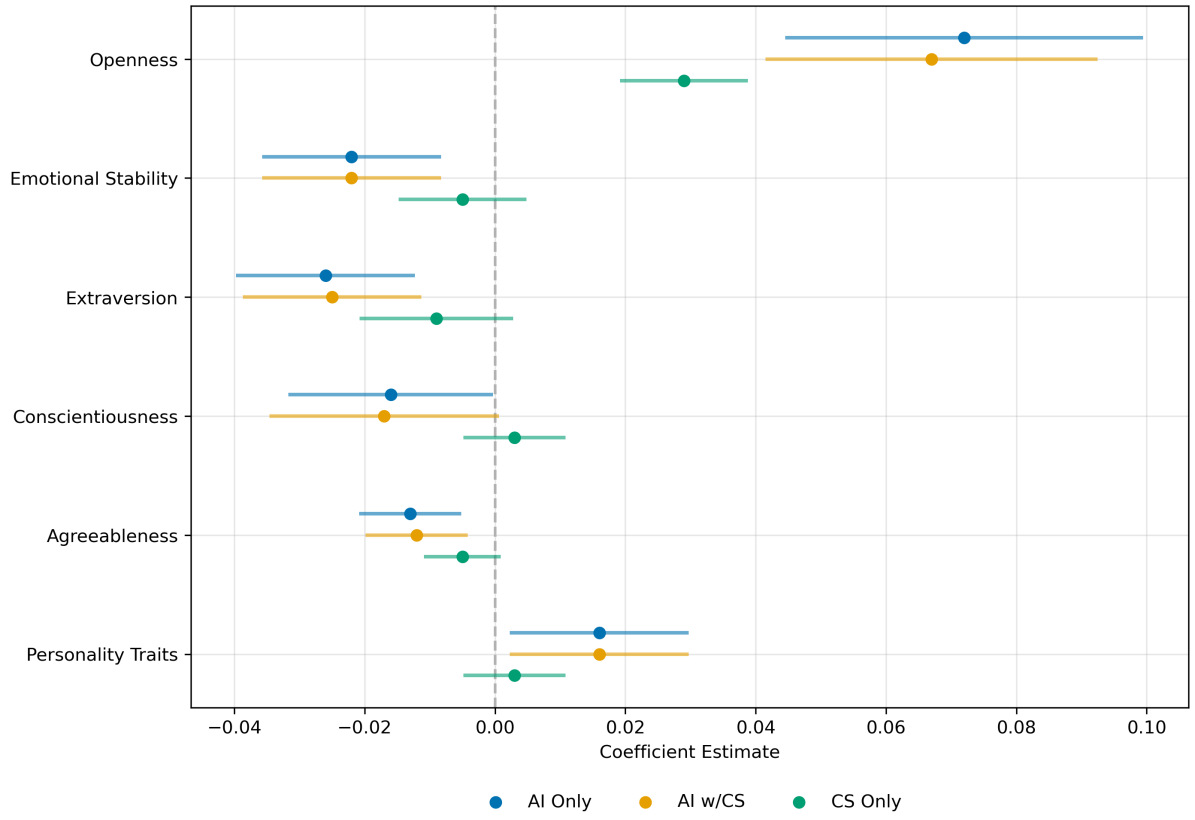
Table A.3 in the Appendix reports estimates excluding computer skills as a control, and Table A.4 in the Appendix shows results using computer skills, rather than AI-related skills, as the key explanatory variable. Excluding computer skills from the model yields an estimated coefficient for AI that is essentially unchanged relative to the estimate obtained in the main specification, where computer skills are included as a control. This stability indicates that the association between AI-related skill demand and personality-trait requirements is not driven by the inclusion of software-development and data-analysis controls.

Figure 3 plots the estimated coefficients from three separate specifications: one where the AI indicator is estimated without computer skills as a control, one where AI and computer skills are included jointly, and one where only computer skills are used instead of the AI indicator. In all regressions, we include the full set of fixed effects. Both AI-related skills and computer skills show a positive association with openness, but the magnitude of the AI association, 7.2 percentage points, is more than twice that of computer skills, 2.9 percentage points. This difference is statistically significant, as shown in Table A.5 in the Appendix. These results indicate that the openness pattern is stronger for AI-related

vacancies than for vacancies requiring general computer skills.

The remaining personality traits show no statistically significant effects in the specification of computer skills only. Furthermore, Table A.6 in the Appendix reports tests of the stability of AI coefficients when we compare specifications with and without computer skills as control. In specifications that include occupation and firm fixed effects, we cannot reject the null hypothesis of equal coefficients, suggesting that the AI estimates are robust to the inclusion of these controls. This placebo exercise reinforces the main findings by indicating that the observed relationships between AI-related skill demand and personality-trait requirements, particularly openness, are not simply capturing software-development or data-analysis requirements.

Figure 3: Personality-trait demand: AI-related skills versus general computer skills



4.2 AI and Personality Traits within the Firm

Next, we shift the unit of analysis from individual vacancies to the firm level. The analysis proceeds in three steps. First, we document the firm-level characteristics associated with

demanding AI-related knowledge and assess whether entropy balancing achieves comparability between treated and control firms on observable pre-treatment characteristics. Second, we estimate whether firms' first AI-related vacancy is followed by changes in personality-trait demand across their overall hiring portfolio, using a weighted staggered difference-in-differences specification. Third, we examine whether these changes extend beyond AI-specific positions by focusing on non-AI vacancies and we examine whether the changes observed among non-AI vacancies reflect shifts in the occupational composition of hiring or changes in openness demand within occupations.

Determinants of firms' demand for AI knowledge and covariate balance. Before analysing subsequent hiring responses, we first examine which types of firms are more likely to seek AI-related knowledge and assess covariate balance after applying the entropy balancing algorithm. The first two columns of Table OD2 in the Online Appendix compare key pre-adoption characteristics of AI-demanding firms, our treated group, and non-AI-demanding firms, our control group, before balancing.

Before balancing, AI-demanding firms differ systematically from non-demanding firms. They exhibit higher pre-treatment demand for openness, conscientiousness, and extraversion, with standardised differences of 0.452, 0.184, and 0.191, respectively. They also post more vacancies requiring hard skills and soft skills, with standardised differences of 0.473 and 0.268. In addition, AI-demanding firms tend to be larger, more profitable, and more likely to be publicly listed. Their occupational and industry composition also differs, with greater concentration in professional occupations and technology-intensive industries. These patterns suggest that AI demand is not random: firms that demand AI are systematically different from other firms before the AI demand takes place.

To address these pre-treatment differences, we use entropy balancing to reweight the control group so that treated and control firms are comparable on observable characteristics. After balancing, shown in the last two columns of Table OD2 in the Online Appendix, all standardised differences fall below 0.05 in absolute value, indicating excellent balance across the included covariates. This means that, after reweighting, treated and control

firms are similar in terms of observable pre-treatment characteristics, including their prior demand for personality traits, skill requirements, firm size, profitability, listed status, and occupational and industry composition. Balance diagnostics for the half-yearly aggregation are reported in Table OD3 in the Online Appendix, and show similar patterns.

Table 6: ATT estimates for staggered regressions

	(1) Half-year aggregation	(2) Yearly aggregation
Panel A: Dependent variable. Any personality trait		
$AI\ demand_{jt}$	0.021***	0.011
	(0.008)	(0.008)
Panel B: Dependent variable. Agreeableness		
$AI\ demand_{jt}$	0.007	0.003
	(0.005)	(0.006)
Panel C: Dependent variable. Conscientiousness		
$AI\ demand_{jt}$	0.005	-0.003
	(0.008)	(0.010)
Panel D: Dependent variable. Emotional Stability		
$AI\ demand_{jt}$	0.002	-0.008
	(0.005)	(0.006)
Panel E: Dependent variable. Extraversion		
$AI\ demand_{jt}$	0.013	0.007
	(0.009)	(0.010)
Panel F: Dependent variable. Openness		
$AI\ demand_{jt}$	0.027***	0.024**
	(0.009)	(0.010)
Observations	243,334	167,634

Notes: AI demand is a dummy variable that takes the value one from the first period in which we observe a firm posting a vacancy requiring AI-related skills. The estimator absorbs time-invariant unobserved firm heterogeneity through within-firm variation over time, playing the role of firm fixed effects. The estimation method is Sant'Anna and Zhao (2020)'s improved doubly robust DiD estimator based on inverse probability of tilting and weighted least squares (`drimp` in `csdid` in Stata). We use the entropy balance algorithm of Hainmueller (2012) to weight the sample. The standard errors are presented in parentheses and are clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Personality traits after AI-demand: Table 6 reports the average treatment effects from the weighted staggered difference-in-differences regressions at the firm level, based on Equation (2). In column (1), we show aggregations at the half-year level, and in column (2), the aggregation is yearly. When considering each personality trait separately, the estimated coefficients are generally small and not statistically significant, with the

exception of openness, which is positive and statistically significant in both columns. The estimates for openness are the largest and most precisely estimated. For example, column 2 provides the most conservative estimate: AI demand is associated with a 2.4 percentage-point increase in the share of vacancies requiring openness.

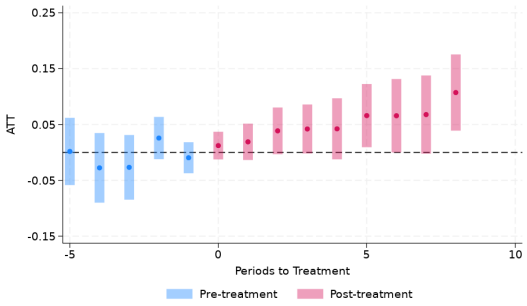
This means that, relative to the average baseline share of 31.9% (Table A.9 in the Appendix), the share of openness-demanding vacancies would increase to approximately 34.3%, corresponding to a 7.6% increase. In terms of vacancy counts, for a firm posting the average number of vacancies per year, 46, this relationship corresponds to about one additional vacancy requiring openness. For treated firms, which are larger and post an average of 284 vacancies per year, the same 2.4 percentage point increase corresponds to about seven additional vacancies requiring openness. Since treated firms have a higher baseline share of openness-demanding vacancies, 47.6%, this represents a 5% increase relative to their baseline.

Next, in Figure 4, we study the dynamic relationship between AI demand and personality traits using an event study design for the half-year estimations. The event-study plots indicate that prior to AI-demand, personality traits were not statistically significantly different for AI-demanding and non-demanding firms, as expected after the balance.[13]

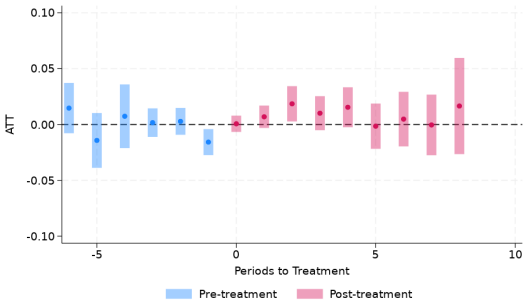
[13]We formally test for pre-trends across all traits and find no evidence against parallel trends. For example, for openness, as the outcome of primary interest, the p-value is equal to 0.842.

Figure 4: Staggered DID ATT. Half-year-level aggregation

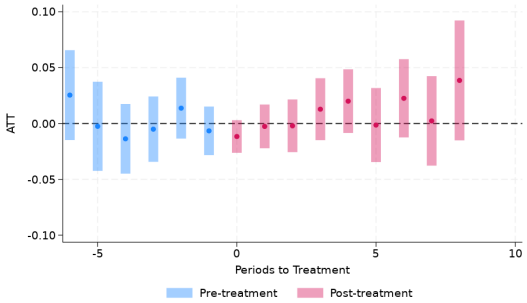
(a) Any Personality Trait



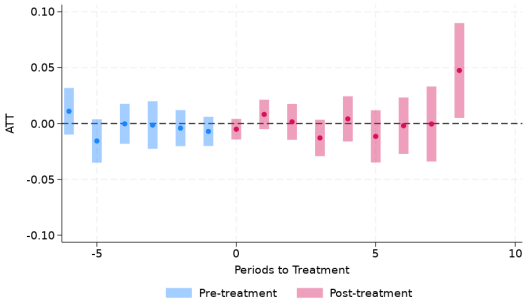
(b) Agreeableness



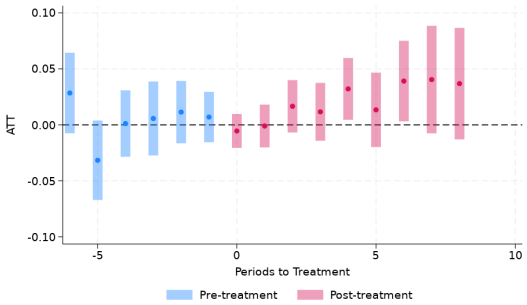
(c) Conscientiousness



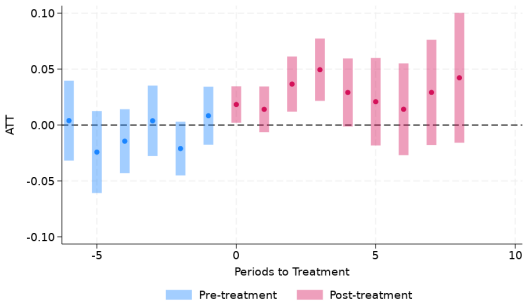
(d) Emotional Stability



(e) Extraversion



(f) Openness



From event time 0 onwards, the estimates show that, for most individual personality traits, the coefficients are small and statistically indistinguishable from zero. There are some exceptions: emotional stability is significantly higher four and a half years after first observed AI-related demand, while extraversion is significantly higher two and a half and three and a half years after first observed AI-related demand. The clearest pattern is for openness, for which the estimates become positive in the year of first observed AI-related demand and remain statistically significant for at least two years. These results indicate that first observed AI-related demand is associated with a medium-term

increase in firms' demand for openness across their overall hiring portfolio. For the aggregate indicator of any personality trait, we also observe a gradual and persistent increase. This pattern suggests that, although first observed AI-related demand is not accompanied by systematic increases in every individual personality trait, it is associated with a broader rise in personality-related requirements in hiring. This aggregate pattern may reflect heterogeneity across traits, with the overall measure capturing a general shift in advertised personality requirements that is less visible in the trait-specific estimates. The corresponding yearly results are presented in Figure A.5 in the Appendix and show a similar pattern.

Table A.7 in the Appendix reports a robustness check using propensity score matching, rather than entropy balancing, to construct the weights for the staggered difference-in-differences estimates. The results are consistent with the main pattern. The association between AI-skill demand and openness remains positive and statistically significant in both the half-year and yearly aggregations, with estimates of 0.019 and 0.017, respectively. By contrast, the estimates for the other personality traits are generally small and statistically insignificant. These results suggest that the positive relationship between firms' first AI-related vacancy and subsequent demand for openness is not specific to the entropy-balancing procedure.

Do changes extend beyond AI-related vacancies? Organisational scope and decomposition: The firm-level estimates above are calculated across all vacancies posted by the firm. A natural concern is that the openness pattern may reflect the inclusion of AI-related vacancies themselves, rather than broader differences in firms' hiring portfolios. To examine this, we recalculate personality-trait demand using only non-AI vacancies. This allows us to assess whether first observed demand for AI-related skills is also associated with trait requirements in positions that are not explicitly AI-related.

Table A.8 in the Appendix reports the results when AI-related vacancies are excluded from the outcome. The openness pattern remains visible among non-AI vacancies. At the firm level, starting to post AI-related vacancies is associated with an increase of

approximately 2 percentage points in the share of non-AI vacancies mentioning openness. Relative to treated firms' baseline openness rate, this corresponds to an increase of around 4–5 percent. Estimates for the remaining personality traits are generally small and statistically insignificant, with the exception of extraversion in the half-year aggregation. The non-AI estimate is very close to the corresponding within-firm estimate obtained when all vacancies are included. This similarity indicates that the firm-level openness result is largely accounted for by changes in non-AI vacancies, rather than by the direct inclusion of AI-related postings in the outcome.

We next examine whether this broader pattern reflects changes in the occupational composition of non-AI hiring or changes in openness demand within occupations. To do so, we conduct a shift-share decomposition for AI-demanding firms, separating within-firm pre-to-post changes in openness demand among non-AI vacancies into a composition component, an intensity component, and their interaction.[14] Table A.10 in the Appendix reports the results. The components are expressed in percentage points and describe pre-to-post changes within AI-demanding firms, rather than a decomposition of the difference-in-differences estimate.

The decomposition points to two offsetting patterns. The composition component is negative and statistically significant (-12.5 pp, $p < 0.01$), indicating that AI-demanding firms shift their non-AI hiring toward occupation groups with lower baseline openness requirements. By contrast, the intensity component is positive and statistically significant ($+1.7$ pp, $p < 0.05$), indicating that non-AI vacancies become more likely to mention openness within major occupation groups. Figure A.6 in the Appendix shows that this positive within-occupation pattern appears in eight of the nine major groups, with the largest contributions from associate professional and technical occupations ($+2.6$ pp), professional occupations ($+1.8$ pp), and managers and directors ($+1.1$ pp).

The decomposition is based on raw pre-to-post changes within AI-demanding firms.

[14]The decomposition is conducted at the SOC 1-digit level to obtain stable estimates for each of the nine major occupation groups. Overall results do not change using the SOC 2-digit level, as used in the main DiD specification.

Therefore, the components are not directly comparable to the DiD estimates reported above. They show that an increase in openness demand among non-AI vacancies comes from greater mention of openness within occupations, not from firms hiring more in occupations where openness is already common. This within-occupation pattern is consistent with the interpretation that firms might hire AI workers to develop new products and services, make greater use of proprietary data, support strategic decision-making, move toward more data-driven production or just new organisational routines, then openness may be relevant not only for AI specialists who design or implement AI systems, but also for workers who need to engage with experimentation, new information, and changing organisational practices. The positive intensity effects among managers and directors, professional occupations, and associate professional and technical occupations are consistent with this interpretation. They also align with recent evidence in Alekseeva et al. (2026) that AI adoption changes the skill requirements of managerial roles. Taken together, the non-AI estimates and decomposition results suggest that observed AI-related demand is associated with broader differences in firms' hiring strategies, rather than only with the composition of AI-specific vacancies.

5 Summary and concluding remarks

This paper examines how firms' demand for AI-related skills is associated with the types of workers they seek to hire, focusing on personality traits. Using 11.7 million UK job advertisements from 2017 to 2022, we document systematic differences in the demand for Big Five personality traits at both the vacancy and firm levels. We extract personality requirements from job descriptions using dictionaries grounded in the psychology literature and identify AI-related roles through detailed skill measures.

We report two main findings. First, comparing AI and non-AI vacancies within firms, occupations, locations, and time periods, we find that AI-related roles are more likely to demand openness—the trait associated with creativity, adaptability, and curiosity—and less likely to mention the other personality traits. Second, using weighted staggered difference-in-differences estimates based on the timing of firms' first AI-related vacancy,

we find that the association between AI-related skill demand and openness extends beyond AI-specific jobs. Firms that become AI-demanding display higher demand for openness across their vacancy portfolio, including in non-AI positions, with no corresponding systematic declines in other traits.

A shift-share decomposition of non-AI vacancies shows that this broader pattern is not driven by firms shifting hiring toward occupations with higher baseline openness requirements. In fact, AI-demanding firms shift non-AI hiring toward occupation groups with lower baseline openness demand, while within most broad occupation groups their non-AI vacancies become more likely to mention openness. Together, these findings suggest that firms seeking AI-related capabilities place greater emphasis on openness to experience, both within AI-related roles and more broadly across the organisation.

Our findings contribute to the literature on technological change and labour demand by highlighting a previously unexplored channel: shifts in the personality of workers that employers demand. In this sense, AI demand appears to be associated with not only what workers need to know, but also which behavioural attributes are most valued in the workplace. This perspective complements recent evidence that the productivity effects of AI depend on heterogeneity in workers' abilities, beliefs, and capacity to adapt to new technologies (D. Deming & Weidmann, 2021). Our results are consistent with the idea that firms screen for these complementarities at the hiring stage through the personality traits they request in job advertisements.

The distinction between skill requirements and personality demands has implications for how firms adjust to new technologies. While technical skills can be developed through training and education, personality traits are less easily changed. As firms begin demanding AI-related capabilities, they may place greater emphasis on hiring workers whose behavioural characteristics fit the way work is organised around these technologies, rather than relying only on retraining their existing workforce. This greater emphasis on selection may affect how workers sort across jobs and firms, and may shape the distribution of opportunities for individuals with different personality profiles. More broadly, the

results suggest that technological change is associated with labour-demand patterns along dimensions that are less responsive to standard education and training policies.

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Appendix

Additional Figures and Tables

Table A.1: Occupation Characteristics at 1-digit SOC level

SOC	Name	Vacancies	Share Adzuna (%)	Census (%)
1	Managers & Directors	170131	5.9	10.4
2	Professional	584393	20.3	16.9
3	Ass. Prof. & Technical	495760	17.2	12.7
4	Administrative	159602	5.5	11.4
5	Skilled Trades	225693	7.8	11.6
6	Caring & Leisure	534602	18.6	9.4
7	Sales & Customer Service	228892	8.0	8.6
8	Process & Machine Ops.	164996	5.7	7.2
9	Elementary	314300	10.9	11.9

Notes: Share Adzuna represents percentage of all Adzuna vacancies. Census shows employment shares from the 2021 Census.

Figure A.1: Frequency of Job Requirements

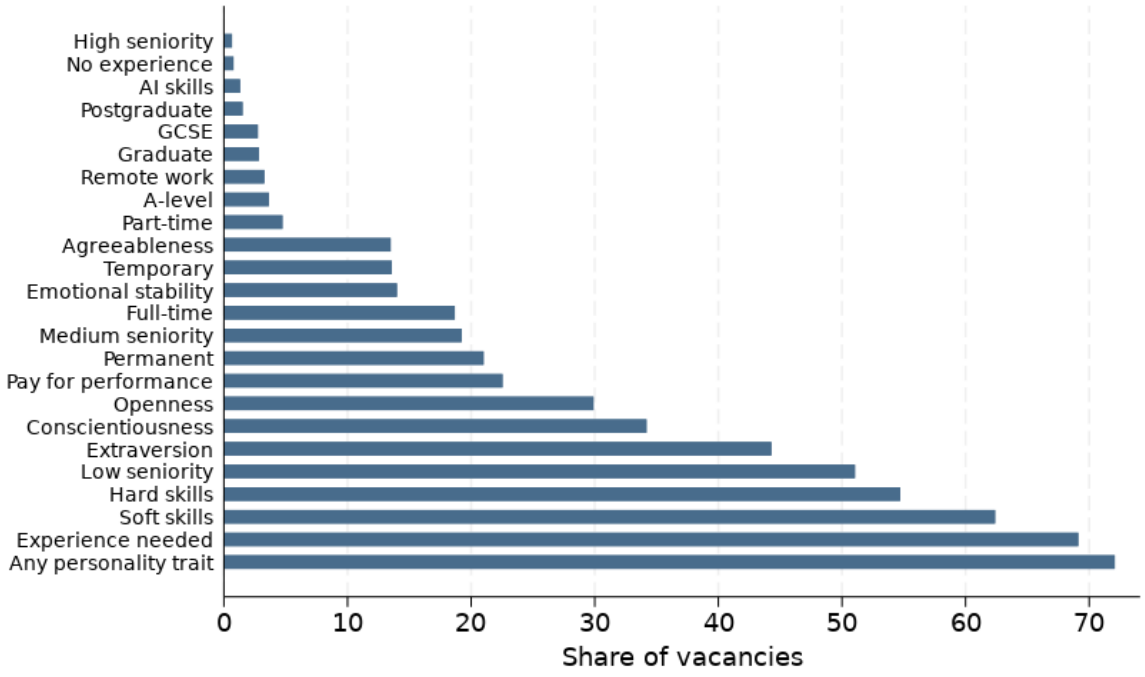
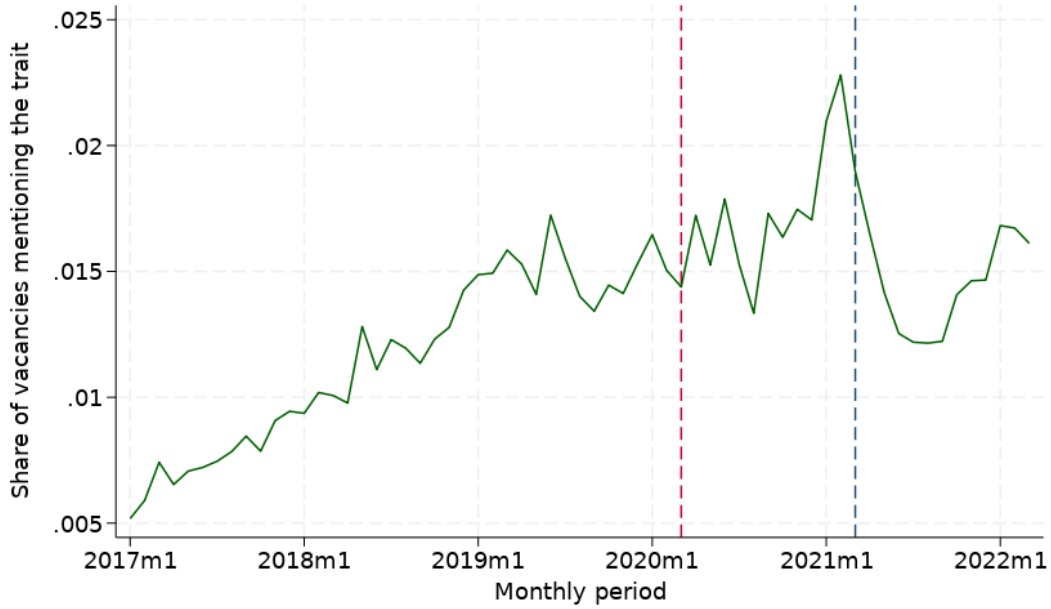


Table A.2: **Dictionary of Personality Traits**

Agreeableness	**Extraversion**	**Conscientiousness**	**Emotional Stability**	**Openness**
affection (4, 5)	adventurous (3, 5, 7)	ambitious (2, 7)	at ease (3)	artistic (3 - 7)
agreeable (all)	assertive (1, 3 - 7)	careful (6)	calm (2, 3, 5)	bright (3, 6)
altruist (5, 7)	bold (3, 6)	competence (5, 7)	emotional (all)	clever (5)
amiable (1)	candour (1)	conscientious (all)	hostil (5, 7)	creative (1 - 3, 6)
considerate (3, 6)	confident (7)	decisive (1)	fearful (6)	curious (1, 2 - 5, 7)
cooperative (1, 3, 5 - 7)	courage (1)	dependable (1, 5)	tempered (2, 5, 6)	ideas (5, 7)
courteous (1)	daring (3, 6)	dutiful (7)	fortitude (ads)	imaginative (2 - 7)
empathy (1)	energetic (1, 3 - 7)	efficient (1, 4 - 7)	pressure (ads)	independent-mind (5)
forgiving (4, 5, 7)	enthusiastic (4, 5, 7)	hardwork (2, 3)	resilience (ads)	ingenious (5)
gentle (5)	expressive (1)	logic (1)	tense (4, 5)	innovative (3, 6)
good-nature (2, 5)	extraverted (all)	neat (3, 7)	unflappable (ads)	insightful (1, 4, 5)
helpful (3, 5, 6)	forceful (5)	organised (1, 3 - 7)	patience (ads)	intellectual (1, 5, 6)
kind (3 - 6)	friendly (7)	painstaking (5)	pressure (ads)	intelligent (1, 3, 5)
naturalness (1)	fun (2)	persevering (2)	-	introspective (3, 6)
pleasant (3, 5, 6)	gregarious (1, 5, 7)	persistence (1)	-	inventive (5)
sensitive (5)	humour (1)	planful (4, 5)	-	openness (all)
straight-forward (5, 7)	optimist (1)	precise (1, 5)	-	original (2, 4)
sympathetic (3 - 7)	outgoing (4, 5)	prompt (3, 6)	-	prefer variety (2)
trust (2, 3 - 7)	outspoken (5)	punctual (1, 2)	-	reflective (3)
unselfish (3, 5, 7)	passionate (2)	reliable (4, 5)	-	sharp (5)
warm (1, 3, 5 - 7)	playful (1)	responsible (3 - 5)	-	unconventional (5, 7)
	pro-active (2 - 7)	striving (5, 7)	-	wide interests (4, 5, 7)
	sociable (5, 7)	systematic (3, 7)	-	witted (5)
	spontaneous (1)	thorough (3 - 5, 7)	-	-
	talkative (1 - 6)		- -	-

Notes: The numbers denote which dictionaries the words are mentioned in. The numbers correspond to the following dictionaries: (1) Goldberg (1990); (2) Costa Jr and McCrae (1992); (3) Goldberg (1992); (4) McCrae and John (1992); (5) O. P. John et al. (1999); (6) Saucier (1994); (7) Heckman and Kautz (2012)

Figure A.2: Change over time of AI-skills demand



Dashed lines show the beginning (red) and end (blue) of Covid-19 restrictions.

Figure A.3: Share of Hard-Skills demanded in AI Vacancies

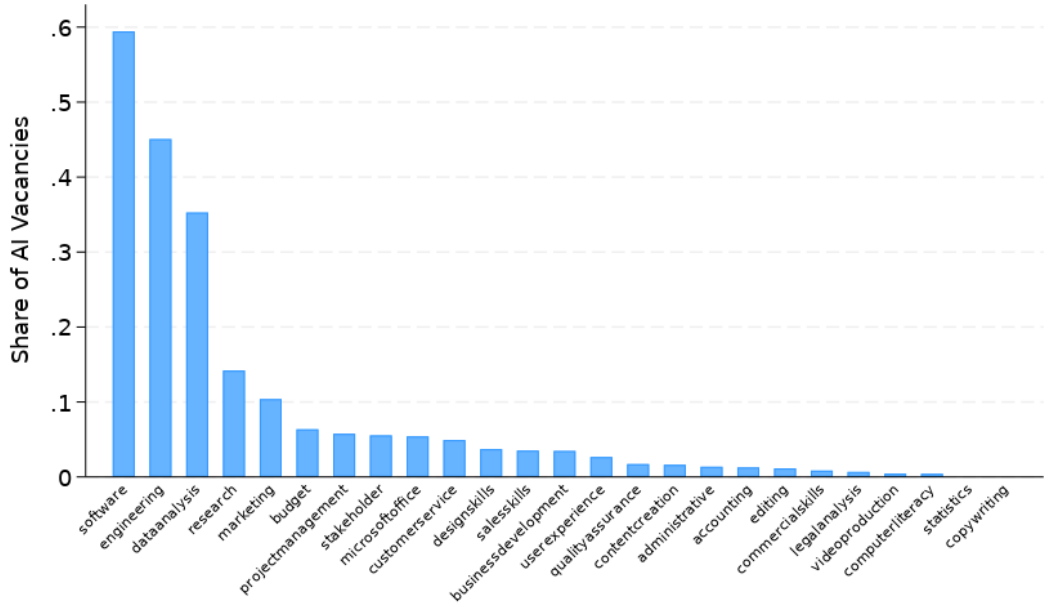


Figure A.4: Share of Soft-Skills demanded in AI Vacancies

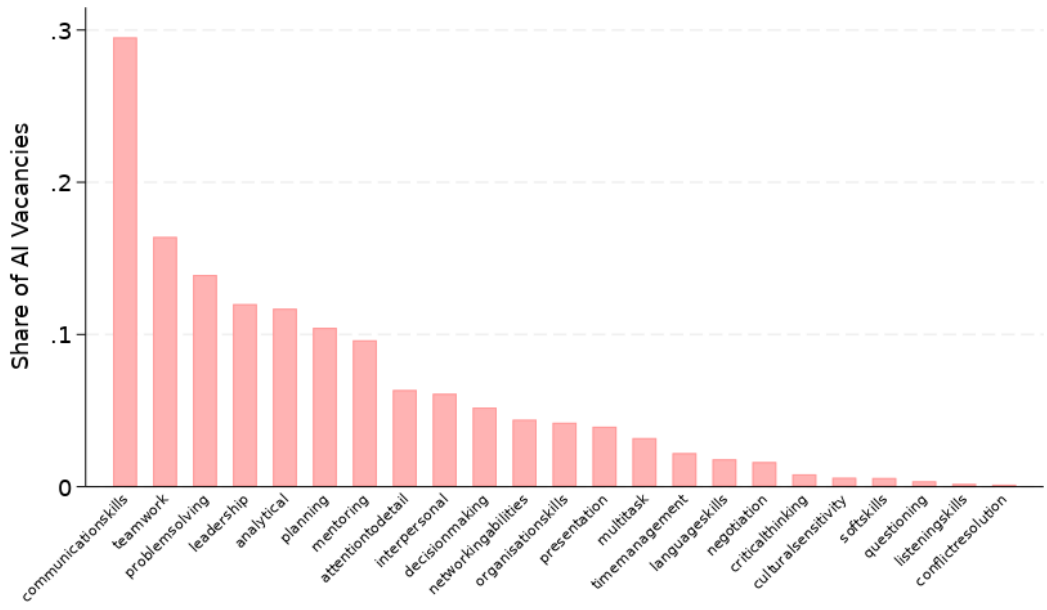
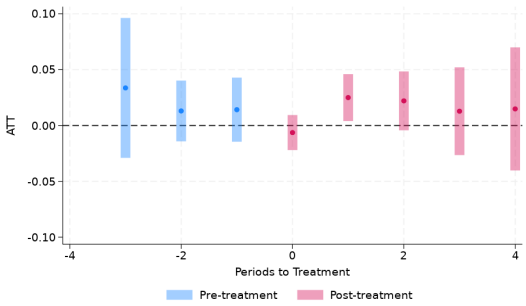
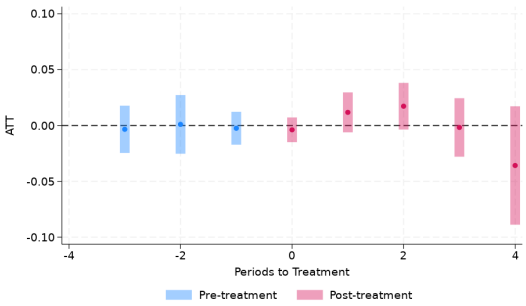


Figure A.5: Staggered DID ATT. Year-level aggregation, Entropy Balance

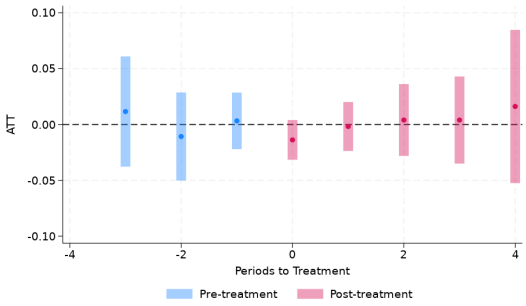
(a) Any Personality Trait



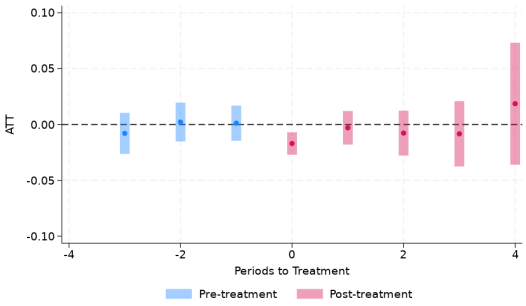
(b) Agreeableness



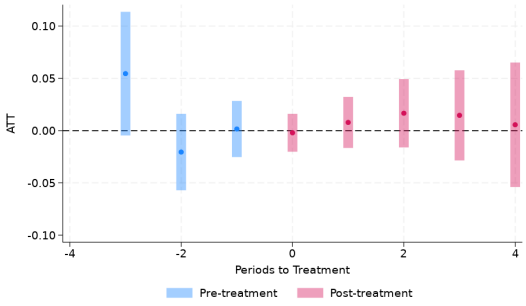
(c) Conscientiousness



(d) Emotional Stability



(e) Extraversion



(f) Openness

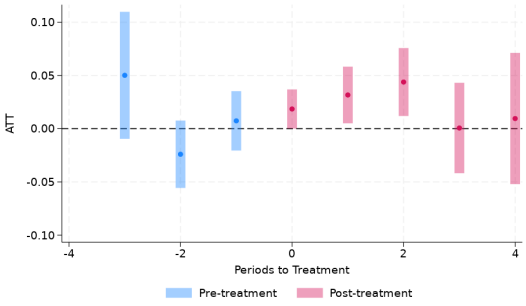


Table A.3: AI and Personality Traits excluding Computer Science skills

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. Any personality trait							
AI_{it}	0.010***	-0.032***	-0.032***	-0.029***	0.012*	0.021***	0.016**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.007)	(0.005)	(0.007)
Panel B: Dependent variable. Agreeableness							
AI_{it}	-0.074***	-0.047***	-0.050***	-0.052***	-0.017***	-0.012***	-0.013***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.004)	(0.003)	(0.004)
Panel C: Dependent variable. Conscientiousness							
AI_{it}	-0.043***	-0.073***	-0.072***	-0.071***	-0.023***	-0.037***	-0.016**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.007)	(0.008)	(0.008)
Panel D: Emotional Stability							
AI_{it}	-0.063***	-0.056***	-0.057***	-0.058***	-0.029***	-0.026***	-0.022***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.005)	(0.009)	(0.007)
Panel E: Dependent variable. Extraversion							
AI_{it}	-0.092***	-0.114***	-0.113***	-0.112***	-0.038***	-0.041***	-0.026***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.008)	(0.009)	(0.007)
Panel F: Dependent variable. Openness							
AI_{it}	0.225***	0.171***	0.173***	0.172***	0.081***	0.120***	0.072***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.008)	(0.016)	(0.014)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes
R-squared range	[0.000, 0.003]	[0.078, 0.119]	[0.080, 0.120]	[0.084, 0.121]	[0.273, 0.351]	[0.105, 0.169]	[0.281, 0.364]
N observations	11662124	11662124	11662124	9923701	9888315	9923701	9888315
N firms	128559	128559	128559	117229	81843	117229	81843

Notes: The R-squared range shows the minimum and maximum values across the six trait regressions for each specification. Controls include: traits and skills demand, remote work requirements, length of advertisement, contract type, work schedule, bonus offerings, education, and negotiable salary option. Standard errors are clustered at the occupation and firm level in specifications (5)-(7). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Impact of Computer Skills on Personality Traits

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. Any personality trait							
Computer skills$_{it}$	0.022***	-0.041***	-0.041***	-0.038***	0.003	0.001	0.003
	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)	(0.005)	(0.004)
Panel B: Dependent variable. Agreeableness							
Computer skills$_{it}$	-0.053***	-0.041***	-0.041***	-0.042***	-0.009**	-0.005	-0.005
	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)	(0.005)	(0.003)
Panel C: Dependent variable. Conscientiousness							
Computer skills$_{it}$	0.028***	-0.020***	-0.019***	-0.018***	0.000	0.005	0.003
	(0.000)	(0.000)	(0.000)	(0.000)	(0.005)	(0.006)	(0.004)
Panel D: Dependent variable. Emotional Stability							
Computer skills$_{it}$	-0.016***	-0.026***	-0.026***	-0.024***	-0.009**	-0.001	-0.005
	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)	(0.005)	(0.005)
Panel E: Dependent variable. Extraversion							
Computer skills$_{it}$	-0.029***	-0.070***	-0.070***	-0.065***	-0.014***	-0.015*	-0.009
	(0.000)	(0.000)	(0.000)	(0.000)	(0.005)	(0.009)	(0.006)
Panel F: Dependent variable. Openness							
Computer skills$_{it}$	0.124***	0.070***	0.070***	0.068***	0.034***	0.037***	0.029***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.005)	(0.006)	(0.005)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes
R-squared range	[0.000, 0.010]	[0.080, 0.121]	[0.082, 0.122]	[0.084, 0.123]	[0.273, 0.351]	[0.105, 0.169]	[0.281, 0.364]
N observations	11662124	11662124	11662124	9923701	9888315	9923701	9888315
N firms	128559	128559	128559	117229	81843	117229	90

Notes: The R-squared range shows the minimum and maximum values across the six trait regressions for each specification. Controls include: traits and skills demand, remote work requirements, length of advertisement, contract type, work schedule, bonus offerings, education, and negotiable salary option. Standard errors are clustered at the occupation and firm level in specifications (5)-(7). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Difference of means between AI only and Computer Skills only on Personality Traits

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. All personality traits							
AI_{it}	-0.012***	0.009***	0.009***	0.009***	0.009	0.020***	0.013*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.264)	(0.007)	(0.097)*
Panel B: Dependent variable. Agreeableness							
AI_{it}	-0.021***	-0.006***	-0.009***	-0.010***	-0.008	-0.007	-0.008
	(0.000)	(0.000)	(0.000)	(0.000)	(0.138)	(0.247)	(0.116)
Panel C: Dependent variable. Conscientiousness							
AI_{it}	-0.071***	-0.053***	-0.053***	-0.053***	-0.023***	-0.041***	-0.020**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.007)	(0.000)	(0.029)
Panel D: Dependent variable. Emotional Stability							
AI_{it}	-0.046***	-0.030***	-0.031***	-0.033***	-0.020***	-0.025**	-0.018**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.016)	(0.039)
Panel E: Dependent variable. Extraversion							
AI_{it}	-0.063***	-0.044***	-0.043***	-0.046***	-0.023**	-0.026**	-0.017*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.015)	(0.041)	(0.061)
Panel F: Dependent variable. Openness							
AI_{it}	0.101***	0.101***	0.103***	0.104***	0.046***	0.083***	0.042***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes

Notes: We present the difference between the effect of AI and Computer Skills on personality traits when estimated separately. We show in parenthesis the p-value corresponding to the z-statistics for these differences, calculated as $(_{AI} - _{CS})/\sqrt{SE(_{AI})^2 + SE(_{CS})^2}$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Difference of means between estimated coefficient for AI only and AI with Computer Science skills on Personality Traits

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. Any Personality Trait							
AI_{it}	0.013***	-0.020***	-0.020***	-0.018***	0.001	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.955)	(0.999)	(0.960)
Panel B: Dependent variable. Agreeableness							
AI_{it}	-0.029***	-0.019***	-0.019***	-0.019***	-0.002	-0.001	-0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.705)	(0.780)	(0.887)
Panel C: Dependent variable. Conscientiousness							
AI_{it}	0.018***	-0.008***	-0.008***	-0.007***	0.000	0.001	0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.983)	(0.903)	(0.953)
Panel D: Dependent variable. Emotional Stability							
AI_{it}	-0.008***	-0.011***	-0.011***	-0.011***	-0.002	-0.000	-0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.772)	(0.998)	(0.934)
Panel E: Dependent variable. Extraversion							
AI_{it}	-0.014***	-0.031***	-0.031***	-0.029***	-0.003	-0.003	-0.002
	(0.000)	(0.000)	(0.000)	(0.000)	(0.785)	(0.791)	(0.872)
Panel F: Dependent variable. Openness							
AI_{it}	0.066***	0.030***	0.030***	0.029***	0.008	0.009	0.005
	(0.000)	(0.000)	(0.000)	(0.000)	(0.481)	(0.693)	(0.786)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes

Notes: We show the difference between AI coefficients estimated without and with computer skills controls. The p-values from the z-statistics for these differences, calculated as $(_{AI} - _{AI|CS})/\sqrt{SE(_{AI})^2 + SE(_{AI|CS})^2}$. are in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: ATT Estimates from staggered specifications using propensity score weights

	(1) Half-year aggregation	(2) Yearly aggregation
Panel A: Dependent variable. Any personality trait		
$AI\ demand_{jt}$	0.007	0.005
	(0.006)	(0.006)
Panel B: Dependent variable. Agreeableness		
$AI\ demand_{jt}$	0.001	-0.002
	(0.004)	(0.005)
Panel C: Dependent variable. Conscientiousness		
$AI\ demand_{jt}$	0.001	0.000
	(0.007)	(0.007)
Panel D: Dependent variable. Emotional Stability		
$AI\ demand_{jt}$	-0.007	-0.013**
	(0.005)	(0.005)
Panel E: Dependent variable. Extraversion		
$AI\ demand_{jt}$	0.003	0.002
	(0.008)	(0.007)
Panel F: Dependent variable. Openness		
$AI\ demand_{jt}$	0.019**	0.017***
	(0.008)	(0.007)

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: ATT Estimates Non-AI Positions: Entropy Balance

	(1) Half-year aggregation	(2) Yearly aggregation
Panel A: Dependent variable. Any personality trait		
$AI\ demand_{jt}$	0.018**	0.011
	(0.008)	(0.008)
Panel B: Dependent variable. Agreeableness		
$AI\ demand_{jt}$	0.007	0.000
	(0.004)	(0.006)
Panel C: Dependent variable. Conscientiousness		
$AI\ demand_{jt}$	0.007	0.003
	(0.008)	(0.009)
Panel D: Dependent variable. Emotional Stability		
$AI\ demand_{jt}$	0.004	-0.006
	(0.006)	(0.006)
Panel E: Dependent variable. Extraversion		
$AI\ demand_{jt}$	0.014*	0.007
	(0.009)	(0.009)
Panel F: Dependent variable. Openness		
$AI\ demand_{jt}$	0.020**	0.019*
	(0.009)	(0.010)
Observations	242,653	167,311

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. ATT estimates for non-AI positions only. Personality traits calculated from vacancies that do not require AI skills. Treatment (AI demand) based on first AI vacancy posted by firm. Half-year and yearly aggregation, mean model. Standard errors in parentheses.

Table A.9: Average Firm-Level Shares of Vacancies Demanding Personality Traits (yearly aggregation)

Trait	Mean	Std.Dev.	Min	Max
Any personality trait	0.7309	0.3742	0	1
Openness	0.3191	0.3916	0	1
Agreeableness	0.1074	0.2576	0	1
Conscientiousness	0.3820	0.4017	0	1
Emotional Stability	0.1334	0.2766	0	1
Extraversion	0.4597	0.4186	0	1

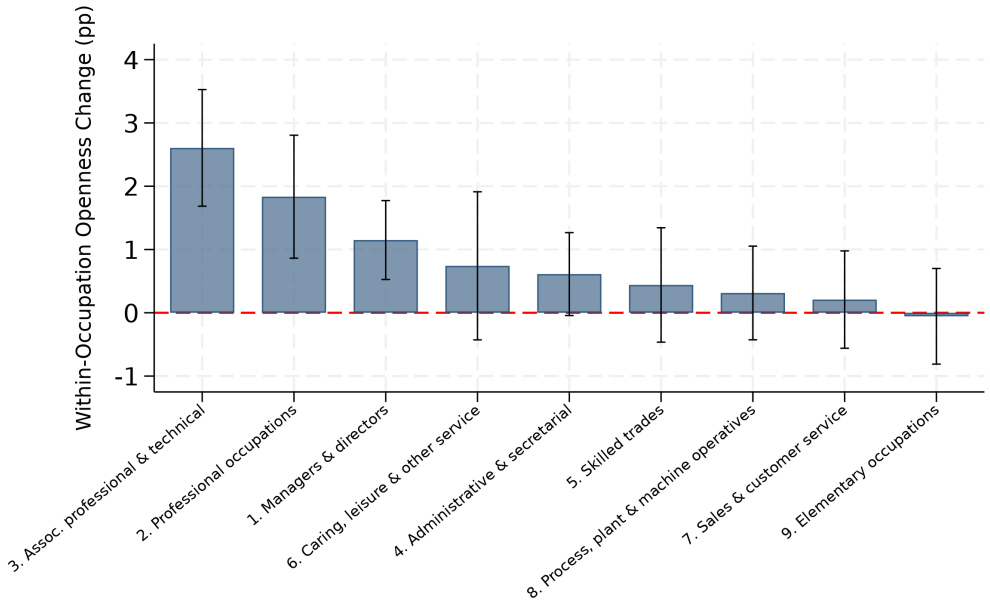
Notes: Statistics calculated at firm-year level. Mean share represents the average proportion of vacancies demanding each trait across all firm-year observations.

Table A.10: Shift-Share Decomposition of Openness Demand in Non-AI Vacancies

Component	Mean (pp)	(SD)
Composition effect	-12.470***	(20.191)
Intensity effect	1.711**	(39.270)
Cross term	2.283***	(27.130)
Total	-8.476***	(36.687)
Firms	3039	

Notes: Shift-share decomposition of the pre-to-post change in openness demand among non-AI vacancies for AI-adopting firms. Each observation is one adopting firm. Values in percentage points. The composition effect equals the change in major occupation group shares (SOC 1-digit) weighted by the sample-wide pre-adoption openness rate per major group. The intensity effect equals the pre-adoption major group share weighted by the within-group change in openness demand. The decomposition uses SOC 1-digit groups for clarity; the main DiD specification controls for occupation at the SOC 2-digit level. Stars from one-sample t-tests against zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A.6: Within-Occupation Openness Change in Non-AI Vacancies by Major Occupation Group



Notes: Each bar is the intensity component of the shift-share decomposition for one SOC 1-digit major occupation group: the pre-adoption share of non-AI vacancies in that group, multiplied by the within-group change in openness demand between the pre- and post-adoption periods, averaged across AI-adopting firms ($N = 3{,}039$). Values are in percentage points. Error bars show 95% confidence intervals. Groups are sorted from highest to lowest mean contribution.

Online Appendix

Contents of Online Appendix

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B. Matching of the datasets

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E. Personality Traits. Evolution over time

A. Datasets used

The datasets we used in this study, in addition to Adzuna, are the following: **Companies House:**

Companies House is an executive agency of the UK government that provides information about registered and dissolved companies in the UK. Companies House includes 5,166,124 companies registered in the UK since 1885. It provides information for most companies' addresses, postcode of registered office, and a 4 to 6 digit Standard Industrial Code (SIC). After removing companies with the same name (hence impossible to differentiate), 4,832,015 unique companies remained. Since SIC codes are only provided for private companies, public institutions like the NHS, Universities, or Charities were not considered. Institutions and dormant companies whose SIC codes are generic (99999) were also not considered. The final Companies House dataset consists of 4,581,872 private companies.

FAME: It is a financial database covering company-level information. It is part of the Bureau van Dijk suite of business databases, which also includes international platforms like Orbis. It covers 3.4 million companies in the UK and the Republic of Ireland. The dataset contains information on company accounts, activities, ownership and management. We retrieve information on whether companies are publicly listed and balance sheet information.

UK Household Labour Survey: Also known as Understanding Society, the survey is a large-scale longitudinal study that aims to provide valuable insights into the social and economic aspects of individuals and households over time. Surveys a wide range of information from participants, including demographic data, employment status, income, health, and subjective well-being. Data were collected with the same methodology from 2009 until 2022.

In Wave C, performed during 2011 and 2012, the survey requested individuals to complete a short questionnaire on self-perceived attitudes, where the Big Five traits were inferred. The module presented three affirmations for each personality, and the individuals

had to choose a number that best described how they saw themselves *as persons*, using a scale from 1 to 7, where 1 meant *does not apply to me at all* and 7 meant *applies to me perfectly.*[15] Then, an average score of the three responses for each trait was calculated.

- **Agreeableness:**
  - I see myself as someone who is sometimes rude to others.
  - I see myself as someone who has a forgiving nature.
  - I see myself as someone considerate and kind to almost everyone.
- **Conscientiousness:**
  - I see myself as someone who does a thorough job.
  - I see myself as someone who tends to be lazy.
  - I see myself as someone who does things efficiently.
- **Extraversion:**
  - I see myself as someone talkative.
  - I see myself as someone who is outgoing, sociable.
  - I see myself as someone who is reserved.
- **Neuroticism:**
  - I see myself as someone who worries a lot.
  - I see myself as someone who gets nervous easily.
  - I see myself as someone who is relaxed, handles stress well.
- **Openness:**
  - I see myself as someone who is original, comes up with new ideas.
  - I see myself as someone who values artistic, aesthetic experiences.
  - I see myself as someone who has an active imagination.

[15] UKHLS creates a score for the *Neuroticism* trait. For comparison purposes with the demand side study, the values had been inverted (1/x) to infer *Emotional Stability.*

B. Matching of the datasets

The main purpose of the matching strategy was to standardise text entries, in particular the name of the company, location, and descriptive variable of the advertisement, in order to be able to match the different datasets we use. To create an occupation variable, the Turrell et al. (2019) algorithm was used. The code requires as inputs advertisement text, job title, and industry. For the latter, companies' variables were matched with Companies House. Cleaned text was later used as input to retrieve job requirements, wages, and location.

Since the entire dataset was too large for the cleaning process to be conducted in one attempt (646 GB), it was first divided into 272 weekly snapshots, and each snapshot was subdivided into 1% samples using the last 2 digits of the vacancy identifier. Each of these 27,200 datasets contained around 10,000 observations.

Companies names cleaning process

Adzuna and Companies House companies lists were cleaned using the same code. The process, in STATA, involved the use of regular expressions to identify characters and words. The process intended to minimise the potential discrepancies between the same name in both datasets, therefore eliminating all non-relevant information to the name and homogenising formats. It included lowering the case; removing consecutive internal blank spaces, quotes, and special characters; changing symbol characters for their word counterpart (& for *and*), dropping endings such as "*.sol*" or "*.com*"; removing "*The*" at the beginning, and replacing text not encoded as American Standard Code for Information Interchange (ASCII) into ASCII.

In Adzuna, some companies' names identify the place where the job is located, such as *KPMG (Liverpool)*, or *Accenture (U.K.)*. This information is also available in the "location" variable, so its presence in the company name stops the algorithm from matching the company with Companies House. In consequence, the information inside the brackets at the end of a company was also removed. A considerable difference between Adzuna and

Companies House is that the former frequently does not incorporate companies' categories, such as *Private Limited Company, Public Limited Company, Limited Liability Company* or its diverse acronyms and synonyms *(plc, llc, ltd, plc, lp, pcc, plc, scio, cio, cic, ukeig, limited, srl, inc, co, sp, group)*. These were removed if they appeared at the end of the company name.The cleaning process was repeated 3 times to ensure that companies with several inaccuracies in their names could be exhaustively cleaned ("*K.F.C (Liverpool) LTD++*" into "*kfc*").

Matching companies with Companies House

Since companies' names in Adzuna could have diverse variations, the cleaning process consisted of generating a dataset with all the existing entries of companies in Adzuna and applying a cleaning code. This *matching dataset* was then associated with the Companies House cleaned list of companies before removing public and dormant companies, merging into a unique data set both Adzuna's raw company name, the clean name common to Companies House, and all Company information matched from Companies House. Nearly half of the unique company entries in Adzuna (337,461) were matched. The remaining companies either present a fantasy name different from the legal name, which means they were not listed in Companies House or have missing company names. From the

Interestingly, out of the 4.5 million active and inactive companies listed in Companies House, less than 10% were found in online job vacancies. This is explained by the high prevalence of Recruitment Agencies, whose growth since the global crisis in 2008 reshaped companies' strategy to recruit employees. Low barriers and increased digitalisation have benefited the industry over the past years. According to the Recruitment and Employment Confederation (REC), by 2021, more than 30,000 recruitment enterprises were operating across the UK. In the same year, nearly 550,000 permanent placements and 22.4 million temporary contracts were made by the recruitment industry (REC, 2022).

In addition, half of the companies in the Adzuna vacancy dataset could not be matched with a company in the CH data. The main reason is that companies do not necessarily publish their legal registration name when posting a job opening. These may appear with

a shorter version, an acronym or a more known fantasy name. Even although the cleaning algorithm aims to unify criteria in terms of format and accessory wording, if a company publishes job advertisements using a highly different name than its legal name, then it is impossible to match it with Companies House. As an example, a company may appear in online postings as *The Berliner*, but their legal name in Companies House incorporates their complete identification *The Berliner Beer House*.

Finally, some companies may share a similar part of their legal name, (*The Berliner Services, The Berliner International, The Berliner Central*). When companies have an almost identical name, it is impossible to systematically identify which version of the name in Companies House corresponds to the clean name in Adzuna. To avoid false-positive matching, whenever a company's clean name was identical to another, none of them was incorporated.

Finally, those vacancies with company names not matched and therefore not identifiable were removed. Therefore, all the analysis done was on vacancy data which contained a matched firm name.

Recruitment agencies and companies not matched removal

In the Adzuna database, 45% of vacancies were posted by recruitment agencies, hence making it impossible to identify either the company behind the search nor the job's sector. It is possible that many enterprises, in particular small businesses, subcontract the human resource task and pay recruitment agencies to be in charge of the first recruiting stages. In addition, many existing enterprises may still not use online job boards to advertise their search, still relying on traditional means such as newspapers, word-of-mouth, or other types of referrals. Robustness checks on published wages show no significant differences between jobs posted through Recruitment agencies and those published by companies themselves.

Recruitment agencies were identified and removed following three strategies; firstly, company names containing words such as "*recruit*", "*employ*,", "*staffing*", "*personnel*",

"*resourcing*", "*careers*", "*talent*" or "*placement*" were removed. These words were mostly associated with companies in SIC codes 62090 (other information technology service activities), 7810 (activities of employment placement agencies), 78200 (temporary employment agency activities), 78300 (human resources provision) and 82990 (other business support service activities), so all vacancies in these industries were removed as well. A manual inspection of companies that had a high rate of job advertisements led to the removal of another 33 recruitment agencies, including CV Library, Reed and Brook Street.

To identify companies' sectors, Standard Industrial Classification (SIC) is needed. Only private companies in the Companies House are assigned this information. Consequently, only vacancies with the company were found in both Adzuna and Companies House, and with no missing SIC, codes were kept. This led to a total of 64.6 million vacancies with associated company names and sectors not published by recruitment agencies.

Duplicates removal

Once companies' names were cleaned and matched with Companies House, and recruitment agencies were deleted, databases with the same 2 last digits of the ID variable were appended, to proceed with the removal of the duplicates.

One issue with vacancies aggregators is that an advertisement remaining open on a website does not necessarily reflect that the position has not been filled, nor vice-versa. While some companies have their advertisements posted for months or even years, constantly receiving applications, other companies may pay for a fixed advertisement time but fill the position before it ends, or not fill it at all. It becomes necessary then to deduplicate the database to only work with unique vacancies.

Using the ID variable provided by Adzuna, we kept the first observation of each vacancy that appears in the dataset. Two more rounds of duplicate removals were performed with other methods. First, by grouping variables with the same job title, company name, date created and source-id variable. In case some of the vacancies posted on different web pages were not initially captured by the Adzuna ID vacancy variable, a last duplication removal

round was performed by deleting repeated observations with the same company, title, date of creation, description, and location. Since only a third of the total observations belong to unique vacancies, the number of vacancies was reduced to 20.7 million.

Cleaning text descriptions and generating new wage variables

The salary variable provided by Adzuna is coded from complementary information on web pages rather than from the advertisement text directly. Around 65% of observations in the raw data have a non-missing salary. After removing recruitment agencies and unmatched companies, this falls to 46% of vacancies, which is still higher than comparable studies using single job boards.[16]

Some entries in Adzuna's wage variable contained errors or were coded at different frequencies (daily, weekly, or monthly rather than annual). To address this, we retrieved salary information directly from advertisement text, converting all figures to annual equivalents. Wages in vacancies appear in a wide variety of formats, as illustrated in Table OB1. The homogenisation process involved standardising number formats, inferring payment frequencies from surrounding text, and applying part-time or full-time adjustments where applicable. When wage information could not be retrieved from the text, Adzuna's original salary variable was used. This procedure recovered an additional 2,446,300 observations with non-missing wages. The final dataset, after a last round of duplicate removal, contains 11,430,457 observations.

[16]For reference, posted wages appear in 24.8% of advertisements on *monster.com* (Brenčič, 2012), 13.3% on *trabajando.com* (Banfi & Villena-Roldan, 2019), 20% on *careerbuilder.com* (Marinescu & Wolthoff, 2020), and 24.3% on *xmrc.com* (Kuhn & Shen, 2023).

Table OB1: **Examples of wages differences in formats**

Differences	**Unique salaries**	**Range of salaries**
Punctuation and spaces	£35,000 £ 35000 £35, 000	£30,000-£40,000 £ 30.000 - £40000
The use of k for thousands or the lack of £ sign	£35k 35000 35k	£30k-£40,000 £ 30.000 - 40k 80 - 120000
Range separators		£30,000 to £40,000 £ 30000 - 40k £35. 000 and 40000
Monthly date	£1300 per month	
Weekly date	£500 per week	
Daily date	£80.5 per day	
Hourly date	£9 9 per hour	£8.46-10

Author's creation based on Adzuna

Creating NUTS location variable

Adzuna presented two location variables: *location_raw*, which could contain entries such as "Newcastle, Staffordshire" and *location_path*, containing long entries such as "UK ∼ North West England ∼ Greater Manchester ∼ Manchester ∼ The Trafford Centre". The first variable is the raw information extracted from the job vacancy, sometimes only incorporating the town where the job is located and, in a few cases, it also incorporates the postcode. The second is coded by Adzuna, but on some occasions, it does not incorporate the full path but only "UK", or "UK∼London". When it does follow the path, the information is always accurate. The approach followed was to first use *location_path*, then complement those observations with a missing detailed location variable with *location_raw*. Ultimately, they were complemented using the information from postcodes. Table OB2 is an example of *location_path* and the resulting subdivision[17]:

Table OB2: **Location example**

Loc. 1	Loc. 2	Loc. 3	Loc. 4	Loc. 5	Loc. 6
UK	North West England	Greater Manchester	Manchester	The Trafford Center	-

Author's creation based on Adzuna

[17]All examples are illustrative and do not represent any actual vacancy information

Similar to the process for matching the companies, the strategy was to match locations with a nomenclature that allows to use known location categories. Locations were mapped with the Nomenclature of Territorial Units for Statistics (NUTS). The NUTS identification at one digit (UK + letter) corresponds to 12 UK regions, as shown in Table OB3. A list containing the 3,102 unique locations found in Adzuna's location variable was created and assigned a NUTS code at 2 digits (UK+letter+number). Then, each job vacancy was associated with this list, to obtain a matching as the one in Table OB4.

Table OB3: **Nuts 1**

NUTS	**Code**
North East	UKC
North West	UKD
Yorkshire and The Humber	UKE
East Midlands	UKF
West Midlands	UKG
East of England	UKH
London	UKI
South East	UKJ
South West	UKK
Wales	UKL
Scotland	UKM
Northern Ireland	UKN

Table OB4: **Nuts 2**

NUTS	**Code**
Belper	UKF1
Ely	UKH1
Mortlake	UKI7
Porthmadog	UKL1
Newton-Le-Willows	UKD7
Barnacle	UKG3
Sea Mills	UKK1
Cross Keys	UKL2
Ickenham	UKI7
Maghera	UKN0
Darlington	UKC1

Author's creation based on ONS

A similar approach was taken for all the components of *location_raw*: they were split and matched with their corresponding NUTS codes at 2 digits. For those observations with missing information on Location 3 onwards, postcodes were used to retrieve the corresponding NUTS. These belong to four sources: first, Adzuna provided a variable with postcodes, although it had 70% missing values. For those missing, postcodes in the location variable or the title of the job were retrieved. Finally, Companies House provides the location and postcode of the companies' headquarters. For those observations with job locations set in London, Companies House postcodes were used. Postcodes were ultimately matched with NUTS. In total, 96% of observations have non-missing values of NUTS at 1 digit and 85% at 2 digits.

Table OB5: Cleaning of the database

Process	N vacancies	N no missing W*	Mean W*	SD W*	Δ
Adzuna raw database	276,325,386	178,229,590	31,120	18,524	-
Without recruitment agencies	150,918,209	83,727,680	29,882	18,558	-45%
Without missing companies match	66,833,773	32,647,950	28,365	18,392	-56%
Keep only Private companies with SIC	64,642,205	31,619,913	28,303	18,468	-3%
Remove duplicates vacancy-id, source-id, same description-location	20,783,111	10,490,006	28,342	18,830	-68%
Remove salaries missing	9,657,106	9,657,106	29,454	18,964	-54%
Incorporate new salaries from cleaned description	12,103,405	11,866,183	31,930	18,873	25%
Remove outliers in wages and text size	11,659,923	11,430,457	32,018	18,944	-4%

Author's creation based on Adzuna. ∗W = Salary considered after trimming 1% tails

Adding Standard Occupational Classification (SOC)

Adzuna's database does not come with official classification labels. We implemented a Natural Language Processing (NLP) matching algorithm that maps the free-form text of job descriptions into UK Standard Occupational Classification (SOC) at three digits, following Turrell et al. (2019). Their algorithm has an accuracy rate of 76%. Python codes are available on GitHub. This algorithm was also implemented by Bellatin and Galassi (2022), who group occupations according to their involvement in the production and use of digital technologies.

By identifying occupations SOC categories, it is possible to provide a granular analysis, identifying the change in personality trait demand within similar occupations. The algorithm requires as inputs a job title, a text description, and a sector identification. For that, being able to match Adzuna's company's name with Companies House was fundamental.

Classification Algorithm of Turrell et al. (2019)

Authors' contribution uses the job title, description and sector for job postings from the recruitment website Reed.com. In addition, they build dictionaries using the information published by the Office for National Statistics (ONS), for the four-digit UK SOC job titles and job descriptions. Inspired in Bellatin and Galassi (2022) description, the algorithm

follows these steps:

1. Cleans the job posting text data using standard text analysis tools.

2. Creates a vector space for the dictionary of UK SOC text data provided by the ONS, using the term frequency-inverse document frequency (IF-IDF).

3. Searches for exact matches between the job titles in the job postings and those in the UK SOC list.

4. Combines the job posting title, the description and the sector into one string and expresses this string as a vector in the IF-IDF matrix of the UK SOC data.

5. For jobs that were not matched, it calculates the cosine similarity between the string obtained in the previous step, and the information in the UK SOC dictionary, and selects the five categories in the UK SOC list with the highest cosine similarity.

6. If the job posting title is empty, it returns the job posting with the highest cosine similarity. Instead, if there is text in the job posting title, the fuzzy-wuzzy Python package is used to identify the best fuzzy match out of the top five ONS categories, following Levenshtein distance calculations.

Duplication of cleaning

Even if Adzuna provides a database without duplicate advertisements for the same week from the same online source, such as job.uk, jobsincare.co.uk or cv-library.co.uk, we also cleaned potential duplicate vacancies published in other sources or with different vacancy IDs but identical company, location and descriptive text. A step-by-step explanation of this procedure is in Online Appendix A. According to Adzuna's representatives, their point-in-time estimates are more comprehensive than other aggregators such as *Burning Glass Technologies (BGT), now Lightcast*. While Adzuna collects data from job boards, BGT's collection focuses on individual company websites, resulting in fewer vacancies.

We removed advertisements from recruitment agencies (reducing the sample by 45%) and those without SIC codes (further reducing the sample by 56%). We only kept private

companies whose names in Adzuna uniquely matched the Companies House database. This study only uses vacancies with posted wages. A common criticism of wage data from online job advertisements is that they often target lower-skilled workers or positions that need to be filled urgently (Banfi & Villena-Roldan, 2019; Brenčič, 2012). Batra et al. (2023)'s meta-analysis highlights the scarcity of wage information, which is often only available in broad ranges, limiting its reliability as a proxy for actual wages. This scarcity is more evident in high-wage occupations and large private firms. After removing recruitment agencies and advertisements not matching Companies House, the dataset contained wage information for 46% of vacancies. This is twice the proportion found in studies using job boards that don't aggregate sources (24.8% in *monster.com* (Brenčič, 2012); 13.3% in *trabajando.com* (Banfi & Villena-Roldan, 2019); 20% in *careerbuilder.com* (Marinescu & Wolthoff, 2020); and 24.3% in *xmrc.com* (Kuhn & Shen, 2023)).

C. Construction of variables and dictionaries

Construction of variables We built dictionaries of job requirements, that were later searched on job advertisements. The thesaurus of both personality traits and requirements was expanded with germane variations, including words containing spelling variations and errors. To identify potential false positives, we did manual checks in 8 rounds of 100 advertisements, for each dictionary. By looking at the 10 closed words in the proximity of the text retrieved, we ensure to be grasping sentences with the correct meaning. When the searched word retrieved a different connotation, we modified the false-positive retrieval. For instance, *flexible hours* were replace by *flx hours* to avoid it being retrieved under the *flexible* matching. Terms with multiple words, such as *machine learning*, were joint (*machinelearning*), and the code only matched these when surrounded by blank spaces, commas or stops. In total, 1702 word replacements were done.

For the personality traits dictionary, we followed the *Knowledge engineering* approach, where a set of explicit rules to classify text are manually defined. A job posting is identified as requiring personality traits as long as it lists at least one keyword in its text description. The strategy of identifying a requirement by pinpointing its appearance in the text description and job titles has been used by Deming and Kahn (2018), Acemoglu et al. (2022), Alekseeva et al. (2021), among others. To construct the traits dictionary, adjectives belonging to the Big Five Personality schemes were used. In comparison with previous papers using *character-related* language, the dictionary was constructed using trait-related terms drawn primarily from established Big Five personality measures and lexical taxonomies in psychology (Costa Jr & McCrae, 1992; Goldberg, 1992; John, 1990; McCrae & John, 1992; Saucier, 1994). We also draw on the economics literature on soft skills to motivate their relevance in labour-market contexts Heckman and Kautz (2012).

For education, we use ONS education descriptions complemented with UK qualification levels.[18] We built 8 categories, later reduced to 5 following Nandi and Nicoletti (2014). The variable was constructed to consider the lowest educational level that was made

[18](https://www.gov.uk/what-different-qualification-levels-mean/list-of-qualification-levels)

explicit in the job search. Hence, if the advertisement is looking for someone with 'a bachelor's degree in Social Science, but a master' is preferable', the observation is marked as requiring at least a bachelor's level.

For remote work, we considered Draca et al. (2022) dictionary of words. For seniority, we followed Nandi and Nicoletti (2014) Nyhus and Pons (2005), who claim that the relevance of a personality trait may differ according to the position. The structure of hierarchy in some occupations may be more sensitive to personality characteristics. For the same SOC code, individuals applying for senior or managerial positions, *ceteris paribus* the experience and education, have different traits than for junior placements. We follow hierarchy descriptions available in Blogs, such as Indeed.

Adzuna's raw database incorporated information from the advertisements that were not necessarily available in the text entry. This was the case for part and full-time indicators, and temporary and permanent schedules. We created dictionaries for the four categories and complemented Adzuna's variables whenever they were missing.

Finally, we incorporated a dictionary on experience and bonuses. For the first one, we identify whether advertisements require any type of experience or explicitly say that this is not required. For the latter, we explore different variations in the text that signal applicants that the position could have a bonus or pay-for-performance reward.

Dictionaries

The skills used by Acemoglu et al. (2022) are *machine learning, computer vision, machine vision, deep learning, virtual agents, image recognition, natural language processing, speech recognition, pattern recognition, object recognition, neural networks, ai chatbot, supervised learning, text mining, unsupervised learning, image processing, mahout, recommender systems, support vector machines, random forests, latent semantic analysis (lsa), sentiment analysis, opinion mining, latent dirichlet allocation (lda), predictive models, kernel methods, keras, gradient boosting, openCV, xgboost, libsvm, word2vec, machine translation, and sentiment classification.*

The skills used by Babina et al. (2024) that were not in Acemoglu et al. (2022) that were incorporated are *apache uima, artificial intelligence, autoencoders, bayesian networks/methods, caffe deep learning framework, classification algorithms, clustering algorithms, computational linguistics, convolutional neural network (cnn), dbscan (density-based spatial clustering), deeplearning4j, dimensionality reduction, dlib, expectation-maximization algorithm, hidden markov model, information extraction, jung framework, k-means, long short-term memory (lstm), matrix factorization, maximum entropy classifier, microsoft cognitive toolkit, mlpack, mxnet, naive bayes, natural language toolkit (nltk), nd4j, opennlp, pybrain, recurrent neural network (rnn), scikit-learn, semi-supervised learning, stochastic gradient descent (sgd), theano, unstructured information management architecture, vowpal, wabbit, weka.*

We expanded the list with the following terms: *a b testing, ai, bidirectional encoder (bert), chaid, cloud technology, cntk, cnns, convolutional, convolutional networks, cross domain learning, data augmentation, decision trees, deep embedding, discriminant analysis, dnn techniques, document embeddings, emotion detection, ensemble learning, ensemble methods, entity linking, fastai, feature engineering, federated learning, ffmpeg, gaussian processes, generative adversarial networks (gans), generative models, generative pre-trained transformer (gpt), hierarchical clustering, huggingface, hyperparameter tuning, image fusion, image segmentation, intelligent automation, k-nearest method, kalman filtering, knowledge distillation, kubernetes, mlflow, model compression, multi-task learning, named entity recognition, natural language generation (nlg), natural language understanding (nlu), object tracking, optical character recognition, pattern detection, pcl, predictive churn, predictive modelling, pytorch, quantum computing, recurrent network, reinforcement learning, resnets vaes, robotic process automation, rnns, sequence-to-sequence modelling, shot learning, siamese networks, similarity models, sound recognition, text classification, topic detection, topic modelling, tree based lerners, tree-based classifiers, t-sne, umap, uniform manifold, vision processing.*

Category	Words searched
Traits	
Agreeableness	affection, agreeableness, altruist, amiability, be gentle, considerate, cooperative, courteous, empathetic, forgiving, good nature, helpful, kind, naturalness, pleasant personality, sensitive, soft hearted, straightforward, sympathetic, trust, unselfish, warm
Extraversion	adventurous, assertive, bold, candour, confident, courage, daring, energetic, enthusiastic, expressive, extraverted, forceful, friendly, funny, gregarious, humour, optimist, outgoing, outspoke, passion, playful, proactive, sociable, spontaneous, talkative
Conscientiousness	ambitious, be careful, competent, conscientious, decisive, dependable, dutiful, efficient, hard work, be logic, be neat, organised, painstaking, perseverance, persistent, planful, precise, prompt, punctual, reliable, responsible, striving, systematic, be thorough
Emotional Stability	at ease, calm, emotional, hostil, not fearful, tempered, fortitude, resilience, tense situations, unflappable, patience, pressure
Openness	artistic, be bright, be clever, creative, curious, flexible, ideas, imaginative, independent mind, ingenious, innovative, insightful, intellectual, intelligent, introspective, inventive, openness, be original, prefer variety, reflective, be sharp, unconventional, wide interests, witted
Skills	
Soft Skills	analytical, communication skills, critical thinking, cultural sensitivity, decision making, language skills, leadership, listening skills, mentoring, negotiation, networking abilities, presentation, problem solving, interpersonal, teamwork, time management, attention to detail, organisation skills, multitask, conflict resolution, soft skills, planning, questioning
Hard Skills	accounting, administrative, content creation, copywriting, data analysis, design skills, editing, engineering, legal analysis, marketing, microsoft office, project management, quality assurance, sales skills, software, statistics, user experience, video production, customer service, business development, computer literacy, commercial skills, budget, research, stakeholder

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Table OC1 – continued from previous page

Category	Words searched
AI Skills	machine learning, computer vision, machine vision, deep learning, virtual agents, image recognition, speech recognition, pattern recognition, object recognition, neural networks, supervised learning, text mining, support vector machines, unsupervised learning, image processing, image fusion, mahout, recommender systems, random forests, latent semantic analysis, sentiment analysis, opinion mining, latent dirichlet allocation, predictive models, kernel methods, keras, gradient boosting, gradient descent, opencv, xgboost, libsvm, word2vec, ai chatbot, machine translation, sentiment classification, generative adversarial networks, reinforcement learning, quantum computing, federated learning, natural language understanding, natural language processing, natural language generation, mlops, artificial intelligence, cloud technology, tensorflow, pytorch, scikit learn, natural language toolkit, bidirectional encoder, shot learning, ensemble methods, decision trees, k means clustering, autoencoders, robotic process automation, cnns, rnns, optical character recognition, named entity recognition, topic modelling, ab testing, dimensionality reduction, feature engineering, hyperparameter tuning, data augmentation, model compression, knowledge distillation, kalman filtering, kubernetes, azure ml, weka, clustering algorithms, classification algorithms, density based spatial clustering, caffe deep learning framework, generative pretrained transformer, expectation maximization, jung framework, theano, hugging face, ffmpeg, dlib, mxnet, vowpal wabbit, mlflow, bayesian methods, kubeflow, spark ml, discriminant analysis, long short term memory, hidden markov model, computational linguistics
Work Location	
Remote Work	conducted remote, fully remote, home based, homework, homeworking, job is remote, job is virtual, location is remote, location remote, position is remote, position remote, remote work, role will be remote, telecommuting, telework, virtual job, work from home, work home, work remote, working remote, flexible with office, remote position, role is remote, remote basis, hybrid work, hybrid remote
Seniority	
Level 1	chief, director, executive, vice president
Level 2 high	associate, coordinator, project lead, senior, supervisor, team lead, supervising a team, manager
Level 2 low	administrator, agent role, analyst, consultant, generalist, representative role, technician, secretary
Level 3	apprentice, assistant, clerk, internship, junior, receptionist, trainee, trial period, entry level
Education	

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Category	Words searched
No Formal Education	basic functional skills, basic skills qualification, entry level certificate, entry level functional skills, entry level qualifications, entry nqf level, essential skills wales, level 1 scqf, level 1 svq, level 2 scqf, level 2 svq, level 3 scqf, level 3 svq, no education requirements, no esol, no formal qualification, no previous experience or qualification, no previous qualification, no qualification, scottish national 1, scottish national 2, scottish national 3, scqf level 2, scqf level 3, skills for life, svq level 2, svq level 3, no formal education
GCSE Level	gcse, gnvq, btec award, btec first, btec level 1, certificate of secondary education, first certificate, foundation welsh baccalaureate, key skills level 1, level 1 award, level 1 certificate, level 1 diploma, level 1 esol, level 1 essential skills, level 1 functional skills, level 1 key skills, level 1 national vocational qualification, level 1 nvq, nqf level 1, nvq level 1, rqf level 1, scqf level 4, diploma level 1, esol 1, national level 1, qualification level 1, level 1 qualification, qcf level 1, level 1 qcf, functional skills level 1, essential skills level 1, level 1 apprenticeship, cipd level 1, level 1 cipd, level 4 scqf, level 4 svq, svq level 4
O-Level	o level, btec level 2, general certificate of education, intermediate apprenticeship, intermediate welsh baccalaureate, key skills level 2, level 2 award, level 2 certificate, level 2 diploma, level 2 esol, level 2 essential skill, level 2 functional skill, level 2 key skills, level 2 national, level 2 nvq, national level 2, nqf level 2, nvq level 2, rqf level 2, scqf level 5, secondary education, diploma level 2, award level 3, esol 2, qcf level 2, level 2 qcf, esol level 2, functional skills level 2, essential skills level 2, qualification level 2, level 2 qualification, level 2 apprenticeship, cipd level 2, level 2 cipd, level 5 scqf, level 5 svq, svq level 5, certificate of secondary education
A-Level	a level qualification, as level qualification, access to higher education diploma, advanced apprenticeship, advanced diploma, advanced extension, advanced welsh baccalaureate, baccalaureate, btec award level 3, btec certificate level 3, btec diploma level 3, btec level 3, btec national, diploma programme, extended certificate, extended diploma, key skills level 3, level 3 award, level 3 certificate, level 3 esol, level 3 national, level 3 nvq, national level 3, nqf level 3, nvq level 3, ocr national, progression diploma, rqf level 3, scottish advanced higher, scottish baccalaureate, scottish higher, scqf level 6, diploma level 3, award level 3, esol 3, nvqs level 3, qcf level 3, level 3 qcf, functional skills level 3, essential skills level 3, qualification level 3, level 3 qualification, level 3 apprenticeship, cipd level 3, level 3 cipd, level 6 scqf, level 6 svq, svq level 6
Higher Certificate	hnc, certhe, certificate of higher education, higher apprenticeship, higher national certificate, level 4 award, level 4 certificate, level 4 diploma, level 4 national, level 4 nvq, national level 4, nqf level 4, nvq level 4, scqf level 7, diploma level 4, award level 4, nvqs level 4, qcf level 4, level 4 qcf, functional skills level 4, essential skills level 4, qualification level 4, level 4 qualification, level 4 apprenticeship, cipd level 4, level 4 cipd, level 7 scqf, level 7 svq, svq level 7

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Category	Words searched
Higher Diploma	hnd, btec professional, diphe, diploma of further education, diploma of higher education, foundation degree, higher diploma, higher national diploma, level 5 award, level 5 certificate, level 5 diploma, level 5 national, level 5 nvq, national level 5, nqf level 5, nvq level 5, rqf level 4, scqf level 8, diploma level 5, award level 5, nvqs level 5, qcf level 5, level 5 qcf, qualification level 5, level 5 qualification, cipd level 5, level 5 cipd, level 8 scqf, level 5 apprenticeship, level 5 professional, level 8 svq, svq level 8
Bachelor's Degree	bachelor, bachelors degree, btec advanced professional, degree apprenticeship, first degree, grad cert, grad dip, bachelor certificate, bachelor diploma, level 6 award, level 6 certificate, level 6 diploma, level 6 national, level 6 nvq, national level 6, nqf level 6, nvq level 6, ordinary degree, oxbridge ma, rqf level 5, scottish ma, scqf level 10, scqf level 9, ugad, undergraduate, higher education, diploma level 6, award level 6, nvqs level 6, qcf level 6, level 6 qcf, qualification level 6, level 6 qualification, cipd level 6, level 6 cipd, level 9 scqf, level 10 scqf, level 6 apprenticeship, level 6 professional, level 10 svq, svq level 10, have a degree, bsc or masters, degree or master, bsc or msc, ba or ma, honours degree
Master's Degree	award level 7, ba and ma, cipd level 7, diploma level 7, have a master, have a masters, level 11 scqf, level 11 svq, level 7 apprenticeship, level 7 award, level 7 certificate, level 7 cipd, level 7 diploma, level 7 national, level 7 nvq, level 7 professional, level 7 qcf, master of arts, master of business administration, master of engineering, master of law, master of mathematics, master of pharmacy, master of philosophy, master of physics, master of professional studies, master of public health, master of research, master of science, master of social work, master of studies, masters, masters by research, masters degree, national level 7, nqf level 7, nvq level 7, nvqs level 7, postgrad, postgraduate certificate, postgraduate certificate in education, postgraduate diploma, professional masters, qcf level 7, rqf level 6, rqf level 7, scqf level 11, svq level 11
Doctorate	doctor of philosophy, doctoral degree, doctorate, doctors degree, level 8 award, level 8 certificate, level 8 diploma, level 8 national, level 8 nvq, national level 8, nqf level 8, nvq level 8, phd, rqf level 8, scqf level 12, diploma level 8, award level 8, nvqs level 8, qcf level 8, level 8 qcf, cipd level 8, level 8 cipd, level 12 scqf, level 8 apprenticeship, level 8 professional, svq level 12, level 12 svq
Schedule	
Full-time	9 to 5, day shift, full time, standard hours, night shift
Part-time	4 hour, 6 hour, part time, 20 hour, 30 hour
Duration	
Temporary	apprenticeship, fixed term, freelance, interim, internship, project base, seasonal contract, self employed, short term contract, temporary, day to day contract, zero hours, casual contract
Permanent	long term, ongoing contract, permanent
Experience	

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Table OC1 – continued from previous page

Category	Words searched
Required	3 to 5 years, experienced, can demonstrate, demonstrable, essential experience, experience a minimum, experience for, experience in, experience is also, experience needed, experience of, experience preferably, experience required, experience working, extensive experience, for experienced, has experience, have experience, have previous experience, have some knowledge, have worked, minimum experience, months experience, need experience, past experience, practical experience, previous, prior experience, proven, recent experience, relevant experience, right experience, solid experience, strong experience, suitable experience, three to five years, two to three years, with experience, work experience, worked in similar, years experience, years of experience, you got experience, required experience, experience essential, experience is preferred
Not Required	a novice, do not need any, do not need experience, don't need experience, don't need previous, experience is not, experience not, lots of experience, need any experience, no essential experience, no experience, no formal experience, no formal qualifications or experience, no past experience, no previous, no prior, no work experience, not essential, not necessary, not need previous, with no experience, without any experience, don't need experience, not needed
Other	
Bonus	bonus, commission, incentive, profit sharing, reward scheme, stock options, variable pay, performance related, rewards packages, pay for performance
Teaching	can teach, teaching
Negotiable	on experience, negotiable salary, salary negotiable, hour negotiable, salary is negotiable, salary up to
Communication	working directly with, reporting, report into, your team leader, working with, working closely, under the direction, speaking to, directly to, work alongside, required by, keeping your, work closely, reports, step to, shadowing, managing the, assistance to the, work with, managed by, approval of, communicate with, supporting

Dictionaries built following (1) Goldberg (1992); (2) Costa Jr and McCrae (1992); (3) Goldberg (1992); (4) McCrae and John (1992); (5) John (1990); (6) Saucier (1994); (7) Heckman and Kautz (2012) were scrutinised as not to retrieve false positive words. Each word's context was analysed by sampling the dataset in 8 rounds of 100 advertisements per word. When the word's context was not related to personality traits and a straightforward amendment of the word was not possible, this was removed from the dictionary. Words that did not retrieve any match were also not considered.

Before searching for words, modifications were made to some words to avoid false positives and account for different variations of words. The list of changes done to the text is the following:

Personality Traits language:

”altruistic” for ”altruist ”, ”altruism” for ”altruist ”, ”amiable” for ”amiability ”, ”empathy” for ”empathetic ”, ”empathic” for ”empathetic ”, ”extraversion” for ”extraverted ”, ”extrovert” for ”extraverted”, ”extrovertion” for ”extraverted”, ”emotion” for ”emotional”, ”emotionally” for ”emotional”, ”reslient” for ”recilience, ”extroversion” for ”extraverted ”, ”extroverted” for ”extraverted ”, ”optimism” for ”optimist ”, ”optimistic” for ”optimist ”, ”outspoken” for ”outspoke ”, ”spontaneity” for ”spontaneous ”, ”ambition” for ”ambitious”, ”competence” for ”competent”, ”efficiency” for ”efficient”, ”persevere” for ”perseverance”, ”persevering” for ”perseverance”, ”persistence” for ”persistent”, ”hostile” for ”hostil ”, ”hostility” for ”hostil ”, ”creativity” for ”creative”, ”creatively” for ”creative”, ”innovation” for ”innovative”, ”innovations” for ”innovative”, ”innovating” for ”innovative”, ”innovator” for ”innovative”, ”intellect” for ”intellectual”, ”intellectually” for ”intellectual”, ”unconvention” for ”unconventional”, ”pro active” for ”proactive”, ”compassion” for ”comps”, ”compassionate” for ”comps”, ”an executive” for ”an exec”, ”analytics” for ”analysis”, ”and executive” for ”and exec”, ”sense of adventure” for ”adventurous”, ”humor” for ”humour”, ”anger” for ”angry”, ”anxiety” for ”anxious”, ”agreeable personality” for ”agreeableness”, ”being on time” for ”punctual”, ”co operative” for ”cooperative”, ”comfortable using” for ”confoble using”, ”appreciative of” for ”appreciativeof”, ”at ease” for ”atease”, ”curios” for ”curious”, ”owned cooperative” for ”owned cop”, ”your energy” for ”energetic”, ”with energy” for ”energetic”, ”and energy” for ”energetic”, ”encourages” for ”encorages”, ”encourage” for ”encorage”, ”encouraged” for ”encoraged”, ”encouragement” for ”encoragement”, ”discourage” for ”discorage”, ”encouraging” for ”encoraging”, ”confidential” for ”confidntial”, ”self confidence” for ”self confident”, ”confident employer” for ”employer”, ”confident in” for ”good in”, ”confidentiality” for ”confid”, ”confidentially” for ”confid”, ”confident at” for ”good at”, ”confidently” for ”good at”, ”originally posted” for ”posted”, ”original proof” for ”proof”, ”daring and apply” for ”apply”, ”darington” for ”daryngton”, ”daring to” for ”daryng to”, ”our daring” for ”our”, ”calendaring” for ”calendar”, ”expressive arts” for ”arts”, ”boldon” for ”bolon”, ”nwebold” for ”newbld”, ”not fearful” for ”notfearful”, ”never fearful” for ”notfearful”, ”be fearful” for ”notfearful”, ”being fearful” for ”notfearful”, ”careful to select” for ”to select”, ” careful ” for ” becareful ”, ”spontaneous holiday” for ”holiday”, ”to spontaneous” for ”to fast”, ”from spontaneous” for ”from fast”, ”spontaneous monthly” for ”monthly”, ”spontaneous moments” for ”moments”, ”various spontaneous” for ”various”, ”spontaneous events” for ”events”, ”are clever” for ”beclever”, ”be clever” for ”beclever”, ”, clever” for ” beclever”, ”clever,” for ”beclever ”, ”with clever” for ”beclever”,”some clever” for ”beclever”, ”clever and” for ”beclever”, ”and clever” for ”beclever”, ”flexible shift” for ”flx shift”, ”flexible shifts” for ”flx shifts”, ”flexible working” for ”flx work”, ”flexible work” for ”flx work”, ”flexible with” for ”flx with”, ”flexible basis” for ”flx basis”, ”flexible hours” for ”flx hours”, ”flexible benefit” for ”flx benefit”, ”flexible benefits” for ”flx benefits”, ”flexible about” for ”flx about”, ”hours are flexible” for ”hours are flx”, ”flexible annual” for ”flx annual”, ”flexible to” for ”flx to”, ”flexible for” for ”flx for”, ”is flexible” for ”is flx”, ”flexible hour” for ”flx hour”, ”fully flexible” for ”fully flx”, ”flexible schedule” for ”flx schedule”, ”flexible salary” for ”flx salary”, ”to flexible” for ”tp flx”, ”flexible new” for ”flx new”, ”flexible holiday” for ”flx holiday”, ”flexible way” for ”flx way”, ”flexible ways” for ”flx ways”, ”pay flexible” for ”flx pay”, ”work flexible” for ”flx work”, ”flexible contract” for ”flx contract”, ”flexible opportunities” for ”flx opportunities”, ”flexible options” for ”flx options”, ”flexible subcontract” for ”flx subcontract”, ”be flexible as” for ”flx as”, ”flexible between” for ”flx between”, ”flexible start” for ”flx start”, ”flexible rota” for ”flx rota”, ”weeks are flexible” for ”flx week”, ”flexible days” for ”flx days”, ”choose flexible” for ”choose flx”, ”entirely flexible” for ”entirely flx”, ”flexible within” for ”flx within”, ”for flexible” for ”for flx”, ”flexible additional” for ”flx additional”, ”flexible fixed” for ”flx fixed”, ”times flexible” for ”times flx”, ”location flexible” for ”location flx”, ”flexible on” for ”flx on”, ”flexible schedules” for ”flx schedules”, ”flexible reward” for ”flx reward”, ”flexible rewards” for ”flx rewards”, ”flexible 40” for ”flx 40”, ”flexible weekend” for ”flx weekend”, ”flexible vacancies” for ”flx vacancies”, ”scheme flexible” for ”scheme flx”, ”flexible in working” for ”flx in working”, ”flexible menu” for ”flx menu”, ”time flexible” for ”time flx”, ”flexible pm” for ”flx pm”, ”flexible saturdays” for ”flx saturdays”, ”flexible home” for ”flx home”, ”flexible team” : ”flx team”, ”cleaner flexible” : ”cleaner flx”, ”flexible and optional” : ”flx and optional”, ”flexible self employed” : ”flx self employed”, ”flexible in relation to” : ”flx in relation to”,”flexible and able to” : ”flx and able to”, ”flexible, full” : ”flx, full”, ”flexible full” : ”flx full”, ”flexible pattern” : ”flx pattern”, ”hours of work can be flexible” : ”flx hours”, ”flexible approach to remote working” : ”flx remote working”, ”flexible work” for ”flx

work", "flexible and guaranteed hours" : "flx hours", "flexible networks" : "flx networks", "flexible job" : "flx job", "benefits flexible" : "flx benefits", "weekdays flexible" : "flx weekdays", "flexible guaranteed hours" : "flx hours", "freelance and flexible" : "freelance and flx", "flexible with office" : "flx with office", "options flexible" : "options flx", "flexible core" : "flx core", "flexiblejob" : "flx job", "workflexible" : "work flx", "flexbile interview" : "flx interview", "flexible temporary" : "flx temporary", "flexible, part" : "flx, part", "flexible part" : "flx part", "flexible and home" : "flx and home", "days are flexible" : "days are flx", "flexible employment" : "flx employment", "that flexible" : "that flx", "flexible location" : "flx location", "flexible rotas" : "flx rotas", "flexible and remote" : "flx and remote", "flexible pension" : "flx pension", "esteemed" for "estemed", "be bright " for "bebright ", " bright and" for " bebright", "and bright" for "bebright", ", bright " for " bebright ", " bright, " for " bebright ", " a bright " for " bebright ", "brightest" for "bebright", "bright welcoming" for "welcoming", "bright future" for "future", "bright sunny" for "sunny", "be helpful" for "be help", "is helpful" for "is help", "experience helpful" for "experience help", "independent minded" for "independentmind", "independent mind" for "independentmind", "prefer variety" for "prefervariety", "straight forward" for "straightforward", "straightforward for you" for "forward for you", "straight forward for you" for "forward for you", "straightforward questionnaire" for "questionnaire", "very straightforward" for "very", "provides straight forward" for "provides", "straightforward design" for "design", "is straight forward" for "is", "friendly policies" for "frend policies", "family friendly" for "family frend", "friendliness" for "friendly", "user friendly" for "user friend", "friendly local" for "local", "dogfriendly" for "dog", "ecofriendly" for "eco", " gentle " for " begentle ", "solver" for "solving", "wide interest" for "wideinterests", "wider interest" for "wideinterests", "otherwise" for "other", "likewise" for "like", "connectwise" for "connect", "deliberateness" for "deliberate", "beneath" for "beneth", " neat " for " beneat ", " be neat " for " beneat ", "helpful tool" for "tool", "effective communication" for "communication skill", "optimisation" for "optimization", "optimisations" for "optimizations", "optimising" for "optimizing", "optimise" for "optimize", "optimised" for "optimized", "optimises" for "optimizes", "optimiser" for "optimizer", "empathy" for "empathetic", "enthusiasm" for "enthusiastic", "open, " for "openness ", "open and" for "openness ","and open " for "openness ","openmind" for "openness", "open mind" for "openness ","open minded" for "openness ", "openness close " for "close", "open and close" for "close", "open source" for "source", "openness source" for "source", "enthusiast" for "enthusiastic", "fun " for "fun ", "fun day" for "fus day", "fun park" for "fus park", "fun " for "funny ", "fun," for "funny ", "fun." for "funny ", "grounded in" for "in", "be grounded in " for "in", "grounded established" for "established", "a grounded" for " a ","grounded knowledge" for "knowledge", "grounded programming" for "programming","grounded perspective" for "perspective","grounded reflected" for "reflected","grounded solutions" for "solutions", "grounded and" for "begrounded", "and grounded" for "begrounded", ", grounded" for " begrounded", "grounded," for "begrounded ", "be grounded" for "begrounded", "grounded analysis" for "analysis", "well grounded" for "well", " a grounded," for " a ", ", grounded in" for " in", "begrounded in" for " in", ", grounded in " for "in ", "hard working" for "hardwork", "sociable hours" for "hours", "sociable working hours" for "working hours", "hard work" for "hardwork", "hardworker" for "hardwork", "hardworking" for "hardwork", "logic apps" for "apps", " logic " for " belogic ," logical " for " belogic ", "thorough knowledge" for "knowledge", "bethorough knowledge" for "knowledge", "and thorough" for "bethorough", ",thorough" for "bethorough", "thorough and" for "bethorough","be thorough" for "bethorough", "home working" for "homeworking", "sensitive information" for "information", "sensitive or contentious information" for "contentious information", "sensitive and restricted information" for "restricted information", "sensitive data" for "data", "sensitive company" for "company", "intellectual property" for "property", "intellectual dis" for "dis", "track record" for "experienced", "background" for "experienced", "outgoing mail" for "mail", "outgoing call" for "call", "pleasant attitude" for "pleasantpersonality", "pleasant personality" for "pleasantpersonality", "pleasant manner" for "pleasantpersonality", "be pleasant" for "pleasantpersonality", "pleasant and" for "pleasantpersonality", "kind of" for "jind of", "kinds of" for "jinds of", "kindly" for "jindly", "kind to" for "jind to", "tender minded" for "tendermind", " tender mind" for "tendermind", "good nature" for "goodnature", "tense situations":"tensesituations", "under control":"undercontrol", "competencies" for "skills", "competence in" for "skilled in", "competence with" for "skilled with", "soft heart" for "warm heart", "soft hearted" for "warm hearted", "competency" for "skilled", "competently" for "skilled", "organized" for "organised","show consideration" for "considerate",

”temperance” for ”tempered”, ”temperament” for ”tempered”, ”original solutions” for ”beoriginal”, ”original thinking” for ”beoriginal”, ”original ideas” for ”beoriginal”, ”original research” for ”beoriginal”, ”be original” for ”beoriginal ”, ”original, ” for ”beoriginal”, ”original writing” for ”beoriginal”, ”original content” for ”beoriginal”, ”is original” for ”beoriginal”, ”think originally” for ”beoriginal”, ”, original” for ” beoriginal”, ”original thought” for ”beoriginal”, ”original choice” for ”beoriginal”, ”originality” for ”beoriginal”, ”creating original” for ”beoriginal”, ”national trust” for ”nationaltr”, ”trusts” for ”trus”, ”trustee” for ”trusee”,”trustees” for ”trusees”, ”cooperative trust” for ”coperative”, ”nhs trust” for ”nhs”, ”trust us” for ”trus us”, ”trusted” for ”trused”, ” our trust” for ” our trus”, ”local trust” for ”local”, ”trust are” for ”are”, ”academy trust” for ”academy”, ”trust schools” for ”schools”, ”the trust” for ”the trus”, ”trust is” for ”is”, ”foundation trust” for ”foundation”, ”at trust” for ”at trus”, ”entrusted” for ”enrusted”, ”wildlife trust” for ”wildlife”, ”castle trust” for ”castle”, ”trust scheme” for ”scheme”, ”trust duties” for ”duties”, ”education trust” for ”education”, ”adventure trust” for ”adventure”, ”disabilities trust” for ”disabilities”, ”affinity trust” for ”affinity”, ”academies trust” for ”academies”, ”whole trust” for ”whole”, ”trust offers” for ”offers”, ”trust within” for ”within”, ”trust and corporate” for ”corporate”, ”trust duties” for ”duties”, ”trust and grants” for ”grants”, ”leisure trust” for ”leisure”, ”we trust” for ”we”, ”trust manages” for ”manages”, ”enterprise trust” for ”enterprise”, ”trust keith” for ”keith”, ”trust flagship” for ”flagship”, ”woodland trust” for ”woodland”, ”trust and fundraising” for ”fundraising”, ”care trust” for ”care”, ”health trust” for ”health”, ”rivers trust” for ”rivers”, ”transform trust” for ”transform”, ”transformation trust” for ”transformation”, ”are a trust ” for ”are ”, ”trust qvd” for ”qvd”, ”educational trust” for ”educational”, ”trust exposure” for ”exposure”, ”trust has” for ”has”, ”dean trust” for ”dean”, ”trust delivers” for ”delivers”, ”as a trust ” for ” ”, ”london trust” for ”london”, ”trustpilot” for ” ”, ”trust pilot” for ” ”, ”trust education” for ”education”, ”trust we believe” for ”we believe”, ”northern trust” for ”northern”, ”trust website” for ”website”, ”trust. org” for ”.org”, ”trust . org” for ”.org”, ”trust.org” for ”.org”, ”trust org” for ”org”, ”persist” for ”persistent”, ”prompt response” for ”prom response”, ”as prompt” for ”as prom”, ”prompting” for ”promting”, ”precisely what” for ”what”, ”as prompt” for ”as prom”, ”warm welcome” for ”welcome”, ”warm you” for ”you”, ”reliable vehicle” for ”rel veh”, ”representative of” for ”represent of”, ”responsible to” for ”res to”, ”responsible of” for ”res of”, ”responsible in” for ”res in”, ”responsible for” for ”res for”, ”reflective of” for ”ref of”, ”a reflective” for ”a ref”, ” sharp ” for ” besharp ”, ”socially responsible” for ”socially res”, ”unsociable” for ”unsoci”, ”are striving” for ”are str”, ”sympathetically converted” for ”converted”, ”sympathetically restored” for ”restored”, ”sympathetic refurbishment” for ”refurbishment”, ”sympathetic donors” for ”donors”.

Soft skills language:

”communication skills” for ”communicationskills”, ”critical thinking” for ”criticalthinking”, ”cultural sensitivity” for ”culturalsensitivity”, ”decision making” for ”decisionmaking”, ”language skills” for ”languageskills”, ”listening skills” for ”listeningskills”, ”problem solving” for ”problemsolving”, ”time management” for ”timemanagement”, ”attention to detail” for ”attentiontodetail”, ”organisation skills” for ”organisationskills”, ”conflict resolution” for ”conflictresolution”, ”soft skills” for ”softskills”,”maintain relationships” for ”networkingabilities”, ”network abilities” for ”networkingabilities”, ”working relationships” for ”networkingabilities”, ”network ability” for ”networkingabilities”, ”relationship building” for ”networkingabilities”, ”headhunting” for ”networkingabilities”, ”networking events” for ”networkingabilities”, ”maintain contacts” for ”networkingabilities”, ”external relationships” for ”networkingabilities”, ”strong relationships” for ”networkingabilities”, ”personal network” for ”networkingabilities”, ”ability to network” for ”networkingabilities”, ”attend events” for ”networkingabilities”, ”business relationships” for ”networkingabilities”, ”positive relationships” for ”networkingabilities”, ”good listening” for ”listening skills”, ”good listener” for ”listening skills”, ”active listening” for ”listening skills”, ”ability to listen” for ”listening skills”, ”resolve issues” for ”problemsolving”, ”problem solver” for ”problemsolving”, ”fluency” for ”fluent”, ”representation” for ”reprsentation”, ”representations” for ”reprsentation”, ”making decisions” for ”decisionmaking”, ”make decisions” for ”decisionmaking”, ”best decisions” for ”decisionmaking”, ”good decisions” for ”decisionmaking”, ”take decisions” for ”decisionmaking”,”informed decisions” for ”decisionmaking”, ”difficult decisions” for ”decisionmaking”, ”appropriate decisions” for ”decisionmaking”, ”resolving the complex problems” for ”problemsolving”, ”resolving complex problems” for ”problemsolv-

ing", "negotiating" for "negotiation", "negotiator" for "negotiation", "good spoken" for "communicationskills", "transparent communication" for "communicationskills", "good communication" for "communicationskills", "strong communication" for "communicationskills", "excellent communication" for "communicationskills", "written and verbal" for "communicationskills", "written and oral" for "communicationskills", "oral and written" for "communicationskills", "communication abilities" for "communicationskills", "great communication" for "communicationskills", "oral" for "communicationskills", "verbal, written" for "communicationskills", "technical writting" for "communicationskills", "verbal and written" for "communicationskills", "written" for "communicationskills", "verbal" for "communicationskills", "oral" for "communicationskills", "ability to communicate" for "communicationskills", "strong communicator" for "communicationskills", "excellent communicator" for "communicationskills", "good communicator" for "communicationskills", "communicator" for "communicationskills", "able to communicate" for "communicationskills", "transparent communication" for "communicationskills", "great communication" for "communicationskills", "can lead" for "leadership", "people management" for "leadership", "able to lead" for "leadership", "team leadership" for "supervisingateam", "team leadership" for "leading a team", "team management" for "leading a team", "cultural awareness" for "culturalsensitivity", "crosscultural" for "culturalsensitivity", "multicultural" for "culturalsensitivity", "multi cultural" for "culturalsensitivity", "culturally diverse" for "culturalsensitivity", "culturally aware" for "culturalsensitivity", "cultural diversity" for "culturalsensitivity", "asking questions" for "questioning", "european language" for "languageskills", "fluent in chinese" for "languageskills", "fluent in german" for "languageskills", "fluent in french" for "languageskills", "fluent in english" for "languageskills", "fluent in mandarin" for "languageskills", "fluent in spanish" for "languageskills", "fluent in portuguese" for "languageskills","fluent in polish" for "languageskills", "fluent speaking" for "languageskills", "fluent in dutch" for "languageskills", "fluent in russian" for "languageskills", "two languages" for "languageskills", "fluent in both" for "languageskills", "fluent in hebrew" for "languageskills", "fluent in speaking" for "languageskills", "fluent in italian" for "languageskills", "fluent in welsh" for "languageskills", "fluent in japanese" for "languageskills", "fluent in danish" for "languageskills", "fluent in english and" for "languageskills", "communicating in two langauges" for "languageskills", "fluent both english and" for "languageskills", "fluent in english as well as" for "languageskills", "fluent in another" for "languageskills", "fluent in the german" for "languageskills", "number of languages" for "languageskills", "fluent in turkish" for "languageskills", "coaching" for "mentoring", "provide guidance" for "mentoring", "multiple projects" for "multitask", "multiple tasks" for "multitask", "training and mentoring" for "training", "mentoring and training" for "training", "multi projects" for "multitask", "multitasker" for "multitask", "multitasking" for "multitask", "ability to motivate" for "mentoring","able to motivate" for "mentoring", "couching" for "mentoring", "problem resolution" for "problemsolving", "complex problems" for "problemsolving", "other languages" for "languageskills", "additional languages" for "languageskills", "multilingual" for "languageskills", "office skills" for "microsoftoffice", "excel skills" for "microsoftoffice", "ms packages" for "microsoftoffice", "ms package" for "microsoftoffice", "organisational skills" for "organisationskills", "organizational skills" for "organisationskills", "organisational capacity" for "organisationskills", "organisational abilities" for "organisationskills", "organisational ability" for "organisationskills", "excellent organisation" for "organisationskills", "organisation, coordination" for "organisationskills", "organisational and managements skills" for "organisationskills", "excellent organisational" for "organisationskills", "good organisational" for "organisationskills", "strong organisational" for "organisationskills", "organisational, " for "organisationskills ", "organisational and " for "organisationskills ", "strong managements skills" for "organisationskills", "excellent managements skills" for "organisationskills", "competent managements skills" for "organisationskills", "strong organisational" for "organisationskills", "prioritisation" for "organisationskills", "able to prioritise" for "organisationskills", "excellent organisation" for "organisationskills", "mentoring of the junior" for "mentoring", "mentoring junior" for "mentoring", "able to mentor the more junior members" for "mentoring", "mentoring our junior" for "mentoring", "mentoring our more junior" for "mentoring", "able to mentor junior" for "mentoring", "especially junior" for "mentoring", "mentor junior" for "mentoring", "mentoring and training of more junior" for "mentoring", "mentoring to junior" for "mentoring", "mentoring of junior" for "mentoring", "mentoring and coaching junior" for "mentoring", "mentor more junior" for "mentoring", "training of junior" for "mentoring", "eye for detail" for "attentiontodetail", "detail oriented" for "attentiontodetail", "detailed approach" for "attentiontodetail", "mentoring more

junior" for "mentoring", "support junior" for "mentoring", "managing junior" for "mentoring", "developing junior" for "mentoring", "developing more junior" for "mentoring", "support to more junior" for "mentoring", "support more junior" for "mentoring", "development of junior" for "mentoring", "work of junior" for "mentoring", "coaching more junior" for "mentoring", "coaching of junior" for "mentoring", "training junior" for "mentoring", "supervising junior" for "mentoring", "work as a team" for "teamwork", "work in a team" for "teamwork ", "work in teams" for "teamwork", "work well in a team" for "teamwork", "within a team" for "teamwork", " intern " for " internship ", "team work" for "teamwork", "team working" for "teamwork", "team environment" for "teamwork", "able to relate" for "teamwork", "team player" for "teamwork", "collaborative work" for "teamwork".

Hard skills language:

"content creation" for "contentcreation", "design skills" for "designskills", "data analysis" for "dataanalysis", "legal analysis" for "legalanalysis", "microsoft office" for "microsoftoffice", "project management" for "projectmanagement", "quality assurance" for "qualityassurance", "sales skills" for "salesskills", "user experience" for "userexperience", "video production" for "videoproduction", "customer service" for "customerservice", "business development" for "businessdevelopment", "computer literacy" for "computerliteracy", "commercial skills" for "commercialskills", "business opportunities" for "businessdevelopment", "database management" for "dataanalysis", "data analytics" for "dataanalysis", "commercial abilities" for "commercialskills", "commercially aware" for "commercialskills", "commercially awareness" for "commercialskills", "focus on commercial" for "commercialskills", "commercially astute" for "commercialskills", "timekeeper" for "timemanagement", "time keeping" for "timemanagement", "commercially minded" for "commercialskills", "commercially focused" for "commercialskills", "commercially astute" for "commercialskills", "commercially driven" for "commercialskills" , "commercially focussed" for "commercialskills", "word outlook" for "microsoftoffice", "excel and outlook" for "microsoftoffice", "microsoft outlook" for "microsoftoffice", "profficient in word" for "microsoftoffice","microsoft word" for "microsoftoffice", "proficient in excel" for "microsoftoffice", "microsoft excel" for "microsoftoffice", "power point" for "microsoftoffice", "powerpoint" for "microsoftoffice", "competent in microsoft" for "microsoftoffice", "microsoft packages" for "microsoftoffice", "office 365" for "microsoftoffice", "ms excel" for "microsoftoffice", "command of microsoft" for "microsoftoffice", "within budget" for "within budg", "word excel" for "microsoftoffice", "word and excel" for "microsoftoffice", "excel and word" for "microsoftoffice", "word, excel" for "microsoftoffice", "excel, word" for "microsoftoffice", "outlook and" for "microsoftoffice", "and outlook" for "microsoftoffice", "using outlook" for "microsoftoffice", "ms office" for "microsoftoffice", "microsoft teams" for "call", "microsoft programmes" for "microsoftoffice", "positive outlook" for "positive", "working with team" for "teamwork", "statistical" for "statistics", "pc literate" for "computerliteracy", "computer literate" for "computerliteracy", "using a computer" for "computerliteracy", "computer skills" for "computerliteracy", "pc skills" for "computerliteracy", "analytics" for "analysis", "investigative skills" for "analytical skills", "research" for "research skills", "cancer research" for "cancer","cancer research" for "cancer", "research organisations" for "organisations", "researching" for "research skills", "research organisation" for "organisations", "research programmes" for "programmes", "researcher" for "research", "good numeracy" for "analytical skills", "numeracy skills" for "analytical skills", "numeracy litarate" for "analytical skills", "conflict management" for "conflictresolution", "collaborative working" for "teamwork", "supervising a" for "supervisingateam", "managing a team" for "supervisingateam", "team management" for "supervisingateam", "part of a team" for "teamwork", "sunrise senior" for "sunrise", "seniors" for "old", "associated" for "asociated", "associates" for "asociates", "their representatives" for "their clients", "client representatives" for "client", "sales representative" for "representativerole salesskills", "sales representatives" for "representativerole salesskills", "customerservice" for "representativerole customerservice", "customer" for "representativerole customer", "marketing representative" for "marketing representativerole", "representative role" for "representativerole", "our consultants" for "our consult", "a consultant" for "a consult", "consultants available" for "consult available", "our qualified consultants" for "our consult", "agent role" for "agentrole", "sales agent" for "agentrole", "support agent" for "agentrole", "customerservice agent" for "agentrole", "customerservice agent" for "agentrole", "service agent" for "agentrole", "accountant" for "accounting", "directory" for "directo", "directorate" for "directo", "sales experience" for

”salesskills experienced”, ”sales person” for ”salesskills”, ”experience in sales” for ”salesskills experienced”, ”identify key sales” for ”salesskills”, ”sales oriented” for ”salesskills”, ”sales culture” for ”salesskills”, ”sales representative” for ”sales representative salesskills”, ”of sales” for ”salesskills”, ”sales plan” for ”salesskills”, ”sales plans” for ”salesskills”, ”sales associate” for ”salesskills”, ”sales role” for ”salesskills”, ”sales team” for ”salesskills team”, ”sales manager” for ”salesskills manager”, ”sales job” for ”salesskills”, ”sales target” for ”salesskills”, ”direct sales” for ”salesskills”, ”success in sales” for ”salesskills”, ”success as sales” for ”salesskills”, ”develop sales” for ”salesskills”, ”sales strategies” for ”salesskills”, ”sales targets” for ”salesskills”, ”target driven sales” for ”salesskills”, ”digital design” for ”designskills”, ”design engineer” for ”designskills”, ”graphic design” for ”designskills”, ”motion design” for ”designskills”, ”design tools” for ”designskills”, ”photoshop” for ”designskills”, ”motion graphics” for ”designskills”, ”digital art” for ”designskills”, ”adobe” for ”designskills”, ”design management” for ”designskills”, ”technical designs” for ”designskills”, ”photo shop” for ”designskills”, ”creating content” for ”contentcreation”, ”generate content” for ”contentcreation”, ”content generation” for ”contentcreation”, ”management of content” for ”contentcreation”, ”illustrator” for ”designskills”, ”content strategy” for ”contentcreation”, ”content development” for ”contentcreation”, ”content management” for ”contentcreation”, ”content creators” for ”contentcreation”, ”social content” for ”contentcreation”, ”managing content” for ”contentcreation”, ”website content” for ”contentcreation”, ”content writer” for ”contentcreation”, ”editorial management” for ”editing”, ”editorial skills” for ”editing”, ”editorial tasks” for ”editing”, ”editorial role” for ”editing”, ”editorial roles” for ”editing”, ”publishing skills” for ”editing”, ”editorial judgment” for ”editing”, ”editorial experience” for ”editing experienced”, ”editorial style” for ”editing”, ”editorial work” for ”editing”, ”proofreading” for ”editing”, ”proof reading” for ”editing”, ”drafting” for ”editing”, ”copy writing” for ”editing”, ”copywriting” for ”editing”, ”copywriter” for ”editing”, ”copywrite” for ”editing”, ”good editorial” for ”editing”, ”excellent editorial” for ”editing”, ”editorial assistant” for ”editing assistant”, ”editorial ability” for ”editing”, ”editorial judgment” for ”editing”, ”creating engaging content” for ”contentcreation”, ”develop content” for ”contentcreation”, ”developing content” for ”contentcreation”, ”writing, editorial” for ”editing”, ”legal text” for ”legalanalysis”, ”legal role” for ”legalanalysis”, ”legal abilities” for ”legalanalysis”, ”paralegal” for ”legalanalysis”, ”legal background” for ”legalanalysis”, ”legal qualifications” for ”legalanalysis”, ”legal qualification” for ”legalanalysis”, ”legal excecutive” for ”legal excecutive legalanalysis”, ”lawyer” for ”lawyer legalanalysis”, ”reading leases” for ”legalanalysis”, ”lawyers” for ”lawyers legalanalysis”, ”reading leases” for ”legalanalysis”, ”legal clerk” for ”legalanalysis”, ”provide legal” for ”legalanalysis”, ”legal billing” for ”legalanalysis”, ” engineer ” for ” engineering ”, ”legal documentation” for ”legalanalysis”, ”legal approach” for ”legalanalysis”, ”caseload” for ”legalanalysis”, ”understand our customers” for ”customerservice”, ”customer focused” for ”customerservice”, ”customer focus” for ”customerservice”, ”identify customer needs” for ”customerservice”, ”understanding the customers” for ”customerservice”, ”needs of the customers” for ”customerservice”, ”manage customer” for ”customerservice”, ”customer orders” for ”customerservice”, ”interact with customers” for ”customerservice”, ”quality checking” for ”qualityassurance”, ”quality checks” for ”qualityassurance”, ”quality check” for ”qualityassurance”, ”ensuring quality” for ”qualityassurance”, ”search engine” for ”software”, ”sem ” for ”software”, ”seo ” for ”software ”, ”web development” for ”software”, ”programming” for ”software”, ”mobile app development” for ”software”, ”coding” for ”software”, ”web design” for ”software”, ”cybersecurity” for ”software”, ”data mining” for ”dataanalysis”, ”data visualization” for ”dataanalysis”, ”database skills” for ”dataanalysis”, ”data to” for ”dataanalysis”, ”quantitative analysis” for ”dataanalysis ”, ”data modelling” for ”dataanalysis”, ”datasets” for ”dataanalysis”, ”interpret data” for ”dataanalysis”, ”data reporting” for ”dataanalysis”, ”it skills” for ”dataanalysis”, ”it literate” for ”dataanalysis”, ”it literacy” for ”dataanalysis”, ”variety of data” for ”dataanalysis”, ”data skills” for ”dataanalysis”, ”data models” for ”dataanalysis”, ”data sets” for ”dataanalysis”, ”data analyst” for ”dataanalysis analyst”, ”data monitoring” for ”dataanalysis”, ”data scientist” for ”dataanalysis”, ”data skill” for ”dataanalysis”, ”handling data” for ”dataanalysis”, ”complex data” for ”dataanalysis”, ”big data” for ”dataanalysis”, ”data science” for ”dataanalysis”, ”data engineer” for ”dataanalysis”, ”collate data” for ”dataanalysis”, ”data interpretation” for ”dataanalysis”, ”interpretation of data” for ”dataanalysis”, ”creation of data” for ”dataanalysis”, ”relational databases” for ”dataanalysis”, ”database management” for ”dataanalysis”, ”database utilisation” for ”dataanalysis”, ”data management” for ”dataanalysis”, ”database administration” for ”dataanalysis”, ”comparative data” for ”dataanalysis”, ”process data” for ”dataanalysis”, ”raw data” for

”dataanalysis”, ”data insight” for ”dataanalysis”, ”customer data” for ”dataanalysis”, ”customer databases” for ”dataanalysis”, ”company databases” for ”dataanalysis”, ”accurate databases” for ”dataanalysis”, ”accurate data” for ”dataanalysis”, ”data driven” for ”dataanalysis”, ”data entry” for ”dataanalysis”, ”data reviews” for ”dataanalysis”, ”review data” for ”dataanalysis”, ”project coordination” for ”projectmanagement”, ”project planning ” for ”projectmanagement”, ”contract drafting” for ”projectmanagement”, ”producing video” for ”videoproduction”, ”photography” for ”videoproduction”, ”video editing” for ”videoproduction”, ”video creation” for ”videoproduction”, ”videographer” for ”videoproduction”, ”film maker” for ”videoproduction”, ”creating videos” for ”videoproduction”, ”video experience” for ”videoproduction experienced”, ”filming” for ”videoproduction”, ”social interaction” for ”interpersonal”, ”social skills” for ”interpersonal”, ”people skills” for ”interpersonal”, ”people person” for ”interpersonal”, ”clerical” for ”administrative”,” ux ” for ”userexperience”, ” ux, ” for ”userexperience”, ” ux.” for ”userexperience”, ”statistics” for ”statistical”, ”stake holder” for ”stakeholder”, ”client management” for ”stakeholder management”, ”supply chain management” for ”product management”.

AI language:

” ai ” for ” artificialintelligence ”, ” xai ” for ” artificialintelligence ”, ” ai,” for ” artificialintelligence ”, ”, ai ” for ” artificialintelligence ”, ”artificial intelligence” for ” artificialintelligence ”, ” gpt ” for ”generativepretrainedtransformer”, ”generative models” for ”generativepretrainedtransformer”, ”generative pretrained” for ”generativepretrainedtransformer”, ”generative pre” for ”generativepretrainedtransformer”, ”ai chatbox” for ”aichatbox”, ”artificialintelligence chatbox” for ”aichatbox”, ”machine learning” for ”machinelearning”, ” ml ” for ” machinelearning ”, ”, ml ” for ” machinelearning ”, ” ml,” for ” machinelearning ”, ”cross domain learning” for ”machinelearning”, ”cross - domain learning” for ”machinelearning”, ”reinforcement learning” for ”machinelearning”, ”mlpack” for ”machinelearning”, ”federated learning” for ”machinelearning”, ”pybrain” for ”machinelearning”, ”shot learning” for ”machinelearning”, ”hugging face” for ”huggingface”, ”multi task learning” for ”machinelearning”, ”multi - task learning” for ”machinelearning”, ”residual learning” for ”machinelearning”, ”mediapipe” for ”machinelearning”, ”media pipe” for ”machinelearning”, ”computer vision” for ”computervision”, ”image segmentation” for ”computervision”, ”vision processing” for ”computervision”, ”robotic process automation” for ”roboticprocessautomation”, ” rpa ” for ” roboticprocessautomation ”, ”autonomous robotics” for ”roboticprocessautomation”, ”intelligent automation” for ”roboticprocessautomation”, ”machine vision” for ”machinevision”, ” bert ” for ”bidirectionalencoder”, ”bidirectional encoder” for ”bidirectionalencoder”, ”, bert” for ” bidirectionalencoder”, ” bert,” for ” bidirectionalencoder”, ”deep learning” for ”deeplearning”, ”residual network” for ”deeplearning”, ” resnet ” for ”deeplearning”, ”fastai” for ”deeplearning”, ”nd4j” for ”deeplearning”, ”microsoft cognitive toolkit” for ”deeplearning”, ” cntk ” for ”deeplearning”, ”deep embeddings” for ”deeplearning”, ”deep embedding” for ”deeplearning”, ”kalman filtering” for ”kalmanfiltering”, ”kalman filter” for ”kalmanfiltering”, ”virtual agents” for ”virtualagents”, ”support vector machines” for ”supportvectormachines”, ” svm ” for ” supportvectormachines ”, ” svm,” for ” supportvectormachines ”, ”, svm ” for ” supportvectormachines ”, ”image recognition” for ”imagerecognition”, ”point cloud library” for ”imagerecognition”, ”image fusion” for ”imagefusion”, ”natural language toolkit” : ”naturallanguagetoolkit”, ”nltk” : ”naturallanguagetoolkit”, ” maximum entropy ” for ” naturallanguageprocessing ”, ” nlp ” for ”naturallanguageprocessing”, ”, nlp ” for ” naturallanguageprocessing ”, ” nlp, ” for ” naturallanguageprocessing ”, ” corenlp” for ” naturallanguageprocessing ”, ” opennlp” for ” naturallanguageprocessing ”,” apache uima” for ” naturallanguageprocessing ”, ”natural language processing” for ”naturallanguageprocessing”, ” gensim ” for ” naturallanguageprocessing ”, ”document embeddings” for ”naturallanguageprocessing”, ”doc2vec” for ”naturallanguageprocessing”, ”speech recognition” for ”speechrecognition”, ”sound recognition” for ”speechrecognition”, ”scikit learn” for ”scikitlearn”, ”scikit - learn” for ”scikitlearn”, ”pattern recognition” for ”patternrecognition”, ”pattern detection” for ”patternrecognition”,”discriminant” for ”discriminantanalysis”, ”object recognition” for ”objectrecognition”, ”object tracking” for ”objectrecognition”, ”neural networks” for ”neuralnetworks”, ”convolutional” for ”neuralnetworks”, ” lstm ” for ”longshorttermmemory”, ” lstms ” for ”longshorttermmemory”, ” , lstm ” for ”longshorttermmemory”, ” , lstms ” for ”longshorttermmemory”, ” lstm, ” for ”longshorttermmemory”, ” lstms, ” for ”longshorttermmemory”, ”recurrent network” for ”neuralnetworks”, ”recurrent networks” for ”neuralnetworks”, ”siamese networks” for ”neuralnetworks”, ” rnn ” for ” rnns ”, ”dnn techniques” for ”neural-

networks", "supervised learning" for "supervisedlearning", "gaussian processes" for "supervisedlearning", "text mining" for "textmining", "information extraction" for "textmining", "unstructured information management" for "textmining", "support vector machines" for "supportvectormachines", "unsupervised learning" for "unsupervisedlearning", "image processing" for "imageprocessing","jung framework" for "jungframework", "recommender systems" for "recommendersystems", "random forests" for "randomforests", "random forest" for "randomforests", "latent semantic analysis" for "latentsemanticanalysis", "sentiment analysis" for "sentimentanalysis", "emotion detection" for "sentimentanalysis", "tensor flow" for "tensorflow", "opinion mining" for "opinionmining", "hidden markov model":"hiddenmarkovmodel", "latent dirichlet allocation" for "latentdirichletallocation", " lda " for " latentdirichletallocation ", " lda," for " latentdirichletallocation ", ", lda" for " latentdirichletallocation ", "predictive models" for "predictivemodels", "predictive modelling" for "predictivemodels", "kernel methods" for "kernelmethods", "kernel method" for "kernelmethods", "gradient descent" for "gradientdescent", " sgd " for "gradientdescent", "gradient boosting" for "gradientboosting", "gradient boost" for "gradientboosting", "gradient boosted" for "gradientboosting", "machine translation" for "machinetranslation", "seq2seq" for "machinetranslation", "sequence - to - sequence" for "machinetranslation", "sequence to sequence" for "machinetranslation", "sentiment classification" for "classificationalgorithms", "classification algorithms" for "classificationalgorithms", "classification algorithm" for "classificationalgorithms", " gans " for "generativeadversarialnetworks", ", gans " for "generativeadversarialnetworks", " gans, " for "generativeadversarialnetworks", " generative adversarial networks" for "generativeadversarialnetworks", "quantum computing" for "quantumcomputing", " nlu " for "naturallanguageunderstanding", ", nlu " for " naturallanguageunderstanding ", " nlu, " for " naturallanguageunderstanding ", "natural language understanding" for "naturallanguageunderstanding", " nlg " for "naturallanguagegeneration", ", nlg " for " naturallanguagegeneration ", " nlg, " for " naturallanguagegeneration ", " natural language generation " for "naturallanguagegeneration", "nlg health " for "health", "nlg are" for "are", "cloud technology" for "cloudtechnology", "cloud technologies" for "cloudtechnology", "ensemble methods" for "ensemblemethods", "ensemble learning" for "ensemblemethods", "decision trees" for "decisiontrees", " chaid" for "decisiontrees", "tree based lerners" for "decisiontrees", "tree - based lerners" for "decisiontrees", "tree - based classifiers" for "decisiontrees", "tree based classifiers" for "decisiontrees", "k means clustering" for "clusteringalgorithms", "hierarchical clustering" for "clusteringalgorithms", "k - means clustering" for "clusteringalgorithms", "clustering algorithms" for "clusteringalgorithms", "optical character recognition" for "opticalcharacterrecognition", "optical flow" for "opticalcharacterrecognition", "named entity recognition" for "namedentityrecognition", "entity linking" for " naturallanguageprocessing ", "topic modeling" for "topicmodelling", "topic modelling" for "topicmodelling","topic detection" for "topicmodelling", "text classification" for "classificationalgorithms", "similarity models" for "topicmodelling", "a b testing" for "abtesting", "a . b testing" for "abtesting", "dimensionality reduction" for "dimensionalityreduction", " umap " for "dimensionalityreduction", "uniform manifold" for "dimensionalityreduction", "dimension reduction" for "dimensionalityreduction", " tsne " for "dimensionalityreduction", "t - sne" for "dimensionalityreduction", " t sne " for "dimensionalityreduction", "feature engineering" for "featureengineering", "hyperparameter tuning" for "hyperparametertuning", "hyper parameter tuning" for "hyperparametertuning", "data augmentation" for "dataaugmentation", "model compression" for "modelcompression", "naive bayes" for "bayesianmethods", "naïve bayes" for "bayesianmethods", "naï ve bayes" for "bayesianmethods", "bayesian" for "bayesianmethods", "knowledge distillation" for "knowledgedistillation", "density based spatial clustering" for "densitybasedspatialclustering","density - based spatial clustering" for "densitybasedspatialclustering", " dbscan " for "densitybasedspatialclustering", "computational linguistics" : "computationallinguistics", "expectation maximization" : "expectationmaximization", "caffe deep learning framework":"caffedeeplearningframework".

Contract language:

"conducted remote" for "conductedremote", "fully remote" for "fullyremote", "home based" for "homebased", "home work" for "homework", "job is virtual" for "jobisvirtual", "location is remote" for "locationisremote", "location remote" for "locationremote", "position is remote" for "positionisremote", "position remote" for "positionremote", "remote work" for "remotework", "role will be remote" for "rolewillberemote", "virtual job" for "virtualjob", "work from home" for

”workfromhome”, ”work home” for ”workhome”, ”work remote” for ”workremote”, ”working remote” for ”workingremote”, ”flx with office” for ”flxwithoffice”, ”remote position” for ”remoteposition”, ”role is remote” for ”roleisremote”, ”remote basis” for ”remotebasis”, ”hybrid work” for ”hybridwork”, ”hybrid remote” for ”hybridremote”, ”vice president” for ”vicepresident”, ”project lead” for ”projectlead”, ”team lead” for ”teamlead”, ”supervising a team” for ”supervisingateam”, ”trial period” for ”trialperiod”, ”entry level” for ”entrylevel”, ”tele commuting” for ”telecommuting”, ”tele work” for ”telework ”, ”no telecommuting” for ”no”, ”does not offer teleworker” for ”no”, ”no taking work home” for ”no”, ”take work home” for ”no”, ”teaching assistant” for ”academic assistant”, ”secure position” for ”permanent position”, ”temporary to permanent” for ”temporary”, ”not permanent” for ”temporary”, ”lead to permanent” for ”temporary”, ”not temporary” for ”permanent”, ”getting a permanent” for ”temporary”, ”seasonal contract” for ”seasonal contract”, ”seasonal hours” for ”seasonal contract”, ”decommissions” for ”decom”, ”commissioning” for ”comis”, ”rewards and benefits” for ”rewards packages”, ”pay - for - performance” for ”pay for performance”, ”share options” for ”stock options”,”based on performance” for ”pay for performance”, ”short term, long term” for ”short term contract, long term contract”, ”short term , long term” for ”short term contract, long term contract”, ”short term and long term” for ”short term contract and long term contract”, ”short term basis” for ”short term contract”,”short term or long term” for ”short term contract or long term contract”, ”contract is short term” for ”short term contract”, ”or short term” for ”or short term contract”, ” temp ” for ” temporary ”, ”long term conditions” for ”conditions”, ”long term condition” for ”condition”, ”long term strategy” for ”strategy”, ”long term strategic” for ”strategic”, ”long term satisfaction” for ”satisfaction”, ”long term residential” for ”residential”, ”long term financial” for ”financial”, ”long term happiness” for ”happiness”, ”long term need” for ”need”, ”long term health” for ”health”, ”temporary ongoing” for ”temporary contract”, ”ongoing contract subject” for ”temporary contract”, ” perm ” for ” permanent ”, ” perm, ” for ” permanent ”, ” ,perm ” for ” permanent ”, ” temp, ” for ” temporary ”, ” ,temp ” for ” temporary ”, ”experience is needed” for ”experience needed”, ”years of experience” for ”years experience”, ” yrs ” for ” years ”, ”don” t need experience” for ”dont need experience”, ”don’ t need previous” for ”dont need experience” , ”experience is essential” for ”experience essential”, ”not essential” for ”no essential”, ”not essential” for ”no essential”, ”no formal qualifications” for ”no formal qualification”, ”casual work” for ”casual contract”, ”casual basis” for ”casual contract”, ”casual work” for ”casual contract”, ”casual hours” for ”part time”, ”casual shifts” for ”casual contract”, ”40 hours” for ”full time”, ”40 hour” for ”full time”, ”38.5 hour” for ”full time”, ”38.5 hours” for ”full time”,”38 hours” for ”full time”, ”38 hour” for ”full time”,”37 hours” for ”full time”, ”37 hour” for ”full time”, ”9 to 17” for ”9 to 5”, ”9 17” for ”9 to 5”, ”9 . 17” for ”9 to 5”, ”profit sharing” for ”profitsharing”, ”reward scheme” for ”rewardscheme”, ”stock options” for ”stockoptions”, ”variable pay” for ”variablepay”, ”performance related” for ”performancerelated”, ”rewards packages” for ”rewardspackages”, ”pay for performance” for ”payforperformance”, ”fixed term” for ”fixedterm”, ”short term contract” for ”shorttermcontract”, ”day to day contract” for ”daytodaycontract”, ”zero hours” for ”zerohours”, ”casual contract” for ”casualcontract”, ”long term” for ”longterm”, ”ongoing contract” for ”ongoingcontract”, ”4 hour” for ”4hour”, ”4 hours” for ”4hour”,”6 hour” for ”6hour”, ”6 hours” for ”6hour”,”20 hour” for ”20hour”, ”20 hours” for ”20hour”,”30 hour” for ”30hour”, ”9 to 5” for ”9to5”, ”full time” for ”fulltime”, ”standard hours” for ”standardhours”, ”day shift” for ”dayshift”, ”night shift” for ”nightshift”, ”3 to 5 years” for ”3to5years”, ”experience a minimum” for ”experienceaminimum”, ”experience for” for ”experiencefor”, ”experience in” for ”experiencein”, ”experience is also” for ”experienceisalso”, ”experience needed” for ”experienceneeded”, ”experience of” for ”experienceof”, ”experience preferably” for ”experiencepreferably”, ”experience required” for ”experiencerequired”, ”experience working” for ”experienceworking”, ”extensive experience” for ”extensiveexperience”, ”have experience” for ”haveexperience”, ”have previous experience” for ”havepreviousexperience”, ”have some knowledge” for ”havesomeknowledge”, ”have worked” for ”haveworked”, ”can teach” for ”canteach”, ”minimum experience” for ”minimumexperience”, ”months experience” for ”monthsexperience”, ”need experience” for ”needexperience”, ”past experience” for ”pastexperience”, ”practical experience” for ”practicalexperience”, ”prior experience” for ”priorexperience”, ”recent experience” for ”recentexperience”, ”relevant experience” for ”relevantexperience”, ”right experience” for ”rightexperience”, ”solid experience” for ”solidexperience”, ”strong experience” for ”strongexperience”, ”suitable experience” for ”suitableexperience”, ”three to five years” for ”threetoFiveyears”, ”two to three years” for ”twotothreeyears”, ”work experience” for ”workexperience”, ”worked in similar” for ”workedinsimilar”,

”required experience” for ”requiredexperience”, ”experience essential” for ”experienceessential”, ”experience is preferred” for ”experienceispreferred”, ”do not need any” for ”donotneedany”, ”do not need experience” for ”donotneedexperience”, ”experience is not” for ”experienceisnot”, ”no essential experience” for ”noessentialexperience”, ”no formal experience” for ”noformalexperience”, ”no formal qualifications or experience” for ”noformalqualificationsorexperience”, ”no past experience” for ”nopastexperience”, ”no work experience” for ”noworkexperience”, ”with no experience” for ”withnoexperience”, ”without any experience” for ”withoutanyexperience”, ”working directly with” for ”workingdirectly”, ”don t need experience” for ”dontneedexperience”, ”not needed” for ”notneeded”, ”working directly with” for ”workingdirectlywith”, ”report into” for ”reportinto”, ”your team leader” for ”yourteamleader”, ”working with” for ”workingwith”, ”working closely” for ”workingclosely”, ”under the direction” for ”underthedirection”, ”speaking to” for ”speakingto”, ”directly to” for ”directlyto”, ”work alongside” for ”workalongside”, ”required by” for ”requiredby”, ”keeping your” for ”keepingyour”, ”work closely” for ”workclosely”, ”step up to” for ”stepto”, ”managing the” for ”managingthe”, ”assistance to the” for ”assistancetothe”, ”work with” for ”workwith”, ”managed by” for ”managedby”, ”approval of” for ”approvalof”, ”communicate with” for ”communicatewith”, ”on experience” for ”onexperience”, ”negotiable salary” for ”negotiablesalary”, ”salary negotiable” for ”salarynegotiable”, ”hour negotiable” for ”hournegotiable”, ”salary is negotiable” for ”salaryisnegotiable”, ”salary up to” for ”salaryupto”.

Education language:

”at level” for ”level”, ”at either level” for ”level”, ” masters of ” for ” masterof ”,” master in ” for ” masterof ”, ” masters in ” for ” masterof ”, ”hour ” for ”hours”, ”skill ” for ”skills ”, ”cipd qualified” for ”cipd”, ”qualified to level” for ”level”, ”qualifications” for ”qualification”, ”mcipd” for ”cipd”, ”scqf level 1 ” for ”scqflevel2 ”, ”svq level 1 ” for ”svqlevel2 ”, ” elc ” for ”entrylevelcertificate”, ” cse ”:” certificateofsecondaryeducation ”, ”a levels” for ”alevelqualification”, ”as levels” for ”aslevelqualification”, ”a level science” for ”alevelqualification”, ”a level in math” for ”alevelqualification”, ”a level in english” for ”alevelqualification”, ”as level in math” for ”aslevelqualification”, ”as level in english” for ”aslevelqualification”, ”as level in science” for ”aslevelqualification”, ” certhe ” for ” certificateofhighereducation ”, ”post graduate” for ”postgraduate”, ” graduate ” for ” bachelor ”, ” bsc ” for ” bachelor ”, ” b . ed ” for ” bachelor ”, ” b. ed ” for ” bachelor ”, ” ba hons ” for ” bachelor ”, ” b eng ” for ” bachelor ”, ” b. eng ” for ” bachelorofengineering ”, ” beng ” for ” bachelorofengineering”, ” mba ” for ” masterofbusinessadministration ”, ” ll. b ” for ” bacheloroflaws”, ” meng ” for ” mastersofengineering ”, ” mmath ” for ” mastersofmathematics ”, ” mphil ” for ”masterofphilosophy ”, ” mpsych ” for ” masterofphysics ”, ” mprof ” for ” masterofprofessionalstudies ”, ” mres ” for ” masterofresearch ”, ” msc ” for ” masterofscience ”, ” msc, ” for ” masterofscience ”, ” msci ” for ” masterofscience ”, ” msci, ” for ” masterofscience ”, ” mst ” for ” masterofstudies ”, ” m. sc ” for ” masterofscience ”, ” m . sc ” for ” masterofscience ”, ” llm ” for ” masteroflaw ”, ” ll . m ” for ” masteroflaw ”, ” mpharm ” for ” mastersofpharmacy ”, ” mchem ” for ” masterofchemistry ”, ” ms ” for ” masterofscience ”, ” msw ” for ” masterofsocialwork ”, ” pcge ” for ” postgraduatecertificateineducation ”, ” pcg ” for ” postgraduatecertificate ”, ” pgd ” for ” postgraduatediploma ”, ” pgcert ” for ” postgraduatecertificate ”, ” pg cert ” for ” postgraduatecertificate ”, ”ph. d” for ” phd ”, ”ph d” for ” phd ”, ” ma programme ” for ” masterofart ”, ” ma programme ” for ” masterofart ”, ” ma in english ” for ” masterofart ”, ”hold an ma ” for ” masterofart ”, ” ma degree ” for ” masterofart ”, ”honors degree” for ”honoursdegree”, ”self employed” for ”selfemployed”, ”self employment” for ”selfemployed”, ”seasonal contract” for ”seasonalcontract”, ”basic functional skills” for ”basicfunctionalskills”, ”basic skills qualification” for ”basicskillsqualification”, ”entry level certificate” for ”entrylevelcertificate”, ”entry level functional skills” for ”entrylevelfunctionalskills”, ”entry level qualifications” for ”entrylevelqualifications”, ”entry nqf level” for ”entrynqflevel”, ”essential skills wales” for ”essentialskillswales”, ”level 1 scqf” for ”level1scqf”, ”level 1 svq” for ”level1svq”, ”level 2 scqf” for ”level2scqf”, ”level 2 svq” for ”level2svq”, ”level 3 scqf” for ”level3scqf”, ”level 3 svq” for ”level3svq”, ”no education requirements” for ”noeducationrequirements”, ”no formal qualification” for ”noformalqualification”, ”no previous experience or qualification” for ”nopreviousexperienceorqualification”, ”no previous qualification” for ”nopreviousqualification”, ”no qualification” for ”noqualification”, ”scottish national 1” for ”scottishnational1”, ”scottish national 2” for ”scottishnational2”,

”scottish national 3” for ”scottishnational3”, ”scqf level 2” for ”scqflevel2”, ”scqf level 3” for ”scqflevel3”, ”skills for life” for ”skillsforlife”, ”svq level 2” for ”svqlevel2”, ”svq level 3” for ”svqlevel3”, ”no formal education” for ”noformaleducation”, ”btec award” for ”btecaward”, ”btec first” for ”btecfirst”, ”btec level 1” for ”bteclevel1”, ”certificate of secondary education” for ”certificateofsecondaryeducation”, ”first certificate” for ”firstcertificate”, ”foundation welsh baccalaureate” for ”foundationwelshbaccalaureate”, ”key skills level 1” for ”keyskillslevel1”, ”level 1 award” for ”level1award”, ”level 1 certificate” for ”level1certificate”, ”level 1 diploma” for ”level1diploma”, ”level 1 esol” for ”level1esol”, ”level 1 essential skills” for ”level1essentialskills”, ”level 1 functional skills” for ”level1functionalskills”, ”level 1 key skills” for ”level1keyskills”, ”level 1 national vocational qualification” for ”level1nationalvocationalqualification”, ”level 1 nvq” for ”level1nvq”, ”nqf level 1” for ”nqflevel1”, ”nvq level 1” for ”nvqlevel1”, ”rqf level 1” for ”rqflevel1”, ”scqf level 4” for ”scqflevel4”, ”diploma level 1” for ”diplomalevel1”, ”national level 1” for ”nationallevel1”, ”qualification level 1” for ”qualificationlevel1”, ”level 1 qualification” for ”level1qualification”, ”level 1 qcf” for ”level1qcf”, ”functional skills level 1” for ”functionalskillslevel1”, ”essential skills level 1” for ”essentialskillslevel1”, ”level 1 apprenticeship” for ”level1apprenticeship”, ”cipd level 1” for ”cipdlevel1”, ”level 1 cipd” for ”level1cipd”, ”level 4 scqf” for ”level4scqf”, ”level 4 svq” for ”level4svq”, ”svq level 4” for ”svqlevel4”, ”btec level 2” for ”bteclevel2”, ”general certificate of education” for ”generalcertificateofeducation”, ”intermediate apprenticeship” for ”intermediateapprenticeship”, ”intermediate welsh baccalaureate” for ”intermediatewelshbaccalaureate”, ”key skills level 2” for ”keyskillslevel2”, ”level 2 award” for ”level2award”, ”level 2 certificate” for ”level2certificate”, ”level 2 diploma” for ”level2diploma”, ”level 2 esol” for ”level2esol”, ”level 2 essential skill” for ”level2essentialskill”, ”level 2 functional skill” for ”level2functionalskill”, ”level 2 key skills” for ”level2keyskills”, ”level 2 national” for ”level2national”, ”level 2 nvq” for ”level2nvq”, ”national level 2” for ”nationallevel2”, ”nqf level 2” for ”nqflevel2”, ”nvq level 2” for ”nvqlevel2”, ”rqf level 2” for ”rqflevel2”, ”scqf level 5” for ”scqflevel5”, ”secondary education” for ”secondaryeducation”, ”diploma level 2” for ”diplomalevel2”, ”award level 3” for ”awardlevel3”, ”esol level 2” for ”esollevel2”, ”functional skills level 2” for ”functionalskillslevel2”, ”essential skills level 2” for ”essentialskillslevel2”, ”qualification level 2” for ”qualificationlevel2”, ”level 2 qualification” for ”level2qualification”, ”level 2 apprenticeship” for ”level2apprenticeship”, ”cipd level 2” for ”cipdlevel2”, ”level 2 cipd” for ”level2cipd”, ”level 5 scqf” for ”level5scqf”, ”level 5 svq” for ”level5svq”, ”svq level 5” for ”svqlevel5”, ”certificate of secondary education” for ”certificateofsecondaryeducation”, ”a level qualification” for ”alevelqualification”, ”as level qualification” for ”aslevelqualification”, ”access to higher education diploma” for ”accesstohighereducationdiploma”, ”advanced apprenticeship” for ”advancedapprenticeship”, ”advanced diploma” for ”advanceddiploma”, ”advanced extension” for ”advancedextension”, ”advanced welsh baccalaureate” for ”advancedwelshbaccalaureate”, ”btec award level 3” for ”btecawardlevel3”, ”btec certificate level 3” for ”bteccertificatelevel3”, ”btec diploma level 3” for ”btecdiplomalevel3”, ”btec level 3” for ”bteclevel3”, ”btec national” for ”btecnational”, ”diploma programme” for ”diplomaprogramme”, ”extended certificate” for ”extendedcertificate”, ”extended diploma” for ”extendeddiploma”, ”key skills level 3” for ”keyskillslevel3”, ”level 3 award” for ”level3award”, ”level 3 certificate” for ”level3certificate”, ”level 3 esol” for ”level3esol”, ”level 3 national” for ”level3national”, ”level 3 nvq” for ”level3nvq”, ”national level 3” for ”nationallevel3”, ”nqf level 3” for ”nqflevel3”, ”nvq level 3” for ”nvqlevel3”, ”ocr national” for ”ocrnational”, ”progression diploma” for ”progressiondiploma”, ”rqf level 3” for ”rqflevel3”, ”scottish advanced higher” for ”scottishadvancedhigher”, ”scottish baccalaureate” for ”scottishbaccalaureate”, ”scottish higher” for ”scottishhigher”, ”scqf level 6” for ”scqflevel6”, ”diploma level 3” for ”diplomalevel3”, ”award level 3” for ”awardlevel3”, ”functional skills level 3” for ”functionalskillslevel3”, ”essential skills level 3” for ”essentialskillslevel3”, ”qualification level 3” for ”qualificationlevel3”, ”level 3 qualification” for ”level3qualification”, ”level 3 apprenticeship” for ”level3apprenticeship”, ”cipd level 3” for ”cipdlevel3”, ”level 3 cipd” for ”level3cipd”, ”level 6 scqf” for ”level6scqf”, ”level 6 svq” for ”level6svq”, ”svq level 6” for ”svqlevel6”, ”certificate of higher education” for ”certificateofhighereducation”, ”higher apprenticeship” for ”higherapprenticeship”, ”higher national certificate” for ”highernationalcertificate”, ”level 4 award” for ”level4award”, ”level 4 certificate” for ”level4certificate”, ”level 4 diploma” for ”level4diploma”, ”level 4 national” for ”level4national”, ”level 4 nvq” for ”level4nvq”, ”national level 4” for ”nationallevel4”, ”nqf level 4” for ”nqflevel4”, ”nvq level 4” for ”nvqlevel4”, ”scqf level 7” for ”scqflevel7”, ”diploma level 4” for ”diplomalevel4”, ”award level 4” for ”awardlevel4”, ”functional skills level 4” for ”functionalskillslevel4”, ”essential skills level 4” for ”essentialskillslevel4”, ”qualification level 4”

for ”qualificationlevel4”, ”level 4 qualification” for ”level4qualification”, ”level 4 apprenticeship” for ”level4apprenticeship”, ”cipd level 4” for ”cipdlevel4”, ”level 4 cipd” for ”level4cipd”, ”level 7 scqf” for ”level7scqf”, ”level 7 svq” for ”level7svq”, ”svq level 7” for ”svqlevel7”, ”btec professional” for ”btecprofessional”, ”diploma of further education” for ”diplomaoffurthereducation”, ”diploma of higher education” for ”diplomaofhighereducation”, ”foundation degree” for ”foundationdegree”, ”higher diploma” for ”higherdiploma”, ”higher national diploma” for ”highernationaldiploma”, ”level 5 award” for ”level5award”, ”level 5 certificate” for ”level5certificate”, ”level 5 diploma” for ”level5diploma”, ”level 5 national” for ”level5national”, ”level 5 nvq” for ”level5nvq”, ”national level 5” for ”nationallevel5”, ”nqf level 5” for ”nqflevel5”, ”nvq level 5” for ”nvqlevel5”, ”scqf level 8” for ”scqflevel8”, ”diploma level 5” for ”diplomalevel5”, ”award level 5” for ”awardlevel5”, ”qualification level 5” for ”qualificationlevel5”, ”level 5 qualification” for ”level5qualification”, ”cipd level 5” for ”cipdlevel5”, ”level 5 cipd” for ”level5cipd”, ”level 8 scqf” for ”level8scqf”, ”level 5 apprenticeship” for ”level5apprenticeship”, ”level 5 professional” for ”level5professional”, ”level 8 svq” for ”level8svq”, ”svq level 8” for ”svqlevel8”, ”bachelors degree” for ”bachelorsdegree”, ”btec advanced professional” for ”btecadvancedprofessional”, ”degree apprenticeship” for ”degreeapprenticeship”, ”first degree” for ”firstdegree”, ”bachelor certificate” for ”bachelorcertificate”, ”bachelor diploma” for ”bachelordiploma”, ”level 6 award” for ”level6award”, ”level 6 certificate” for ”level6certificate”, ”level 6 diploma” for ”level6diploma”, ”level 6 national” for ”level6national”, ”level 6 nvq” for ”level6nvq”, ”national level 6” for ”nationallevel6”, ”nqf level 6” for ”nqflevel6”, ”nvq level 6” for ”nvqlevel6”, ”ordinary degree” for ”ordinarydegree”, ”oxbridge ma” for ”oxbridgema”, ”scottish ma” for ”scottishma”, ”scqf level 10” for ”scqflevel10”, ”scqf level 9” for ”scqflevel9”, ”higher education” for ”highereducation”, ”diploma level 6” for ”diplomalevel6”, ”award level 6” for ”awardlevel6”, ”qualification level 6” for ”qualificationlevel6”, ”level 6 qualification” for ”level6qualification”, ”cipd level 6” for ”cipdlevel6”, ”level 6 cipd” for ”level6cipd”, ”level 9 scqf” for ”level9scqf”, ”level 10 scqf” for ”level10scqf”, ”level 6 apprenticeship” for ”level6apprenticeship”, ”level 6 professional” for ”level6professional”, ”level 10 svq” for ”level10svq”, ”svq level 10” for ”svqlevel10”, ”have a degree” for ”haveadegree”, ”bsc or masters” for ”bscormasters”, ”degree or master” for ”degreeormaster”, ”bsc or msc” for ”bscormsc”, ”ba or ma” for ”baorma”, ”honours degree” for ”honoursdegree”, ”award level 7” for ”awardlevel7”, ”ba and ma” for ”baandma”, ”cipd level 7” for ”cipdlevel7”, ”diploma level 7” for ”diplomalevel7”, ”have a master” for ”haveamaster”, ”have a masters” for ”haveamasters”, ”level 11 scqf” for ”level11scqf”, ”level 11 svq” for ”level11svq”, ”level 7 apprenticeship” for ”level7apprenticeship”, ”level 7 award” for ”level7award”, ”level 7 certificate” for ”level7certificate”, ”level 7 cipd” for ”level7cipd”, ”level 7 diploma” for ”level7diploma”, ”level 7 national” for ”level7national”, ”level 7 nvq” for ”level7nvq”, ”level 7 professional” for ”level7professional”, ”level 7 qcf” for ”level7qcf”, ”master of arts” for ”masterofarts”, ”master of business administration” for ”masterofbusinessadministration”, ”master of engineering” for ”masterofengineering”, ”master of law” for ”masteroflaw”, ”master of mathematics” for ”masterofmathematics”, ”master of pharmacy” for ”masterofpharmacy”, ”master of philosophy” for ”masterofphilosophy”, ”master of physics” for ”masterofphysics”, ”master of professional studies” for ”masterofprofessionalstudies”, ”master of public health” for ”masterofpublichealth”, ”master of research” for ”masterofresearch”, ”master of science” for ”masterofscience”, ”master of social work” for ”masterofsocialwork”, ”master of studies” for ”masterofstudies”, ”masters by research” for ”mastersbyresearch”, ”masters degree” for ”mastersdegree”, ”national level 7” for ”nationallevel7”, ”nqf level 7” for ”nqflevel7”, ”nvq level 7” for ”nvqlevel7”, ”postgraduate certificate” for ”postgraduatecertificate”, ”postgraduate certificate in education” for ”postgraduatecertificateineducation”, ”postgraduate diploma” for ”postgraduatediploma”, ”professional masters” for ”professionalmasters”, ”scqf level 11” for ”scqflevel11”, ”svq level 11” for ”svqlevel11”, ”doctor of philosophy” for ”doctorofphilosophy”, ”doctoral degree” for ”doctoraldegree”, ”doctors degree” for ”doctorsdegree”, ”level 8 award” for ”level8award”, ”level 8 certificate” for ”level8certificate”, ”level 8 diploma” for ”level8diploma”, ”level 8 national” for ”level8national”, ”level 8 nvq” for ”level8nvq”, ”national level 8” for ”nationallevel8”, ”nqf level 8” for ”nqflevel8”, ”nvq level 8” for ”nvqlevel8”, ”rqf level 8” for ”rqflevel8”, ”scqf level 12” for ”scqflevel12”, ”diploma level 8” for ”diplomalevel8”, ”award level 8” for ”awardlevel8”, ”qcf level 8” for ”qcflevel8”, ”level 8 qcf” for ”level8qcf”, ”cipd level 8” for ”cipdlevel8”, ”level 8 cipd” for ”level8cipd”, ”level 12 scqf” for ”level12scqf”, ”level 8 apprenticeship” for ”level8apprenticeship”, ”level 8 professional” for ”level8professional”, ”svq level 12” for ”svqlevel12”, ”level 12 svq” for ”level12svq”.

D. Aditional Figures and Tables

Table OD1: Personality Traits by 3-digit Occupation group

SOC	Name	PT	Agree.	Consc.	Emot.	Extra.	Open.
111	Chief Executives and Senior Officials	61.1%	11.0%	29.8%	10.6%	30.8%	24.4%
112	Production Managers and Directors	59.4%	4.2%	35.0%	7.8%	27.9%	23.0%
113	Functional Managers and Directors	68.7%	7.3%	33.6%	10.4%	34.6%	37.9%
115	Financial Institution Managers and Directors	63.7%	7.5%	35.5%	11.6%	33.0%	21.0%
116	Managers and Directors in Transport and Logistics	79.3%	5.9%	49.9%	10.2%	51.6%	26.1%
117	Senior Officers in Protective Services	68.2%	7.9%	30.2%	11.7%	40.4%	24.8%
118	Health and Social Services Managers and Directors	82.1%	19.2%	41.5%	16.8%	47.5%	35.0%
119	Managers and Directors in Retail and Wholesale	77.3%	6.0%	39.5%	15.7%	59.0%	33.0%
121	Managers and Proprietors in Agriculture Related S.	74.1%	10.3%	44.6%	8.3%	42.1%	29.1%
122	Managers and Proprietors in Hospitality and Leisure S.	81.5%	13.9%	40.8%	13.4%	64.0%	20.3%
124	Managers and Proprietors in Health and Care S.	75.2%	21.6%	36.3%	17.5%	40.6%	25.6%
125	Managers and Proprietors in Other Services	69.9%	9.8%	36.6%	12.7%	45.7%	24.7%
211	Natural and Social Science Professionals	71.5%	5.3%	26.7%	5.5%	37.6%	45.9%
212	Engineering Professionals	55.6%	3.2%	25.9%	6.3%	24.1%	29.4%
213	I.T. and Telecommunications Professionals	63.2%	4.0%	25.3%	5.7%	30.2%	39.4%
214	Conservation and Environment Professionals	74.7%	11.2%	35.6%	10.5%	49.8%	37.7%
215	Research and Development Managers	70.0%	8.9%	29.0%	5.9%	36.5%	48.1%
221	Health Professionals	69.7%	13.6%	23.3%	10.1%	42.7%	28.8%
222	Therapy Professionals	70.2%	10.2%	26.8%	9.6%	46.7%	21.1%
223	Nursing and Midwifery Professionals	75.1%	23.1%	39.3%	24.7%	41.1%	23.8%
231	Teaching and Educational Professionals	77.6%	11.1%	31.8%	13.2%	52.2%	32.5%
241	Legal Professionals	59.7%	10.4%	29.7%	6.4%	32.9%	18.9%
242	Business, Research and Administrative Professionals	64.5%	6.4%	32.9%	9.8%	31.8%	35.3%
243	Architects, Town Planners and Surveyors	51.3%	4.4%	28.7%	5.5%	24.4%	20.2%
244	Welfare Professionals	71.1%	15.1%	29.1%	17.2%	38.7%	37.7%
245	Librarians and Related Professionals	79.2%	12.9%	33.9%	14.0%	48.2%	46.2%
246	Quality and Regulatory Professionals	62.3%	5.1%	33.6%	8.5%	29.5%	29.5%
247	Media Professionals	82.9%	8.4%	31.9%	11.6%	50.2%	61.9%
311	Science, Engineering and Production Technicians	67.1%	4.9%	39.1%	10.6%	35.4%	26.5%
312	Draughtspersons and Related Architectural Technicians	57.7%	3.7%	27.3%	8.0%	27.2%	29.7%
313	Information Technology Technicians	69.5%	8.5%	34.1%	13.7%	41.4%	30.1%

Continued on next page

Table OD1 – continued from previous page

SOC	Name	PT	Agree.	Consc.	Emot.	Extra.	Open.
321	Health Associate Professionals	69.2%	9.4%	30.7%	7.2%	43.5%	19.6%
323	Welfare and Housing Associate Professionals	79.4%	18.0%	31.5%	24.4%	49.4%	39.3%
331	Protective Service Occupations	76.1%	11.2%	33.2%	26.4%	38.3%	31.8%
341	Artistic, Literary and Media Occupations	77.4%	6.2%	27.9%	10.2%	45.8%	51.8%
342	Design Occupations	84.7%	4.4%	25.0%	7.6%	48.0%	72.0%
344	Sports and Fitness Occupations	79.4%	16.2%	29.2%	5.5%	70.4%	24.4%
351	Transport Associate Professionals	50.9%	1.5%	31.6%	14.5%	26.4%	15.0%
352	Legal Associate Professionals	61.8%	8.2%	32.8%	10.7%	33.6%	20.6%
353	Business, Finance and Related Associate Prof.	65.3%	7.6%	35.2%	10.2%	34.2%	29.1%
354	Sales, Marketing and Related Associate Prof.	77.6%	7.8%	36.6%	11.5%	49.9%	39.7%
355	Conservation and Environmental Associate Prof.	79.8%	15.9%	32.3%	5.9%	59.1%	24.4%
356	Public Services and Other Associate Prof.	77.9%	9.8%	41.4%	15.8%	50.4%	32.0%
411	Admin. Occ.: Government and Related Organisations	61.5%	10.8%	24.8%	8.2%	38.9%	26.8%
412	Admin. Occ.: Finance	62.4%	6.2%	32.0%	10.7%	37.8%	19.9%
413	Admin. Occ.: Records	72.9%	7.2%	43.1%	15.2%	42.7%	23.7%
415	Other Administrative Occupations	76.8%	10.6%	44.7%	15.9%	47.8%	26.8%
416	Administrative Occupations: Office Managers and Sup.	73.8%	11.1%	43.7%	15.6%	50.0%	24.2%
421	Secretarial and Related Occupations	75.5%	17.5%	41.8%	18.3%	51.0%	24.9%
511	Agricultural and Related Trades	76.1%	9.1%	46.2%	4.2%	52.3%	27.3%
521	Metal Forming, Welding and Related Trades	33.2%	0.8%	20.6%	4.1%	11.0%	8.4%
522	Metal Machining, Fitting and Instrument Making Trades	49.7%	2.8%	29.2%	6.8%	22.0%	16.1%
523	Vehicle Trades	66.2%	5.8%	43.3%	12.6%	35.2%	16.0%
524	Electrical and Electronic Trades	53.4%	3.6%	27.5%	5.2%	23.1%	25.7%
525	Skilled Metal, Electrical and Electronic Trades Sup.	41.3%	3.0%	28.6%	5.9%	17.1%	10.0%
531	Construction and Building Trades	46.3%	7.6%	29.1%	3.1%	19.9%	12.9%
532	Building Finishing Trades	33.9%	3.6%	21.8%	3.0%	13.8%	8.7%
533	Construction and Building Trades Sup.	42.3%	2.6%	28.3%	4.2%	17.4%	12.1%
541	Textiles and Garments Trades	67.8%	8.1%	31.3%	8.5%	47.6%	25.3%
542	Printing Trades	59.4%	2.8%	28.0%	12.2%	27.9%	29.6%
543	Food Preparation and Hospitality Trades	81.2%	9.4%	41.6%	19.7%	58.7%	33.1%
544	Other Skilled Trades	65.8%	2.2%	41.4%	13.7%	33.9%	16.8%
612	Childcare and Related Personal Services	78.8%	17.5%	22.7%	18.4%	59.7%	34.4%
613	Animal Care and Control Services	81.2%	9.3%	28.6%	6.6%	61.6%	45.8%
614	Caring Personal Services	82.6%	39.1%	26.6%	27.4%	50.2%	29.3%
621	Leisure and Travel Services	83.6%	9.8%	51.7%	22.3%	67.6%	20.6%
622	Hairdressers and Related Services	78.8%	7.6%	33.1%	7.4%	61.8%	22.0%
623	Housekeeping and Related Services	72.2%	28.1%	33.8%	11.0%	37.9%	16.4%

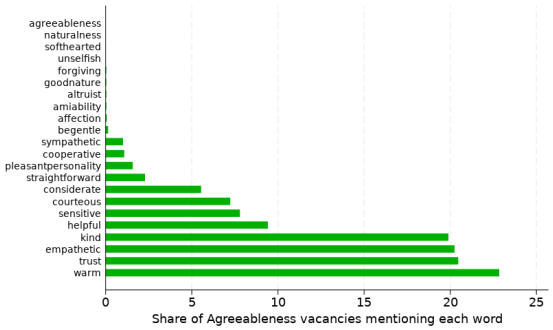
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Table OD1 – continued from previous page

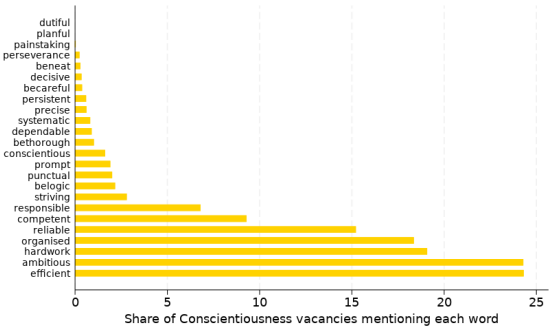
SOC	Name	PT	Agree.	Consc.	Emot.	Extra.	Open.
624	Cleaning and Housekeeping Managers and Sup.	72.1%	20.5%	52.0%	15.4%	39.2%	15.8%
711	Sales Assistants and Retail Cashiers	87.3%	16.0%	42.8%	9.5%	76.4%	31.9%
712	Sales Related Occupations	82.1%	7.5%	35.2%	11.7%	49.3%	39.5%
713	Sales Supervisors	79.0%	13.0%	41.7%	4.7%	60.3%	43.8%
721	Customer Service Occupations	85.4%	17.8%	39.6%	17.7%	67.8%	30.3%
722	Customer Service Managers and Sup.	84.9%	13.0%	43.8%	20.9%	60.5%	30.1%
811	Process Operatives	49.5%	2.0%	29.6%	6.2%	21.4%	16.7%
812	Plant and Machine Operatives	49.2%	1.7%	29.8%	5.7%	24.5%	17.1%
813	Assemblers and Routine Operatives	52.0%	2.6%	29.3%	6.5%	26.5%	18.6%
814	Construction Operatives	62.6%	9.5%	39.7%	6.5%	31.6%	20.6%
821	Road Transport Drivers	52.6%	8.8%	33.5%	5.8%	27.9%	19.0%
822	Mobile Machine Drivers and Operatives	40.4%	5.2%	23.7%	4.0%	18.0%	10.9%
823	Other Drivers and Transport Operatives	61.9%	4.8%	45.3%	17.1%	30.8%	23.0%
911	Elementary Agricultural Occupations	66.9%	5.6%	38.3%	7.9%	32.4%	32.0%
912	Elementary Construction Occupations	20.2%	3.2%	13.9%	1.1%	6.6%	2.6%
913	Elementary Process Plant Occupations	55.6%	1.8%	29.8%	6.3%	29.2%	14.6%
921	Elementary Administration Occupations	88.6%	34.4%	48.5%	35.6%	42.4%	75.5%
923	Elementary Cleaning Occupations	63.9%	7.6%	39.4%	6.7%	33.5%	23.4%
924	Elementary Security Occupations	71.5%	9.4%	27.1%	11.0%	44.7%	34.0%
925	Elementary Sales Occupations	69.3%	10.8%	27.8%	4.6%	51.5%	17.1%
926	Elementary Storage Occupations	70.5%	6.3%	44.1%	6.2%	45.5%	19.5%
927	Other Elementary Services Occupations	82.6%	21.1%	38.5%	17.9%	67.0%	22.6%

Figure OD1: Frequency of words by personality trait

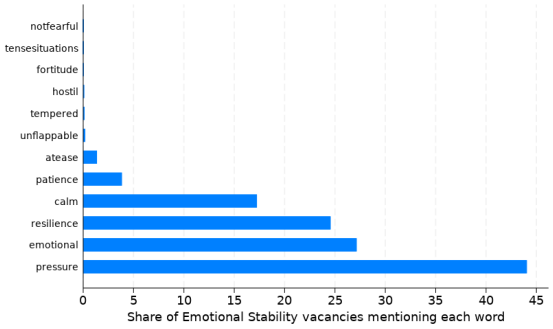
(a) Agreeableness



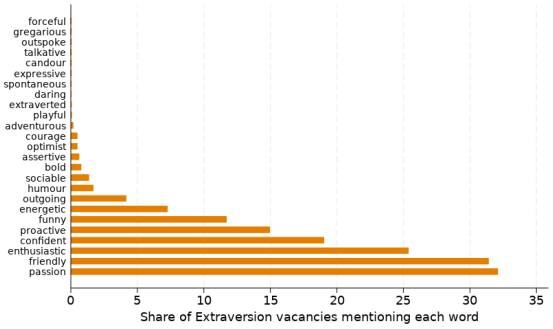
(b) Conscientiousness



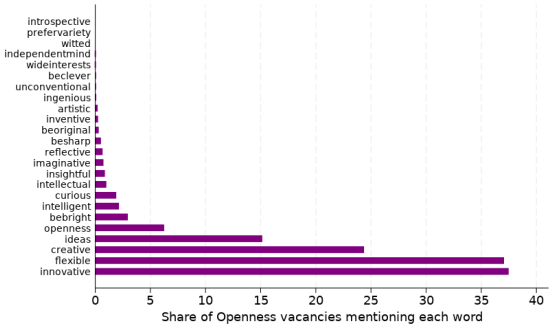
(c) Emotional Stability



(d) Extraversion



(e) Openness



Author's creation using Adzuna

Table OD2: Covariate of main variables

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
Panel A: Personality Traits and Skills					
	Agreeableness	0.105	0.104	0.105	0.105
	Conscientiousness	0.406	0.378	0.406	0.406
	Emotional stability	0.132	0.131	0.132	0.132
	Extraversion	0.460	0.455	0.460	0.460
	Openness	0.476	0.302	0.476	0.476
	Soft Skills	0.744	0.629	0.743	0.743
	Hard Skills	0.834	0.571	0.832	0.832
Panel B: Location Distribution (NUTS1)					
UKC	North East	0.014	0.019	0.014	0.014
UKD	North West	0.077	0.093	0.077	0.077
UKE	Yorkshire and The Humber	0.044	0.060	0.044	0.044
UKF	East Midlands	0.035	0.058	0.035	0.035
UKG	West Midlands	0.057	0.072	0.057	0.057
UKH	East of England	0.070	0.098	0.070	0.070
UKI	London	0.404	0.258	0.404	0.404
UKJ	South East	0.155	0.172	0.155	0.155
UKK	South West	0.076	0.084	0.076	0.076
UKL	Wales	0.020	0.025	0.020	0.020
UKM	Scotland	0.041	0.053	0.041	0.041
Panel C: Occupation Distribution (SOC 2-digit)					
11	Corporate managers and directors	0.039	0.048	0.039	0.039
12	Other managers and proprietors	0.009	0.024	0.009	0.009

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Table OD2 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
21	Science, research, engineering and technology professionals	0.274	0.057	0.271	0.271
22	Health professionals	0.007	0.015	0.007	0.007
23	Teaching and educational professionals	0.008	0.018	0.008	0.008
24	Business, media and public service professionals	0.132	0.072	0.132	0.132
31	Science, engineering and technology associate professionals	0.048	0.033	0.049	0.049
32	Health and social care associate professionals	0.009	0.022	0.009	0.009
33	Protective service occupations	0.001	0.000	0.001	0.001
34	Culture, media and sports occupations	0.017	0.016	0.017	0.017
35	Business and public service associate professionals	0.218	0.165	0.219	0.219
41	Administrative occupations	0.052	0.093	0.052	0.052
42	Secretarial and related occupations	0.007	0.013	0.007	0.007
51	Skilled agricultural and related trades	0.000	0.004	0.000	0.000
52	Skilled metal, electrical and electronic trades	0.033	0.050	0.034	0.034
53	Skilled construction and building trades	0.005	0.020	0.005	0.005
54	Textiles, printing and other skilled trades	0.007	0.052	0.007	0.007
61	Caring personal service occupations	0.017	0.087	0.017	0.017
62	Leisure, travel and related personal service occupations	0.006	0.016	0.006	0.006
71	Sales occupations	0.016	0.023	0.016	0.016
72	Customer service occupations	0.053	0.042	0.054	0.054
81	Process, plant and machine operatives	0.005	0.013	0.005	0.005
82	Transport and mobile machine drivers and operatives	0.010	0.035	0.010	0.010
91	Elementary trades and related occupations	0.002	0.007	0.002	0.002
92	Elementary administration and service occupations	0.046	0.071	0.046	0.046
Panel D: Industry Distribution (SIC 2-digit)					
01	Crop and animal production	0.003	0.005	0.003	0.003

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Table OD2 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
02	Forestry and logging	0.000	0.001	0.000	0.000
03	Fishing and aquaculture	0.000	0.001	0.000	0.000
05	Mining of coal and lignite	0.000	0.000	0.000	0.000
07	Extraction of crude petroleum and natural gas	0.000	0.001	0.000	0.000
08	Mining of metal ores	0.001	0.001	0.001	0.001
09	Other mining and quarrying	0.004	0.010	0.004	0.004
10	Manufacture of food products	0.001	0.002	0.001	0.001
12	Manufacture of tobacco products	0.001	0.002	0.001	0.001
13	Manufacture of textiles	0.002	0.002	0.002	0.002
15	Manufacture of leather and related products	0.001	0.002	0.001	0.001
16	Manufacture of wood and products of wood	0.000	0.002	0.000	0.000
17	Manufacture of paper and paper products	0.003	0.004	0.003	0.003
19	Manufacture of coke and refined petroleum products	0.001	0.004	0.001	0.001
20	Manufacture of chemicals and chemical products	0.001	0.001	0.001	0.001
21	Manufacture of basic pharmaceutical products	0.000	0.004	0.000	0.000
22	Manufacture of rubber and plastic products	0.001	0.002	0.001	0.001
23	Manufacture of other non-metallic mineral products	0.001	0.002	0.001	0.001
24	Manufacture of basic metals	0.004	0.012	0.004	0.004
25	Manufacture of fabricated metal products	0.007	0.005	0.007	0.007
26	Manufacture of computer, electronic and optical products	0.003	0.004	0.003	0.003
27	Manufacture of electrical equipment	0.004	0.007	0.004	0.004
28	Manufacture of machinery and equipment	0.001	0.002	0.001	0.001
29	Manufacture of motor vehicles, trailers and semi-trailers	0.002	0.001	0.002	0.002
30	Manufacture of other transport equipment	0.001	0.003	0.001	0.001
31	Manufacture of furniture	0.008	0.012	0.008	0.008

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Table OD2 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
32	Other manufacturing	0.003	0.008	0.003	0.003
34	Electricity, gas, steam and air conditioning supply	0.006	0.002	0.006	0.006
35	Water collection, treatment and supply	0.002	0.001	0.002	0.002
36	Sewerage	0.000	0.000	0.000	0.000
37	Waste collection, treatment and disposal	0.001	0.004	0.001	0.001
38	Remediation activities and other waste management	0.001	0.001	0.001	0.001
40	Repair and installation of machinery and equipment	0.014	0.022	0.014	0.014
41	Construction of buildings	0.002	0.005	0.002	0.002
42	Civil engineering	0.016	0.045	0.016	0.016
43	Specialised construction activities	0.010	0.019	0.010	0.010
44	Wholesale and retail trade and repair of motor vehicles	0.027	0.045	0.027	0.027
45	Wholesale trade, except of motor vehicles	0.040	0.047	0.041	0.041
46	Retail trade, except of motor vehicles	0.007	0.017	0.007	0.007
47	Land transport and transport via pipelines	0.001	0.001	0.001	0.001
48	Water transport	0.002	0.001	0.002	0.002
49	Air transport	0.004	0.007	0.004	0.004
50	Warehousing and support activities for transportation	0.000	0.003	0.000	0.000
51	Postal and courier activities	0.004	0.020	0.004	0.004
52	Accommodation	0.018	0.060	0.018	0.018
53	Food and beverage service activities	0.022	0.010	0.021	0.021
54	Publishing activities	0.007	0.005	0.007	0.007
55	Motion picture, video and television production	0.001	0.001	0.001	0.001
56	Programming and broadcasting activities	0.019	0.007	0.018	0.018
57	Telecommunications	0.196	0.041	0.194	0.194
58	Computer programming, consultancy and related	0.027	0.011	0.027	0.027

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Table OD2 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
59	Information service activities	0.055	0.027	0.055	0.055
60	Financial service activities	0.004	0.003	0.004	0.004
61	Insurance, reinsurance and pension funding	0.011	0.007	0.011	0.011
63	Activities auxiliary to financial and insurance	0.038	0.039	0.039	0.039
64	Real estate activities	0.012	0.017	0.012	0.012
65	Legal and accounting activities	0.107	0.045	0.108	0.108
66	Activities of head offices	0.016	0.012	0.016	0.016
67	Architectural and engineering activities	0.017	0.005	0.017	0.017
68	Scientific research and development	0.024	0.012	0.024	0.024
69	Other professional, scientific and technical	0.056	0.036	0.056	0.056
70	Advertising and market research	0.000	0.001	0.000	0.000
71	Veterinary activities	0.004	0.005	0.004	0.004
72	Rental and leasing activities	0.006	0.005	0.006	0.006
73	Employment activities	0.004	0.008	0.004	0.004
74	Travel agency and tour operator activities	0.002	0.019	0.002	0.002
75	Security and investigation activities	0.007	0.006	0.007	0.007
76	Services to buildings and landscape activities	0.002	0.002	0.002	0.002
77	Office administrative and other business support	0.025	0.062	0.025	0.025
78	Public administration and defence	0.018	0.042	0.018	0.018
79	Education	0.004	0.030	0.004	0.004
80	Human health activities	0.007	0.031	0.007	0.007
81	Residential care activities	0.006	0.009	0.006	0.006
82	Social work activities without accommodation	0.001	0.003	0.001	0.001
83	Creative, arts and entertainment activities	0.001	0.001	0.001	0.001
84	Libraries, archives, museums and other cultural	0.007	0.015	0.007	0.007

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Table OD2 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
85	Gambling and betting activities	0.008	0.014	0.008	0.008
86	Sports activities and amusement and recreation	0.001	0.001	0.001	0.001
87	Activities of membership organisations	0.037	0.039	0.037	0.037
88	Repair of computers and personal household goods	0.000	0.000	0.000	0.000
89	Other personal service activities	0.003	0.006	0.003	0.003
90	Other service activities	0.036	0.021	0.036	0.036
Panel E: Firm Characteristics					
	Publicly Listed	0.013	0.003	0.014	0.014
Panel F: Sample Sizes					
	Observations	17,196	237,255	17,196	237,255

Notes: Table shows mean values of covariates for AI-adopters and non-adopters before and after entropy balancing. After weighting columns show reweighted control group means. Yearly aggregation, mean model. Covariates include all personality traits, skills, location dummies (NUTS1), occupation dummies (SOC 2-digit), industry dummies (SIC 2-digit), and firm characteristics used in entropy balancing.

Table OD3: Covariate Balance - Half-Year model

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
Panel A: Personality Traits and Skills					
	Agreeableness	0.104	0.108	0.104	0.104
	Conscientiousness	0.401	0.380	0.401	0.401
	Emotional Stability	0.132	0.134	0.132	0.132
	Extraversion	0.457	0.460	0.457	0.457
	Openness	0.468	0.307	0.467	0.467
	Soft Skills	0.740	0.634	0.740	0.740
	Hard Skills	0.828	0.572	0.827	0.827
Panel B: Location Distribution (NUTS1)					
UKC	North East	0.015	0.018	0.015	0.015
UKD	North West	0.079	0.093	0.079	0.079
UKE	Yorkshire and The Humber	0.045	0.059	0.045	0.045
UKF	East Midlands	0.036	0.058	0.036	0.036
UKG	West Midlands	0.058	0.072	0.058	0.058
UKH	East of England	0.070	0.098	0.070	0.070
UKI	London	0.396	0.262	0.396	0.396
UKJ	South East	0.157	0.172	0.157	0.157
UKK	South West	0.077	0.084	0.077	0.077
UKL	Wales	0.020	0.024	0.020	0.020
UKM	Scotland	0.042	0.052	0.042	0.042
Panel C: Occupation Distribution (SOC 2-digit)					
11	Corporate managers and directors	0.040	0.049	0.040	0.040
12	Other managers and proprietors	0.009	0.024	0.009	0.009

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Table OD3 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
21	Science, research, engineering and technology professionals	0.266	0.057	0.263	0.262
22	Health professionals	0.007	0.016	0.007	0.007
23	Teaching and educational professionals	0.008	0.019	0.008	0.008
24	Business, media and public service professionals	0.134	0.074	0.133	0.133
31	Science, engineering and technology associate professionals	0.049	0.033	0.049	0.049
32	Health and social care associate professionals	0.009	0.024	0.009	0.009
33	Protective service occupations	0.001	0.000	0.001	0.001
34	Culture, media and sports occupations	0.017	0.015	0.017	0.017
35	Business and public service associate professionals	0.217	0.165	0.218	0.218
41	Administrative occupations	0.053	0.090	0.053	0.053
42	Secretarial and related occupations	0.008	0.012	0.008	0.008
51	Skilled agricultural and related trades	0.000	0.004	0.000	0.000
52	Skilled metal, electrical and electronic trades	0.034	0.047	0.034	0.034
53	Skilled construction and building trades	0.005	0.018	0.005	0.005
54	Textiles, printing and other skilled trades	0.007	0.052	0.007	0.007
61	Caring personal service occupations	0.017	0.091	0.017	0.017
62	Leisure, travel and related personal service occupations	0.006	0.016	0.006	0.006
71	Sales occupations	0.016	0.023	0.016	0.016
72	Customer service occupations	0.055	0.043	0.055	0.055
81	Process, plant and machine operatives	0.005	0.012	0.005	0.005
82	Transport and mobile machine drivers and operatives	0.010	0.034	0.010	0.010
91	Elementary trades and related occupations	0.002	0.007	0.002	0.002
92	Elementary administration and service occupations	0.047	0.073	0.047	0.047
Panel D: Industry Distribution (SIC 2-digit)					
01	Crop and animal production	0.003	0.004	0.003	0.003

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Table OD3 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
02	Forestry and logging	0.000	0.001	0.000	0.000
03	Fishing and aquaculture	0.000	0.001	0.000	0.000
05	Mining of coal and lignite	0.000	0.000	0.000	0.000
07	Extraction of crude petroleum and natural gas	0.000	0.001	0.000	0.000
08	Mining of metal ores	0.001	0.001	0.001	0.001
09	Other mining and quarrying	0.004	0.010	0.004	0.004
10	Manufacture of food products	0.001	0.002	0.001	0.001
12	Manufacture of tobacco products	0.001	0.002	0.001	0.001
13	Manufacture of textiles	0.002	0.002	0.002	0.002
15	Manufacture of leather and related products	0.001	0.002	0.001	0.001
16	Manufacture of wood and products of wood	0.000	0.002	0.000	0.000
17	Manufacture of paper and paper products	0.003	0.003	0.003	0.003
19	Manufacture of coke and refined petroleum products	0.001	0.003	0.001	0.001
20	Manufacture of chemicals and chemical products	0.001	0.001	0.001	0.001
21	Manufacture of basic pharmaceutical products	0.000	0.004	0.000	0.000
22	Manufacture of rubber and plastic products	0.001	0.002	0.001	0.001
23	Manufacture of other non-metallic mineral products	0.001	0.002	0.001	0.001
24	Manufacture of basic metals	0.004	0.012	0.004	0.004
25	Manufacture of fabricated metal products	0.007	0.005	0.007	0.007
26	Manufacture of computer, electronic and optical products	0.003	0.004	0.003	0.003
27	Manufacture of electrical equipment	0.004	0.007	0.004	0.004
28	Manufacture of machinery and equipment	0.001	0.002	0.001	0.001
29	Manufacture of motor vehicles, trailers and semi-trailers	0.002	0.001	0.002	0.002
30	Manufacture of other transport equipment	0.001	0.003	0.001	0.001
31	Manufacture of furniture	0.007	0.012	0.007	0.007

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Table OD3 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
32	Other manufacturing	0.003	0.007	0.003	0.003
34	Electricity, gas, steam and air conditioning supply	0.006	0.002	0.006	0.006
35	Water collection, treatment and supply	0.002	0.001	0.002	0.002
36	Sewerage	0.000	0.000	0.000	0.000
37	Waste collection, treatment and disposal	0.001	0.004	0.001	0.001
38	Remediation activities and other waste management	0.001	0.001	0.001	0.001
40	Repair and installation of machinery and equipment	0.014	0.021	0.014	0.014
41	Construction of buildings	0.002	0.005	0.002	0.002
42	Civil engineering	0.016	0.042	0.016	0.016
43	Specialised construction activities	0.011	0.018	0.011	0.011
44	Wholesale and retail trade and repair of motor vehicles	0.026	0.044	0.026	0.026
45	Wholesale trade, except of motor vehicles	0.041	0.047	0.041	0.041
46	Retail trade, except of motor vehicles	0.007	0.017	0.007	0.007
47	Land transport and transport via pipelines	0.000	0.001	0.000	0.000
48	Water transport	0.002	0.001	0.002	0.002
49	Air transport	0.004	0.007	0.004	0.004
50	Warehousing and support activities for transportation	0.000	0.003	0.000	0.000
51	Postal and courier activities	0.004	0.021	0.004	0.004
52	Accommodation	0.018	0.060	0.018	0.018
53	Food and beverage service activities	0.021	0.010	0.020	0.020
54	Publishing activities	0.006	0.005	0.006	0.006
55	Motion picture, video and television production	0.001	0.001	0.001	0.001
56	Programming and broadcasting activities	0.018	0.007	0.018	0.018
57	Telecommunications	0.187	0.040	0.184	0.184
58	Computer programming, consultancy and related	0.026	0.010	0.026	0.026

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Table OD3 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
59	Information service activities	0.057	0.028	0.057	0.057
60	Financial service activities	0.004	0.002	0.004	0.004
61	Insurance, reinsurance and pension funding	0.011	0.006	0.011	0.011
63	Activities auxiliary to financial and insurance	0.040	0.039	0.040	0.040
64	Real estate activities	0.012	0.016	0.012	0.012
65	Legal and accounting activities	0.112	0.047	0.113	0.113
66	Activities of head offices	0.016	0.012	0.016	0.016
67	Architectural and engineering activities	0.016	0.005	0.016	0.016
68	Scientific research and development	0.024	0.012	0.024	0.024
69	Other professional, scientific and technical	0.057	0.037	0.057	0.057
70	Advertising and market research	0.000	0.001	0.000	0.000
71	Veterinary activities	0.004	0.005	0.004	0.004
72	Rental and leasing activities	0.006	0.004	0.006	0.006
73	Employment activities	0.004	0.008	0.004	0.004
74	Travel agency and tour operator activities	0.002	0.018	0.002	0.002
75	Security and investigation activities	0.007	0.006	0.007	0.007
76	Services to buildings and landscape activities	0.002	0.003	0.002	0.002
77	Office administrative and other business support	0.026	0.065	0.026	0.026
78	Public administration and defence	0.017	0.045	0.017	0.017
79	Education	0.004	0.031	0.004	0.004
80	Human health activities	0.008	0.033	0.008	0.008
81	Residential care activities	0.006	0.009	0.006	0.006
82	Social work activities without accommodation	0.001	0.003	0.001	0.001
83	Creative, arts and entertainment activities	0.002	0.001	0.002	0.002
84	Libraries, archives, museums and other cultural	0.007	0.015	0.007	0.007

Continued on next page

Table OD3 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-adopters	Non-adopters	AI-adopters	Non-adopters
85	Gambling and betting activities	0.009	0.014	0.008	0.008
86	Sports activities and amusement and recreation	0.001	0.001	0.001	0.001
87	Activities of membership organisations	0.039	0.038	0.039	0.039
88	Repair of computers and personal household goods	0.000	0.000	0.000	0.000
89	Other personal service activities	0.003	0.006	0.003	0.003
90	Other service activities	0.037	0.022	0.038	0.038
Panel E: Firm Characteristics					
	Publicly Listed	0.014	0.003	0.015	0.015
Panel F: Sample Sizes					
	Observations	25,143	305,899	25,143	305,899

Notes: Table shows mean values of covariates for AI-adopters and non-adopters before and after entropy balancing. After weighting columns show reweighted control group means. Half-yearly aggregation, mean model. Covariates include all personality traits, skills, location dummies (NUTS1), occupation dummies (SOC 2-digit), industry dummies (SIC 2-digit), and firm characteristics used in entropy balancing.

E. Personality traits. Evolution over time

In this section, we examine whether personality traits change over our sample period. In order to do so, we regress personality traits against year dummies in the following way:

$$P(Trait_i = 1) = \beta_0 + \beta_1 X_i + \sum_{t=2018}^{2022} \alpha_t Year_t + \gamma_k + \phi_j + \epsilon_i$$

where X_i is a vector of job posting characteristics including the demand for other personality traits, δ_l are location fixed effects, $Year_t$ are dummy variables (with 2017 as the base year), ϕ_j are firm fixed effects and γ_k are occupation fixed effects. The coefficients α_t capture the evolution of personality trait demands over 2018-2022, after accounting for changes in job requirements, occupational composition, geographical variation, and firm-specific practices.

After accounting for compositional changes in location, occupation, and firm, Table OE4 shows the estimated year effects relative to 2017 for the full period. Conscientiousness shows a gradual decline, becoming significantly negative by 2022. Extraversion shows a consistent positive trend, with significant increases during 2019-2021 (ranging from 1.7 to 2.8 percentage points). Emotional Stability shows temporary positive effects during 2021 (1.1 percentage points), while Openness and Agreeableness show no significant time trends after controlling for composition.

Table OE4: Year effect on each Personality Trait. Baseline: 2017

	(1) P(All PT)	(2) P(Agree.)	(3) P(Consc.)	(4) P(Emot.)	(5) P(Extra.)	(6) P(Open.)
2018	−0.003	0.002	−0.010	0.004	0.009	0.002
	(0.004)	(0.004)	(0.008)	(0.004)	(0.006)	(0.005)
2019	−0.003	0.003	−0.012	0.011	0.017*	0.006
	(0.006)	(0.007)	(0.010)	(0.007)	(0.009)	(0.007)
2020	0.001	0.012	−0.014	0.009	0.028**	−0.000
	(0.008)	(0.013)	(0.011)	(0.007)	(0.012)	(0.007)
2021	−0.005	0.018	−0.016	0.011*	0.022*	−0.001
	(0.008)	(0.016)	(0.010)	(0.007)	(0.012)	(0.008)
2022	−0.014	0.003	−0.021**	0.006	0.017	−0.003
	(0.009)	(0.009)	(0.011)	(0.008)	(0.012)	(0.009)
R^2	0.349	0.364	0.280	0.286	0.341	0.331

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. $N = 9888315$. Regression controls for traits and skills demand, remote work requirements, length of advertisement, contract type, work schedule, bonus offerings, education, and negotiable salary option. Location, occupation and firm fixed effects. SE clustered at the occupation and firm level. Author's creation using Adzuna

Artificial Intelligence and Personality Traits: Evidence from Online Job Vacancies *

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Abstract

Using text analysis of 11.7 million UK online job postings by 128,559 companies from 2017 to 2022, drawn from a near-universe dataset of online vacancies, we examine how demand for AI-related skills is associated with firms' desired worker personality traits. We construct measures of Big Five traits and AI-related skill requirements from job descriptions. At the vacancy level, AI-related positions are 6.7 percentage points more likely to mention openness to experience and less likely to mention the remaining traits. At the firm level, staggered difference-in-differences estimates show 2.7 percentage points higher openness demand after firms post AI vacancies.

Keywords: Artificial intelligence, labour demand, personality, online job postings
JEL Codes: J23, J24, O33, C55

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1 Introduction

The diffusion of artificial intelligence (AI), from machine learning to large language models, has created concerns about labour displacement and reorganisation of tasks (Acemoglu et al., 2022, 2026; Brynjolfsson, Chandar, & Chen, 2025). Existing research examines how AI affects tasks and the demand for cognitive, technical skills, as well as experience and expertise levels (Alekseeva et al., 2026, 2021; Autor & Thompson, 2025; Brynjolfsson, Li, & Raymond, 2025; Squicciarini & Nachtigall, 2021). Beyond altering task content, AI may modify how tasks are performed, problems approached, and solutions developed. As a result, firms that seek to integrate AI-related capabilities into their workforce may perceive certain personality traits as more relevant in the new work environment. In particular, personality traits such as openness, curiosity, or cooperation may influence workers' ability to effectively interact with and adapt to AI-based processes. These traits are fundamental components of human capital that are not fully captured by conventional measures of education, experience, or technical expertise (Deming & Silliman, 2025; Heckman et al., 2021). This raises a largely unexplored question, which we address in this paper: what personality traits do firms seek in workers as they demand AI-related skills?

Using text analysis of 11.7 million UK online job postings by 128,559 firms from 2017 to 2022, drawn from a near-universe dataset of online vacancies, we construct measures of AI-related skill requirements and employer demand for the Big Five personality traits (openness, conscientiousness, extraversion, agreeableness, and emotional stability). We focus on personality traits because they are relatively stable traits, and not easily changeable in adulthood, that affect labour market outcomes (Alderotti et al., 2023; Bowles et al., 2001; Fitzenberger et al., 2022; Flinn et al., 2025; Heckman et al., 2006; Nyhus & Pons, 2005; Woessmann, 2025). Personality traits are also less directly trainable than many technical skills (Cobb-Clark & Schurer, 2012; Heckman & Kautz, 2012; McCrae & John, 1992). This means that when firms demand particular personality attributes, they may express these requirements through recruitment because they cannot be developed internally (Grugulis & Vincent, 2009; Matteson et al., 2016; McGee & McGee, 2024).

Because firms cannot emphasize every desirable characteristic with equal prominence in a job advertisement, the traits that appear in vacancy text are likely to reflect attributes that employers consider especially salient for the position.[1] Consequently, job postings are a particularly informative source for studying which personality traits firms demand as they seek to integrate AI knowledge into their production processes.

AI may influence the nature of jobs (Babina, 2026) and modify how firms organise production, process information, and make decisions (Barua et al., 2025). In industrial settings, AI knowledge is used to support prediction, pattern recognition, anomaly detection or optimisation.[2] These capabilities are central to firms' use of AI, including more recent generative AI systems. Therefore, AI may influence the personality traits firms value in workers, and potentially the types of workers they seek to hire. On the one hand, AI may increase coordination and interaction requirements within firms. Consequently, it can raise the value of interpersonal traits such as agreeableness and extraversion. Task-specific AI systems are often imperfect substitutes for human interaction in activities involving teamwork, and organisational integration. As Johnson and Acemoglu (2023) observe: "It is therefore not surprising that despite the spread of AI technologies, many companies are increasingly seeking workers with social, rather than mathematical, or technical skills. . . "

On the other hand, task-based models consider that AI can complement non-routine cognitive tasks while substituting for routine cognitive activities (Acemoglu & Restrepo, 2019; Autor, 2014). Therefore, AI tools may be associated with greater relative demand for traits linked to creativity and complex problem-solving, such as openness, while reducing the importance of traits associated with routine cognitive task execution, such as conscientiousness. More broadly, as firms seek to integrate AI knowledge into production

[1] Online job advertisements are not strictly constrained by space the way print ads are, but employers still face incentives to be selective to decide which requirements to feature (Abraham et al., 2024; Flory et al., 2015). Traits that appear in a posting are therefore more likely to reflect attributes employers consider especially salient for the position.

[2] For example, Siemens and GE use predictive analytics to monitor machinery and anticipate maintenance needs; BMW applies AI-based quality-control systems that analyse sensor and image data on production lines; Nike analyses customer behaviour to personalise products; and Bosch uses AI in factories to detect deviations in production processes and improve efficiency.

processes, they may value workers who can adapt to new tools and are open to integrating these technologies.

We retrieve data from Adzuna, a job vacancy aggregator providing near-universe coverage of online job postings in the UK. A key advantage of this dataset is that it includes full job descriptions, which we use to identify personality-trait requirements using dictionaries grounded in the psychology literature (Costa Jr & McCrae, 1992; Goldberg, 1992; O. John, 1990). We manually checked random samples of AI and non-AI vacancies and systematically removed from the whole sample ambiguous word combinations that did not refer to worker personality traits

We identify AI-related postings using an expanded dictionary based on Acemoglu et al. (2022) and Babina et al. (2024), whose sample periods overlap with ours. This measure captures firms seeking to bring internal AI-related capabilities or knowledge (such as machine learning, neural networks, or natural language processing) rather than procuring AI-powered services externally. We use the terms "AI-related skills" and "AI-related knowledge" interchangeably to refer to the technical competencies identified by our dictionary, following the convention in prior work (Acemoglu et al., 2022; Babina et al., 2024). Importantly, we distinguish AI-related demand from broader computer-skill demand.[3] We manually validated the AI dictionary against random job advertisements; the final dictionary includes over 100 AI terms. We combine the vacancy data with firm-level data from FAME, a financial database within the Bureau van Dijk suite of business databases, to examine how personality-trait demand varies among firms seeking AI-related skills. Our analysis is conducted at both the vacancy and firm levels. Our results are as follows:

First, at the vacancy level, we show systematic differences in personality requirements between AI and non-AI vacancies. Positions requiring AI-related skills are 6.7 percentage points more likely to demand openness to experience, equivalent to a relative difference

[3]Our AI measure is based on terms related to artificial intelligence and machine learning, while general computer skills, such as software development and data analysis requirements, are measured separately and used as a comparison in the empirical analysis.

of about 22% compared with the average openness-demand rate, while they are 1.2 to 2.5 percentage points less likely to demand the remaining four personality traits. These patterns remain robust after controlling for detailed job characteristics, as well as firm, occupation, location, and time fixed effects. Importantly, the pattern for AI-related vacancies differs from that for postings requiring general computer skills: while both are positively related to demand for openness, the association is more than twice as large for AI-related skills, and only AI-related vacancies show lower demand for the other four personality traits. This comparison shows that the association between AI and personality-trait demand is not simply capturing broader digital, software, or data-analysis requirements.

Second, we examine whether these vacancy-level differences translate into changes in firms' overall hiring strategies using an event-study design. We define the event date as the first observed vacancy requiring AI-related skills posted by a firm. This timing captures the point at which a firm first appears in our data as demanding AI-related capabilities. Following prior work that uses demand for AI-related skills in job postings as a proxy for firms' engagement with AI technologies (Acemoglu et al., 2022; Alekseeva et al., 2026, 2021; Babina et al., 2024), we refer to firms that post at least one AI-related vacancy as AI-demanding firms.

We then estimate changes in personality-trait demand around the first observed AI-related vacancy using a weighted staggered difference-in-differences estimator (Callaway & Sant'Anna, 2021), where weights are derived from entropy balancing (Hainmueller, 2012). Because AI-related hiring was low before expanding rapidly in the second half of the 2010s (Acemoglu et al., 2022), our analysis focuses on the period from 2017 onward.[4] The estimator exploits within-firm variation over time, implicitly absorbing time-invariant unobserved firm heterogeneity, while the entropy balancing weights ensure that AI-demanding and non-AI-demanding firms are comparable on observed pre-event

[4]Firms already observed demanding AI-related skills in the first sample period are treated as left-censored and are excluded from the identifying comparison. The estimates are therefore identified from firms whose first observed AI-related vacancy occurs after the initial period and from firms in the control group, comprising those that have not yet posted any AI-related vacancies at each point of comparison, including firms that never do so during the sample.

characteristics, including baseline personality requirements, firm attributes, occupational composition, and skill demands. Our estimates document associations between firms' first observed demand for AI-related skills and subsequent changes in their hiring patterns, conditional on observable pre-event characteristics and time-invariant firm characteristics.

When comparing AI-demanding firms with non-AI-demanding firms, we observe 2.7 percentage points higher demand for openness among AI-demanding firms across both AI-related and non-AI-related jobs, equivalent to roughly 9% of the average firm-year openness-demand rate. By contrast, we do not observe significant differences in the demand for other personality traits. This pattern is also present when AI-related vacancies are excluded from the outcome, with a corresponding estimate of 2.0 percentage points for non-AI vacancies. A shift-share decomposition shows that AI-demanding firms shift non-AI hiring toward occupation groups with lower baseline openness requirements, while openness mentions increase within most occupation groups. This suggests that the positive non-AI estimate is not driven by compositional shifts toward occupations that already demand more openness.

Related literature: We contribute to the literature on technological change and labour demand by showing that AI is associated with firms' demand for personality-trait requirements. Existing work documents how AI affects task content and demand for cognitive, technical, expertise, and managerial skills (Acemoglu et al., 2022; Alekseeva et al., 2026, 2021; Autor & Thompson, 2025). A related vacancy-based literature shows that firms differ not only in the occupations they hire for, but also in the bundles of worker attributes they demand within occupations. In particular, Deming and Kahn (2018) show that firms vary in the combinations of cognitive and social skills they request, even for similar jobs, and that these skill bundles are related to wages and firm performance. We bring this insight to the study of AI-related hiring by asking whether firms that demand AI-related skills also place distinctive emphasis on non-cognitive worker characteristics.

Our analysis connects this literature with work on personality traits in labour markets. Prior studies using vacancy data show that personality traits are widely demanded and

vary across occupations (Brenčič & McGee, 2023a, 2023b), but they do not study how trait demands relate to technological change. We examine both whether AI-related job postings differ from otherwise comparable vacancies within firms, and whether firms' first observed AI-related hiring is followed by changes in the personality profile of their broader recruitment portfolio. Our work is also related to vacancy-based evidence on AI and skill demand. Alekseeva et al. (2021) document the growth of demand for AI-related skills across firms, occupations, and industries, while Squicciarini and Nachtigall (2021) show that AI-related jobs combine AI-specific knowledge with complementary requirements such as software, problem-solving, creativity, teamwork, and communication. We show that AI-related hiring is associated not only with technical and complementary skill requirements, but also with demand for workers' stable behavioural characteristics.

We also contribute to the literature on how firms adjust their recruitment following AI demand. Most existing work focuses on the characteristics of AI-related jobs or on aggregate labour market effects and productivity (Acemoglu et al., 2022; Aghion et al., 2025; Autor & Thompson, 2025; Brynjolfsson, Chandar, & Chen, 2025). We analyse both vacancy-level and firm-level responses. We show that AI is linked to substitution across personality traits within AI-intensive roles, while being positively associated with the demand for openness across firms' entire hiring portfolios. This contrast suggests that AI demand is accompanied by organisational adjustments beyond the requirements of specific jobs, consistent with evidence that AI-investing firms experience complementary changes in employment and workforce composition (Babina et al., 2024).

We contribute methodologically to the measurement of employer demand for personality traits. We construct text-based indicators of Big Five trait requirements using dictionaries grounded in the psychology literature (Costa Jr & McCrae, 1992; Goldberg, 1992; O. John, 1990) and apply them to UK job postings. Our approach follows the methodology of Brenčič and McGee (2026), who develop personality demand measures from US job ads using the same lexical tradition. We extend their approach in three respects: we apply it to a substantially larger dataset of UK vacancies; we examine how personality demand varies with firms' AI-related hiring rather than characterising

occupational demand in the cross-section; and we exploit within-firm variation over time to study how firms' overall personality requirements shift as they begin recruiting AI-capable workers. Our approach complements individual-level studies that measure workers' personality endowments (Collischon, 2020; Gensowski, 2018) by directly capturing firms' stated hiring requirements.

The rest of the paper proceeds as follows. Section 2 describes the data sources and construction of the variables. Section 3 presents the empirical methodology for both vacancy-level and firm-level analyses. Section 4 reports the results. Section 5 concludes.

2 Data

This section describes the data sources and construction of key variables. Comprehensive information on the datasets, data cleaning procedures, and variable definitions is provided in the Online Appendix.

2.1 Data Sources

The primary data source comes from Adzuna, a job vacancy aggregator providing near-universe coverage of online job postings in the UK. Adzuna's algorithms collect advertisements from multiple job boards and company websites, offering the most comprehensive collection of online vacancy information available. The UK Office for National Statistics uses this database to produce its weekly vacancy indicator. Our dataset contains weekly snapshots of open vacancies from January 2017 to March 2022, with each snapshot in our final database capturing between 2,500 and 93,000 listings. Each record includes the full text of the job advertisement, the job title, and the employer's name.

One advantage of this dataset is that it provides the raw description text of each advertisement, allowing flexible construction of measures for personality traits and AI-related skills. From job descriptions, we extract information on job schedule and contract type. We recode job locations to the two-digit Nomenclature of Territorial Units for Statistics (NUTS) level. We also create indicators for soft and hard skills requirements,

as well as education, experience, and seniority level.

To determine Standard Occupational Classification (SOC) codes, we implement the machine learning algorithm by Turrell et al. (2019), which classifies advertisements into 3-digit SOCs based on title, description, and industry. To obtain industry classifications, we match companies listed in Adzuna's job postings with data from Companies House, the UK government's official register of companies. Through this process, we retrieve Standard Industrial Classification (SIC) codes for the relevant companies. We drop advertisements from recruitment agencies because they lack information on the user firm, preventing industry code identification. Additional details of the different databases used and the matching process are provided in Online Appendix A and Online Appendix B, respectively.

We complement the job vacancy data with information at both the occupational and firm level. For occupations, we obtain data from the UK Household Longitudinal Study (UKHLS), a household panel survey provided by the Institute for Social and Economic Research. The UKHLS asks individuals to self-assess their personality using the Big Five framework, allowing us to validate our personality trait measures by comparing demands in job vacancies with workers' self-reported traits in different occupations. At the firm level, we match Adzuna data with the FAME database to identify publicly listed companies and obtain balance sheet information. To link vacancies to firm characteristics, we consider non-agency postings with posted salary information that can be uniquely matched to Companies House and to firms with the necessary firm identifiers and industry information.[5] After combining the datasets, our final database contains 11.7 million vacancies from 128,559 companies.

A possible concern is the representativeness of the dataset. Not all UK job advertisements are posted online, and certain types of work, such as casual, freelance, or informal jobs, may be advertised through alternative channels. Table A.1 in the Appendix compares the occupational composition of Adzuna vacancies in 2021 with that of the 2021 UK

[5]Wage structures are used as a control variable in the main specifications and to verify that personality-trait demand does not differ systematically between agency and non-agency postings, providing reassurance that our restriction to direct employer postings does not drive the results.

Census at the 1-digit SOC level. The Adzuna data differ from the Census distribution, with Caring and Leisure occupations over-represented and Administrative and Managers and Directors occupations under-represented. Since part of this difference may reflect the distinction between vacancy flows and employment stocks, as well as online-posting coverage, we include occupational fixed effects in the analysis to mitigate concerns related to broad occupational composition.

2.2 Measuring Personality Traits

Similar to Brenčič and McGee (2026), we measure personality trait demands using the Big Five framework, the most widely used taxonomy for studying personality in psychology and economics. The five dimensions are openness, conscientiousness, extraversion, agreeableness, and emotional stability. Each trait encompasses specific behavioural characteristics (Costa Jr & McCrae, 1992; Goldberg, 1992; Stanek & Ones, 2023): *Conscientiousness* relates to being organised, responsible, and diligent; *openness* refers to interest in new aesthetic, cultural, or intellectual experiences and creativity; *extraversion* involves an outward orientation towards social interactions and activities; *agreeableness* appears in cooperative and unselfish behaviour; and *emotional stability* refers to consistency in emotional responses. It is considered the opposite of *neuroticism*. This framework originated from the lexical hypothesis: important personality characteristics become encoded in everyday language. Its structure has been extensively validated through factor analyses of natural language adjectives and psychometric questionnaires, and across different languages and cultures (Borghans et al., 2008; King et al., 2005). In job advertisements, employers likely use words associated with known traits, and job seekers recognise and respond to these terms when applying (Abraham et al., 2024; Flory et al., 2015).

To identify personality traits in job advertisements, we construct a binary indicator equal to one if a job posting contains at least one keyword from a Big Five trait dictionary, either in the job title or job requirements section. We build these dictionaries using trait-descriptive adjectives from authoritative sources in psychology literature (Costa Jr & McCrae, 1992; Goldberg, 1992; Heckman & Kautz, 2012; O. John, 1990; McCrae &

John, 1992; Saucier, 1994). Since vacancies often mention many traits simultaneously, we also construct a binary indicator for whether the advertisement requires *any* keyword from the Big Five trait dictionary.

As personality-trait retrieval from online vacancies can produce false positives (Brenčič & McGee, 2026), we mitigate this risk through a two-step approach grounded in both the source vocabulary and systematic manual validation. First, the starting vocabulary is drawn exclusively from established Big Five personality measures and lexical taxonomies in psychology (Costa Jr & McCrae, 1992; Goldberg, 1992; Heckman & Kautz, 2012; O. John, 1990; McCrae & John, 1992; Saucier, 1994). Because these sources catalogue adjectives that describe stable individual characteristics, the initial word list already excludes generic task-language or product-related language by construction.

Second, we refine the dictionaries through iterative manual inspection. For each word, we examine up to eight rounds of 100 randomly selected sentences from job advertisements containing that word, reading the ten surrounding words to verify that the term refers to a worker or team characteristic in context. This process identifies terms whose meaning in vacancy text systematically differs from their psychological sense. For instance, "flexible" in vacancy descriptions predominantly refers to work schedules, not to a personality trait. Rather than removing the word entirely, we replace its schedule-related variants with the modified string "flx" before matching, so the dictionary captures only the personality-relevant usage. Words whose vacancy-context meaning cannot be disambiguated are removed from the dictionary entirely. We repeat the inspection cycle until a fresh sample of 100 advertisements yields no spurious matches. In total, 1,702 text replacements were made before the final dictionary matching. The complete list is reported in Online Appendix C. This procedure substantially reduces the risk that the trait indicators capture product or firm descriptive language rather than demand for a personality characteristic.[6]

[6]We also compare our trait terms with the dictionaries used by Brenčič and McGee (2026). For agreeableness, extraversion, conscientiousness, and openness, our dictionaries largely correspond to a narrower subset of their positive-pole Big Five terms, or to close variants adapted to vacancy demand. This comparison is reassuring because it shows that our measures are grounded in the same lexical tradition, while adopting a more parsimonious approach. This is important given that Brenčič and McGee (2026) show that broad dictionary-based searches in job advertisements can generate a large number of false positives, with many trait-descriptive terms used in contexts unrelated to worker personality. We

An example of a job vacancy and the classification using this approach is shown in Table 1. The complete dictionaries are presented in Table A.2 in the Appendix. Further details on the dictionary construction process are provided in Online Appendix C.

Table 1: **Example of a job vacancy with AI skills and personality trait requirements.**

Vacancy: Data Scientist

As a data scientist you will use your expertise in ***machine learning****,* ***neural networks****, and* ***deep learning*** *to develop products. You must have experience with* ***TensorFlow*** *and* ***PyTorch****. You should be* ***creative*** *and* ***imaginative*** *with the ability to develop* ***innovative*** *solutions. We are looking for someone who is* ***persistent*** *and can work independently. Strong analytical skills are required.*

Categorization

Personality Trait	
Agreeableness	0
Conscientiousness	1
Extraversion	0
Openness	1
Emotional Stability	0
AI Skills	1

To examine the validity of our text-based measures, we assess whether personality traits identified in vacancy texts align with self-reported personality characteristics of workers in the same occupations using data from the UKHLS. If our measures capture personality demands in job advertisements, occupations with higher advertised demand for a given trait should tend to employ workers who score higher on that trait. This validation exercise helps assess whether personality-related language in job postiOverall, the evidence suggests that personality-related language in job postings generally aligns with the traits observed among workers in the same occupations, although emotional stability appears to capture a more context-specific dimension of job requirements. While this validation exercise exploits cross-occupation variation, our main specifications include occupational fixed effects to account for systematic differences across occupations.

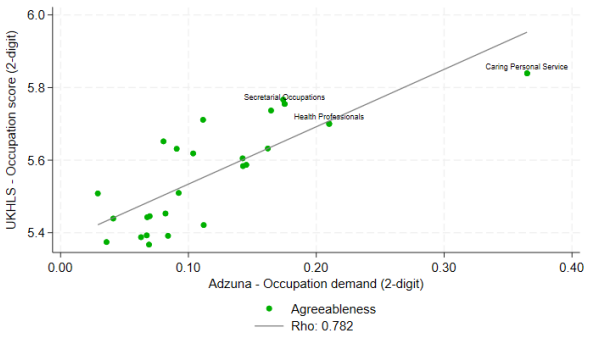
think that our narrower dictionaries reflect a conservative measurement choice: rather than maximizing the number of potentially relevant terms, we prioritize terms that are more likely to capture genuine employer demand for worker attributes in vacancy text.

Figure 1: Scatter-plot between mean standardised trait scores in UKHLS and share of vacancies that mention Personality traits in Adzuna. 2-digit SOC groups

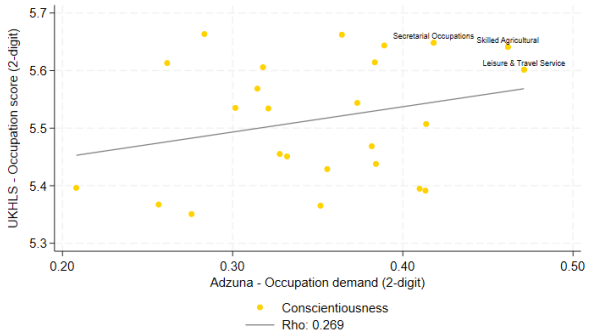
(a) Any Personality Trait



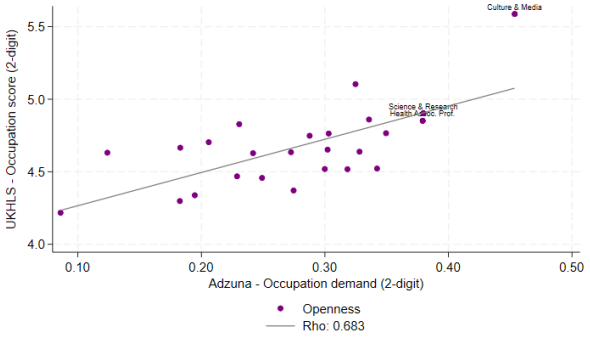
(b) Agreeableness



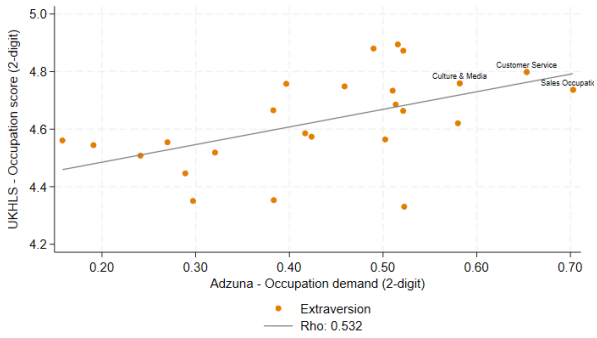
(c) Conscientiousness



(d) Openness



(e) Extraversion



(f) Emotional Stability

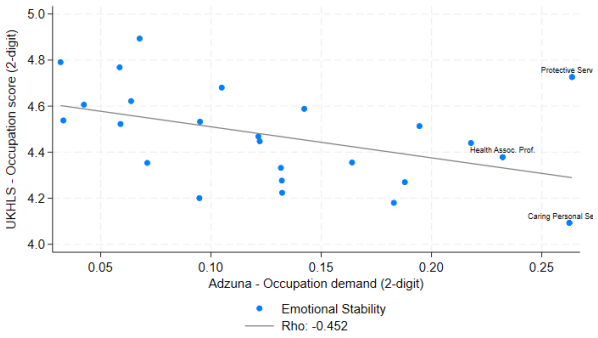


Table 2 presents descriptive statistics at the advertisement level. In the data, 72% of advertisements contain at least one personality trait-related word. Extraversion appears most frequently (44%), followed by conscientiousness (34%), openness (30%), emotional stability (14%) and agreeableness (13%).[7] Personality traits appear more often in job postings than traditional human capital measures such as soft skills (62%), hard skills (55%), or education requirements (11%), as shown in Figure A.1 in the Appendix.

Table 2: Summary statistics

	Category	Mean	SD
Basic Statistics	Number of Ads	11,662,124	
	Mean length characters	2,297	1,236
	Number of firms	128,559	
Baseline	No trait requirements	0.28	0.45
	No traits nor skills requirements	0.09	0.29
	Obs. with no requirements	89,620	
Traits	Any Personality Trait	0.72	0.45
	Conscientiousness	0.34	0.47
	Openness	0.30	0.46
	Emotional Stability	0.14	0.35
	Extraversion	0.44	0.50
	Agreeableness	0.13	0.34
Skills	Soft skills	0.62	0.48
	Hard skills	0.55	0.50
	AI skills	0.01	0.11
Seniority	High Seniority	0.01	0.08
	Middle Seniority	0.19	0.39
	Low Seniority	0.51	0.50
Education	No education specified	0.89	0.31
	GCSE	0.03	0.16
	A-level	0.04	0.19
	Bachelor's degree	0.03	0.17
	Master or Doctoral degree	0.02	0.12
Job schedule	Full time job	0.30	0.46
	Part time job	0.08	0.28
Job duration	Temporary job	0.13	0.34
	Permanent job	0.35	0.48
Experience	Experience requested	0.69	0.46
	No experience requested	0.01	0.09
Other job info	Remote Work	0.03	0.18
	Average Salary	31,940	18,690
	Negotiable salary	0.49	0.50
	Bonus offered	0.23	0.42

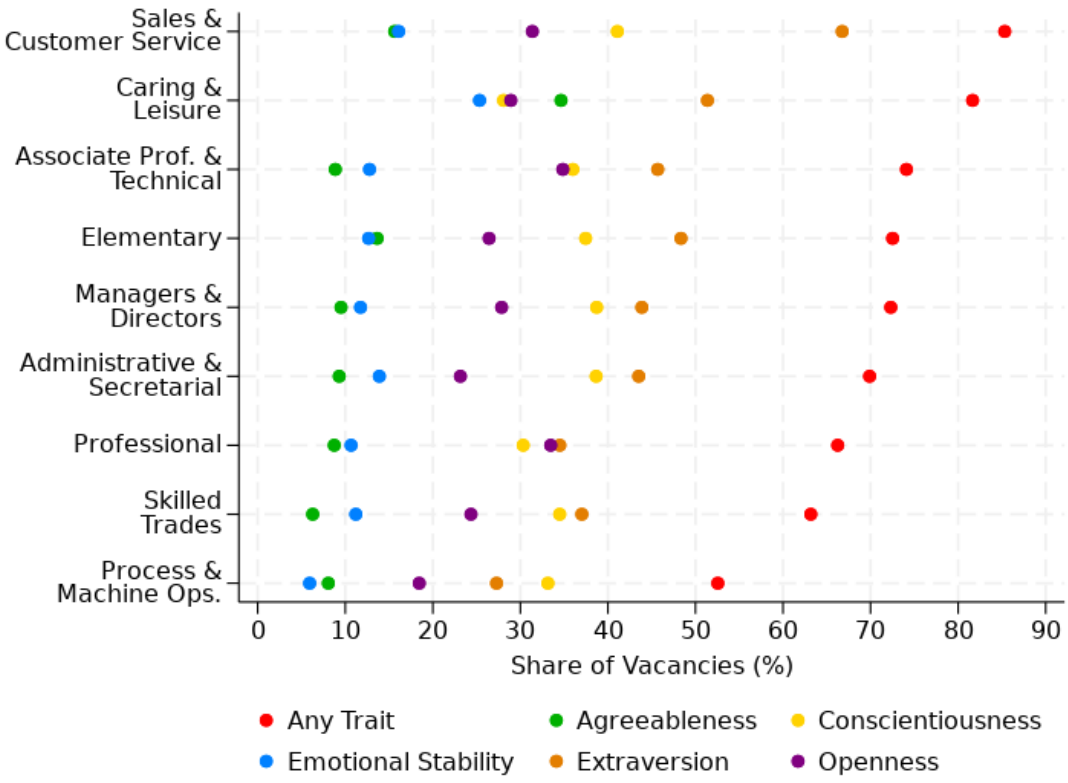
Notes: No trait requirements denotes the share of vacancies that do not contain any words from the personality-trait dictionaries. *No traits nor skills* denotes the share of vacancies that do not mention personality traits or either soft or hard skills. *Obs. with no requirements* refers to observations in which no job requirements were matched in the vacancy text.

[7] The most common terms within each trait dictionary are shown in Figure OD1 in the Online Appendix D.

Personality trait demands vary systematically across occupations. Figure 2 plots trait demands by 1-digit SOC groups. Jobs involving complex social interactions or non-routine cognitive activities show higher demands for personality traits, supporting the idea that these characteristics are increasingly important in the service-oriented economy (Borghans et al., 2014; Deming, 2017). Sales and Customer Service occupations show the highest overall trait demands (85.3%), with particular emphasis on extraversion (66.8%) and conscientiousness (41.1%). Caring and Leisure Services follow with 81.7% overall trait demand and show the highest requirements for agreeableness (34.7%) and emotional stability (25.3%). In contrast, Process and Machine Operatives show the lowest overall trait demands (52.6%). More detailed patterns at the 3-digit SOC level are presented in Table OD1 in the Online Appendix D. The demand for personality traits remains relatively stable during our sample period. As shown in Online Appendix E., after accounting for location, occupation, and firm, most personality traits exhibit limited systematic changes over time.[8]

[8]The exception is Conscientiousness, which declines gradually and becomes significantly negative by 2022. Extraversion shows some positive and statistically significant increases between 2019 and 2021 as compared to 2017. Overall, this suggests that changes in the demand for personality traits over time are modest and that the observed patterns are largely stable across the sample period.

Figure 2: Personality Trait demand by 1-digit SOC group



2.3 Measuring AI Demand

The period covered by our data, 2017–2022, captures demand for established AI, such as machine learning technologies used systematically in business functions for customer analytics, forecasting, optimisation, fraud detection, and decision support. These forms of AI are widely used in firms today. In our measurement, AI demand is defined as demand for AI-related methods, applications, and tools, rather than as general demand for digital or computer skills. We therefore distinguish AI-related skill requirements from broader computer-skill requirements, such as software development, programming, data analysis, and other general IT skills, which are measured separately and used as controls and comparison variables in the empirical analysis.

To identify AI-related vacancies, we build on the AI skill dictionaries developed by Acemoglu et al. (2022) and Babina et al. (2024), which we expand to improve coverage

and consistency in our data.[9] The expanded dictionary is intended to capture terms directly related to artificial intelligence and machine learning, including AI methods, AI applications, and AI-specific software tools.

We manually review 100 job advertisements for each AI-related term from the original dictionaries, examining the surrounding text for related tools, skills, or technologies. This process serves a similar function to the co-occurrence methods in Babina et al. (2024) and Choi and Leigh (2024), identifying contextually related AI terms. Additionally, we review over 2,000 AI-related vacancies to capture new AI tools and software packages that might be absent from structured databases. Throughout this process, we distinguish AI-specific requirements from broader IT, programming, software-development, and data-analysis requirements. Terms that refer to general digital proficiency are not used to classify vacancies as AI-related, while ambiguous terms are assessed in context and retained only when they refer to AI methods, applications, or tools. Based on this dictionary, we construct a binary indicator for AI demand: a job posting is coded as AI-related if it contains at least one keyword from the extended AI dictionary, either in the job title or in the job requirements. We report the complete list of AI skills, sources, and text replacements in Online Appendix C.

In our dataset, 1.3% of job advertisements mention AI-related skills. As shown in Figure A.2 in the Appendix, this demand has increased over time, rising from 0.75% in 2017 to a peak of 2.3% in early 2021, before stabilising around 1.5%. These numbers mirror findings by Schmidt et al. (2024), who show that AI hiring intensity in the UK peaked in late 2020 and then returned to pre-pandemic levels.

Of the 128,559 firms in the dataset, 6,436 (5%) posted at least one vacancy requiring AI-related skills over the sample period, collectively accounting for 153,079 AI-related job advertisements, equivalent to 1.3% of all postings. First-time AI-related postings were spread across the sample window: 1,396 firms posted their first AI-related vacancy in 2017, 1,129 in 2018, 1,173 in 2019, 990 in 2020, 1,385 in 2021, and 363 in the first quarter

[9]For example, we group closely related concepts such as *cross-domain learning*, *federated learning*, and *reinforcement learning* under broader categories such as *machine learning* when searching job postings.

of 2022. The variation in the demand timing across firms forms the basis of our staggered difference-in-differences analysis at the firm level, described in Section 3.

A natural concern is whether a firm's first AI-related vacancy reflects a durable shift in its demand for AI-related capabilities or a one-off change in advertisement language. The evidence suggests the former: 69.6% of the 6,436 firms that ever posted AI-related vacancies did so on more than one occasion, with a median of 3 such vacancies per firm (75th percentile: 8). Among firms whose first AI-related vacancy appeared at least eighteen months before the end of the sample, allowing three consecutive post-demand half-year periods to be observed, 46.9% posted an AI-related vacancy in the first six months after first observed AI-related demand, 40.7% in the following six months, and 38.2% in the subsequent six months. This indicates that firms' demand for AI-related skills was sustained rather than concentrated at the initial demand event. This pattern holds across AI-demand cohorts, with positive persistence rates recorded for all cohorts for which at least twelve post-demand months of data are available.

AI vacancies are strongly associated with technical skill requirements: 91.3% list at least one hard skill, compared to 54.2% of non-AI postings. The most frequently co-demanded technical skills are software proficiency (59.2%), engineering knowledge (45.3%), and data analysis (35.1%). These roles also call for interpersonal abilities such as communication (29.3%), teamwork (16.4%), and problem-solving (13.8%). The patterns of hard and soft skill co-demands with AI are illustrated in Figure A.3 in the Appendix. Despite these overlaps, AI demand remains relatively specialised. Among vacancies requiring software skills, only 8.5% also mention AI; for engineering and data analysis roles, the figures are 7.3% and 5.5% respectively. This suggests that AI complements broader technical profiles, but constitutes a distinct requirement. To ensure our results are not confounded by general programming demand, we separately examine the effect of computer science skills in the robustness analysis.

AI demand is unevenly distributed across occupations. AI-related terms appear in all 90 3-digit SOC occupational categories, but 78% of AI-demanding advertisements are

concentrated in just five: IT and Telecommunications Professionals (49.1%), Business, Research and Administrative Professionals (10.3%), Engineering Professionals (7.2%), Legal Associate Professionals (6.2%), and Electrical and Electronic Trades (5.1%). This concentration aligns with Schmidt et al. (2024), who find that AI hiring is most intense in the Information and Communication, Finance and Insurance, and Professional, Scientific and Technical sectors.

We find distinct differences in personality trait demand between AI and non-AI positions, which we show in Table 3. Openness is more common in AI-related postings (52.1% versus 29.6%), suggesting that employers seek adaptability and creativity alongside technical skills. This is related to Alekseeva et al. (2021), who document that AI-related jobs are associated with higher demand for complementary cognitive and managerial skills, which may reflect a need for broader capabilities beyond purely technical expertise. Conversely, AI postings mention lower levels of all the other personality traits, especially agreeableness and emotional stability. However, the overall incidence of soft skills and personality traits is similar across both groups (62.8% versus 62.4% for soft skills; 73.1% versus 72.1% for personality traits). This suggests that AI demand is related to specific types of personality traits rather than reducing the overall importance of non-technical attributes in hiring.

Table 3: Demand for Personality Traits and Skills in vacancies by their AI demand (percentages)

Demands AI	Hard	Soft	P.T.	Agree	Consc	Emot	Extra	Open
Yes	91.3	62.8	73.1	6.2	29.9	7.8	35.2	52.1
No	54.2	62.4	72.1	13.6	34.2	14.1	44.4	29.6

Notes: *Hard* stands for *Hard-Skill*, *Soft* for *Soft-Skill*, *P.T.* for *Any Personality Traits*, *Agree* for *Agreeableness*, *Consc* for *Conscientiousness*, *Emot* for *Emotional Stability*, *Extra* for *Extraversion*, and *Open* for *Openness*

Table 4 shows descriptive statistics aggregated at the firm-year level. On average, firms in our sample are large in terms of number of employees, posting 46 vacancies per year. In the sample, 5% of companies advertised AI skills. Approximately half of the companies have advertisements mentioning personality traits, hard skills, and soft skills.

Table 4: Summary statistics - Year Aggregation

	Category	Mean	SD
Basic Statistics	Number of observations	254,451	
	Number of vacancies per period	46	
	Number of firms	128,559	
Personality Traits	Any Personality Trait	0.56	0.50
	Conscientiousness	0.38	0.42
	Openness	0.31	0.41
	Emotional Stability	0.13	0.29
	Extraversion	0.46	0.44
	Agreeableness	0.10	0.26
	No trait requirements	0.18	0.38
	No traits nor skills requirements	0.08	0.27
Skills	Soft skills	0.64	0.42
	Hard skills	0.59	0.44
	AI skills	0.05	0.21
Firm Characteristics	Firm size	632.13	8497.23
	Profit	35,422	354,118

Notes: Data aggregated at firm-year level. *No trait requirements* refers to firm-periods with no personality traits. *No traits nor skills* refers to firm-periods with no traits or skills.

3 Methodology

We employ two complementary empirical approaches to examine how AI is associated with personality trait demands. First, we analyse vacancy-level patterns using high-dimensional fixed effects models to compare AI and non-AI positions, controlling for firm, occupation, location and time fixed effects. This analysis provides information on how personality requirements differ between AI-intensive roles and other positions within firm, occupation, location, and time periods. Second, we examine firm-level dynamics using staggered difference-in-differences estimation to study how firms' efforts to demand AI knowledge are associated with changes in their overall hiring strategies. This approach captures whether firms adjust their demand for personality traits across all positions as they begin seeking AI knowledge, rather than only within AI-specific roles.

The advantage of combining these two approaches is that they provide complementary perspectives on the relationship between AI and personality demands. The vacancy-level analysis captures differences between AI and non-AI positions, isolating how personality requirements vary across roles, holding firm, occupation, location, job characteristics, and time constant. The firm-level analysis examines how firms' overall hiring patterns evolve

as they begin seeking AI knowledge, capturing changes in the composition of personality demands across all vacancies. Together, these approaches distinguish between role-specific differences and broader changes in firms' hiring behaviour.

3.1 Vacancy-Level Analysis

Our vacancy-level specification examines the relationship between AI skill requirements and personality trait demands using a linear probability model with high-dimensional fixed effects:

$$P(T_i = 1) = \alpha + \beta AI_i + \theta X_i + \delta_l + \phi_j + \gamma_k + \lambda_t + \epsilon_i \tag{1}$$

where T_i is a binary indicator equal to one if personality trait T is mentioned in vacancy i, AI_i is a binary indicator equal to one if the vacancy requires AI skills, X_i is a vector of job posting characteristics, δ_l are location fixed effects (2-digit NUTS), ϕ_j are firm fixed effects, γ_k are occupation fixed effects (3-digit SOC), λ_t are month-year fixed effects (capturing each specific month-year combination such as September 2019, October 2019, etc.), and ϵ_i is the error term. Standard errors are clustered at both the occupation and firm level to account for potential correlation within these groups. We estimate Equation 1 separately for each personality trait (openness, conscientiousness, extraversion, agreeableness, emotional stability) and for an overall indicator equal to one if the vacancy mentions any personality trait.

The vector X_i includes observable job characteristics that might correlate with both AI requirements and personality demands. In particular, we consider: other personality traits (when the dependent variable is a specific trait), soft skills, hard skills (including computer science skills), remote work, contract type (temporary/permanent), work schedule (full-time/part-time), seniority level, education requirements, experience requirements, performance-based pay, wage negotiability, and advertisement length. Computer science skills are defined as requirements related to software development and data analysis capabilities within the broader hard skills category.[10] Including these controls addresses concerns that the AI-personality relationship reflects correlations with other job require-

[10]This appears in 1,911,086 vacancies, approximately 12 times more than AI-demanding positions.

ments rather than AI-specific demands. The coefficient β identifies how personality trait demands differ between AI and non-AI positions after controlling for occupation, location, firm, time, and other job characteristics.[11]

Several threats could bias our estimates. First, personality trait measures may be endogenous if they correlate with unobserved job requirements. We address this by including extensive observable controls in X_i, as previously explained. Second, personality demands vary systematically across occupations. We include occupation fixed effects to absorb time-invariant, occupation-specific heterogeneity and focus on within-occupation variation. Third, firm-specific HR practices might jointly influence job advertisement phrasing and underlying requirements. We address this with firm fixed effects. Fourth, local labour market conditions may vary across regions and influence both personality and AI demands. We include location fixed effects at the 2-digit NUTS level. Finally, month-year fixed effects account for temporal variation, controlling for any aggregate trends or seasonal patterns in personality or AI demands. This identification strategy isolates within-occupation, within-location, within-firm, and within-time-period variation. The high-dimensional fixed effects absorb systematic differences across these dimensions, allowing us to compare similar positions that differ primarily in AI requirements. While we cannot rule out all sources of endogeneity, the extensive controls and fixed effects address the most plausible confounders. As a robustness check, we examine whether the AI-personality relationship differs from that of general computer skills. By comparing AI and computer skill effects, we assess whether our results reflect AI-specific demands or broader technological requirements. The comparison also helps rule out that we are simply capturing general programming or technical skill requirements.

[11] For example, consider two software developer roles posted by Google in London in September 2019, one requiring AI skills and the other not. Despite having the same firm, occupation, location, and month-year, these positions may differ in other respects such as required experience or contract terms. By controlling for these observable characteristics and including fixed effects, we isolate the association between AI requirements and personality trait demands.

3.2 Firm-Level Analysis

The vacancy-level analysis described above examines how personality requirements differ between AI and non-AI positions within the same firm. When firm fixed effects are included, identification comes from variation across vacancies posted by the same firm, so the estimates can be interpreted as within-firm differences in advertised personality requirements between AI and non-AI vacancies. In addition to these role-specific differences, the emergence of demand for AI-related skills may induce broader organisational adjustments that affect hiring strategies beyond AI-specific roles. To study this possibility, we use a difference-in-differences approach in which the treatment is defined by a firm's first observed demand for AI-related skills. Specifically, we define the event date as the date of the firm's first job advertisement requiring AI-related skills. Given that our dataset begins in 2017, when relatively few firms were demanding AI-related skills, and that job advertisements typically reflect emerging skill needs and strategic shifts in technology use, the first AI-related job posting provides a meaningful indicator of when a firm begins seeking AI-related capabilities and may adjust its workforce accordingly.

Following Callaway and Sant'Anna (2021), we employ a staggered difference-in-differences approach that addresses potential bias from treatment effect heterogeneity across cohorts and time periods. The main building block is the group-time average treatment effect (ATT), defined as:

$$ATT(g,t) = \mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1] \tag{2}$$

where $Y_t(g)$ denotes the outcome at time t for firms whose first observed AI-related demand occurs in period g, and $Y_t(0)$ denotes the counterfactual outcome that would be observed in the absence of AI-related demand. G indicates the period in which a firm first appears in the data as demanding AI-related skills. The group-time average treatment effects are aggregated using non-negative weights that sum to one to obtain

overall treatment effects.[12] The analysis is conducted at the firm-period level, where the outcome variable measures average personality-trait demand across all vacancies posted by a firm within a given time period. Additionally, we separately examine the association with non-AI vacancies to assess whether changes in personality-trait demand extend beyond positions explicitly requiring AI-related skills. We consider yearly and half-yearly aggregations to balance timing precision against the number of vacancies observed per firm-period.

A concern is that AI-demanding firms may differ systematically from non-AI-demanding firms before their first observed AI-related vacancy along dimensions correlated with personality-trait demand. For example, more innovative or forward-looking firms may be both more likely to demand AI-related skills and more inclined to value traits such as openness, independently of the emergence of AI-related demand itself (Acemoglu et al., 2023). In this case, estimated treatment effects could reflect pre-existing differences between firms rather than changes associated with first observed AI-related demand. To address this potential selection issue, we employ an entropy balancing approach (Hainmueller, 2012), which reweights non-AI-demanding firms so that their pre-event characteristics match those of AI-demanding firms along key covariates. This procedure reduces bias from observable selection into AI-related demand.

The balancing is implemented separately for each personality trait and includes: initial demands for other personality traits, firm characteristics (public status, industry at 2-digit SIC), average skill requirements, location distribution indicators, and occupational composition indicators. The weights are estimated targeting the first and second moments (mean and variance) of each covariate, computed as averages over each firm's pre-treatment periods. Standard errors are clustered at the firm level throughout.

Our methodology uses a weighted staggered difference-in-differences estimator. The estimator implicitly accounts for time-invariant unobserved firm heterogeneity by exploiting within-firm variation over time (analogous to the role of firm fixed effects in a two-way

[12]The estimation excludes left-censored observations, that is, firms already advertising AI-related skills in the first observed period.

fixed effects regression). The balancing weights ensure that treated and control firms are comparable on observable pre-treatment characteristics.

4 Results

4.1 AI and Personality Traits within Vacancies

Table 5 presents the relationship between AI demand and personality traits across different specifications. Each row in the table corresponds to a different personality trait used as the dependent variable. In panel A, the dependent variable is a dummy variable that takes the value of one if the advertisement presents at least one personality trait, while in the rest of the panels, the dependent variable is each of the Big Five. Column (1) reports the baseline specification without any controls or fixed effects. Column (2) adds controls for job requirements. Column (3) further includes time fixed effects. Column (4) builds on this by incorporating location fixed effects. Column (5) adds firm fixed effects to the specification with job controls, time, and location fixed effects. Column (6) replaces firm fixed effects with occupation fixed effects. Finally, column (7) includes all controls and fixed effects: time, location, firm, and occupation.

In Panel A, where the dependent variable indicates the presence of any personality trait requirement, AI is positively associated with personality requirements in most specifications. When we examine each personality trait separately, however, the aggregate relationship masks substantial heterogeneity across traits. Panels B to F show that, across specifications, AI is positively and statistically significantly associated with openness, but negatively associated with the other personality traits. Openness captures attributes such as adaptability, creativity, and intellectual curiosity, which are particularly relevant in tasks involving novel technologies and problem-solving. Accordingly, AI-related vacancies are more likely to value these individual characteristics. This might reflect the need for workers who can adapt to evolving tools and engage with complex, non-routine tasks. In contrast, the negative associations with the remaining traits suggest that AI-related roles place relatively less emphasis on the other personality traits. The inclusion of firm fixed

effects is associated with the largest reduction in estimated coefficients, highlighting that firm-specific characteristics account for a substantial part of the relationship between AI and personality trait demands at the vacancy level. In contrast, occupation fixed effects explain less of the variation, suggesting that firm-specific differences in how AI-related capabilities are integrated matter more than occupation-specific technological needs. This is in line with Deming and Kahn (2018), who highlight the importance of firm heterogeneity in skill demands.

In the most comprehensive specifications (columns 6 and 7), the relationship with openness remains substantially high, while the coefficients for the other traits are smaller and negative. Overall, the net association on the aggregate measure of *any* trait is small but positive. For example, in column (7), the estimated coefficient for openness is 6.7 percentage points, a sizeable magnitude equivalent to about 22% of the average openness-demand rate in the sample.[13] The estimated coefficients for the other traits are negative and smaller in absolute terms, ranging from 1.2 to 2.5 percentage points.[14] This pattern is consistent with AI being associated with a reallocation across traits, and it is in line with Johnson and Acemoglu (2023), who argue that AI tends to automate routine cognitive tasks but increases demand for human attributes such as adaptability and creativity. This pattern is also consistent with Autor (2014)'s observations on how technological change tends to amplify the comparative advantage of workers in supplying problem-solving, adaptability, and creativity (p. 2), while substituting for more routine activities.

[13] This is calculated as 6.7 divided by the average openness-demand rate of about 30%.
[14] Roughly 5% to 16% of their respective average demand rates.

Table 5: AI and Personality Traits within vacancies

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. Any personality trait							
AI_{it}	0.010***	-0.012***	-0.012***	-0.011***	0.011	0.021***	0.016**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.007)	(0.005)	(0.007)
Panel B: Dependent variable Agreeableness							
AI_{it}	-0.074***	-0.028***	-0.031***	-0.033***	-0.015***	-0.011***	-0.012***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.004)	(0.003)	(0.004)
Panel C: Dependent variable Conscientiousness							
AI_{it}	-0.043***	-0.065***	-0.064***	-0.064***	-0.023***	-0.038***	-0.017**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.007)	(0.009)	(0.009)
Panel D: Dependent variable: Emotional Stability							
AI_{it}	-0.063***	-0.044***	-0.046***	-0.047***	-0.027***	-0.026***	-0.022***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.004)	(0.010)	(0.007)
Panel E: Dependent variable: Extraversion							
AI_{it}	-0.092***	-0.082***	-0.082***	-0.083***	-0.034***	-0.038***	-0.025***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.008)	(0.009)	(0.007)
Panel F: Dependent variable: Openness							
AI_{it}	0.225***	0.141***	0.143***	0.143***	0.073***	0.111***	0.067***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.008)	(0.014)	(0.013)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes
R-squared range	[0.001, 0.012]	[0.080, 0.121]	[0.082, 0.122]	[0.084, 0.123]	[0.273, 0.351]	[0.105, 0.169]	[0.281, 0.364]
N observations	11662124	11662124	11662124	9923701	9888315	9923701	9888315
N firms	128559	128559	128559	117229	81843	117229	81843

Notes: AI_{it} is AI at the vacancy level. The R-squared range shows the minimum and maximum values across the six trait regressions for each specification. Controls include: computer skills, traits and skills demand, remote work requirements, length of advertisement, contract type, work schedule, bonus offerings, education, and negotiable salary option. Standard errors are clustered at the occupation and firm level in specifications (5)-(7). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Distinguishing AI demand from general computer-skill demand: A potential concern is that the estimated association between AI-related skill demand and personality-trait requirements may reflect hiring for general software-development or data-analysis roles rather than demand for AI-specific capabilities. We address this concern by distinguishing AI-related skill demand from general computer-skill demand. We estimate three complementary specifications. First, we re-estimate our baseline model excluding computer skills as a control variable, allowing us to capture the overall association between AI-related skill demand and personality-trait requirements without conditioning on general computer-skill requirements. Second, we include both AI-related skill demand and computer-skill demand jointly to assess whether the association with AI is distinct from this general computer-skill measure. Third, we replace AI with computer skills as the main explanatory variable. This allows us to compare the personality-trait profile of AI-related vacancies with that of vacancies requiring software-development or data-analysis skills.

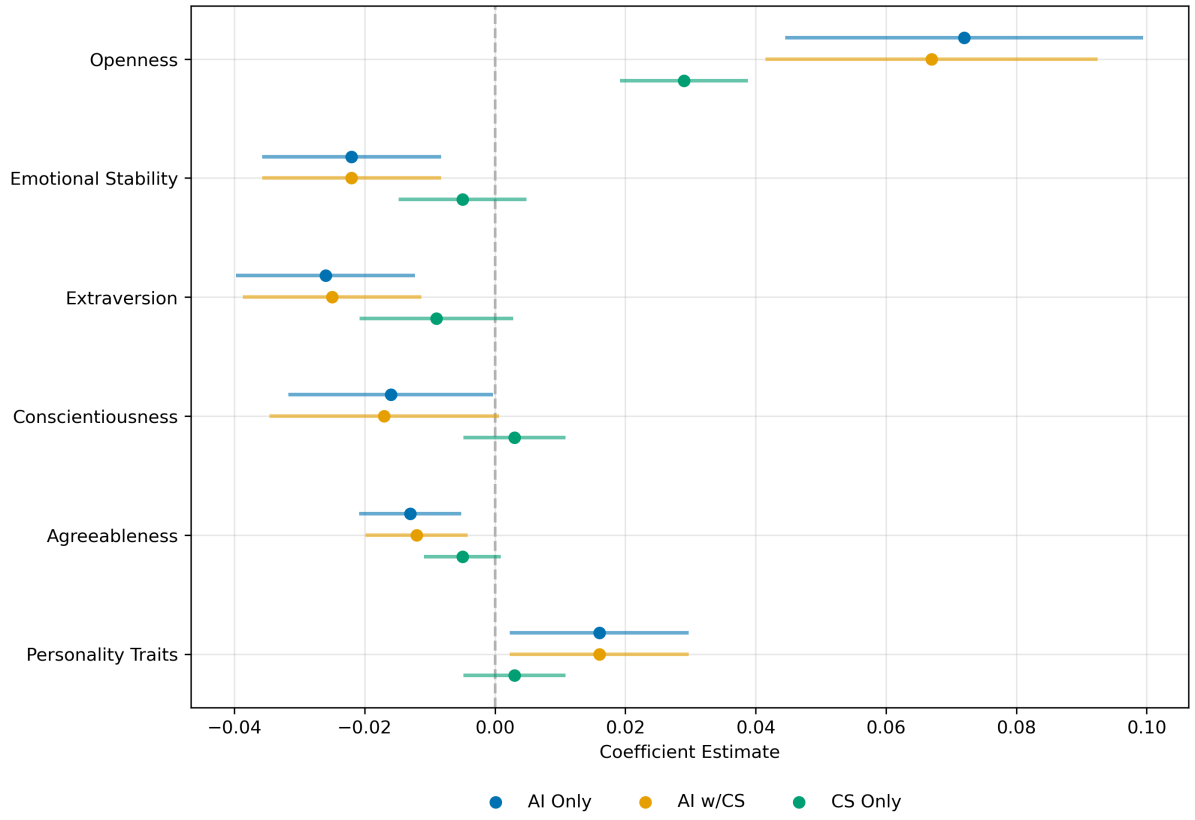
Table A.3 in the Appendix reports estimates excluding computer skills as a control, and Table A.4 in the Appendix shows results using computer skills, rather than AI-related skills, as the key explanatory variable. Excluding computer skills from the model yields an estimated coefficient for AI that is essentially unchanged relative to the estimate obtained in the main specification, where computer skills are included as a control. This stability indicates that the association between AI-related skill demand and personality-trait requirements is not driven by the inclusion of software-development and data-analysis controls.

Figure 3 plots the estimated coefficients from three separate specifications: one where the AI indicator is estimated without computer skills as a control, one where AI and computer skills are included jointly, and one where only computer skills are used instead of the AI indicator. In all regressions, we include the full set of fixed effects. Both AI-related skills and computer skills show a positive association with openness, but the magnitude of the AI association, 7.2 percentage points, is more than twice that of computer skills, 2.9 percentage points. This difference is statistically significant, as shown in Table A.5 in the Appendix. These results indicate that the openness pattern is stronger for AI-related

vacancies than for vacancies requiring general computer skills.

The remaining personality traits show no statistically significant effects in the specification of computer skills only. Furthermore, Table A.6 in the Appendix reports tests of the stability of AI coefficients when we compare specifications with and without computer skills as control. In specifications that include occupation and firm fixed effects, we cannot reject the null hypothesis of equal coefficients, suggesting that the AI estimates are robust to the inclusion of these controls. This placebo exercise reinforces the main findings by indicating that the observed relationships between AI-related skill demand and personality-trait requirements, particularly openness, are not simply capturing software-development or data-analysis requirements.

Figure 3: Personality-trait demand: AI-related skills versus general computer skills



4.2 AI and Personality Traits within the Firm

Next, we shift the unit of analysis from individual vacancies to the firm level. The analysis proceeds in three steps. First, we document the firm-level characteristics associated with

demanding AI-related knowledge and assess whether entropy balancing achieves comparability between treated and control firms on observable pre-treatment characteristics. Second, we estimate whether firms' first AI-related vacancy is followed by changes in personality-trait demand across their overall hiring portfolio, using a weighted staggered difference-in-differences specification. Third, we examine whether these changes extend beyond AI-specific positions by focusing on non-AI vacancies and we examine whether the changes observed among non-AI vacancies reflect shifts in the occupational composition of hiring or changes in openness demand within occupations.

Determinants of firms' demand for AI knowledge and covariate balance. Before analysing subsequent hiring responses, we first examine which types of firms are more likely to seek AI-related knowledge and assess covariate balance after applying the entropy balancing algorithm. The first two columns of Table OD2 in the Online Appendix compare key pre-adoption characteristics of AI-demanding firms, our treated group, and non-AI-demanding firms, our control group, before balancing.

Before balancing, AI-demanding firms differ systematically from non-demanding firms. They exhibit higher pre-treatment demand for openness, conscientiousness, and extraversion, with standardised differences of 0.452, 0.184, and 0.191, respectively. They also post more vacancies requiring hard skills and soft skills, with standardised differences of 0.473 and 0.268. In addition, AI-demanding firms tend to be larger, more profitable, and more likely to be publicly listed. Their occupational and industry composition also differs, with greater concentration in professional occupations and technology-intensive industries. These patterns suggest that AI demand is not random: firms that demand AI are systematically different from other firms before the AI demand takes place.

To address these pre-treatment differences, we use entropy balancing to reweight the control group so that treated and control firms are comparable on observable characteristics. After balancing, shown in the last two columns of Table OD2 in the Online Appendix, all standardised differences fall below 0.05 in absolute value, indicating excellent balance across the included covariates. This means that, after reweighting, treated and control

firms are similar in terms of observable pre-treatment characteristics, including their prior demand for personality traits, skill requirements, listed status, location, and occupational and industry composition. Balance diagnostics for the half-yearly aggregation are reported in Table OD3 in the Online Appendix, and show similar patterns.

Table 6: ATT estimates for staggered regressions

	(1) Half-year aggregation	(2) Yearly aggregation
Panel A: Dependent variable. Any personality trait		
$AI\ demand_{jt}$	0.021***	0.011
	(0.008)	(0.008)
Panel B: Dependent variable. Agreeableness		
$AI\ demand_{jt}$	0.007	0.003
	(0.005)	(0.006)
Panel C: Dependent variable. Conscientiousness		
$AI\ demand_{jt}$	0.005	-0.003
	(0.008)	(0.010)
Panel D: Dependent variable. Emotional Stability		
$AI\ demand_{jt}$	0.002	-0.008
	(0.005)	(0.006)
Panel E: Dependent variable. Extraversion		
$AI\ demand_{jt}$	0.013	0.007
	(0.009)	(0.010)
Panel F: Dependent variable. Openness		
$AI\ demand_{jt}$	0.027***	0.024**
	(0.009)	(0.010)
Observations	245,020	167,634

Notes: AI demand is a dummy variable that takes the value one from the first period in which we observe a firm posting a vacancy requiring AI-related skills. The estimator absorbs time-invariant unobserved firm heterogeneity through within-firm variation over time, playing the role of firm fixed effects. The estimation method is Sant'Anna and Zhao (2020)'s improved doubly robust DiD estimator based on inverse probability of tilting and weighted least squares (`drimp` in `csdid` in Stata). We use the entropy balance algorithm of Hainmueller (2012) to weight the sample. The standard errors are presented in parentheses and are clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Personality traits after AI-demand: Table 6 reports the average treatment effects from the weighted staggered difference-in-differences regressions at the firm level, based on Equation (2). In column (1), we show aggregations at the half-year level, and in column (2), the aggregation is yearly. When considering each personality trait separately, the estimated coefficients are generally small and not statistically significant, with the exception of

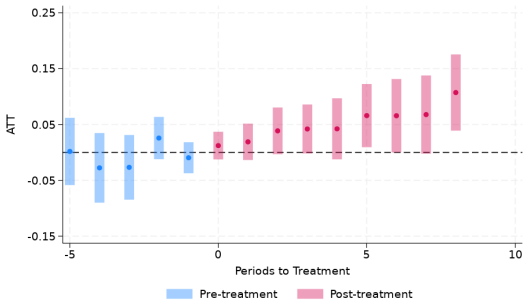
openness, which is positive and statistically significant in both columns. The estimates for openness are the largest and most precisely estimated. For example, in column 1, AI demand is associated with a 2.7 percentage-point increase in the share of vacancies requiring openness. This means that, relative to the average baseline share of 31.29% (Table 4), the share of openness-demanding vacancies would increase to approximately 34%, corresponding to an 8.6% increase. In terms of vacancy counts, for a firm posting the average number of vacancies per year, 46, this relationship corresponds to about 1.2 additional vacancies requiring openness. For treated firms, which are larger and post an average of 284 vacancies per year, the same 2.7 percentage-point increase corresponds to about eight additional vacancies requiring openness. Since treated firms have a higher baseline share of openness-demanding vacancies, 47.6%, this represents a 5.7% increase relative to their baseline.

Next, in Figure 4, we examine the dynamic relationship between AI demand and personality traits using an event study design for the half-year estimations. The event-study plots indicate that prior to AI-demand, personality traits were not statistically significantly different for AI-demanding and non-demanding firms, as expected after the balance.[15]

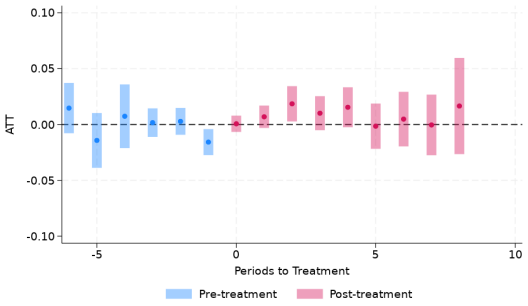
[15]We formally test for pre-trends across all traits and find no evidence against parallel trends. For example, for openness, as the outcome of primary interest, the p-value is equal to 0.842.

Figure 4: Staggered DID ATT. Half-year-level aggregation

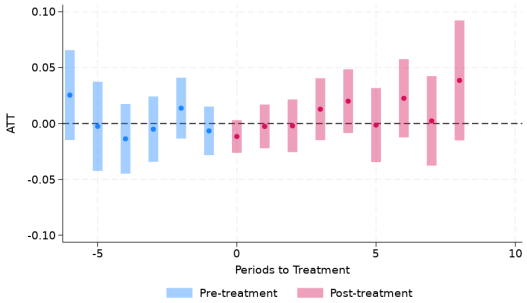
(a) Any Personality Trait



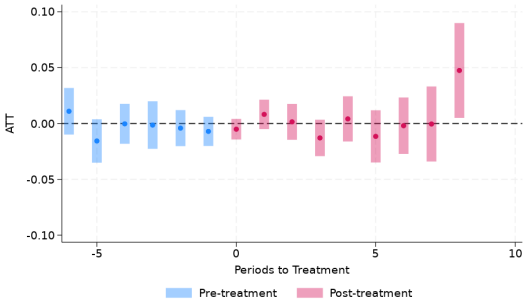
(b) Agreeableness



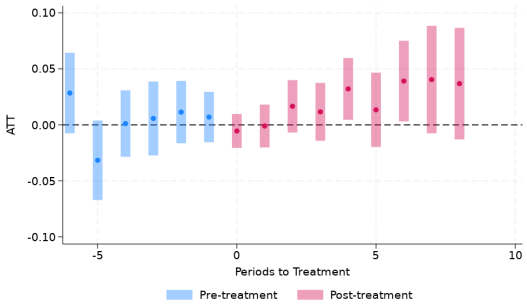
(c) Conscientiousness



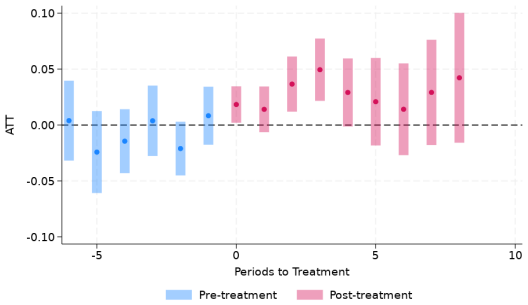
(d) Emotional Stability



(e) Extraversion



(f) Openness



From event time 0 onwards, the estimates show that, for most individual personality traits, the coefficients are small and statistically indistinguishable from zero. There are some exceptions: agreeableness is significantly higher one and a two years after first observed AI-related demand, while extraversion is significantly higher two and a half and three and a half years after first observed AI-related demand. The clearest pattern is for openness, for which the estimates become positive in the year of first observed AI-related demand and remain statistically significant for at least two years. These results indicate that first observed AI-related demand is associated with a medium-term

increase in firms' demand for openness across their overall hiring portfolio. For the aggregate indicator of any personality trait, we also observe a gradual and persistent increase. This pattern suggests that, although first observed AI-related demand is not accompanied by systematic increases in every individual personality trait, it is associated with a broader rise in personality-related requirements in hiring. This aggregate pattern may reflect heterogeneity across traits, with the overall measure capturing a general shift in advertised personality requirements that is less visible in the trait-specific estimates. The corresponding yearly results are presented in Figure A.5 in the Appendix and show a similar pattern.

Table A.7 in the Appendix reports a robustness check using propensity score matching, rather than entropy balancing, to construct the weights for the staggered difference-in-differences estimates. The results are consistent with the main pattern. The association between AI-skill demand and openness remains positive and statistically significant in both the half-year and yearly aggregations, with estimates of 0.019 and 0.017, respectively. By contrast, the estimates for the other personality traits are generally small and statistically insignificant. These results suggest that the positive relationship between firms' first AI-related vacancy and subsequent demand for openness is not specific to the entropy-balancing procedure.

Do changes extend beyond AI-related vacancies? Organisational scope and decomposition: The firm-level estimates above are calculated across all vacancies posted by the firm. A natural concern is that the openness pattern may reflect the inclusion of AI-related vacancies themselves, rather than broader differences in firms' hiring portfolios. To examine this, we recalculate personality-trait demand using only non-AI vacancies. This allows us to assess whether first observed demand for AI-related skills is also associated with trait requirements in positions that are not explicitly AI-related.

Table A.8 in the Appendix reports the results when AI-related vacancies are excluded from the outcome. The openness pattern remains visible among non-AI vacancies. At the firm level, starting to post AI-related vacancies is associated with an increase of

approximately 2 percentage points, relative to comparable control firms, in the share of non-AI vacancies mentioning openness. Relative to treated firms' baseline openness rate, this corresponds to an increase of around 4–5 percent. Estimates for the remaining personality traits are generally small and statistically insignificant, with the exception of extraversion in the half-year aggregation. The non-AI estimate is very close to the corresponding within-firm estimate obtained when all vacancies are included. This similarity indicates that the firm-level openness result is largely accounted for by changes in non-AI vacancies, rather than by the direct inclusion of AI-related postings in the outcome.

We next examine whether this broader pattern reflects changes in the occupational composition of non-AI hiring or changes in openness demand within occupations. To do so, we conduct a shift-share decomposition for AI-demanding firms, separating within-firm pre-to-post changes in openness demand among non-AI vacancies into a composition component, an intensity component, and their interaction.[16] Table A.9 in the Appendix reports the results. The components are expressed in percentage points and describe pre-to-post changes within AI-demanding firms, rather than a decomposition of the difference-in-differences estimate.

The decomposition points to two offsetting patterns. The composition component is negative and statistically significant (-12.5 pp, $p < 0.01$), indicating that AI-demanding firms shift their non-AI hiring toward occupation groups with lower baseline openness requirements. By contrast, the intensity component is positive and statistically significant ($+1.7$ pp, $p < 0.05$), indicating that non-AI vacancies become more likely to mention openness within major occupation groups. Figure A.6 in the Appendix shows that this positive within-occupation pattern appears in eight of the nine major groups, with the largest contributions from associate professional and technical occupations ($+2.6$ pp), professional occupations ($+1.8$ pp), and managers and directors ($+1.1$ pp). Because the decomposition is based on raw pre-to-post changes within AI-demanding firms, the

[16]The decomposition is conducted at the SOC 1-digit level to obtain stable estimates for each of the nine major occupation groups. Overall results do not change using the SOC 2-digit level, as used in the main DiD specification.

components are not directly comparable to the DiD estimates reported above. Instead, they help interpret the source of the non-AI result. The positive DiD estimate for non-AI vacancies is not driven by firms reallocating hiring toward occupations where openness is already more commonly required. If anything, AI-demanding firms shift non-AI hiring toward occupation groups with lower baseline openness requirements. The positive intensity component instead suggests that the relative increase documented in the DiD reflects greater mention of openness within occupations.

This within-occupation pattern is consistent with the interpretation that firms might hire AI workers to develop new products and services, make greater use of proprietary data, support strategic decision-making, move toward more data-driven production or just new organisational routines, then openness may be relevant not only for AI specialists who design or implement AI systems, but also for workers who need to engage with experimentation, new information, and changing organisational practices. The positive intensity effects among managers and directors, professional occupations, and associate professional and technical occupations are consistent with this interpretation. They also align with recent evidence in Alekseeva et al. (2026) that AI adoption changes the skill requirements of managerial roles. Taken together, the non-AI estimates and decomposition results suggest that observed AI-related demand is associated with broader differences in firms' hiring strategies, rather than only with the composition of AI-specific vacancies.

5 Summary and concluding remarks

This paper examines how firms' demand for AI-related skills is associated with the types of workers they seek to hire, focusing on personality traits. Using 11.7 million UK job advertisements by 128,559 companies from 2017 to 2022, we document systematic differences in the demand for Big Five personality traits at both the vacancy and firm levels. We extract personality requirements from job descriptions using dictionaries grounded in the psychology literature and identify AI-related roles through detailed skill measures.

We report two main findings. First, comparing AI and non-AI vacancies within firms,

occupations, locations, and time periods, we find that AI-related roles are 6.7 percentage points more likely to demand openness—the trait associated with creativity, adaptability, and curiosity—and less likely to mention the other personality traits. Second, using weighted staggered difference-in-differences estimates based on the timing of firms' first AI-related vacancy, we find that the association between AI-related skill demand and openness extends beyond AI-specific jobs. Firms that become AI-demanding display 2.7 percentage points higher demand for openness across their vacancy portfolio, equivalent to roughly 9% of the average firm-year openness-demand rate. The pattern is also present in non-AI positions, with no corresponding systematic declines in other traits.

A shift-share decomposition of non-AI vacancies shows that this broader pattern is not driven by firms shifting hiring toward occupations with higher baseline openness requirements. In fact, AI-demanding firms shift non-AI hiring toward occupation groups with lower baseline openness demand, while within most broad occupation groups their non-AI vacancies become more likely to mention openness. Together, these findings suggest that AI-related hiring is associated with a greater relative emphasis on openness to experience: directly within AI-related roles, and more broadly through higher openness-related language in non-AI vacancies relative to comparable firms and within broad occupation groups.

Our findings contribute to the literature on technological change and labour demand by highlighting a previously unexplored channel: shifts in the personality of workers that employers demand. In this sense, AI demand appears to be associated with not only what workers need to know, but also which behavioural attributes are most valued in the workplace. Our results are consistent with the idea that firms screen for these complementarities at the hiring stage through the personality traits they request in job advertisements.

The distinction between skill requirements and personality demands has implications for how firms adjust to new technologies. While technical skills can be developed through training and education, personality traits are less easily changed. As firms begin

demanding AI-related capabilities, they may place greater emphasis on hiring workers whose behavioural characteristics fit the way work is organised around these technologies, rather than relying only on retraining their existing workforce. This greater emphasis on selection may affect how workers sort across jobs and firms, and may shape the distribution of opportunities for individuals with different personality profiles. More broadly, the results suggest that technological change is associated with labour-demand patterns along dimensions that are less responsive to standard education and training policies.

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Appendix

Additional Figures and Tables

Table A.1: Occupation Characteristics at 1-digit SOC level

SOC	Name	Vacancies	Share Adzuna (%)	Census (%)
1	Managers & Directors	170131	5.9	10.4
2	Professional	584393	20.3	16.9
3	Ass. Prof. & Technical	495760	17.2	12.7
4	Administrative	159602	5.5	11.4
5	Skilled Trades	225693	7.8	11.6
6	Caring & Leisure	534602	18.6	9.4
7	Sales & Customer Service	228892	8.0	8.6
8	Process & Machine Ops.	164996	5.7	7.2
9	Elementary	314300	10.9	11.9

Notes: Share Adzuna represents percentage of all Adzuna vacancies. Census shows employment shares from the 2021 Census.

Figure A.1: Frequency of Job Requirements

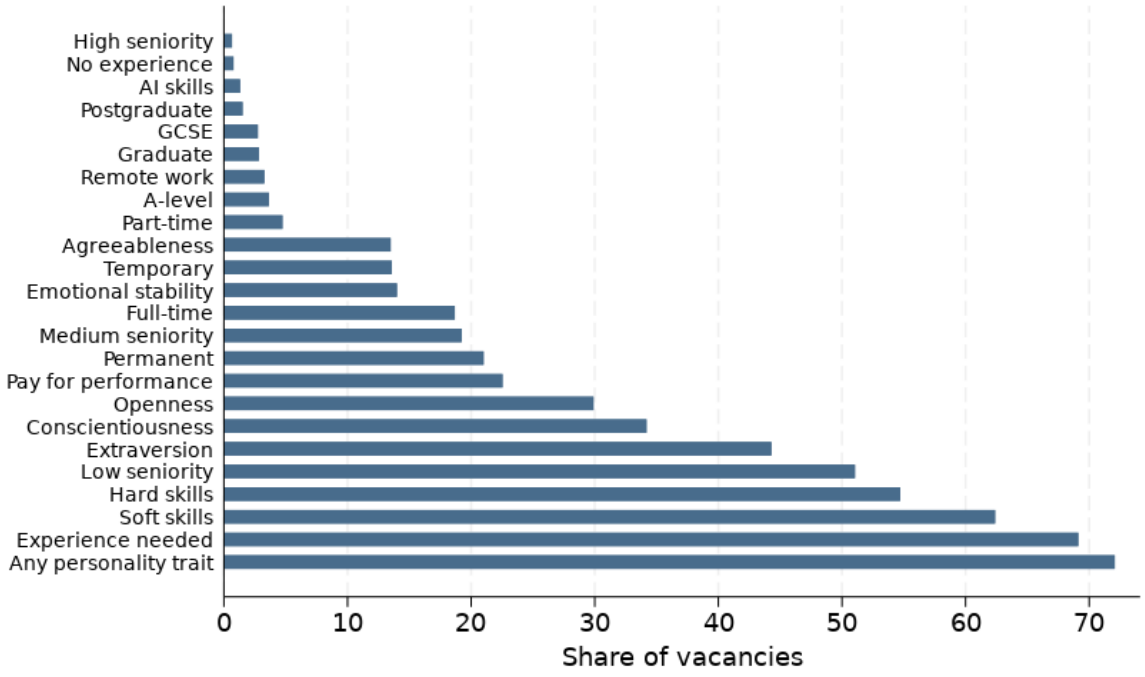
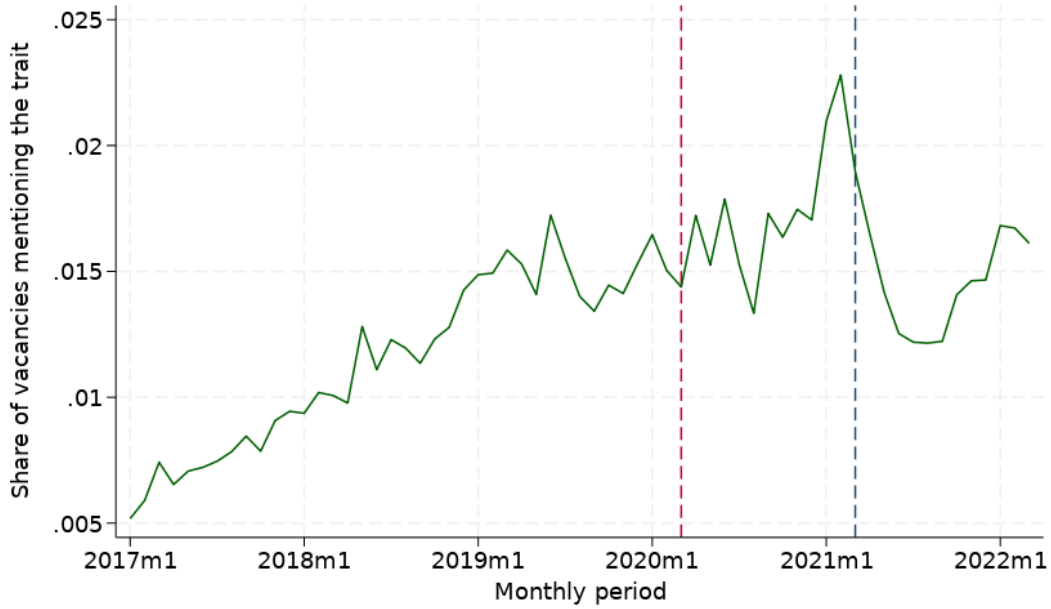


Table A.2: **Dictionary of Personality Traits**

Agreeableness	**Extraversion**	**Conscientiousness**	**Emotional Stability**	**Openness**
affection (4, 5)	adventurous (3, 5, 7)	ambitious (2, 7)	at ease (3)	artistic (3 - 7)
agreeable (all)	assertive (1, 3 - 7)	careful (6)	calm (2, 3, 5)	bright (3, 6)
altruist (5, 7)	bold (3, 6)	competence (5, 7)	emotional (all)	clever (5)
amiable (1)	candour (1)	conscientious (all)	hostil (5, 7)	creative (1 - 3, 6)
considerate (3, 6)	confident (7)	decisive (1)	fearful (6)	curious (1, 2 - 5, 7)
cooperative (1, 3, 5 - 7)	courage (1)	dependable (1, 5)	tempered (2, 5, 6)	ideas (5, 7)
courteous (1)	daring (3, 6)	dutiful (7)	fortitude (ads)	imaginative (2 - 7)
empathy (1)	energetic (1, 3 - 7)	efficient (1, 4 - 7)	pressure (ads)	independent-mind (5)
forgiving (4, 5, 7)	enthusiastic (4, 5, 7)	hardwork (2, 3)	resilience (ads)	ingenious (5)
gentle (5)	expressive (1)	logic (1)	tense (4, 5)	innovative (3, 6)
good-nature (2, 5)	extraverted (all)	neat (3, 7)	unflappable (ads)	insightful (1, 4, 5)
helpful (3, 5, 6)	forceful (5)	organised (1, 3 - 7)	patience (ads)	intellectual (1, 5, 6)
kind (3 - 6)	friendly (7)	painstaking (5)	-	intelligent (1, 3, 5)
naturalness (1)	fun (2)	persevering (2)	-	introspective (3, 6)
pleasant (3, 5, 6)	gregarious (1, 5, 7)	persistence (1)	-	inventive (5)
sensitive (5)	humour (1)	planful (4, 5)	-	openness (all)
straight-forward (5, 7)	optimist (1)	precise (1, 5)	-	original (2, 4)
sympathetic (3 - 7)	outgoing (4, 5)	prompt (3, 6)	-	prefer variety (2)
trust (2, 3 - 7)	outspoken (5)	punctual (1, 2)	-	reflective (3)
unselfish (3, 5, 7)	passionate (2)	reliable (4, 5)	-	sharp (5)
warm (1, 3, 5 - 7)	playful (1)	responsible (3 - 5)	-	unconventional (5, 7)
	pro-active (2 - 7)	striving (5, 7)	-	wide interests (4, 5, 7)
	sociable (5, 7)	systematic (3, 7)	-	witted (5)
	spontaneous (1)	thorough (3 - 5, 7)	-	-
	talkative (1 - 6)		- -	-

Notes: The numbers denote which dictionaries the words are mentioned in. The numbers correspond to the following dictionaries: (1) Goldberg (1990); (2) Costa Jr and McCrae (1992); (3) Goldberg (1992); (4) McCrae and John (1992); (5) O. P. John and Srivastava (1999); (6) Saucier (1994); (7) Heckman and Kautz (2012)

Figure A.2: Change over time of AI-skills demand



Dashed lines show the beginning (red) and end (blue) of Covid-19 restrictions.

Figure A.3: Share of Hard-Skills demanded in AI Vacancies

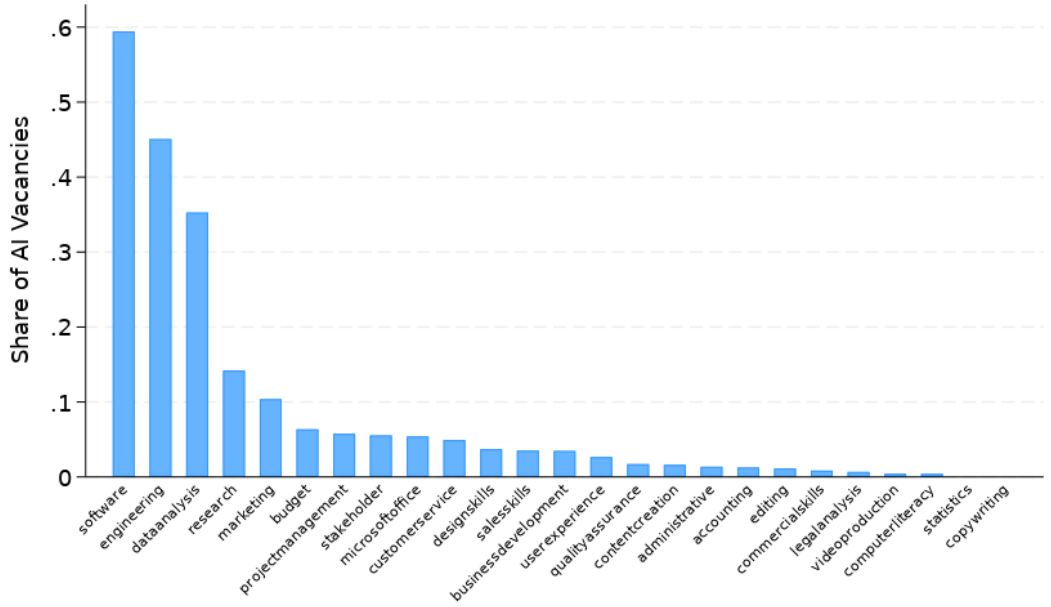


Figure A.4: Share of Soft-Skills demanded in AI Vacancies

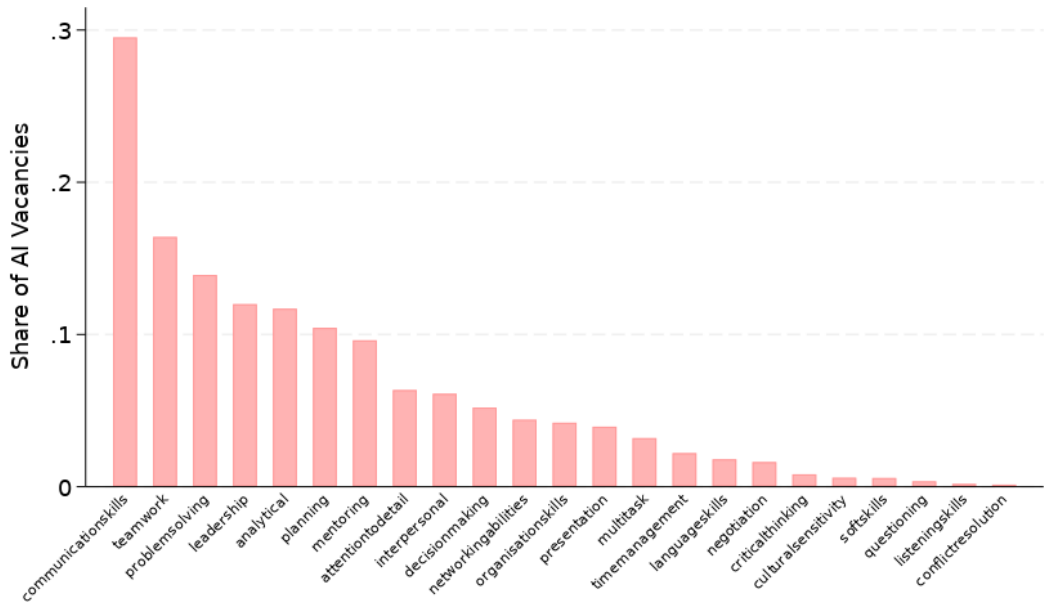
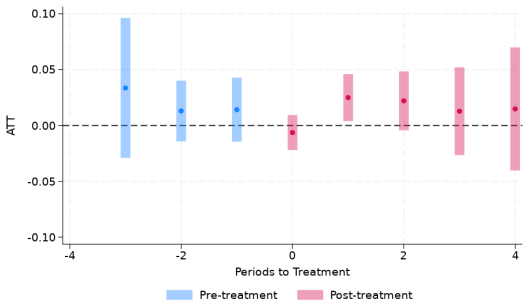
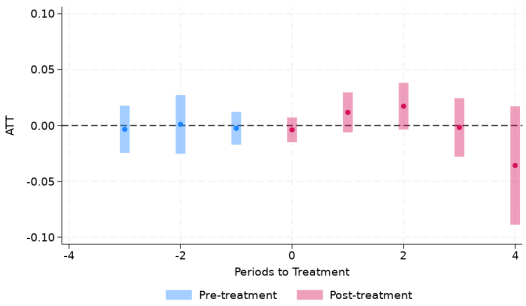


Figure A.5: Staggered DID ATT. Year-level aggregation, Entropy Balance

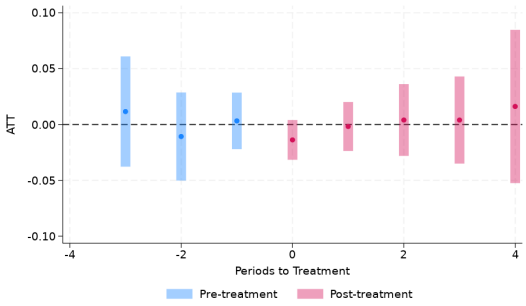
(a) Any Personality Trait



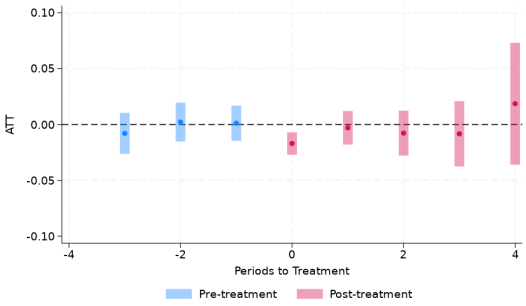
(b) Agreeableness



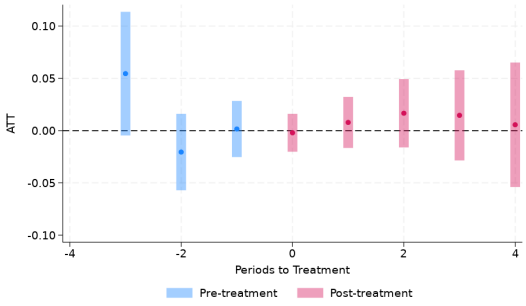
(c) Conscientiousness



(d) Emotional Stability



(e) Extraversion



(f) Openness

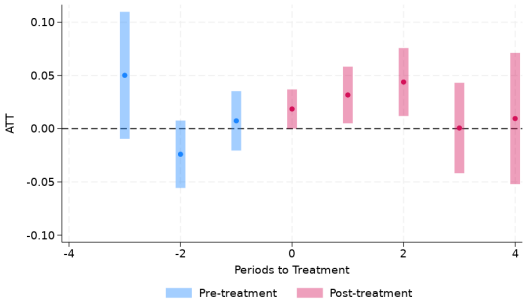


Table A.3: AI and Personality Traits excluding Computer Science skills

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. Any personality trait							
AI_{it}	0.010***	-0.032***	-0.032***	-0.029***	0.012*	0.021***	0.016**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.007)	(0.005)	(0.007)
Panel B: Dependent variable. Agreeableness							
AI_{it}	-0.074***	-0.047***	-0.050***	-0.052***	-0.017***	-0.012***	-0.013***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.004)	(0.003)	(0.004)
Panel C: Dependent variable. Conscientiousness							
AI_{it}	-0.043***	-0.073***	-0.072***	-0.071***	-0.023***	-0.037***	-0.016**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.007)	(0.008)	(0.008)
Panel D: Emotional Stability							
AI_{it}	-0.063***	-0.056***	-0.057***	-0.058***	-0.029***	-0.026***	-0.022***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.005)	(0.009)	(0.007)
Panel E: Dependent variable. Extraversion							
AI_{it}	-0.092***	-0.114***	-0.113***	-0.112***	-0.038***	-0.041***	-0.026***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.008)	(0.009)	(0.007)
Panel F: Dependent variable. Openness							
AI_{it}	0.225***	0.171***	0.173***	0.172***	0.081***	0.120***	0.072***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.008)	(0.016)	(0.014)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes
R-squared range	[0.000, 0.003]	[0.078, 0.119]	[0.080, 0.120]	[0.084, 0.121]	[0.273, 0.351]	[0.105, 0.169]	[0.281, 0.364]
N observations	11662124	11662124	11662124	9923701	9888315	9923701	9888315
N firms	128559	128559	128559	117229	81843	117229	81843

Notes: The R-squared range shows the minimum and maximum values across the six trait regressions for each specification. Controls include: traits and skills demand, remote work requirements, length of advertisement, contract type, work schedule, bonus offerings, education, and negotiable salary option. Standard errors are clustered at the occupation and firm level in specifications (5)-(7). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Impact of Computer Skills on Personality Traits

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. Any personality trait							
Computer skills$_{it}$	0.022***	-0.041***	-0.041***	-0.038***	0.003	0.001	0.003
	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)	(0.005)	(0.004)
Panel B: Dependent variable. Agreeableness							
Computer skills$_{it}$	-0.053***	-0.041***	-0.041***	-0.042***	-0.009**	-0.005	-0.005
	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)	(0.005)	(0.003)
Panel C: Dependent variable. Conscientiousness							
Computer skills$_{it}$	0.028***	-0.020***	-0.019***	-0.018***	0.000	0.005	0.003
	(0.000)	(0.000)	(0.000)	(0.000)	(0.005)	(0.006)	(0.004)
Panel D: Dependent variable. Emotional Stability							
Computer skills$_{it}$	-0.016***	-0.026***	-0.026***	-0.024***	-0.009**	-0.001	-0.005
	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)	(0.005)	(0.005)
Panel E: Dependent variable. Extraversion							
Computer skills$_{it}$	-0.029***	-0.070***	-0.070***	-0.065***	-0.014***	-0.015*	-0.009
	(0.000)	(0.000)	(0.000)	(0.000)	(0.005)	(0.009)	(0.006)
Panel F: Dependent variable. Openness							
Computer skills$_{it}$	0.124***	0.070***	0.070***	0.068***	0.034***	0.037***	0.029***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.005)	(0.006)	(0.005)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes
R-squared range	[0.000, 0.010]	[0.080, 0.121]	[0.082, 0.122]	[0.084, 0.123]	[0.273, 0.351]	[0.105, 0.169]	[0.281, 0.364]
N observations	11662124	11662124	11662124	9923701	9888315	9923701	9888315
N firms	128559	128559	128559	117229	81843	117229	81843

Notes: The R-squared range shows the minimum and maximum values across the six trait regressions for each specification. Controls include: traits and skills demand, remote work requirements, length of advertisement, contract type, work schedule, bonus offerings, education, and negotiable salary option. Standard errors are clustered at the occupation and firm level in specifications (5)-(7). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Tests of whether AI-related skill coefficients differ from computer-skill coefficients

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. All personality traits							
$AI - Computer$	-0.012***	0.009***	0.009***	0.009***	0.009	0.020***	0.013*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.264)	(0.007)	(0.097)*
Panel B: Dependent variable. Agreeableness							
$AI - Computer$	-0.021***	-0.006***	-0.009***	-0.010***	-0.008	-0.007	-0.008
	(0.000)	(0.000)	(0.000)	(0.000)	(0.138)	(0.247)	(0.116)
Panel C: Dependent variable. Conscientiousness							
$AI - Computer$	-0.071***	-0.053***	-0.053***	-0.053***	-0.023***	-0.041***	-0.020**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.007)	(0.000)	(0.029)
Panel D: Dependent variable. Emotional Stability							
$AI - Computer$	-0.046***	-0.030***	-0.031***	-0.033***	-0.020***	-0.025**	-0.018**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.016)	(0.039)
Panel E: Dependent variable. Extraversion							
$AI - Computer$	-0.063***	-0.044***	-0.043***	-0.046***	-0.023**	-0.026**	-0.017*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.015)	(0.041)	(0.061)
Panel F: Dependent variable. Openness							
$AI - Computer$	0.101***	0.101***	0.103***	0.104***	0.046***	0.083***	0.042***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes
N observations	11662124	11662124	11662124	9923701	9888315	9923701	9888315
N firms	128559	128559	128559	117229	81843	117229	81843

Notes: Each column reports the estimated coefficient on AI_{it} from a single regression that includes both AI_{it} and computer-skill demand jointly. The p-values in parentheses correspond to a Wald test of the null hypothesis H_0: $\beta_{AI} = \beta_{CS}$ within the same regression, using the full variance-covariance matrix of the estimator (`test ai_extended = computer_skills` in Stata). This accounts for the covariance between the two estimates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Difference between Estimated AI Coefficients With and Without Computer-Skill Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable. Any Personality Trait							
AI	0.013***	-0.020***	-0.020***	-0.018***	0.001	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.955)	(0.999)	(0.960)
Panel B: Dependent variable. Agreeableness							
AI	-0.029***	-0.019***	-0.019***	-0.019***	-0.002	-0.001	-0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.705)	(0.780)	(0.887)
Panel C: Dependent variable. Conscientiousness							
AI	0.018***	-0.008***	-0.008***	-0.007***	0.000	0.001	0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.983)	(0.903)	(0.953)
Panel D: Dependent variable. Emotional Stability							
AI	-0.008***	-0.011***	-0.011***	-0.011***	-0.002	-0.000	-0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.772)	(0.998)	(0.934)
Panel E: Dependent variable. Extraversion							
AI	-0.014***	-0.031***	-0.031***	-0.029***	-0.003	-0.003	-0.002
	(0.000)	(0.000)	(0.000)	(0.000)	(0.785)	(0.791)	(0.872)
Panel F: Dependent variable. Openness							
AI	0.066***	0.030***	0.030***	0.029***	0.008	0.009	0.005
	(0.000)	(0.000)	(0.000)	(0.000)	(0.481)	(0.693)	(0.786)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	No	Yes
Occupation FE	No	No	No	No	No	Yes	Yes
N observations	11662124	11662124	11662124	9923701	9888315	9923701	9888315
N firms	128559	128559	128559	117229	81843	117229	81843

Notes: Each cell reports the difference between the AI coefficient estimated in a specification without computer-skill controls and the corresponding coefficient estimated with computer-skill controls. Positive values indicate that the estimated AI coefficient is larger when computer-skill controls are omitted. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: ATT Estimates from staggered specifications p-score

	(1) Half-year aggregation	(2) Yearly aggregation
Panel A: Dependent variable. Any personality trait		
$AI\ demand_{jt}$	0.007	0.005
	(0.006)	(0.006)
Panel B: Dependent variable. Agreeableness		
$AI\ demand_{jt}$	0.001	-0.002
	(0.004)	(0.005)
Panel C: Dependent variable. Conscientiousness		
$AI\ demand_{jt}$	0.001	0.000
	(0.007)	(0.007)
Panel D: Dependent variable. Emotional Stability		
$AI\ demand_{jt}$	-0.007	-0.013**
	(0.005)	(0.005)
Panel E: Dependent variable. Extraversion		
$AI\ demand_{jt}$	0.003	0.002
	(0.008)	(0.007)
Panel F: Dependent variable. Openness		
$AI\ demand_{jt}$	0.019**	0.017***
	(0.008)	(0.007)
N Observations	242,666	167,634

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. ATT estimates for AI and non-AI positions. Treatment (AI demand) based on first AI vacancy posted by firm. Half-year and yearly aggregation. Standard errors in parentheses.

Table A.8: ATT Estimates Non-AI Positions: Entropy Balance

	(1) Half-year aggregation	(2) Yearly aggregation
Panel A: Dependent variable. Any personality trait		
$AI\ demand_{jt}$	0.018**	0.011
	(0.008)	(0.008)
Panel B: Dependent variable. Agreeableness		
$AI\ demand_{jt}$	0.007	0.000
	(0.004)	(0.006)
Panel C: Dependent variable. Conscientiousness		
$AI\ demand_{jt}$	0.007	0.003
	(0.008)	(0.009)
Panel D: Dependent variable. Emotional Stability		
$AI\ demand_{jt}$	0.004	-0.006
	(0.006)	(0.006)
Panel E: Dependent variable. Extraversion		
$AI\ demand_{jt}$	0.014*	0.007
	(0.009)	(0.009)
Panel F: Dependent variable. Openness		
$AI\ demand_{jt}$	0.020**	0.019*
	(0.009)	(0.010)
Observations	244,591	167,311

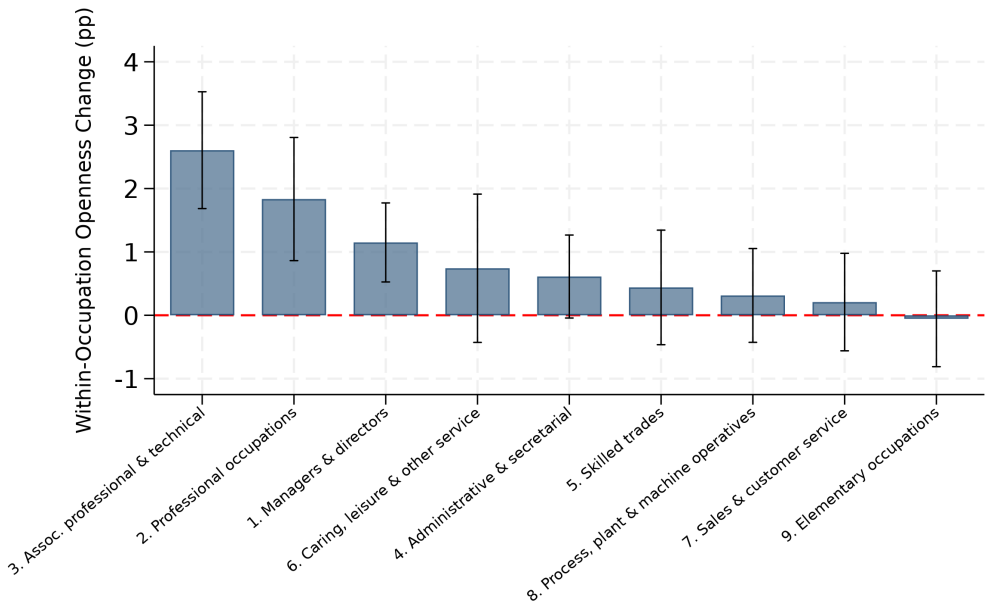
Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. ATT estimates for non-AI positions only. Personality traits calculated from vacancies that do not require AI skills. Treatment (AI demand) based on first AI vacancy posted by firm. Half-year and yearly aggregation. Standard errors in parentheses.

Table A.9: Shift-Share Decomposition of Openness Demand in Non-AI Vacancies

Component	Mean (pp)	(SD)
Composition effect	-12.470***	(20.191)
Intensity effect	1.711**	(39.270)
Cross term	2.283***	(27.130)
Total	-8.476***	(36.687)
Firms	3039	

Notes: Shift-share decomposition of the pre-to-post change in openness demand among non-AI vacancies for AI-demanding firms. Each observation is one demanding firm. Values in percentage points. The composition effect equals the change in major occupation group shares (SOC 1-digit) weighted by the sample-wide pre-treatment openness rate per major group. The intensity effect equals the pre-treatment major group share weighted by the within-group change in openness demand. The decomposition uses SOC 1-digit groups for clarity; the main DiD specification controls for occupation at the SOC 2-digit level. Stars from one-sample t-tests against zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A.6: Within-Occupation Openness Change in Non-AI Vacancies by Major Occupation Group



Notes: Each bar is the intensity component of the shift-share decomposition for one SOC 1-digit major occupation group: the pre-treatment share of non-AI vacancies in that group, multiplied by the within-group change in openness demand between the pre- and post-treatment periods, averaged across AI-demanding firms (N = 3,039). Values are in percentage points. Error bars show 95% confidence intervals. Groups are sorted from highest to lowest mean contribution.

Online Appendix

Contents of Online Appendix

A. Datasets used

B. Matching of the datasets

C. Construction of variables and dictionaries

D. Additional Figures and Tables

E. Personality Traits. Evolution over time

A. Datasets used

The datasets we used in this study, in addition to Adzuna, are the following: **Companies House:**

Companies House is an executive agency of the UK government that provides information about registered and dissolved companies in the UK. Companies House includes 5,166,124 companies registered in the UK since 1885. It provides information for most companies' addresses, postcode of registered office, and a 4 to 6 digit Standard Industrial Code (SIC). After removing companies with the same name (hence impossible to differentiate), 4,832,015 unique companies remained. Since SIC codes are only provided for private companies, public institutions like the NHS, Universities, or Charities were not considered. Institutions and dormant companies whose SIC codes are generic (99999) were also not considered. The final Companies House dataset consists of 4,581,872 private companies.

FAME: It is a financial database covering company-level information. It is part of the Bureau van Dijk suite of business databases, which also includes international platforms like Orbis. It covers 3.4 million companies in the UK and the Republic of Ireland. The dataset contains information on company accounts, activities, ownership and management. We retrieve information on whether companies are publicly listed and balance sheet information.

UK Household Longitudinal Study: Also known as Understanding Society, the survey is a large-scale longitudinal study that aims to provide valuable insights into the social and economic aspects of individuals and households over time. Surveys a wide range of information from participants, including demographic data, employment status, income, health, and subjective well-being. Data were collected with the same methodology from 2009 until 2022.

In wave C, conducted during 2011 and 2012, the survey asked individuals to complete a short questionnaire on self-perceived attitudes from which the Big Five traits were inferred. The module presented three statements for each personality trait, and individuals

selected the number that best described how they saw themselves *as persons*, using a scale from 1 to 7, where 1 meant *does not apply to me at all* and 7 meant *applies to me perfectly*. The scores were constructed by averaging the three responses for each trait after reverse-coding items phrased in the opposite direction of the trait. For consistency with our demand-side measure, which captures *emotional stability* rather than neuroticism, the neuroticism score was transformed into emotional stability using $8 - x$, so that higher values indicate greater emotional stability.

- **Agreeableness:**
  - I see myself as someone who is sometimes rude to others.
  - I see myself as someone who has a forgiving nature.
  - I see myself as someone considerate and kind to almost everyone.
- **Conscientiousness:**
  - I see myself as someone who does a thorough job.
  - I see myself as someone who tends to be lazy.
  - I see myself as someone who does things efficiently.
- **Extraversion:**
  - I see myself as someone talkative.
  - I see myself as someone who is outgoing, sociable.
  - I see myself as someone who is reserved.
- **Neuroticism:**
  - I see myself as someone who worries a lot.
  - I see myself as someone who gets nervous easily.
  - I see myself as someone who is relaxed, handles stress well.
- **Openness:**
  - I see myself as someone who is original, comes up with new ideas.
  - I see myself as someone who values artistic, aesthetic experiences.
  - I see myself as someone who has an active imagination.

B. Matching of the datasets

The main purpose of the matching strategy was to standardise text entries, in particular the name of the company, location, and descriptive variable of the advertisement, in order to be able to match the different datasets we use. To create an occupation variable, the Turrell et al. (2019) algorithm was used. The code requires as inputs advertisement text, job title, and industry. For the latter, companies' variables were matched with Companies House. Cleaned text was later used as input to retrieve job requirements, wages, and location.

Since the entire dataset was too large for the cleaning process to be conducted in one attempt (646 GB), it was first divided into 272 weekly snapshots, and each snapshot was subdivided into 1% samples using the last 2 digits of the vacancy identifier. Each of these 27,200 datasets contained around 10,000 observations.

Companies names cleaning process

Adzuna and Companies House companies lists were cleaned using the same code. The process, in STATA, involved the use of regular expressions to identify characters and words. The process intended to minimise the potential discrepancies between the same name in both datasets, therefore eliminating all non-relevant information to the name and homogenising formats. It included lowering the case; removing consecutive internal blank spaces, quotes, and special characters; changing symbol characters for their word counterpart (& for *and*), dropping endings such as "*.sol*" or "*.com*"; removing "*The*" at the beginning, and replacing text not encoded as American Standard Code for Information Interchange (ASCII) into ASCII.

In Adzuna, some companies' names identify the place where the job is located, such as *KPMG (Liverpool)*, or *Accenture (U.K.)*. This information is also available in the "location" variable, so its presence in the company name stops the algorithm from matching the company with Companies House. In consequence, the information inside the brackets at the end of a company was also removed. A considerable difference between Adzuna and

Companies House is that the former frequently does not incorporate companies' categories, such as *Private Limited Company, Public Limited Company, Limited Liability Company* or its diverse acronyms and synonyms *(plc, llc, ltd, plc, lp, pcc, plc, scio, cio, cic, ukeig, limited, srl, inc, co, sp, group)*. These were removed if they appeared at the end of the company name.The cleaning process was repeated 3 times to ensure that companies with several inaccuracies in their names could be exhaustively cleaned ("*K.F.C (Liverpool) LTD++*" into "*kfc*").

Matching companies with Companies House

Since companies' names in Adzuna could have diverse variations, the cleaning process consisted of generating a dataset with all the existing entries of companies in Adzuna and applying a cleaning code. This *matching dataset* was then associated with the Companies House cleaned list of companies before removing public and dormant companies, merging into a unique data set both Adzuna's raw company name, the clean name common to Companies House, and all Company information matched from Companies House. Nearly half of the unique company entries in Adzuna (337,461) were matched. The remaining companies either present a fantasy name different from the legal name, which means they were not listed in Companies House or have missing company names. From the

Interestingly, out of the 4.5 million active and inactive companies listed in Companies House, less than 10% were found in online job vacancies. This is explained by the high prevalence of Recruitment Agencies, whose growth since the global crisis in 2008 reshaped companies' strategy to recruit employees. Low barriers and increased digitalisation have benefited the industry over the past years. According to the Recruitment and Employment Confederation (REC), by 2021, more than 30,000 recruitment enterprises were operating across the UK. In the same year, nearly 550,000 permanent placements and 22.4 million temporary contracts were made by the recruitment industry (REC, 2022).

In addition, half of the companies in the Adzuna vacancy dataset could not be matched with a company in the CH data. The main reason is that companies do not necessarily publish their legal registration name when posting a job opening. These may appear with

a shorter version, an acronym or a more known fantasy name. Even although the cleaning algorithm aims to unify criteria in terms of format and accessory wording, if a company publishes job advertisements using a highly different name than its legal name, then it is impossible to match it with Companies House. As an example, a company may appear in online postings as *The Berliner*, but their legal name in Companies House incorporates their complete identification *The Berliner Beer House*.

Finally, some companies may share a similar part of their legal name, (*The Berliner Services, The Berliner International, The Berliner Central*). When companies have an almost identical name, it is impossible to systematically identify which version of the name in Companies House corresponds to the clean name in Adzuna. To avoid false-positive matching, whenever a company's clean name was identical to another, none of them was incorporated.

Finally, those vacancies with company names not matched and therefore not identifiable were removed. Therefore, all the analysis done was on vacancy data which contained a matched firm name.

Recruitment agencies and companies not matched removal

In the Adzuna database, 45% of vacancies were posted by recruitment agencies, hence making it impossible to identify either the company behind the search nor the job's sector. It is possible that many enterprises, in particular small businesses, subcontract the human resource task and pay recruitment agencies to be in charge of the first recruiting stages. In addition, many existing enterprises may still not use online job boards to advertise their search, still relying on traditional means such as newspapers, word-of-mouth, or other types of referrals. Robustness checks on published wages show no significant differences between jobs posted through Recruitment agencies and those published by companies themselves.

Recruitment agencies were identified and removed using three complementary criteria. First, company names containing words such as "*recruit*", "*employ*", "*staffing*", "*person-*

nel", "*resourcing*", "*careers*", "*talent*", or "*placement*" were flagged. Second, all companies registered under the official employment intermediary SIC codes—781xx (activities of employment placement agencies), 782xx (temporary employment agency activities), and 783xx (human resources provision)—were removed directly. Third, all companies registered under SIC 62090 (other information technology service activities) and SIC 82990 (other business support service activities) were also removed in their entirety. Although these categories are not limited to intermediaries by definition, a manual inspection of companies with these SIC codes in the Adzuna data confirmed that they were predominantly occupied by staffing and placement intermediaries rather than direct employers; all vacancies in these two categories were therefore excluded. Finally, a manual inspection of companies with a high rate of job advertisements led to the additional removal of 33 recruitment agencies, including CV Library, Reed, and Brook Street.

To identify companies' sectors, Standard Industrial Classification (SIC) is needed. Only private companies in the Companies House are assigned this information. Consequently, only vacancies with the company were found in both Adzuna and Companies House, and with no missing SIC, codes were kept. This led to a total of 64.6 million vacancies with associated company names and sectors not published by recruitment agencies.

Duplicates removal

Once companies' names were cleaned and matched with Companies House, and recruitment agencies were deleted, databases with the same 2 last digits of the ID variable were appended, to proceed with the removal of the duplicates.

One issue with vacancies aggregators is that an advertisement remaining open on a website does not necessarily reflect that the position has not been filled, nor vice-versa. While some companies have their advertisements posted for months or even years, constantly receiving applications, other companies may pay for a fixed advertisement time but fill the position before it ends, or not fill it at all. It becomes necessary then to deduplicate the database to only work with unique vacancies.

Using the ID variable provided by Adzuna, we kept the first observation of each vacancy that appears in the dataset. Two more rounds of duplicate removals were performed with other methods. First, by grouping variables with the same job title, company name, date created and source-id variable. In case some of the vacancies posted on different web pages were not initially captured by the Adzuna ID vacancy variable, a last duplication removal round was performed by deleting repeated observations with the same company, title, date of creation, description, and location. Since only a third of the total observations belong to unique vacancies, the number of vacancies was reduced to 20.7 million.

Cleaning text descriptions and generating new wage variables

The salary variable provided by Adzuna is coded from complementary information on web pages rather than from the advertisement text directly. Around 65% of observations in the raw data have a non-missing salary. After removing recruitment agencies and unmatched companies, this falls to 46% of vacancies, which is still higher than comparable studies using single job boards.[17]

Some entries in Adzuna's wage variable contained errors or were coded at different frequencies (daily, weekly, or monthly rather than annual). To address this, we retrieved salary information directly from advertisement text, converting all figures to annual equivalents. Wages in vacancies appear in a wide variety of formats, as illustrated in Table OB1. The homogenisation process involved standardising number formats, inferring payment frequencies from surrounding text, and applying part-time or full-time adjustments where applicable. When wage information could not be retrieved from the text, Adzuna's original salary variable was used. This procedure recovered an additional 2,446,300 observations with non-missing wages. The final dataset, after a last round of duplicate removal, contains 11,662,124 vacancies.

[17]For reference, posted wages appear in 24.8% of advertisements on *monster.com* (Brenčič, 2012), 13.3% on *trabajando.com* (Banfi & Villena-Roldan, 2019), 20% on *careerbuilder.com* (Marinescu & Wolthoff, 2020), and 24.3% on *xmrc.com* (Kuhn & Shen, 2023).

Table OB1: **Examples of wages differences in formats**

Differences	**Unique salaries**	**Range of salaries**
Punctuation and spaces	£35,000 £ 35000 £35, 000	£30,000-£40,000 £ 30.000 - £40000
The use of k for thousands or the lack of £ sign	£35k 35000 35k	£30k-£40,000 £ 30.000 - 40k 80 - 120000
Range separators		£30,000 to £40,000 £ 30000 - 40k £35. 000 and 40000
Monthly rate	£1300 per month	
Weekly rate	£500 per week	
Daily rate	£80.5 per day	
Hourly rate	£9 9 per hour	£8.46-10

Authors' creation based on Adzuna

Creating NUTS location variable

Adzuna presented two location variables: *location_raw*, which could contain entries such as "Newcastle, Staffordshire" and *location_path*, containing long entries such as "UK ∼ North West England ∼ Greater Manchester ∼ Manchester ∼ The Trafford Centre". The first variable is the raw information extracted from the job vacancy, sometimes only incorporating the town where the job is located and, in a few cases, it also incorporates the postcode. The second is coded by Adzuna, but on some occasions, it does not incorporate the full path but only "UK", or "UK∼London". When it does follow the path, the information is always accurate. The approach followed was to first use *location_path*, then complement those observations with a missing detailed location variable with *location_raw*. Ultimately, they were complemented using the information from postcodes. Table OB2 is an example of *location_path* and the resulting subdivision:[18]

Table OB2: **Location example**

Loc. 1	Loc. 2	Loc. 3	Loc. 4	Loc. 5	Loc. 6
UK	North West England	Greater Manchester	Manchester	The Trafford Center	-

Author's creation based on Adzuna

[18] All examples are illustrative and do not represent any actual vacancy information

Similar to the process for matching the companies, the strategy was to match locations with a nomenclature that allows to use known location categories. Locations were mapped with the Nomenclature of Territorial Units for Statistics (NUTS). The NUTS identification at one digit (UK + letter) corresponds to 12 UK regions, as shown in Table OB3. A list containing the 3,102 unique locations found in Adzuna's location variable was created and assigned a NUTS code at 2 digits (UK+letter+number). Then, each job vacancy was associated with this list, to obtain a matching as the one in Table OB4.

Table OB3: **NUTS 1**

NUTS	**Code**
North East	UKC
North West	UKD
Yorkshire and The Humber	UKE
East Midlands	UKF
West Midlands	UKG
East of England	UKH
London	UKI
South East	UKJ
South West	UKK
Wales	UKL
Scotland	UKM
Northern Ireland	UKN

Table OB4: **NUTS 2**

NUTS	**Code**
Belper	UKF1
Ely	UKH1
Mortlake	UKI7
Porthmadog	UKL1
Newton-Le-Willows	UKD7
Barnacle	UKG3
Sea Mills	UKK1
Cross Keys	UKL2
Ickenham	UKI7
Maghera	UKN0
Darlington	UKC1

Authors' creation based on ONS

A similar approach was taken for all the components of *location_raw*: they were split and matched with their corresponding NUTS codes at 2 digits. For those observations with missing information on Location 3 onwards, postcodes were used to retrieve the corresponding NUTS. These belong to four sources: first, Adzuna provided a variable with postcodes, although it had 70% missing values. For those missing, postcodes in the location variable or the title of the job were retrieved. Finally, Companies House provides the location and postcode of the companies' headquarters. For those observations with job locations set in London, Companies House postcodes were used. Postcodes were ultimately matched with NUTS. In total, 96% of observations have non-missing values of NUTS at 1 digit and 85% at 2 digits.

Table OB5: Cleaning of the database

Process	N vacancies	N no missing W*	Mean W*	SD W*	Δ
Adzuna raw database	276,325,386	178,229,590	31,120	18,524	-
Without recruitment agencies	150,918,209	83,727,680	29,882	18,558	-45%
Without missing companies match	66,833,773	32,647,950	28,365	18,392	-56%
Keep only Private companies with SIC	64,642,205	31,619,913	28,303	18,468	-3%
Remove duplicates vacancy-id, source-id, same description-location	20,783,111	10,490,006	28,342	18,830	-68%
Remove salaries missing	9,657,106	9,657,106	29,454	18,964	-54%
Incorporate new salaries from cleaned description	12,103,405	12,103,405	31,930	18,873	25%
Remove outliers in wages and text size	11,662,124	11,662,124	32,018	18,944	-4%

Authors' creation based on Adzuna. ∗W = Salary considered after trimming 1% tails

Adding Standard Occupational Classification (SOC)

Adzuna's database does not come with official classification labels. We implemented a Natural Language Processing (NLP) matching algorithm that maps the free-form text of job descriptions into UK Standard Occupational Classification (SOC) at three digits, following Turrell et al. (2019). Their algorithm has an accuracy rate of 76%. Python codes are available on GitHub. This algorithm was also implemented by Bellatin and Galassi (2022), who group occupations according to their involvement in the production and use of digital technologies.

By identifying occupations SOC categories, it is possible to provide a granular analysis, identifying the change in personality trait demand within similar occupations. The algorithm requires as inputs a job title, a text description, and a sector identification. For that, being able to match Adzuna's company's name with Companies House was fundamental.

Classification Algorithm of Turrell et al. (2019)

Authors' contribution uses the job title, description and sector for job postings from the recruitment website Reed.com. In addition, they build dictionaries using the information published by the Office for National Statistics (ONS), for the four-digit UK SOC job titles and job descriptions. Inspired in Bellatin and Galassi (2022) description, the algorithm

follows these steps:

1. Cleans the job posting text data using standard text analysis tools.

2. Creates a vector space for the dictionary of UK SOC text data provided by the ONS, using the term frequency-inverse document frequency (TF-IDF).

3. Searches for exact matches between the job titles in the job postings and those in the UK SOC list.

4. Combines the job posting title, the description and the sector into one string and expresses this string as a vector in the TF-IDF matrix of the UK SOC data.

5. For jobs that were not matched, it calculates the cosine similarity between the string obtained in the previous step, and the information in the UK SOC dictionary, and selects the five categories in the UK SOC list with the highest cosine similarity.

6. If the job posting title is empty, it returns the job posting with the highest cosine similarity. Instead, if there is text in the job posting title, the fuzzy-wuzzy Python package is used to identify the best fuzzy match out of the top five ONS categories, following Levenshtein distance calculations.

Duplication of cleaning

Even if Adzuna provides a database without duplicate advertisements for the same week from the same online source, such as job.uk, jobsincare.co.uk or cv-library.co.uk, we also cleaned potential duplicate vacancies published in other sources or with different vacancy IDs but identical company, location and descriptive text. A step-by-step explanation of this procedure is in Online Appendix A. According to Adzuna's representatives, their point-in-time estimates are more comprehensive than other aggregators such as *Burning Glass Technologies (BGT), now Lightcast*. While Adzuna collects data from job boards, BGT's collection focuses on individual company websites, resulting in fewer vacancies.

We removed advertisements from recruitment agencies (reducing the sample by 45%) and those without SIC codes (further reducing the sample by 56%). We only kept private

companies whose names in Adzuna uniquely matched the Companies House database. This study only uses vacancies with posted wages. A common criticism of wage data from online job advertisements is that they often target lower-skilled workers or positions that need to be filled urgently (Banfi & Villena-Roldan, 2019; Brenčič, 2012). Batra et al. (2023)'s meta-analysis highlights the scarcity of wage information, which is often only available in broad ranges, limiting its reliability as a proxy for actual wages. This scarcity is more evident in high-wage occupations and large private firms. After removing recruitment agencies and advertisements not matching Companies House, the dataset contained wage information for 46% of vacancies. This is twice the proportion found in studies using job boards that don't aggregate sources (24.8% in *monster.com* (Brenčič, 2012); 13.3% in *trabajando.com* (Banfi & Villena-Roldan, 2019); 20% in *careerbuilder.com* (Marinescu & Wolthoff, 2020); and 24.3% in *xmrc.com* (Kuhn & Shen, 2023)).

C. Construction of variables and dictionaries

Construction of variables We built dictionaries of job requirements, that were later searched on job advertisements. The thesaurus of both personality traits and requirements was expanded with germane variations, including words containing spelling variations and errors. To identify potential false positives, we did manual checks in 8 rounds of 100 advertisements, for each dictionary. By looking at the 10 closest words in the proximity of the text retrieved, we ensure to be grasping sentences with the correct meaning. When the searched word retrieved a different connotation, we modified the false-positive retrieval. For instance, *flexible hours* was replace by *flx hours* to avoid it being retrieved under the *flexible* matching. Terms with multiple words, such as *machine learning*, were joined (*machinelearning*), and the code only matched these when surrounded by blank spaces, commas or stops. In total, 1702 word replacements were done.

For the personality traits dictionary, we followed the *Knowledge engineering* approach, where a set of explicit rules to classify text are manually defined. A job posting is identified as requiring personality traits as long as it lists at least one keyword in its text description. The strategy of identifying a requirement by pinpointing its appearance in the text description and job titles has been used by Deming and Kahn (2018), Acemoglu et al. (2022), Alekseeva et al. (2021), among others. To construct the traits dictionary, adjectives belonging to the Big Five Personality schemes were used. In comparison with previous papers using *character-related* language, the dictionary was constructed using trait-related terms drawn primarily from established Big Five personality measures and lexical taxonomies in psychology (Costa Jr & McCrae, 1992; Goldberg, 1992; O. John, 1990; McCrae & John, 1992; Saucier, 1994). We also draw on the economics literature on soft skills to motivate their relevance in labour-market contexts (Heckman & Kautz, 2012).

For education, we use ONS education descriptions complemented with UK qualification levels.[19] We built 9 raw qualification-level dictionaries, covering the full range from entry-

[19] https://www.gov.uk/what-different-qualification-levels-mean/list-of-qualification

level qualifications to doctoral degrees. Following Nandi and Nicoletti (2014), these were then collapsed into 5 analytical categories for use in the regressions: (1) none or not mentioned; (2) GCSE or equivalent (raw levels 1–2); (3) A-level or sub-degree higher education (raw levels 3–5, which includes HNC and HND qualifications); (4) bachelor's degree; and (5) master's or doctoral degree. The variable was constructed to consider the lowest educational level that was made explicit in the job search. Hence, if the advertisement is looking for someone with 'a bachelor's degree in Social Science, but a master' is preferable', the observation is marked as requiring at least a bachelor's level.

For remote work, we considered Draca et al. (2022) dictionary of words. For seniority, we followed Nandi and Nicoletti (2014) and Nyhus and Pons (2005), who claim that the relevance of a personality trait may differ according to the position. The structure of hierarchy in some occupations may be more sensitive to personality characteristics. For the same SOC code, individuals applying for senior or managerial positions, *ceteris paribus* the experience and education, have different traits than for junior placements. We follow hierarchy descriptions available in Blogs, such as Indeed.

Adzuna's raw database incorporated information from the advertisements that were not necessarily available in the text entry. This was the case for part and full-time indicators, and temporary and permanent schedules. We created dictionaries for the four categories and complemented Adzuna's variables whenever they were missing.

Finally, we incorporated a dictionary on experience and bonuses. For the first one, we identify whether advertisements require any type of experience or explicitly say that this is not required. For the latter, we explore different variations in the text that signal applicants that the position could have a bonus or pay-for-performance reward.

Dictionaries

The skills used by Acemoglu et al. (2022) are *machine learning, computer vision, machine vision, deep learning, virtual agents, image recognition, natural language processing, speech recognition, pattern recognition, object recognition, neural networks, ai chatbot,*

`-levels`

supervised learning, text mining, unsupervised learning, image processing, mahout, recommender systems, support vector machines, random forests, latent semantic analysis (lsa), sentiment analysis, opinion mining, latent dirichlet allocation (lda), predictive models, kernel methods, keras, gradient boosting, openCV, xgboost, libsvm, word2vec, machine translation, and sentiment classification.

The skills used by Babina et al. (2024) that were not in Acemoglu et al. (2022) that were incorporated are *apache uima, artificial intelligence, autoencoders, bayesian networks/methods, caffe deep learning framework, classification algorithms, clustering algorithms, computational linguistics, convolutional neural network (cnn), dbscan (density-based spatial clustering), deeplearning4j, dimensionality reduction, dlib, expectation-maximization algorithm, hidden markov model, information extraction, jung framework, k-means, long short-term memory (lstm), matrix factorization, maximum entropy classifier, microsoft cognitive toolkit, mlpack, mxnet, naive bayes, natural language toolkit (nltk), nd4j, opennlp, pybrain, recurrent neural network (rnn), scikit-learn, semi-supervised learning, stochastic gradient descent (sgd), theano, unstructured information management architecture, vowpal, wabbit, weka.*

We expanded the list with the following terms: *a b testing, ai, bidirectional encoder (bert), chaid, cloud technology, cntk, cnns, convolutional, convolutional networks, cross domain learning, data augmentation, decision trees, deep embedding, discriminant analysis, dnn techniques, document embeddings, emotion detection, ensemble learning, ensemble methods, entity linking, fastai, feature engineering, federated learning, ffmpeg, gaussian processes, generative adversarial networks (gans), generative models, generative pre-trained transformer (gpt), hierarchical clustering, huggingface, hyperparameter tuning, image fusion, image segmentation, intelligent automation, k-nearest method, kalman filtering, knowledge distillation, kubernetes, mlflow, model compression, multi-task learning, named entity recognition, natural language generation (nlg), natural language understanding (nlu), object tracking, optical character recognition, pattern detection, pcl, predictive churn, predictive modelling, pytorch, quantum computing, recurrent network, reinforcement learning, resnets vaes, robotic process automation, rnns, sequence-to-sequence modelling,*

shot learning, siamese networks, similarity models, sound recognition, text classification, topic detection, topic modelling, tree based learners, tree-based classifiers, t-sne, umap, uniform manifold, vision processing.

Category	Words searched
Traits	
Agreeableness	affection, agreeableness, altruist, amiability, be gentle, considerate, cooperative, courteous, empathetic, forgiving, good nature, helpful, kind, naturalness, pleasant personality, sensitive, soft hearted, straightforward, sympathetic, trust, unselfish, warm
Extraversion	adventurous, assertive, bold, candour, confident, courage, daring, energetic, enthusiastic, expressive, extraverted, forceful, friendly, funny, gregarious, humour, optimist, outgoing, outspoke, passion, playful, proactive, sociable, spontaneous, talkative
Conscientiousness	ambitious, be careful, competent, conscientious, decisive, dependable, dutiful, efficient, hard work, be logic, be neat, organised, painstaking, perseverance, persistent, planful, precise, prompt, punctual, reliable, responsible, striving, systematic, be thorough
Emotional Stability	at ease, calm, emotional, hostil, not fearful, tempered, fortitude, resilience, tense situations, unflappable, patience, pressure
Openness	artistic, be bright, be clever, creative, curious, flexible, ideas, imaginative, independent mind, ingenious, innovative, insightful, intellectual, intelligent, introspective, inventive, openness, be original, prefer variety, reflective, be sharp, unconventional, wide interests, witted
Skills	
Soft Skills	analytical, communication skills, critical thinking, cultural sensitivity, decision making, language skills, leadership, listening skills, mentoring, negotiation, networking abilities, presentation, problem solving, interpersonal, teamwork, time management, attention to detail, organisation skills, multitask, conflict resolution, soft skills, planning, questioning
Hard Skills	accounting, administrative, content creation, copywriting, data analysis, design skills, editing, engineering, legal analysis, marketing, microsoft office, project management, quality assurance, sales skills, software, statistics, user experience, video production, customer service, business development, computer literacy, commercial skills, budget, research, stakeholder

Continued on next page

Table OC1 – continued from previous page

Category	Words searched
AI Skills	machine learning, computer vision, machine vision, deep learning, virtual agents, image recognition, speech recognition, pattern recognition, object recognition, neural networks, supervised learning, text mining, support vector machines, unsupervised learning, image processing, image fusion, mahout, recommender systems, random forests, latent semantic analysis, sentiment analysis, opinion mining, latent dirichlet allocation, predictive models, kernel methods, keras, gradient boosting, gradient descent, opencv, xgboost, libsvm, word2vec, ai chatbot, machine translation, sentiment classification, generative adversarial networks, reinforcement learning, quantum computing, federated learning, natural language understanding, natural language processing, natural language generation, mlops, artificial intelligence, cloud technology, tensorflow, pytorch, scikit learn, natural language toolkit, bidirectional encoder, shot learning, ensemble methods, decision trees, k means clustering, autoencoders, robotic process automation, cnns, rnns, optical character recognition, named entity recognition, topic modelling, ab testing, dimensionality reduction, feature engineering, hyperparameter tuning, data augmentation, model compression, knowledge distillation, kalman filtering, kubernetes, azure ml, weka, clustering algorithms, classification algorithms, density based spatial clustering, caffe deep learning framework, generative pretrained transformer, expectation maximization, jung framework, theano, hugging face, ffmpeg, dlib, mxnet, vowpal wabbit, mlflow, bayesian methods, kubeflow, spark ml, discriminant analysis, long short term memory, hidden markov model, computational linguistics
Work Location	
Remote Work	conducted remote, fully remote, home based, homework, homeworking, job is remote, job is virtual, location is remote, location remote, position is remote, position remote, remote work, role will be remote, telecommuting, telework, virtual job, work from home, work home, work remote, working remote, flexible with office, remote position, role is remote, remote basis, hybrid work, hybrid remote
Seniority	
Level 1	chief, director, executive, vice president
Level 2 high	associate, coordinator, project lead, senior, supervisor, team lead, supervising a team, manager
Level 2 low	administrator, agent role, analyst, consultant, generalist, representative role, technician, secretary
Level 3	apprentice, assistant, clerk, internship, junior, receptionist, trainee, trial period, entry level
Education	

Continued on next page

Table OC1 – continued from previous page

Category	Words searched
No Education required	basic functional skills, basic skills qualification, entry level certificate, entry level functional skills, entry level qualifications, entry nqf level, essential skills wales, level 1 scqf, level 1 svq, level 2 scqf, level 2 svq, level 3 scqf, level 3 svq, no education requirements, no esol, no formal qualification, no previous experience or qualification, no previous qualification, no qualification, scottish national 1, scottish national 2, scottish national 3, scqf level 2, scqf level 3, skills for life, svq level 2, svq level 3, no formal education
GCSE Level	gcse, gnvq, btec award, btec first, btec level 1, certificate of secondary education, first certificate, foundation welsh baccalaureate, key skills level 1, level 1 award, level 1 certificate, level 1 diploma, level 1 esol, level 1 essential skills, level 1 functional skills, level 1 key skills, level 1 national vocational qualification, level 1 nvq, nqf level 1, nvq level 1, rqf level 1, scqf level 4, diploma level 1, esol 1, national level 1, qualification level 1, level 1 qualification, qcf level 1, level 1 qcf, functional skills level 1, essential skills level 1, level 1 apprenticeship, cipd level 1, level 1 cipd, level 4 scqf, level 4 svq, svq level 4
O-Level	o level, btec level 2, general certificate of education, intermediate apprenticeship, intermediate welsh baccalaureate, key skills level 2, level 2 award, level 2 certificate, level 2 diploma, level 2 esol, level 2 essential skill, level 2 functional skill, level 2 key skills, level 2 national, level 2 nvq, national level 2, nqf level 2, nvq level 2, rqf level 2, scqf level 5, secondary education, diploma level 2, award level 3, esol 2, qcf level 2, level 2 qcf, esol level 2, functional skills level 2, essential skills level 2, qualification level 2, level 2 qualification, level 2 apprenticeship, cipd level 2, level 2 cipd, level 5 scqf, level 5 svq, svq level 5, certificate of secondary education
A-Level	a level qualification, as level qualification, access to higher education diploma, advanced apprenticeship, advanced diploma, advanced extension, advanced welsh baccalaureate, baccalaureate, btec award level 3, btec certificate level 3, btec diploma level 3, btec level 3, btec national, diploma programme, extended certificate, extended diploma, key skills level 3, level 3 award, level 3 certificate, level 3 esol, level 3 national, level 3 nvq, national level 3, nqf level 3, nvq level 3, ocr national, progression diploma, rqf level 3, scottish advanced higher, scottish baccalaureate, scottish higher, scqf level 6, diploma level 3, award level 3, esol 3, nvqs level 3, qcf level 3, level 3 qcf, functional skills level 3, essential skills level 3, qualification level 3, level 3 qualification, level 3 apprenticeship, cipd level 3, level 3 cipd, level 6 scqf, level 6 svq, svq level 6
Higher Certificate	hnc, certhe, certificate of higher education, higher apprenticeship, higher national certificate, level 4 award, level 4 certificate, level 4 diploma, level 4 national, level 4 nvq, national level 4, nqf level 4, nvq level 4, scqf level 7, diploma level 4, award level 4, nvqs level 4, qcf level 4, level 4 qcf, functional skills level 4, essential skills level 4, qualification level 4, level 4 qualification, level 4 apprenticeship, cipd level 4, level 4 cipd, level 7 scqf, level 7 svq, svq level 7

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Category	Words searched
Higher Diploma	hnd, btec professional, diphe, diploma of further education, diploma of higher education, foundation degree, higher diploma, higher national diploma, level 5 award, level 5 certificate, level 5 diploma, level 5 national, level 5 nvq, national level 5, nqf level 5, nvq level 5, rqf level 4, scqf level 8, diploma level 5, award level 5, nvqs level 5, qcf level 5, level 5 qcf, qualification level 5, level 5 qualification, cipd level 5, level 5 cipd, level 8 scqf, level 5 apprenticeship, level 5 professional, level 8 svq, svq level 8
Bachelor's Degree	bachelor, bachelors degree, btec advanced professional, degree apprenticeship, first degree, grad cert, grad dip, bachelor certificate, bachelor diploma, level 6 award, level 6 certificate, level 6 diploma, level 6 national, level 6 nvq, national level 6, nqf level 6, nvq level 6, ordinary degree, oxbridge ma, rqf level 5, scottish ma, scqf level 10, scqf level 9, ugad, undergraduate, higher education, diploma level 6, award level 6, nvqs level 6, qcf level 6, level 6 qcf, qualification level 6, level 6 qualification, cipd level 6, level 6 cipd, level 9 scqf, level 10 scqf, level 6 apprenticeship, level 6 professional, level 10 svq, svq level 10, have a degree, bsc or masters, degree or master, bsc or msc, ba or ma, honours degree
Master's Degree	award level 7, ba and ma, cipd level 7, diploma level 7, have a master, have a masters, level 11 scqf, level 11 svq, level 7 apprenticeship, level 7 award, level 7 certificate, level 7 cipd, level 7 diploma, level 7 national, level 7 nvq, level 7 professional, level 7 qcf, master of arts, master of business administration, master of engineering, master of law, master of mathematics, master of pharmacy, master of philosophy, master of physics, master of professional studies, master of public health, master of research, master of science, master of social work, master of studies, masters, masters by research, masters degree, national level 7, nqf level 7, nvq level 7, nvqs level 7, postgrad, postgraduate certificate, postgraduate certificate in education, postgraduate diploma, professional masters, qcf level 7, rqf level 6, rqf level 7, scqf level 11, svq level 11
Doctorate	doctor of philosophy, doctoral degree, doctorate, doctors degree, level 8 award, level 8 certificate, level 8 diploma, level 8 national, level 8 nvq, national level 8, nqf level 8, nvq level 8, phd, rqf level 8, scqf level 12, diploma level 8, award level 8, nvqs level 8, qcf level 8, level 8 qcf, cipd level 8, level 8 cipd, level 12 scqf, level 8 apprenticeship, level 8 professional, svq level 12, level 12 svq
Schedule	
Full-time	9 to 5, day shift, full time, standard hours, night shift
Part-time	4 hour, 6 hour, part time, 20 hour, 30 hour
Duration	
Temporary	apprenticeship, fixed term, freelance, interim, internship, project base, seasonal contract, self employed, short term contract, temporary, day to day contract, zero hours, casual contract
Permanent	long term, ongoing contract, permanent
Experience	

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Table OC1 – continued from previous page

Category	Words searched
Required	3 to 5 years, experienced, can demonstrate, demonstrable, essential experience, experience a minimum, experience for, experience in, experience is also, experience needed, experience of, experience preferably, experience required, experience working, extensive experience, for experienced, has experience, have experience, have previous experience, have some knowledge, have worked, minimum experience, months experience, need experience, past experience, practical experience, previous, prior experience, proven, recent experience, relevant experience, right experience, solid experience, strong experience, suitable experience, three to five years, two to three years, with experience, work experience, worked in similar, years experience, years of experience, you got experience, required experience, experience essential, experience is preferred
Not Required	a novice, do not need any, do not need experience, don't need experience, don't need previous, experience is not, experience not, need any experience, no essential experience, no experience, no formal experience, no formal qualifications or experience, no past experience, no previous, no prior, no work experience, not essential, not necessary, not need previous, with no experience, without any experience, don't need experience, not needed
Other	
Bonus	bonus, commission, incentive, profit sharing, reward scheme, stock options, variable pay, performance related, rewards packages, pay for performance
Teaching	can teach, teaching
Negotiable	on experience, negotiable salary, salary negotiable, hour negotiable, salary is negotiable, salary up to
Communication	working directly with, reporting, report into, your team leader, working with, working closely, under the direction, speaking to, directly to, work alongside, required by, keeping your, work closely, reports, step to, shadowing, managing the, assistance to the, work with, managed by, approval of, communicate with, supporting

The 9 raw education categories are collapsed into 5 analytical categories for the regressions: *None or not mentioned*; *GCSE*; *A-level or below degree*; *Bachelor's degree*; *Master's or Doctoral degree* .

Dictionaries built following (1) Goldberg (1990); (2) Costa Jr and McCrae (1992); (3) Goldberg (1992); (4) McCrae and John (1992); (5) O. P. John and Srivastava (1999); (6) Saucier (1994); (7) Heckman and Kautz (2012) were scrutinised as not to retrieve false positive words. Each word's context was analysed by sampling the dataset in 8 rounds of 100 advertisements per word. When the word's context was not related to personality traits and a straightforward amendment of the word was not possible, this was removed from the dictionary. Words that did

not retrieve any match were also not considered.

Before searching for words, modifications were made to some words to avoid false positives and account for different variations of words. The list of changes done to the text is the following:

Personality Traits language:

”altruistic” for ”altruist ”, ”altruism” for ”altruist ”, ”amiable” for ”amiability ”, ”empathy” for ”empathetic ”, ”empathic” for ”empathetic ”, ”extraversion” for ”extraverted ”, ”extrovert” for ”extraverted”, ”extrovertion” for ”extraverted”, ”emotion” for ”emotional”, ”emotionally” for ”emotional”, ”resilient” for ”resilience, ”extroversion” for ”extraverted ”, ”extroverted” for ”extraverted ”, ”optimism” for ”optimist ”, ”optimistic” for ”optimist ”, ”outspoken” for ”outspoke ”, ”spontaneity” for ”spontaneous ”, ”ambition” for ”ambitious”, ”competence” for ”competent”, ”efficiency” for ”efficient”, ”persevere” for ”perseverance”, ”persevering” for ”perseverance”, ”persistence” for ”persistent”, ”hostile” for ”hostil ”, ”hostility” for ”hostil ”, ”creativity” for ”creative”, ”creatively” for ”creative”, ”innovation” for ”innovative”, ”innovations” for ”innovative”, ”innovating” for ”innovative”, ”innovator” for ”innovative”, ”intellect” for ”intellectual”, ”intellectually” for ”intellectual”, ”unconvention” for ”unconventional”, ”pro active” for ”proactive”, ”compassion” for ”comps”, ”compassionate” for ”comps”, ”an executive” for ”an exec”, ”analytics” for ”analysis”, ”and executive” for ”and exec”, ”sense of adventure” for ”adventurous”, ”humor” for ”humour”, ”anger” for ”angry”, ”anxiety” for ”anxious”, ”agreeable personality” for ”agreeableness”, ”being on time” for ”punctual”, ”co operative” for ”cooperative”, ”comfortable using” for ”confoble using”, ”appreciative of” for ”appreciativeof”, ”at ease” for ”atease”, ”curios” for ”curious”, ”owned cooperative” for ”owned cop”, ”your energy” for ”energetic”, ”with energy” for ”energetic”, ”and energy” for ”energetic”, ”encourages” for ”encorages”, ”encourage” for ”encorage”, ”encouraged” for ”encoraged”, ”encouragement” for ”encoragement”, ”discourage” for ”discorage”, ”encouraging” for ”encoraging”, ”confidential” for ”confidntial”, ”self confidence” for ”self confident”, ”confident employer” for ”employer”, ”confident in” for ”good in”, ”confidentiality” for ”confid”, ”confidentially” for ”confid”, ”confident at” for ”good at”, ”confidently” for ”good at”, ”originally posted” for ”posted”, ”original proof” for ”proof”, ”daring and apply” for ”apply”, ”darington” for ”daryngton”, ”daring to” for ”daryng to”, ”our daring” for ”our”, ”calendaring” for ”calendar”, ”expressive arts” for ”arts”, ”boldon” for ”bolon”, ”nwebold” for ”newbld”, ”not fearful” for ”notfearful”, ”never fearful” for ”notfearful”, ”be fearful” for ”notfearful”, ”being fearful” for ”notfearful”, ”careful to select” for ”to select”, ” careful ” for ” becareful ”, ”spontaneous holiday” for ”holiday”, ”to spontaneous” for ”to fast”, ”from spontaneous” for ”from fast”, ”spontaneous monthly” for ”monthly”, ”spontaneous moments” for ”moments”, ”various spontaneous” for ”various”, ”spontaneous events” for ”events”, ”are clever” for ”beclever”, ”be clever” for ”beclever”, ”, clever” for ” beclever”, ”clever,” for ”beclever ”, ”with clever” for ”beclever”,”some clever” for ”beclever”, ”clever and” for ”beclever”, ”and clever” for ”beclever”, ”flexible shift” for ”flx shift”, ”flexible shifts” for ”flx shifts”, ”flexible working” for ”flx work”, ”flexible work” for ”flx work”, ”flexible with” for ”flx with”, ”flexible basis” for ”flx basis”, ”flexible hours” for ”flx hours”, ”flexible benefit” for ”flx benefit”, ”flexible benefits” for ”flx benefits”, ”flexible about” for ”flx about”, ”hours are flexible” for ”hours are flx”, ”flexible annual” for ”flx annual”, ”flexible to” for ”flx to”, ”flexible for” for ”flx for”, ”is flexible” for ”is flx”, ”flexible hour” for ”flx hour”, ”fully flexible” for ”fully flx”, ”flexible schedule” for ”flx schedule”, ”flexible salary” for ”flx salary”, ”to flexible” for ”tp flx”, ”flexible new” for ”flx new”, ”flexible holiday” for ”flx holiday”, ”flexible way” for ”flx way”, ”flexible ways” for ”flx ways”, ”pay flexible” for ”flx pay”, ”work flexible” for ”flx work”, ”flexible contract” for ”flx contract”, ”flexible opportunities” for ”flx opportunities”, ”flexible options” for ”flx options”, ”flexible subcontract” for ”flx subcontract”, ”be flexible as” for ”flx as”, ”flexible between” for ”flx between”, ”flexible start” for ”flx start”, ”flexible rota” for ”flx rota”, ”weeks are flexible” for ”flx week”, ”flexible days” for ”flx days”, ”choose flexible” for ”choose flx”, ”entirely flexible” for ”entirely flx”, ”flexible within” for ”flx within”, ”for flexible” for ”for flx”, ”flexible additional” for ”flx additional”, ”flexible fixed” for ”flx fixed”, ”times flexible” for ”times flx”, ”location flexible” for ”location flx”, ”flexible on” for ”flx on”, ”flexible schedules” for ”flx schedules”, ”flexible reward” for ”flx reward”, ”flexible rewards” for ”flx rewards”, ”flexible 40” for ”flx 40”, ”flexible weekend” for ”flx weekend”, ”flexible vacancies” for ”flx vacancies”, ”scheme flexible” for ”scheme flx”, ”flexible in working” for ”flx in working”, ”flexible menu”

for ”flx menu”, ”time flexible” for ”time flx”, ”flexible pm” for ”flx pm”, ”flexible saturdays” for ”flx saturdays”, ”flexible home” for ”flx home”, ”flexible team” : ”flx team”, ”cleaner flexible” : ”cleaner flx”, ”flexible and optional” : ”flx and optional”, ”flexible self employed” : ”flx self employed”, ”flexible in relation to” : ”flx in relation to”,”flexible and able to” : ”flx and able to”, ”flexible, full” : ”flx, full”, ”flexible full” : ”flx full”, ”flexible pattern” : ”flx pattern”, ”hours of work can be flexible” : ”flx hours”, ”flexible approach to remote working” : ”flx remote working”, ”flexible work” for ”flx work”, ”flexible and guaranteed hours” : ”flx hours”, ”flexible networks” : ”flx networks”, ”flexible job” : ”flx job”, ”benefits flexible” : ”flx benefits”, ”weekdays flexible” : ”flx weekdays”, ”flexible guaranteed hours” : ”flx hours”, ”freelance and flexible” : ”freelance and flx”, ”flexible with office” : ”flx with office”, ”options flexible” : ”options flx”, ”flexible core” : ”flx core”, ”flexiblejob” : ”flx job”, ”workflexible” : ”work flx”, ”flexbile interview” : ”flx interview”, ”flexible temporary” : ”flx temporary”, ”flexible, part” : ”flx, part”, ”flexible part” : ”flx part”, ”flexible and home” : ”flx and home”, ”days are flexible” : ”days are flx”, ”flexible employment” : ”flx employment”, ”that flexible” : ”that flx”, ”flexible location” : ”flx location”, ”flexible rotas” : ”flx rotas”, ”flexible and remote” : ”flx and remote”, ”flexible pension” : ”flx pension”, ”esteemed” for ”estemed”, ”be bright ” for ”bebright ”, ” bright and” for ” bebright”, ”and bright” for ”bebright”, ”, bright ” for ” bebright ”, ” bright, ” for ” bebright ”, ” a bright ” for ” bebright ”, ”brightest” for ”bebright”, ”bright welcoming” for ”welcoming”, ”bright future” for ”future”, ”bright sunny” for ”sunny”, ”be helpful” for ”be help”, ”is helpful” for ”is help”, ”experience helpful” for ”experience help”, ”independent minded” for ”independentmind”, ”independent mind” for ”independentmind”, ”prefer variety” for ”prefervariety”, ”straight forward” for ”straightforward”, ”straightforward for you” for ”forward for you”, ”straight forward for you” for ”forward for you”, ”straightforward questionnaire” for ”questionnaire”, ”very straightforward” for ”very”, ”provides straight forward” for ”provides”, ”straightforward design” for ”design”, ”is straight forward” for ”is”, ”friendly policies” for ”frend policies”, ”family friendly” for ”family frend”, ”friendliness” for ”friendly”, ”user friendly” for ”user friend”, ”friendly local” for ”local”, ”dogfriendly” for ”dog”, ”ecofriendly” for ”eco”, ” gentle ” for ” begentle ”, ”solver” for ”solving”, ”wide interest” for ”wideinterests”, ”wider interest” for ”wideinterests”, ”otherwise” for ”other”, ”likewise” for ”like”, ”connectwise” for ”connect”, ”deliberateness” for ”deliberate”, ”beneath” for ”beneth”, ” neat ” for ” beneat ”, ” be neat ” for ” beneat ”, ”helpful tool” for ”tool”, ”effective communication” for ”communication skill”, ”optimisation” for ”optimization”, ”optimisations” for ”optimizations”, ”optimising” for ”optimizing”, ”optimise” for ”optimize”, ”optimised” for ”optimized”, ”optimises” for ”optimizes”, ”optimiser” for ”optimizer”, ”empathy” for ”empathetic”, ”enthusiasm” for ”enthusiastic”, ”open, ” for ”openness ”, ”open and” for ”openness ”,”and open ” for ”openness ”,”openmind” for ”openness”, ”open mind” for ”openness ”,”open minded” for ”openness ”, ”openness close ” for ”close”, ”open and close” for ”close”, ”open source” for ”source”, ”openness source” for ”source”, ”enthusiast” for ”enthusiastic”, ”fun ” for ”fun ”, ”fun day” for ”fus day”, ”fun park” for ”fus park”, ”fun ” for ”funny ”, ”fun,” for ”funny ”, ”fun.” for ”funny ”, ”grounded in” for ”in”, ”be grounded in ” for ”in”, ”grounded established” for ”established”, ”a grounded” for ” a ”,”grounded knowledge” for ”knowledge”, ”grounded programming” for ”programming”,”grounded perspective” for ”perspective”,”grounded reflected” for ”reflected”,”grounded solutions” for ”solutions”, ”grounded and” for ”begrounded”, ”and grounded” for ”begrounded”, ”, grounded” for ” begrounded”, ”grounded,” for ”begrounded ”, ”be grounded” for ”begrounded”, ”grounded analysis” for ”analysis”, ”well grounded” for ”well”, ” a grounded,” for ” a ”, ”, grounded in” for ” in”, ”begrounded in” for ” in”, ”, grounded in ” for ”in ”, ”hard working” for ”hardwork”, ”sociable hours” for ”hours”, ”sociable working hours” for ”working hours”, ”hard work” for ”hardwork”, ”hardworker” for ”hardwork”, ”hardworking” for ”hardwork”, ”logic apps” for ”apps”, ” logic ” for ” belogic ”,” logical ” for ” belogic ”, ”thorough knowledge” for ”knowledge”, ”bethorough knowledge” for ”knowledge”, ”and thorough” for ”bethorough”, ”,thorough” for ”bethorough”, ”thorough and” for ”bethorough”,”be thorough” for ”bethorough”, ”home working” for ”homeworking”, ”sensitive information” for ”information”, ”sensitive or contentious information” for ”contentious information”, ”sensitive and restricted information” for ”restricted information”, ”sensitive data” for ”data”, ”sensitive company” for ”company”, ”intellectual property” for ”property”, ”intellectual dis” for ”dis”, ”track record” for ”experienced”, ”background” for ”experienced”, ”outgoing mail” for ”mail”, ”outgoing call” for ”call”, ”pleasant attitude” for ”pleasantpersonality”, ”pleasant personality” for ”pleasantpersonality”, ”pleasant manner” for ”pleasantpersonality”, ”be pleasant” for ”pleasantpersonality”,

”pleasant and” for ”pleasantpersonality”, ”kind of” for ”jind of”, ”kinds of” for ”jinds of”, ”kindly” for ”jindly”, ”kind to” for ”jind to”, ”tender minded” for ”tendermind”, ” tender mind” for ”tendermind”, ”good nature” for ”goodnature”, ”tense situations”:”tensesituations”, ”under control”:”undercontrol”, ”competencies” for ”skills”, ”competence in” for ”skilled in”, ”competence with” for ”skilled with”, ”soft heart” for ”warm heart”, ”soft hearted” for ”warm hearted”, ”competency” for ”skilled”, ”competently” for ”skilled”, ”organized” for ”organised”,”show consideration” for ”considerate”, ”temperance” for ”tempered”, ”temperament” for ”tempered”, ”original solutions” for ”beoriginal”, ”original thinking” for ”beoriginal”, ”original ideas” for ”beoriginal”, ”original research” for ”beoriginal”, ”be original” for ”beoriginal ”, ”original, ” for ”beoriginal”, ”original writing” for ”beoriginal”, ”original content” for ”beoriginal”, ”is original” for ”beoriginal”, ”think originally” for ”beoriginal”, ”, original” for ” beoriginal”, ”original thought” for ”beoriginal”, ”original choice” for ”beoriginal”, ”originality” for ”beoriginal”, ”creating original” for ”beoriginal”, ”national trust” for ”nationaltr”, ”trusts” for ”trus”, ”trustee” for ”trusee”,”trustees” for ”trusees”, ”cooperative trust” for ”coperative”, ”nhs trust” for ”nhs”, ”trust us” for ”trus us”, ”trusted” for ”trused”, ” our trust” for ” our trus”, ”local trust” for ”local”, ”trust are” for ”are”, ”academy trust” for ”academy”, ”trust schools” for ”schools”, ”the trust” for ”the trus”, ”trust is” for ”is”, ”foundation trust” for ”foundation”, ”at trust” for ”at trus”, ”entrusted” for ”enrusted”, ”wildlife trust” for ”wildlife”, ”castle trust” for ”castle”, ”trust scheme” for ”scheme”, ”trust duties” for ”duties”, ”education trust” for ”education”, ”adventure trust” for ”adventure”, ”disabilities trust” for ”disabilities”, ”affinity trust” for ”affinity”, ”academies trust” for ”academies”, ”whole trust” for ”whole”, ”trust offers” for ”offers”, ”trust within” for ”within”, ”trust and corporate” for ”corporate”, ”trust duties” for ”duties”, ”trust and grants” for ”grants”, ”leisure trust” for ”leisure”, ”we trust” for ”we”, ”trust manages” for ”manages”, ”enterprise trust” for ”enterprise”, ”trust keith” for ”keith”, ”trust flagship” for ”flagship”, ”woodland trust” for ”woodland”, ”trust and fundraising” for ”fundraising”, ”care trust” for ”care”, ”health trust” for ”health”, ”rivers trust” for ”rivers”, ”transform trust” for ”transform”, ”transformation trust” for ”transformation”, ”are a trust ” for ”are ”, ”trust qvd” for ”qvd”, ”educational trust” for ”educational”, ”trust exposure” for ”exposure”, ”trust has” for ”has”, ”dean trust” for ”dean”, ”trust delivers” for ”delivers”, ”as a trust ” for ” ”, ”london trust” for ”london”, ”trustpilot” for ” ”, ”trust pilot” for ” ”, ”trust education” for ”education”, ”trust we believe” for ”we believe”, ”northern trust” for ”northern”, ”trust website” for ”website”, ”trust. org” for ”.org”, ”trust . org” for ”.org”, ”trust.org” for ”.org”, ”trust org” for ”org”, ”persist” for ”persistent”, ”prompt response” for ”prom response”, ”as prompt” for ”as prom”, ”prompting” for ”promting”, ”precisely what” for ”what”, ”as prompt” for ”as prom”, ”warm welcome” for ”welcome”, ”warm you” for ”you”, ”reliable vehicle” for ”rel veh”, ”representative of” for ”represent of”, ”responsible to” for ”res to”, ”responsible of” for ”res of”, ”responsible in” for ”res in”, ”responsible for” for ”res for”, ”reflective of” for ”ref of”, ”a reflective” for ”a ref”, ” sharp ” for ” besharp ”, ”socially responsible” for ”socially res”, ”unsociable” for ”unsoci”, ”are striving” for ”are str”, ”sympathetically converted” for ”converted”, ”sympathetically restored” for ”restored”, ”sympathetic refurbishment” for ”refurbishment”, ”sympathetic donors” for ”donors”.

Soft skills language:

”communication skills” for ”communicationskills”, ”critical thinking” for ”criticalthinking”, ”cultural sensitivity” for ”culturalsensitivity”, ”decision making” for ”decisionmaking”, ”language skills” for ”languageskills”, ”listening skills” for ”listeningskills”, ”problem solving” for ”problemsolving”, ”time management” for ”timemanagement”, ”attention to detail” for ”attentiontodetail”, ”organisation skills” for ”organisationskills”, ”conflict resolution” for ”conflictresolution”, ”soft skills” for ”softskills”,”maintain relationships” for ”networkingabilities”, ”network abilities” for ”networkingabilities”, ”working relationships” for ”networkingabilities”, ”network ability” for ”networkingabilities”, ”relationship building” for ”networkingabilities”, ”headhunting” for ”networkingabilities”, ”networking events” for ”networkingabilities”, ”maintain contacts” for ”networkingabilities”, ”external relationships” for ”networkingabilities”, ”strong relationships” for ”networkingabilities”, ”personal network” for ”networkingabilities”, ”ability to network” for ”networkingabilities”, ”attend events” for ”networkingabilities”, ”business relationships” for ”networkingabilities”, ”positive relationships” for ”networkingabilities”, ”good listening” for ”listening skills”, ”good listener” for ”listening skills”, ”active listening” for ”listening skills”, ”ability to listen” for ”listening

skills", "resolve issues" for "problemsolving", "problem solver" for "problemsolving", "fluency" for "fluent", "representation" for "reprsentation", "representations" for "reprsentation", "making decisions" for "decisionmaking", "make decisions" for "decisionmaking", "best decisions" for "decisionmaking", "good decisions" for "decisionmaking", "take decisions" for "decisionmaking","informed decisions" for "decisionmaking", "difficult decisions" for "decisionmaking", "appropriate decisions" for "decisionmaking", "resolving the complex problems" for "problemsolving", "resolving complex problems" for "problemsolving", "negotiating" for "negotiation", "negotiator" for "negotiation", "good spoken" for "communicationskills", "transparent communication" for "communicationskills", "good communication" for "communicationskills", "strong communication" for "communicationskills", "excellent communication" for "communicationskills", "written and verbal" for "communicationskills", "written and oral" for "communicationskills", "oral and written" for "communicationskills", "communication abilities" for "communicationskills", "great communication" for "communicationskills", "oral" for "communicationskills", "verbal, written" for "communicationskills", "technical writting" for "communicationskills", "verbal and written" for "communicationskills", "written" for "communicationskills", "verbal" for "communicationskills", "oral" for "communicationskills", "ability to communicate" for "communicationskills", "strong communicator" for "communicationskills", "excellent communicator" for "communicationskills", "good communicator" for "communicationskills", "communicator" for "communicationskills", "able to communicate" for "communicationskills", "transparent communication" for "communicationskills", "great communication" for "communicationskills", "can lead" for "leadership", "people management" for "leadership", "able to lead" for "leadership", "team leadership" for "supervisingateam", "team leadership" for "leading a team", "team management" for "leading a team", "cultural awareness" for "culturalsensitivity", "crosscultural" for "culturalsensitivity", "multicultural" for "culturalsensitivity", "multi cultural" for "culturalsensitivity", "culturally diverse" for "culturalsensitivity", "culturally aware" for "culturalsensitivity", "cultural diversity" for "culturalsensitivity", "asking questions" for "questioning", "european language" for "languageskills", "fluent in chinese" for "languageskills", "fluent in german" for "languageskills", "fluent in french" for "languageskills", "fluent in english" for "languageskills", "fluent in mandarin" for "languageskills", "fluent in spanish" for "languageskills", "fluent in portuguese" for "languageskills","fluent in polish" for "languageskills", "fluent speaking" for "languageskills", "fluent in dutch" for "languageskills", "fluent in russian" for "languageskills", "two languages" for "languageskills", "fluent in both" for "languageskills", "fluent in hebrew" for "languageskills", "fluent in speaking" for "languageskills", "fluent in italian" for "languageskills", "fluent in welsh" for "languageskills", "fluent in japanese" for "languageskills", "fluent in danish" for "languageskills", "fluent in english and" for "languageskills", "communicating in two langauges" for "languageskills", "fluent both english and" for "languageskills", "fluent in english as well as" for "languageskills", "fluent in another" for "languageskills", "fluent in the german" for "languageskills", "number of languages" for "languageskills", "fluent in turkish" for "languageskills", "coaching" for "mentoring", "provide guidance" for "mentoring", "multiple projects" for "multitask", "multiple tasks" for "multitask", "training and mentoring" for "training", "mentoring and training" for "training", "multi projects" for "multitask", "multitasker" for "multitask", "multitasking" for "multitask", "ability to motivate" for "mentoring","able to motivate" for "mentoring", "couching" for "mentoring", "problem resolution" for "problemsolving", "complex problems" for "problemsolving", "other languages" for "languageskills", "additional languages" for "languageskills", "multilingual" for "languageskills", "office skills" for "microsoftoffice", "excel skills" for "microsoftoffice", "ms packages" for "microsoftoffice", "ms package" for "microsoftoffice", "organisational skills" for "organisationskills", "organizational skills" for "organisationskills", "organisational capacity" for "organisationskills", "organisational abilities" for "organisationskills", "organisational ability" for "organisationskills", "excellent organisation" for "organisationskills", "organisation, coordination" for "organisationskills", "organisational and managements skills" for "organisationskills", "excellent organisational" for "organisationskills", "good organisational" for "organisationskills", "strong organisational" for "organisationskills", "organisational, " for "organisationskills ", "organisational and " for "organisationskills ", "strong managements skills" for "organisationskills", "excellent managements skills" for "organisationskills", "competent managements skills" for "organisationskills", "strong organisational" for "organisationskills", "prioritisation" for "organisationskills", "able to prioritise" for "organisationskills", "excellent organisation" for "organisationskills", "mentoring of the junior" for "mentoring", "mentoring junior" for "mentoring", "able to mentor the more junior members" for "mentor-

ing", "mentoring our junior" for "mentoring", "mentoring our more junior" for "mentoring", "able to mentor junior" for "mentoring", "especially junior" for "mentoring", "mentor junior" for "mentoring", "mentoring and training of more junior" for "mentoring", "mentoring to junior" for "mentoring", "mentoring of junior" for "mentoring", "mentoring and coaching junior" for "mentoring", "mentor more junior" for "mentoring", "training of junior" for "mentoring", "eye for detail" for "attentiontodetail", "detail oriented" for "attentiontodetail", "detailed approach" for "attentiontodetail", "mentoring more junior" for "mentoring", "support junior" for "mentoring", "managing junior" for "mentoring", "developing junior" for "mentoring", "developing more junior" for "mentoring", "support to more junior" for "mentoring", "support more junior" for "mentoring", "development of junior" for "mentoring", "work of junior" for "mentoring", "coaching more junior" for "mentoring", "coaching of junior" for "mentoring", "training junior" for "mentoring", "supervising junior" for "mentoring", "work as a team" for "teamwork", "work in a team" for "teamwork ", "work in teams" for "teamwork", "work well in a team" for "teamwork", "within a team" for "teamwork", " intern " for " internship ", "team work" for "teamwork", "team working" for "teamwork", "team environment" for "teamwork", "able to relate" for "teamwork", "team player" for "teamwork", "collaborative work" for "teamwork".

Hard skills language:

"content creation" for "contentcreation", "design skills" for "designskills", "data analysis" for "dataanalysis", "legal analysis" for "legalanalysis", "microsoft office" for "microsoftoffice", "project management" for "projectmanagement", "quality assurance" for "qualityassurance", "sales skills" for "salesskills", "user experience" for "userexperience", "video production" for "videoproduction", "customer service" for "customerservice", "business development" for "businessdevelopment", "computer literacy" for "computerliteracy", "commercial skills" for "commercialskills", "business opportunities" for "businessdevelopment", "database management" for "dataanalysis", "data analytics" for "dataanalysis", "commercial abilities" for "commercialskills", "commercially aware" for "commercialskills", "commercially awareness" for "commercialskills", "focus on commercial" for "commercialskills", "commercially astute" for "commercialskills", "timekeeper" for "timemanagement", "time keeping" for "timemanagement", "commercially minded" for "commercialskills", "commercially focused" for "commercialskills", "commercially astute" for "commercialskills", "commercially driven" for "commercialskills" , "commercially focussed" for "commercialskills", "word outlook" for "microsoftoffice", "excel and outlook" for "microsoftoffice", "microsoft outlook" for "microsoftoffice", "profficient in word" for "microsoftoffice","microsoft word" for "microsoftoffice", "proficient in excel" for "microsoftoffice", "microsoft excel" for "microsoftoffice", "power point" for "microsoftoffice", "powerpoint" for "microsoftoffice", "competent in microsoft" for "microsoftoffice", "microsoft packages" for "microsoftoffice", "office 365" for "microsoftoffice", "ms excel" for "microsoftoffice", "command of microsoft" for "microsoftoffice", "within budget" for "within budg", "word excel" for "microsoftoffice", "word and excel" for "microsoftoffice", "excel and word" for "microsoftoffice", "word, excel" for "microsoftoffice", "excel, word" for "microsoftoffice", "outlook and" for "microsoftoffice", "and outlook" for "microsoftoffice", "using outlook" for "microsoftoffice", "ms office" for "microsoftoffice", "microsoft teams" for "call", "microsoft programmes" for "microsoftoffice", "positive outlook" for "positive", "working with team" for "teamwork", "statistical" for "statistics", "pc literate" for "computerliteracy", "computer literate" for "computerliteracy", "using a computer" for "computerliteracy", "computer skills" for "computerliteracy", "pc skills" for "computerliteracy", "analytics" for "analysis", "investigative skills" for "analytical skills", "research" for "research skills", "cancer research" for "cancer","cancer research" for "cancer", "research organisations" for "organisations", "researching" for "research skills", "research organisation" for "organisations", "research programmes" for "programmes", "researcher" for "research", "good numeracy" for "analytical skills", "numeracy skills" for "analytical skills", "numeracy litarate" for "analytical skills", "conflict management" for "conflictresolution", "collaborative working" for "teamwork", "supervising a" for "supervisingateam", "managing a team" for "supervisingateam", "team management" for "supervisingateam", "part of a team" for "teamwork", "sunrise senior" for "sunrise", "seniors" for "old", "associated" for "asociated", "associates" for "asociates", "their representatives" for "their clients", "client representatives" for "client", "sales representative" for "representativerole salesskills", "sales representatives" for "representativerole salesskills", "customerservice" for "representativerole customerservice", "customer"

for ”representativerole customer”, ”marketing representative” for ”marketing representativerole”, ”representative role” for ”representativerole”, ”our consultants” for ”our consult”, ”a consultant” for ”a consult”, ”consultants available” for ”consult available”, ”our qualified consultants” for ”our consult”, ”agent role” for ”agentrole”, ”sales agent” for ”agentrole”, ”support agent” for ”agentrole”, ”customerservice agent” for ”agentrole”, ”customerservice agent” for ”agentrole”, ”service agent” for ”agentrole”, ”accountant” for ”accounting”, ”directory” for ”directo”, ”directorate” for ”directo”, ”sales experience” for ”salesskills experienced”, ”sales person” for ”salesskills”, ”experience in sales” for ”salesskills experienced”, ”identify key sales” for ”salesskills”, ”sales oriented” for ”salesskills”, ”sales culture” for ”salesskills”, ”sales representative” for ”sales representative salesskills”, ”of sales” for ”salesskills”, ”sales plan” for ”salesskills”, ”sales plans” for ”salesskills”, ”sales associate” for ”salesskills”, ”sales role” for ”salesskills”, ”sales team” for ”salesskills team”, ”sales manager” for ”salesskills manager”, ”sales job” for ”salesskills”, ”sales target” for ”salesskills”, ”direct sales” for ”salesskills”, ”success in sales” for ”salesskills”, ”success as sales” for ”salesskills”, ”develop sales” for ”salesskills”, ”sales strategies” for ”salesskills”, ”sales targets” for ”salesskills”, ”target driven sales” for ”salesskills”, ”digital design” for ”designskills”, ”design engineer” for ”designskills”, ”graphic design” for ”designskills”, ”motion design” for ”designskills”, ”design tools” for ”designskills”, ”photoshop” for ”designskills”, ”motion graphics” for ”designskills”, ”digital art” for ”designskills”, ”adobe” for ”designskills”, ”design management” for ”designskills”, ”technical designs” for ”designskills”, ”photo shop” for ”designskills”, ”creating content” for ”contentcreation”, ”generate content” for ”contentcreation”, ”content generation” for ”contentcreation”, ”management of content” for ”contentcreation”, ”illustrator” for ”designskills”, ”content strategy” for ”contentcreation”, ”content development” for ”contentcreation”, ”content management” for ”contentcreation”, ”content creators” for ”contentcreation”, ”social content” for ”contentcreation”, ”managing content” for ”contentcreation”, ”website content” for ”contentcreation”, ”content writer” for ”contentcreation”, ”editorial management” for ”editing”, ”editorial skills” for ”editing”, ”editorial tasks” for ”editing”, ”editorial role” for ”editing”, ”editorial roles” for ”editing”, ”publishing skills” for ”editing”, ”editorial judgment” for ”editing”, ”editorial experience” for ”editing experienced”, ”editorial style” for ”editing”, ”editorial work” for ”editing”, ”proofreading” for ”editing”, ”proof reading” for ”editing”, ”drafting” for ”editing”, ”copy writing” for ”editing”, ”copywriting” for ”editing”, ”copywriter” for ”editing”, ”copywrite” for ”editing”, ”good editorial” for ”editing”, ”excellent editorial” for ”editing”, ”editorial assistant” for ”editing assistant”, ”editorial ability” for ”editing”, ”editorial judgment” for ”editing”, ”creating engaging content” for ”contentcreation”, ”develop content” for ”contentcreation”, ”developing content” for ”contentcreation”, ”writing, editorial” for ”editing”, ”legal text” for ”legalanalysis”, ”legal role” for ”legalanalysis”, ”legal abilities” for ”legalanalysis”, ”paralegal” for ”legalanalysis”, ”legal background” for ”legalanalysis”, ”legal qualifications” for ”legalanalysis”, ”legal qualification” for ”legalanalysis”, ”legal excecutive” for ”legal excecutive legalanalysis”, ”lawyer” for ”lawyer legalanalysis”, ”reading leases” for ”legalanalysis”, ”lawyers” for ”lawyers legalanalysis”, ”reading leases” for ”legalanalysis”, ”legal clerk” for ”legalanalysis”, ”provide legal” for ”legalanalysis”, ”legal billing” for ”legalanalysis”, ” engineer ” for ” engineering ”, ”legal documentation” for ”legalanalysis”, ”legal approach” for ”legalanalysis”, ”caseload” for ”legalanalysis”, ”understand our customers” for ”customerservice”, ”customer focused” for ”customerservice”, ”customer focus” for ”customerservice”, ”identify customer needs” for ”customerservice”, ”understanding the customers” for ”customerservice”, ”needs of the customers” for ”customerservice”, ”manage customer” for ”customerservice”, ”customer orders” for ”customerservice”, ”interact with customers” for ”customerservice”, ”quality checking” for ”qualityassurance”, ”quality checks” for ”qualityassurance”, ”quality check” for ”qualityassurance”, ”ensuring quality” for ”qualityassurance”, ”search engine” for ”software”, ”sem ” for ”software”, ”seo ” for ”software ”, ”web development” for ”software”, ”programming” for ”software”, ”mobile app development” for ”software”, ”coding” for ”software”, ”web design” for ”software”, ”cybersecurity” for ”software”, ”data mining” for ”dataanalysis”, ”data visualization” for ”dataanalysis”, ”database skills” for ”dataanalysis”, ”data to” for ”dataanalysis”, ”quantitative analysis” for ”dataanalysis ”, ”data modelling” for ”dataanalysis”, ”datasets” for ”dataanalysis”, ”interpret data” for ”dataanalysis”, ”data reporting” for ”dataanalysis”, ”it skills” for ”dataanalysis”, ”it literate” for ”dataanalysis”, ”it literacy” for ”dataanalysis”, ”variety of data” for ”dataanalysis”, ”data skills” for ”dataanalysis”, ”data models” for ”dataanalysis”, ”data sets” for ”dataanalysis”, ”data analyst” for ”dataanalysis analyst”, ”data monitoring” for ”dataanalysis”, ”data scientist” for ”dataanalysis”, ”data skill” for ”dataanalysis”, ”handling data” for

”dataanalysis”, ”complex data” for ”dataanalysis”, ”big data” for ”dataanalysis”, ”data science” for ”dataanalysis”, ”data engineer” for ”dataanalysis”, ”collate data” for ”dataanalysis”, ”data interpretation” for ”dataanalysis”, ”interpretation of data” for ”dataanalysis”, ”creation of data” for ”dataanalysis”, ”relational databases” for ”dataanalysis”, ”database management” for ”dataanalysis”, ”database utilisation” for ”dataanalysis”, ”data management” for ”dataanalysis”, ”database administration” for ”dataanalysis”, ”comparative data” for ”dataanalysis”, ”process data” for ”dataanalysis”, ”raw data” for ”dataanalysis”, ”data insight” for ”dataanalysis”, ”customer data” for ”dataanalysis”, ”customer databases” for ”dataanalysis”, ”company databases” for ”dataanalysis”, ”accurate databases” for ”dataanalysis”, ”accurate data” for ”dataanalysis”, ”data driven” for ”dataanalysis”, ”data entry” for ”dataanalysis”, ”data reviews” for ”dataanalysis”, ”review data” for ”dataanalysis”, ”project coordination” for ”projectmanagement”, ”project planning ” for ”projectmanagement”, ”contract drafting” for ”projectmanagement”, ”producing video” for ”videoproduction”, ”photography” for ”videoproduction”, ”video editing” for ”videoproduction”, ”video creation” for ”videoproduction”, ”videographer” for ”videoproduction”, ”film maker” for ”videoproduction”, ”creating videos” for ”videoproduction”, ”video experience” for ”videoproduction experienced”, ”filming” for ”videoproduction”, ”social interaction” for ”interpersonal”, ”social skills” for ”interpersonal”, ”people skills” for ”interpersonal”, ”people person” for ”interpersonal”, ”clerical” for ”administrative”,” ux ” for ”userexperience”, ” ux, ” for ”userexperience”, ” ux.” for ”userexperience”, ”statistics” for ”statistical”, ”stake holder” for ”stakeholder”, ”client management” for ”stakeholder management”, ”supply chain management” for ”product management”.

AI language:

” ai ” for ” artificialintelligence ”, ” xai ” for ” artificialintelligence ”, ” ai,” for ” artificialintelligence ”, ”, ai ” for ” artificialintelligence ”, ”artificial intelligence” for ” artificialintelligence ”, ” gpt ” for ”generativepretrainedtransformer”, ”generative models” for ”generativepretrainedtransformer”, ”generative pretrained” for ”generativepretrainedtransformer”, ”generative pre” for ”generativepretrainedtransformer”, ”ai chatbox” for ”aichatbox”, ”artificialintelligence chatbox” for ”aichatbox”, ”machine learning” for ”machinelearning”, ” ml ” for ” machinelearning ”, ”, ml ” for ” machinelearning ”, ” ml,” for ” machinelearning ”, ”cross domain learning” for ”machinelearning”, ”cross - domain learning” for ”machinelearning”, ”reinforcement learning” for ”machinelearning”, ”mlpack” for ”machinelearning”, ”federated learning” for ”machinelearning”, ”pybrain” for ”machinelearning”, ”shot learning” for ”machinelearning”, ”hugging face” for ”huggingface”, ”multi task learning” for ”machinelearning”, ”multi - task learning” for ”machinelearning”, ”residual learning” for ”machinelearning”, ”mediapipe” for ”machinelearning”, ”media pipe” for ”machinelearning”, ”computer vision” for ”computervision”, ”image segmentation” for ”computervision”, ”vision processing” for ”computervision”, ”robotic process automation” for ”roboticprocessautomation”, ” rpa ” for ” roboticprocessautomation ”, ”autonomous robotics” for ”roboticprocessautomation”, ”intelligent automation” for ”roboticprocessautomation”, ”machine vision” for ”machinevision”, ” bert ” for ”bidirectionalencoder”, ”bidirectional encoder” for ”bidirectionalencoder”, ”, bert” for ” bidirectionalencoder”, ” bert,” for ” bidirectionalencoder”, ”deep learning” for ”deeplearning”, ”residual network” for ”deeplearning”, ” resnet ” for ”deeplearning”, ”fastai” for ”deeplearning”, ”nd4j” for ”deeplearning”, ”microsoft cognitive toolkit” for ”deeplearning”, ” cntk ” for ”deeplearning”, ”deep embeddings” for ”deeplearning”, ”deep embedding” for ”deeplearning”, ”kalman filtering” for ”kalmanfiltering”, ”kalman filter” for ”kalmanfiltering”, ”virtual agents” for ”virtualagents”, ”support vector machines” for ”supportvectormachines”, ” svm ” for ” supportvectormachines ”, ” svm,” for ” supportvectormachines ”, ”, svm ” for ” supportvectormachines ”, ”image recognition” for ”imagerecognition”, ”point cloud library” for ”imagerecognition”, ”image fusion” for ”imagefusion”, ”natural language toolkit” : ”naturallanguagetoolkit”, ”nltk” : ”naturallanguagetoolkit”, ” maximum entropy ” for ” naturallanguageprocessing ”, ” nlp ” for ”naturallanguageprocessing”, ”, nlp ” for ” naturallanguageprocessing ”, ” nlp, ” for ” naturallanguageprocessing ”, ” corenlp” for ” naturallanguageprocessing ”, ” opennlp” for ” naturallanguageprocessing ”,” apache uima” for ” naturallanguageprocessing ”, ”natural language processing” for ”naturallanguageprocessing”, ” gensim ” for ” naturallanguageprocessing ”, ”document embeddings” for ”naturallanguageprocessing”, ”doc2vec” for ”naturallanguageprocessing”, ”speech recognition” for ”speechrecognition”, ”sound recognition” for ”speechrecognition”, ”scikit learn” for ”scikitlearn”, ”scikit - learn” for ”scikitlearn”, ”pattern recognition” for ”patternrecognition”, ”pattern detection” for ”patternrecognition”,”discriminant”

for ”discriminantanalysis”, ”object recognition” for ”objectrecognition”, ”object tracking” for ”objectrecognition”, ”neural networks” for ”neuralnetworks”, ”convolutional” for ”neuralnetworks”, ” lstm ” for ”longshorttermmemory”, ” lstms ” for ”longshorttermmemory”, ” , lstm ” for ”longshorttermmemory”, ” , lstms ” for ”longshorttermmemory”, ” lstm, ” for ”longshorttermmemory”, ” lstms, ” for ”longshorttermmemory”, ”recurrent network” for ”neuralnetworks”, ”recurrent networks” for ”neuralnetworks”, ”siamese networks” for ”neuralnetworks”, ” rnn ” for ” rnns ”, ”dnn techniques” for ”neuralnetworks”, ”supervised learning” for ”supervisedlearning”, ”gaussian processes” for ”supervisedlearning”, ”text mining” for ”textmining”, ”information extraction” for ”textmining”, ”unstructured information management” for ”textmining”, ”support vector machines” for ”supportvectormachines”, ”unsupervised learning” for ”unsupervisedlearning”, ”image processing” for ”imageprocessing”,”jung framework” for ”jungframework”, ”recommender systems” for ”recommendersystems”, ”random forests” for ”randomforests”, ”random forest” for ”randomforests”, ”latent semantic analysis” for ”latentsemanticanalysis”, ”sentiment analysis” for ”sentimentanalysis”, ”emotion detection” for ”sentimentanalysis”, ”tensor flow” for ”tensorflow”, ”opinion mining” for ”opinionmining”, ”hidden markov model”:”hiddenmarkovmodel”, ”latent dirichlet allocation” for ”latentdirichletallocation”, ” lda ” for ” latentdirichletallocation ”, ” lda,” for ” latentdirichletallocation ”, ”, lda” for ” latentdirichletallocation ”, ”predictive models” for ”predictivemodels”, ”predictive modelling” for ”predictivemodels”, ”kernel methods” for ”kernelmethods”, ”kernel method” for ”kernelmethods”, ”gradient descent” for ”gradientdescent”, ” sgd ” for ”gradientdescent”, ”gradient boosting” for ”gradientboosting”, ”gradient boost” for ”gradientboosting”, ”gradient boosted” for ”gradientboosting”, ”machine translation” for ”machinetranslation”, ”seq2seq” for ”machinetranslation”, ”sequence - to - sequence” for ”machinetranslation”, ”sequence to sequence” for ”machinetranslation”, ”sentiment classification” for ”classificationalgorithms”, ”classification algorithms” for ”classificationalgorithms”, ”classification algorithm” for ”classificationalgorithms”, ” gans ” for ”generativeadversarialnetworks”, ”, gans ” for ”generativeadversarialnetworks”, ” gans, ” for ”generativeadversarialnetworks”, ” generative adversarial networks” for ”generativeadversarialnetworks”, ”quantum computing” for ”quantumcomputing”, ” nlu ” for ”naturallanguageunderstanding”, ”, nlu ” for ” naturallanguageunderstanding ”, ” nlu, ” for ” naturallanguageunderstanding ”, ”natural language understanding” for ”naturallanguageunderstanding”, ” nlg ” for ”naturallanguagegeneration”, ”, nlg ” for ” naturallanguagegeneration ”, ” nlg, ” for ” naturallanguagegeneration ”, ” natural language generation ” for ”naturallanguagegeneration”, ”nlg health ” for ”health”, ”nlg are” for ”are”, ”cloud technology” for ”cloudtechnology”, ”cloud technologies” for ”cloudtechnology”, ”ensemble methods” for ”ensemblemethods”, ”ensemble learning” for ”ensemblemethods”, ”decision trees” for ”decisiontrees”, ” chaid” for ”decisiontrees”, ”tree based lerners” for ”decisiontrees”, ”tree - based lerners” for ”decisiontrees”, ”tree - based classifiers” for ”decisiontrees”, ”tree based classifiers” for ”decisiontrees”, ”k means clustering” for ”clusteringalgorithms”, ”hierarchical clustering” for ”clusteringalgorithms”, ”k - means clustering” for ”clusteringalgorithms”, ”clustering algorithms” for ”clusteringalgorithms”, ”optical character recognition” for ”opticalcharacterrecognition”, ”optical flow” for ”opticalcharacterrecognition”, ”named entity recognition” for ”namedentityrecognition”, ”entity linking” for ” naturallanguageprocessing ”, ”topic modeling” for ”topicmodelling”, ”topic modelling” for ”topicmodelling”,”topic detection” for ”topicmodelling”, ”text classification” for ”classificationalgorithms”, ”similarity models” for ”topicmodelling”, ”a b testing” for ”abtesting”, ”a . b testing” for ”abtesting”, ”dimensionality reduction” for ”dimensionalityreduction”, ” umap ” for ”dimensionalityreduction”, ”uniform manifold” for ”dimensionalityreduction”, ”dimension reduction” for ”dimensionalityreduction”, ” tsne ” for ”dimensionalityreduction”, ”t - sne” for ”dimensionalityreduction”, ” t sne ” for ”dimensionalityreduction”, ”feature engineering” for ”featureengineering”, ”hyperparameter tuning” for ”hyperparametertuning”, ”hyper parameter tuning” for ”hyperparametertuning”, ”data augmentation” for ”dataaugmentation”, ”model compression” for ”modelcompression”, ”naive bayes” for ”bayesianmethods”, ”naïve bayes” for ”bayesianmethods”, ”naï ve bayes” for ”bayesianmethods”, ”bayesian” for ”bayesianmethods”, ”knowledge distillation” for ”knowledgedistillation”, ”density based spatial clustering” for ”densitybasedspatialclustering”,”density - based spatial clustering” for ”densitybasedspatialclustering”, ” dbscan ” for ”densitybasedspatialclustering”, ”computational linguistics” : ”computationallinguistics”, ”expectation maximization” : ”expectationmaximization”, ”caffe deep learning framework”:”caffedeeplearningframework”.

Contract language:

”conducted remote” for ”conductedremote”, ”fully remote” for ”fullyremote”, ”home based” for ”homebased”, ”home work” for ”homework”, ”job is virtual” for ”jobisvirtual”, ”location is remote” for ”locationisremote”, ”location remote” for ”locationremote”, ”position is remote” for ”positionisremote”, ”position remote” for ”positionremote”, ”remote work” for ”remotework”, ”role will be remote” for ”rolewillberemote”, ”virtual job” for ”virtualjob”, ”work from home” for ”workfromhome”, ”work home” for ”workhome”, ”work remote” for ”workremote”, ”working remote” for ”workingremote”, ”flx with office” for ”flxwithoffice”, ”remote position” for ”remoteposition”, ”role is remote” for ”roleisremote”, ”remote basis” for ”remotebasis”, ”hybrid work” for ”hybridwork”, ”hybrid remote” for ”hybridremote”, ”vice president” for ”vicepresident”, ”project lead” for ”projectlead”, ”team lead” for ”teamlead”, ”supervising a team” for ”supervisingateam”, ”trial period” for ”trialperiod”, ”entry level” for ”entrylevel”, ”tele commuting” for ”telecommuting”, ”tele work” for ”telework ”, ”no telecommuting” for ”no”, ”does not offer teleworker” for ”no”, ”no taking work home” for ”no”, ”take work home” for ”no”, ”teaching assistant” for ”academic assistant”, ”secure position” for ”permanent position”, ”temporary to permanent” for ”temporary”, ”not permanent” for ”temporary”, ”lead to permanent” for ”temporary”, ”not temporary” for ”permanent”, ”getting a permanent” for ”temporary”, ”seasonal contract” for ”seasonal contract”, ”seasonal hours” for ”seasonal contract”, ”decommissions” for ”decom”, ”commissioning” for ”comis”, ”rewards and benefits” for ”rewards packages”, ”pay - for - performance” for ”pay for performance”, ”share options” for ”stock options”,”based on performance” for ”pay for performance”, ”short term, long term” for ”short term contract, long term contract”, ”short term , long term” for ”short term contract, long term contract”, ”short term and long term” for ”short term contract and long term contract”, ”short term basis” for ”short term contract”,”short term or long term” for ”short term contract or long term contract”, ”contract is short term” for ”short term contract”, ”or short term” for ”or short term contract”, ” temp ” for ” temporary ”, ”long term conditions” for ”conditions”, ”long term condition” for ”condition”, ”long term strategy” for ”strategy”, ”long term strategic” for ”strategic”, ”long term satisfaction” for ”satisfaction”, ”long term residential” for ”residential”, ”long term financial” for ”financial”, ”long term happiness” for ”happiness”, ”long term need” for ”need”, ”long term health” for ”health”, ”temporary ongoing” for ”temporary contract”, ”ongoing contract subject” for ”temporary contract”, ” perm ” for ” permanent ”, ” perm, ” for ” permanent ”, ” ,perm ” for ” permanent ”, ” temp, ” for ” temporary ”, ” ,temp ” for ” temporary ”, ”experience is needed” for ”experience needed”, ”years of experience” for ”years experience”, ” yrs ” for ” years ”, ”don” t need experience” for ”dont need experience”, ”don’ t need previous” for ”dont need experience” , ”experience is essential” for ”experience essential”, ”not essential” for ”no essential”, ”not essential” for ”no essential”, ”no formal qualifications” for ”no formal qualification”, ”casual work” for ”casual contract”, ”casual basis” for ”casual contract”, ”casual work” for ”casual contract”, ”casual hours” for ”part time”, ”casual shifts” for ”casual contract”, ”40 hours” for ”full time”, ”40 hour” for ”full time”, ”38.5 hour” for ”full time”, ”38.5 hours” for ”full time”,”38 hours” for ”full time”, ”38 hour” for ”full time”,”37 hours” for ”full time”, ”37 hour” for ”full time”, ”9 to 17” for ”9 to 5”, ”9 17” for ”9 to 5”, ”9 . 17” for ”9 to 5”, ”profit sharing” for ”profitsharing”, ”reward scheme” for ”rewardscheme”, ”stock options” for ”stockoptions”, ”variable pay” for ”variablepay”, ”performance related” for ”performancerelated”, ”rewards packages” for ”rewardspackages”, ”pay for performance” for ”payforperformance”, ”fixed term” for ”fixedterm”, ”short term contract” for ”shorttermcontract”, ”day to day contract” for ”daytodaycontract”, ”zero hours” for ”zerohours”, ”casual contract” for ”casualcontract”, ”long term” for ”longterm”, ”ongoing contract” for ”ongoingcontract”, ”4 hour” for ”4hour”, ”4 hours” for ”4hour”,”6 hour” for ”6hour”, ”6 hours” for ”6hour”,”20 hour” for ”20hour”, ”20 hours” for ”20hour”,”30 hour” for ”30hour”, ”9 to 5” for ”9to5”, ”full time” for ”fulltime”, ”standard hours” for ”standardhours”, ”day shift” for ”dayshift”, ”night shift” for ”nightshift”, ”3 to 5 years” for ”3to5years”, ”experience a minimum” for ”experienceaminimum”, ”experience for” for ”experiencefor”, ”experience in” for ”experiencein”, ”experience is also” for ”experienceisalso”, ”experience needed” for ”experienceneeded”, ”experience of” for ”experienceof”, ”experience preferably” for ”experiencepreferably”, ”experience required” for ”experiencerequired”, ”experience working” for ”experienceworking”, ”extensive experience” for ”extensiveexperience”, ”have experience” for ”haveexperience”, ”have previous experience” for ”havepreviousexperience”, ”have some knowledge” for ”havesomeknowledge”, ”have worked” for ”haveworked”, ”can teach” for ”canteach”, ”minimum experience” for ”minimumexperience”, ”months experience” for

”monthsexperience”, ”need experience” for ”needexperience”, ”past experience” for ”pastexperience”, ”practical experience” for ”practicalexperience”, ”prior experience” for ”priorexperience”, ”recent experience” for ”recentexperience”, ”relevant experience” for ”relevantexperience”, ”right experience” for ”rightexperience”, ”solid experience” for ”solidexperience”, ”strong experience” for ”strongexperience”, ”suitable experience” for ”suitableexperience”, ”three to five years” for ”threetoFiveyears”, ”two to three years” for ”twotothreeyears”, ”work experience” for ”workexperience”, ”worked in similar” for ”workedinsimilar”, ”required experience” for ”requiredexperience”, ”experience essential” for ”experienceessential”, ”experience is preferred” for ”experienceispreferred”, ”do not need any” for ”donotneedany”, ”do not need experience” for ”donotneedexperience”, ”experience is not” for ”experienceisnot”, ”no essential experience” for ”noessentialexperience”, ”no formal experience” for ”noformalexperience”, ”no formal qualifications or experience” for ”noformalqualificationsorexperience”, ”no past experience” for ”nopastexperience”, ”no work experience” for ”noworkexperience”, ”with no experience” for ”withnoexperience”, ”without any experience” for ”withoutanyexperience”, ”working directly with” for ”workingdirectly”, ”don t need experience” for ”dontneedexperience”, ”not needed” for ”notneeded”, ”working directly with” for ”workingdirectlywith”, ”report into” for ”reportinto”, ”your team leader” for ”yourteamleader”, ”working with” for ”workingwith”, ”working closely” for ”workingclosely”, ”under the direction” for ”underthedirection”, ”speaking to” for ”speakingto”, ”directly to” for ”directlyto”, ”work alongside” for ”workalongside”, ”required by” for ”requiredby”, ”keeping your” for ”keepingyour”, ”work closely” for ”workclosely”, ”step up to” for ”stepto”, ”managing the” for ”managingthe”, ”assistance to the” for ”assistancetothe”, ”work with” for ”workwith”, ”managed by” for ”managedby”, ”approval of” for ”approvalof”, ”communicate with” for ”communicatewith”, ”on experience” for ”onexperience”, ”negotiable salary” for ”negotiablesalary”, ”salary negotiable” for ”salarynegotiable”, ”hour negotiable” for ”hournegotiable”, ”salary is negotiable” for ”salaryisnegotiable”, ”salary up to” for ”salaryupto”.

Education language:

”at level” for ”level”, ”at either level” for ”level”, ” masters of ” for ” masterof ”,” master in ” for ” masterof ”, ” masters in ” for ” masterof ”, ”hour ” for ”hours”, ”skill ” for ”skills ”, ”cipd qualified” for ”cipd”, ”qualified to level” for ”level”, ”qualifications” for ”qualification”, ”mcipd” for ”cipd”, ”scqf level 1 ” for ”scqflevel2 ”, ”svq level 1 ” for ”svqlevel2 ”, ” elc ” for ”entrylevelcertificate”, ” cse ”:” certificateofsecondaryeducation ”, ”a levels” for ”alevelqualification”, ”as levels” for ”aslevelqualification”, ”a level science” for ”alevelqualification”, ”a level in math” for ”alevelqualification”, ”a level in english” for ”alevelqualification”, ”as level in math” for ”aslevelqualification”, ”as level in english” for ”aslevelqualification”, ”as level in science” for ”aslevelqualification”, ” certhe ” for ” certificateofhighereducation ”, ”post graduate” for ”postgraduate”, ” graduate ” for ” bachelor ”, ” bsc ” for ” bachelor ”, ” b . ed ” for ” bachelor ”, ” b. ed ” for ” bachelor ”, ” ba hons ” for ” bachelor ”, ” b eng ” for ” bachelor ”, ” b. eng ” for ” bachelorofengineering ”, ” beng ” for ” bachelorofengineering”, ” mba ” for ” masterofbusinessadministration ”, ” ll. b ” for ” bacheloroflaws”, ” meng ” for ” mastersofengineering ”, ” mmath ” for ” mastersofmathematics ”, ” mphil ” for ”masterofphilosophy ”, ” mpsych ” for ” masterofphysics ”, ” mprof ” for ” masterofprofessionalstudies ”, ” mres ” for ” masterofresearch ”, ” msc ” for ” masterofscience ”, ” msc, ” for ” masterofscience ”, ” msci ” for ” masterofscience ”, ” msci, ” for ” masterofscience ”, ” mst ” for ” masterofstudies ”, ” m. sc ” for ” masterofscience ”, ” m . sc ” for ” masterofscience ”, ” llm ” for ” masteroflaw ”, ” ll . m ” for ” masteroflaw ”, ” mpharm ” for ” mastersofpharmacy ”, ” mchem ” for ” masterofchemistry ”, ” ms ” for ” masterofscience ”, ” msw ” for ” masterofsocialwork ”, ” pcge ” for ” postgraduatecertificateineducation ”, ” pcg ” for ” postgraduatecertificate ”, ” pgd ” for ” postgraduatediploma ”, ” pgcert ” for ” postgraduatecertificate ”, ” pg cert ” for ” postgraduatecertificate ”, ”ph. d” for ” phd ”, ”ph d” for ” phd ”, ” ma programme ” for ” masterofart ”, ” ma programme ” for ” masterofart ”, ” ma in english ” for ” masterofart ”, ”hold an ma ” for ” masterofart ”, ” ma degree ” for ” masterofart ”, ”honors degree” for ”honoursdegree”, ”self employed” for ”selfemployed”, ”self employment” for ”selfemployed”, ”seasonal contract” for ”seasonalcontract”, ”basic functional skills” for ”basicfunctionalskills”, ”basic skills qualification” for ”basicskillsqualification”, ”entry level certificate” for ”entrylevelcertificate”, ”entry level functional skills” for ”entrylevelfunctionalskills”, ”entry level qualifications” for ”entrylevelqualifications”, ”entry nqf level” for ”entrynqflevel”,

”essential skills wales” for ”essentialskillswales”, ”level 1 scqf” for ”level1scqf”, ”level 1 svq” for ”level1svq”, ”level 2 scqf” for ”level2scqf”, ”level 2 svq” for ”level2svq”, ”level 3 scqf” for ”level3scqf”, ”level 3 svq” for ”level3svq”, ”no education requirements” for ”noeducationrequirements”, ”no formal qualification” for ”noformalqualification”, ”no previous experience or qualification” for ”nopreviousexperienceorqualification”, ”no previous qualification” for ”nopreviousqualification”, ”no qualification” for ”noqualification”, ”scottish national 1” for ”scottishnational1”, ”scottish national 2” for ”scottishnational2”, ”scottish national 3” for ”scottishnational3”, ”scqf level 2” for ”scqflevel2”, ”scqf level 3” for ”scqflevel3”, ”skills for life” for ”skillsforlife”, ”svq level 2” for ”svqlevel2”, ”svq level 3” for ”svqlevel3”, ”no formal education” for ”noformaleducation”, ”btec award” for ”btecaward”, ”btec first” for ”btecfirst”, ”btec level 1” for ”bteclevel1”, ”certificate of secondary education” for ”certificateofsecondaryeducation”, ”first certificate” for ”firstcertificate”, ”foundation welsh baccalaureate” for ”foundationwelshbaccalaureate”, ”key skills level 1” for ”keyskillslevel1”, ”level 1 award” for ”level1award”, ”level 1 certificate” for ”level1certificate”, ”level 1 diploma” for ”level1diploma”, ”level 1 esol” for ”level1esol”, ”level 1 essential skills” for ”level1essentialskills”, ”level 1 functional skills” for ”level1functionalskills”, ”level 1 key skills” for ”level1keyskills”, ”level 1 national vocational qualification” for ”level1nationalvocationalqualification”, ”level 1 nvq” for ”level1nvq”, ”nqf level 1” for ”nqflevel1”, ”nvq level 1” for ”nvqlevel1”, ”rqf level 1” for ”rqflevel1”, ”scqf level 4” for ”scqflevel4”, ”diploma level 1” for ”diplomalevel1”, ”national level 1” for ”nationallevel1”, ”qualification level 1” for ”qualificationlevel1”, ”level 1 qualification” for ”level1qualification”, ”level 1 qcf” for ”level1qcf”, ”functional skills level 1” for ”functionalskillslevel1”, ”essential skills level 1” for ”essentialskillslevel1”, ”level 1 apprenticeship” for ”level1apprenticeship”, ”cipd level 1” for ”cipdlevel1”, ”level 1 cipd” for ”level1cipd”, ”level 4 scqf” for ”level4scqf”, ”level 4 svq” for ”level4svq”, ”svq level 4” for ”svqlevel4”, ”btec level 2” for ”bteclevel2”, ”general certificate of education” for ”generalcertificateofeducation”, ”intermediate apprenticeship” for ”intermediateapprenticeship”, ”intermediate welsh baccalaureate” for ”intermediatewelshbaccalaureate”, ”key skills level 2” for ”keyskillslevel2”, ”level 2 award” for ”level2award”, ”level 2 certificate” for ”level2certificate”, ”level 2 diploma” for ”level2diploma”, ”level 2 esol” for ”level2esol”, ”level 2 essential skill” for ”level2essentialskill”, ”level 2 functional skill” for ”level2functionalskill”, ”level 2 key skills” for ”level2keyskills”, ”level 2 national” for ”level2national”, ”level 2 nvq” for ”level2nvq”, ”national level 2” for ”nationallevel2”, ”nqf level 2” for ”nqflevel2”, ”nvq level 2” for ”nvqlevel2”, ”rqf level 2” for ”rqflevel2”, ”scqf level 5” for ”scqflevel5”, ”secondary education” for ”secondaryeducation”, ”diploma level 2” for ”diplomalevel2”, ”award level 3” for ”awardlevel3”, ”esol level 2” for ”esollevel2”, ”functional skills level 2” for ”functionalskillslevel2”, ”essential skills level 2” for ”essentialskillslevel2”, ”qualification level 2” for ”qualificationlevel2”, ”level 2 qualification” for ”level2qualification”, ”level 2 apprenticeship” for ”level2apprenticeship”, ”cipd level 2” for ”cipdlevel2”, ”level 2 cipd” for ”level2cipd”, ”level 5 scqf” for ”level5scqf”, ”level 5 svq” for ”level5svq”, ”svq level 5” for ”svqlevel5”, ”certificate of secondary education” for ”certificateofsecondaryeducation”, ”a level qualification” for ”alevelqualification”, ”as level qualification” for ”aslevelqualification”, ”access to higher education diploma” for ”accesstohighereducationdiploma”, ”advanced apprenticeship” for ”advancedapprenticeship”, ”advanced diploma” for ”advanceddiploma”, ”advanced extension” for ”advancedextension”, ”advanced welsh baccalaureate” for ”advancedwelshbaccalaureate”, ”btec award level 3” for ”btecawardlevel3”, ”btec certificate level 3” for ”bteccertificatelevel3”, ”btec diploma level 3” for ”btecdiplomalevel3”, ”btec level 3” for ”bteclevel3”, ”btec national” for ”btecnational”, ”diploma programme” for ”diplomaprogramme”, ”extended certificate” for ”extendedcertificate”, ”extended diploma” for ”extendeddiploma”, ”key skills level 3” for ”keyskillslevel3”, ”level 3 award” for ”level3award”, ”level 3 certificate” for ”level3certificate”, ”level 3 esol” for ”level3esol”, ”level 3 national” for ”level3national”, ”level 3 nvq” for ”level3nvq”, ”national level 3” for ”nationallevel3”, ”nqf level 3” for ”nqflevel3”, ”nvq level 3” for ”nvqlevel3”, ”ocr national” for ”ocrnational”, ”progression diploma” for ”progressiondiploma”, ”rqf level 3” for ”rqflevel3”, ”scottish advanced higher” for ”scottishadvancedhigher”, ”scottish baccalaureate” for ”scottishbaccalaureate”, ”scottish higher” for ”scottishhigher”, ”scqf level 6” for ”scqflevel6”, ”diploma level 3” for ”diplomalevel3”, ”award level 3” for ”awardlevel3”, ”functional skills level 3” for ”functionalskillslevel3”, ”essential skills level 3” for ”essentialskillslevel3”, ”qualification level 3” for ”qualificationlevel3”, ”level 3 qualification” for ”level3qualification”, ”level 3 apprenticeship” for ”level3apprenticeship”, ”cipd level 3” for ”cipdlevel3”, ”level 3 cipd” for ”level3cipd”, ”level 6 scqf” for ”level6scqf”, ”level 6 svq” for ”level6svq”, ”svq level 6” for ”svqlevel6”, ”certificate of higher education” for ”certificateofhighereducation”, ”higher

apprenticeship" for "higherapprenticeship", "higher national certificate" for "highernationalcertificate", "level 4 award" for "level4award", "level 4 certificate" for "level4certificate", "level 4 diploma" for "level4diploma", "level 4 national" for "level4national", "level 4 nvq" for "level4nvq", "national level 4" for "nationallevel4", "nqf level 4" for "nqflevel4", "nvq level 4" for "nvqlevel4", "scqf level 7" for "scqflevel7", "diploma level 4" for "diplomalevel4", "award level 4" for "awardlevel4", "functional skills level 4" for "functionalskillslevel4", "essential skills level 4" for "essentialskillslevel4", "qualification level 4" for "qualificationlevel4", "level 4 qualification" for "level4qualification", "level 4 apprenticeship" for "level4apprenticeship", "cipd level 4" for "cipdlevel4", "level 4 cipd" for "level4cipd", "level 7 scqf" for "level7scqf", "level 7 svq" for "level7svq", "svq level 7" for "svqlevel7", "btec professional" for "btecprofessional", "diploma of further education" for "diplomaoffurthereducation", "diploma of higher education" for "diplomaofhighereducation", "foundation degree" for "foundationdegree", "higher diploma" for "higherdiploma", "higher national diploma" for "highernationaldiploma", "level 5 award" for "level5award", "level 5 certificate" for "level5certificate", "level 5 diploma" for "level5diploma", "level 5 national" for "level5national", "level 5 nvq" for "level5nvq", "national level 5" for "nationallevel5", "nqf level 5" for "nqflevel5", "nvq level 5" for "nvqlevel5", "scqf level 8" for "scqflevel8", "diploma level 5" for "diplomalevel5", "award level 5" for "awardlevel5", "qualification level 5" for "qualificationlevel5", "level 5 qualification" for "level5qualification", "cipd level 5" for "cipdlevel5", "level 5 cipd" for "level5cipd", "level 8 scqf" for "level8scqf", "level 5 apprenticeship" for "level5apprenticeship", "level 5 professional" for "level5professional", "level 8 svq" for "level8svq", "svq level 8" for "svqlevel8", "bachelors degree" for "bachelorsdegree", "btec advanced professional" for "btecadvancedprofessional", "degree apprenticeship" for "degreeapprenticeship", "first degree" for "firstdegree", "bachelor certificate" for "bachelorcertificate", "bachelor diploma" for "bachelordiploma", "level 6 award" for "level6award", "level 6 certificate" for "level6certificate", "level 6 diploma" for "level6diploma", "level 6 national" for "level6national", "level 6 nvq" for "level6nvq", "national level 6" for "nationallevel6", "nqf level 6" for "nqflevel6", "nvq level 6" for "nvqlevel6", "ordinary degree" for "ordinarydegree", "oxbridge ma" for "oxbridgema", "scottish ma" for "scottishma", "scqf level 10" for "scqflevel10", "scqf level 9" for "scqflevel9", "higher education" for "highereducation", "diploma level 6" for "diplomalevel6", "award level 6" for "awardlevel6", "qualification level 6" for "qualificationlevel6", "level 6 qualification" for "level6qualification", "cipd level 6" for "cipdlevel6", "level 6 cipd" for "level6cipd", "level 9 scqf" for "level9scqf", "level 10 scqf" for "level10scqf", "level 6 apprenticeship" for "level6apprenticeship", "level 6 professional" for "level6professional", "level 10 svq" for "level10svq", "svq level 10" for "svqlevel10", "have a degree" for "haveadegree", "bsc or masters" for "bscormasters", "degree or master" for "degreeormaster", "bsc or msc" for "bscormsc", "ba or ma" for "baorma", "honours degree" for "honoursdegree", "award level 7" for "awardlevel7", "ba and ma" for "baandma", "cipd level 7" for "cipdlevel7", "diploma level 7" for "diplomalevel7", "have a master" for "haveamaster", "have a masters" for "haveamasters", "level 11 scqf" for "level11scqf", "level 11 svq" for "level11svq", "level 7 apprenticeship" for "level7apprenticeship", "level 7 award" for "level7award", "level 7 certificate" for "level7certificate", "level 7 cipd" for "level7cipd", "level 7 diploma" for "level7diploma", "level 7 national" for "level7national", "level 7 nvq" for "level7nvq", "level 7 professional" for "level7professional", "level 7 qcf" for "level7qcf", "master of arts" for "masterofarts", "master of business administration" for "masterofbusinessadministration", "master of engineering" for "masterofengineering", "master of law" for "masteroflaw", "master of mathematics" for "masterofmathematics", "master of pharmacy" for "masterofpharmacy", "master of philosophy" for "masterofphilosophy", "master of physics" for "masterofphysics", "master of professional studies" for "masterofprofessionalstudies", "master of public health" for "masterofpublichealth", "master of research" for "masterofresearch", "master of science" for "masterofscience", "master of social work" for "masterofsocialwork", "master of studies" for "masterofstudies", "masters by research" for "mastersbyresearch", "masters degree" for "mastersdegree", "national level 7" for "nationallevel7", "nqf level 7" for "nqflevel7", "nvq level 7" for "nvqlevel7", "postgraduate certificate" for "postgraduatecertificate", "postgraduate certificate in education" for "postgraduatecertificateineducation", "postgraduate diploma" for "postgraduatediploma", "professional masters" for "professionalmasters", "scqf level 11" for "scqflevel11", "svq level 11" for "svqlevel11", "doctor of philosophy" for "doctorofphilosophy", "doctoral degree" for "doctoraldegree", "doctors degree" for "doctorsdegree", "level 8 award" for "level8award", "level 8 certificate" for "level8certificate", "level 8 diploma" for "level8diploma", "level 8 national" for "level8national", "level 8 nvq" for "level8nvq", "national level 8" for

”nationallevel8”, ”nqf level 8” for ”nqflevel8”, ”nvq level 8” for ”nvqlevel8”, ”rqf level 8” for ”rqflevel8”, ”scqf level 12” for ”scqflevel12”, ”diploma level 8” for ”diplomalevel8”, ”award level 8” for ”awardlevel8”, ”qcf level 8” for ”qcflevel8”, ”level 8 qcf” for ”level8qcf”, ”cipd level 8” for ”cipdlevel8”, ”level 8 cipd” for ”level8cipd”, ”level 12 scqf” for ”level12scqf”, ”level 8 apprenticeship” for ”level8apprenticeship”, ”level 8 professional” for ”level8professional”, ”svq level 12” for ”svqlevel12”, ”level 12 svq” for ”level12svq”.

D. Additional Figures and Tables

Table OD1: Personality Traits by 3-digit Occupation group

SOC	Name	PT	Agree.	Consc.	Emot.	Extra.	Open.
111	Chief Executives and Senior Officials	61.1%	11.0%	29.8%	10.6%	30.8%	24.4%
112	Production Managers and Directors	59.4%	4.2%	35.0%	7.8%	27.9%	23.0%
113	Functional Managers and Directors	68.7%	7.3%	33.6%	10.4%	34.6%	37.9%
115	Financial Institution Managers and Directors	63.7%	7.5%	35.5%	11.6%	33.0%	21.0%
116	Managers and Directors in Transport and Logistics	79.3%	5.9%	49.9%	10.2%	51.6%	26.1%
117	Senior Officers in Protective Services	68.2%	7.9%	30.2%	11.7%	40.4%	24.8%
118	Health and Social Services Managers and Directors	82.1%	19.2%	41.5%	16.8%	47.5%	35.0%
119	Managers and Directors in Retail and Wholesale	77.3%	6.0%	39.5%	15.7%	59.0%	33.0%
121	Managers and Proprietors in Agriculture Related S.	74.1%	10.3%	44.6%	8.3%	42.1%	29.1%
122	Managers and Proprietors in Hospitality and Leisure S.	81.5%	13.9%	40.8%	13.4%	64.0%	20.3%
124	Managers and Proprietors in Health and Care S.	75.2%	21.6%	36.3%	17.5%	40.6%	25.6%
125	Managers and Proprietors in Other Services	69.9%	9.8%	36.6%	12.7%	45.7%	24.7%
211	Natural and Social Science Professionals	71.5%	5.3%	26.7%	5.5%	37.6%	45.9%
212	Engineering Professionals	55.6%	3.2%	25.9%	6.3%	24.1%	29.4%
213	I.T. and Telecommunications Professionals	63.2%	4.0%	25.3%	5.7%	30.2%	39.4%
214	Conservation and Environment Professionals	74.7%	11.2%	35.6%	10.5%	49.8%	37.7%
215	Research and Development Managers	70.0%	8.9%	29.0%	5.9%	36.5%	48.1%
221	Health Professionals	69.7%	13.6%	23.3%	10.1%	42.7%	28.8%
222	Therapy Professionals	70.2%	10.2%	26.8%	9.6%	46.7%	21.1%
223	Nursing and Midwifery Professionals	75.1%	23.1%	39.3%	24.7%	41.1%	23.8%
231	Teaching and Educational Professionals	77.6%	11.1%	31.8%	13.2%	52.2%	32.5%
241	Legal Professionals	59.7%	10.4%	29.7%	6.4%	32.9%	18.9%
242	Business, Research and Administrative Professionals	64.5%	6.4%	32.9%	9.8%	31.8%	35.3%
243	Architects, Town Planners and Surveyors	51.3%	4.4%	28.7%	5.5%	24.4%	20.2%
244	Welfare Professionals	71.1%	15.1%	29.1%	17.2%	38.7%	37.7%
245	Librarians and Related Professionals	79.2%	12.9%	33.9%	14.0%	48.2%	46.2%
246	Quality and Regulatory Professionals	62.3%	5.1%	33.6%	8.5%	29.5%	29.5%
247	Media Professionals	82.9%	8.4%	31.9%	11.6%	50.2%	61.9%
311	Science, Engineering and Production Technicians	67.1%	4.9%	39.1%	10.6%	35.4%	26.5%
312	Draughtspersons and Related Architectural Technicians	57.7%	3.7%	27.3%	8.0%	27.2%	29.7%
313	Information Technology Technicians	69.5%	8.5%	34.1%	13.7%	41.4%	30.1%

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Table OD1 – continued from previous page

SOC	Name	PT	Agree.	Consc.	Emot.	Extra.	Open.
321	Health Associate Professionals	69.2%	9.4%	30.7%	7.2%	43.5%	19.6%
323	Welfare and Housing Associate Professionals	79.4%	18.0%	31.5%	24.4%	49.4%	39.3%
331	Protective Service Occupations	76.1%	11.2%	33.2%	26.4%	38.3%	31.8%
341	Artistic, Literary and Media Occupations	77.4%	6.2%	27.9%	10.2%	45.8%	51.8%
342	Design Occupations	84.7%	4.4%	25.0%	7.6%	48.0%	72.0%
344	Sports and Fitness Occupations	79.4%	16.2%	29.2%	5.5%	70.4%	24.4%
351	Transport Associate Professionals	50.9%	1.5%	31.6%	14.5%	26.4%	15.0%
352	Legal Associate Professionals	61.8%	8.2%	32.8%	10.7%	33.6%	20.6%
353	Business, Finance and Related Associate Prof.	65.3%	7.6%	35.2%	10.2%	34.2%	29.1%
354	Sales, Marketing and Related Associate Prof.	77.6%	7.8%	36.6%	11.5%	49.9%	39.7%
355	Conservation and Environmental Associate Prof.	79.8%	15.9%	32.3%	5.9%	59.1%	24.4%
356	Public Services and Other Associate Prof.	77.9%	9.8%	41.4%	15.8%	50.4%	32.0%
411	Admin. Occ.: Government and Related Organisations	61.5%	10.8%	24.8%	8.2%	38.9%	26.8%
412	Admin. Occ.: Finance	62.4%	6.2%	32.0%	10.7%	37.8%	19.9%
413	Admin. Occ.: Records	72.9%	7.2%	43.1%	15.2%	42.7%	23.7%
415	Other Administrative Occupations	76.8%	10.6%	44.7%	15.9%	47.8%	26.8%
416	Administrative Occupations: Office Managers and Sup.	73.8%	11.1%	43.7%	15.6%	50.0%	24.2%
421	Secretarial and Related Occupations	75.5%	17.5%	41.8%	18.3%	51.0%	24.9%
511	Agricultural and Related Trades	76.1%	9.1%	46.2%	4.2%	52.3%	27.3%
521	Metal Forming, Welding and Related Trades	33.2%	0.8%	20.6%	4.1%	11.0%	8.4%
522	Metal Machining, Fitting and Instrument Making Trades	49.7%	2.8%	29.2%	6.8%	22.0%	16.1%
523	Vehicle Trades	66.2%	5.8%	43.3%	12.6%	35.2%	16.0%
524	Electrical and Electronic Trades	53.4%	3.6%	27.5%	5.2%	23.1%	25.7%
525	Skilled Metal, Electrical and Electronic Trades Sup.	41.3%	3.0%	28.6%	5.9%	17.1%	10.0%
531	Construction and Building Trades	46.3%	7.6%	29.1%	3.1%	19.9%	12.9%
532	Building Finishing Trades	33.9%	3.6%	21.8%	3.0%	13.8%	8.7%
533	Construction and Building Trades Sup.	42.3%	2.6%	28.3%	4.2%	17.4%	12.1%
541	Textiles and Garments Trades	67.8%	8.1%	31.3%	8.5%	47.6%	25.3%
542	Printing Trades	59.4%	2.8%	28.0%	12.2%	27.9%	29.6%
543	Food Preparation and Hospitality Trades	81.2%	9.4%	41.6%	19.7%	58.7%	33.1%
544	Other Skilled Trades	65.8%	2.2%	41.4%	13.7%	33.9%	16.8%
612	Childcare and Related Personal Services	78.8%	17.5%	22.7%	18.4%	59.7%	34.4%
613	Animal Care and Control Services	81.2%	9.3%	28.6%	6.6%	61.6%	45.8%
614	Caring Personal Services	82.6%	39.1%	26.6%	27.4%	50.2%	29.3%
621	Leisure and Travel Services	83.6%	9.8%	51.7%	22.3%	67.6%	20.6%
622	Hairdressers and Related Services	78.8%	7.6%	33.1%	7.4%	61.8%	22.0%
623	Housekeeping and Related Services	72.2%	28.1%	33.8%	11.0%	37.9%	16.4%

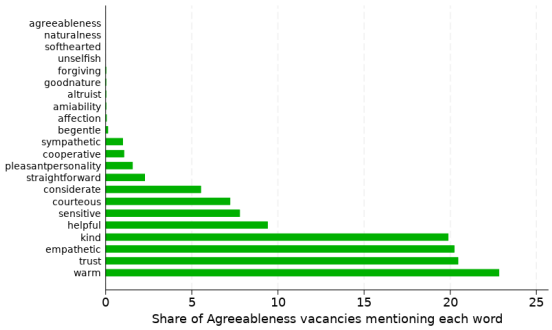
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Table OD1 – continued from previous page

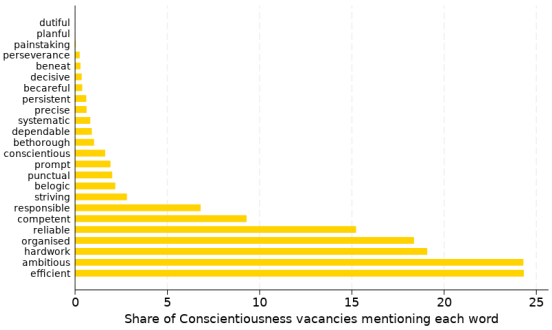
SOC	Name	PT	Agree.	Consc.	Emot.	Extra.	Open.
624	Cleaning and Housekeeping Managers and Sup.	72.1%	20.5%	52.0%	15.4%	39.2%	15.8%
711	Sales Assistants and Retail Cashiers	87.3%	16.0%	42.8%	9.5%	76.4%	31.9%
712	Sales Related Occupations	82.1%	7.5%	35.2%	11.7%	49.3%	39.5%
713	Sales Supervisors	79.0%	13.0%	41.7%	4.7%	60.3%	43.8%
721	Customer Service Occupations	85.4%	17.8%	39.6%	17.7%	67.8%	30.3%
722	Customer Service Managers and Sup.	84.9%	13.0%	43.8%	20.9%	60.5%	30.1%
811	Process Operatives	49.5%	2.0%	29.6%	6.2%	21.4%	16.7%
812	Plant and Machine Operatives	49.2%	1.7%	29.8%	5.7%	24.5%	17.1%
813	Assemblers and Routine Operatives	52.0%	2.6%	29.3%	6.5%	26.5%	18.6%
814	Construction Operatives	62.6%	9.5%	39.7%	6.5%	31.6%	20.6%
821	Road Transport Drivers	52.6%	8.8%	33.5%	5.8%	27.9%	19.0%
822	Mobile Machine Drivers and Operatives	40.4%	5.2%	23.7%	4.0%	18.0%	10.9%
823	Other Drivers and Transport Operatives	61.9%	4.8%	45.3%	17.1%	30.8%	23.0%
911	Elementary Agricultural Occupations	66.9%	5.6%	38.3%	7.9%	32.4%	32.0%
912	Elementary Construction Occupations	20.2%	3.2%	13.9%	1.1%	6.6%	2.6%
913	Elementary Process Plant Occupations	55.6%	1.8%	29.8%	6.3%	29.2%	14.6%
921	Elementary Administration Occupations	88.6%	34.4%	48.5%	35.6%	42.4%	75.5%
923	Elementary Cleaning Occupations	63.9%	7.6%	39.4%	6.7%	33.5%	23.4%
924	Elementary Security Occupations	71.5%	9.4%	27.1%	11.0%	44.7%	34.0%
925	Elementary Sales Occupations	69.3%	10.8%	27.8%	4.6%	51.5%	17.1%
926	Elementary Storage Occupations	70.5%	6.3%	44.1%	6.2%	45.5%	19.5%
927	Other Elementary Services Occupations	82.6%	21.1%	38.5%	17.9%	67.0%	22.6%

Figure OD1: Frequency of words by personality trait

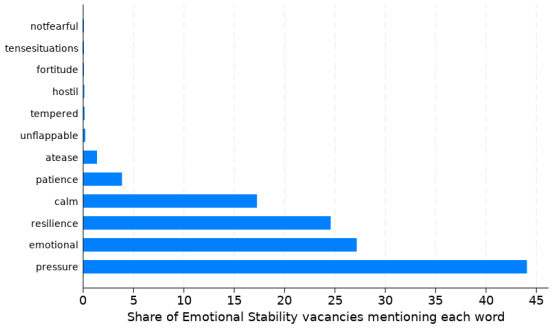
(a) Agreeableness



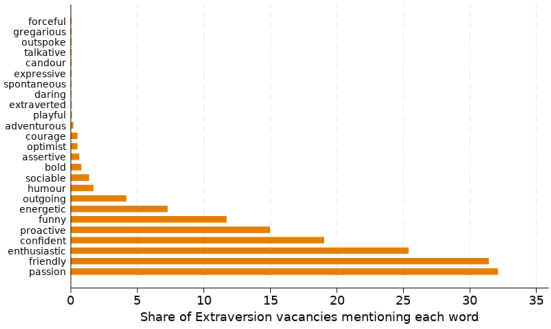
(b) Conscientiousness



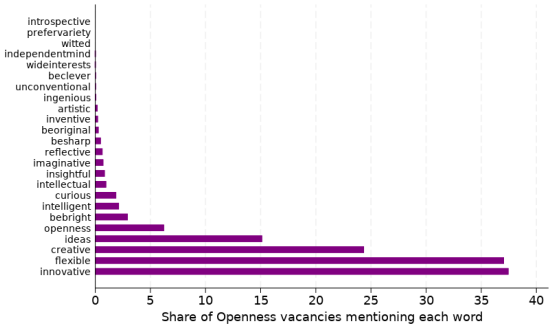
(c) Emotional Stability



(d) Extraversion



(e) Openness



Authors' creation using Adzuna

Table OD2: Covariate of main variables

		Before Weighting		After Weighting	
Code	Variable	AI-demanding	Non-demanding	AI-demanding	Non-demanding
Panel A: Personality Traits and Skills					
	Agreeableness	0.105	0.104	0.105	0.105
	Conscientiousness	0.406	0.378	0.406	0.406
	Emotional stability	0.132	0.131	0.132	0.132
	Extraversion	0.460	0.455	0.460	0.460
	Openness	0.476	0.302	0.476	0.476
	Soft Skills	0.744	0.629	0.743	0.743
	Hard Skills	0.834	0.571	0.832	0.832
Panel B: Location Distribution (NUTS1)					
UKC	North East	0.014	0.019	0.014	0.014
UKD	North West	0.077	0.093	0.077	0.077
UKE	Yorkshire and The Humber	0.044	0.060	0.044	0.044
UKF	East Midlands	0.035	0.058	0.035	0.035
UKG	West Midlands	0.057	0.072	0.057	0.057
UKH	East of England	0.070	0.098	0.070	0.070
UKI	London	0.404	0.258	0.404	0.404
UKJ	South East	0.155	0.172	0.155	0.155
UKK	South West	0.076	0.084	0.076	0.076
UKL	Wales	0.020	0.025	0.020	0.020
UKM	Scotland	0.041	0.053	0.041	0.041
Panel C: Occupation Distribution (SOC 2-digit)					
11	Corporate managers and directors	0.039	0.048	0.039	0.039
12	Other managers and proprietors	0.009	0.024	0.009	0.009

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Table OD2 – continued from previous page

Code	Variable	Before Weighting		After Weighting	
		AI-demanding	Non-demanding	AI-demanding	Non-demanding
21	Science, research, engineering and technology professionals	0.274	0.057	0.271	0.271
22	Health professionals	0.007	0.015	0.007	0.007
23	Teaching and educational professionals	0.008	0.018	0.008	0.008
24	Business, media and public service professionals	0.132	0.072	0.132	0.132
31	Science, engineering and technology associate professionals	0.048	0.033	0.049	0.049
32	Health and social care associate professionals	0.009	0.022	0.009	0.009
33	Protective service occupations	0.001	0.000	0.001	0.001
34	Culture, media and sports occupations	0.017	0.016	0.017	0.017
35	Business and public service associate professionals	0.218	0.165	0.219	0.219
41	Administrative occupations	0.052	0.093	0.052	0.052
42	Secretarial and related occupations	0.007	0.013	0.007	0.007
51	Skilled agricultural and related trades	0.000	0.004	0.000	0.000
52	Skilled metal, electrical and electronic trades	0.033	0.050	0.034	0.034
53	Skilled construction and building trades	0.005	0.020	0.005	0.005
54	Textiles, printing and other skilled trades	0.007	0.052	0.007	0.007
61	Caring personal service occupations	0.017	0.087	0.017	0.017
62	Leisure, travel and related personal service occupations	0.006	0.016	0.006	0.006
71	Sales occupations	0.016	0.023	0.016	0.016
72	Customer service occupations	0.053	0.042	0.054	0.054
81	Process, plant and machine operatives	0.005	0.013	0.005	0.005
82	Transport and mobile machine drivers and operatives	0.010	0.035	0.010	0.010
91	Elementary trades and related occupations	0.002	0.007	0.002	0.002
92	Elementary administration and service occupations	0.046	0.071	0.046	0.046
Panel D: Industry Distribution (SIC 2-digit)					
01	Crop and animal production	0.003	0.005	0.003	0.003

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Table OD2 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-demanding	Non-demanding	AI-demanding	Non-demanding
02	Forestry and logging	0.000	0.001	0.000	0.000
03	Fishing and aquaculture	0.000	0.001	0.000	0.000
05	Mining of coal and lignite	0.000	0.000	0.000	0.000
07	Extraction of crude petroleum and natural gas	0.000	0.001	0.000	0.000
08	Mining of metal ores	0.001	0.001	0.001	0.001
09	Other mining and quarrying	0.004	0.010	0.004	0.004
10	Manufacture of food products	0.001	0.002	0.001	0.001
12	Manufacture of tobacco products	0.001	0.002	0.001	0.001
13	Manufacture of textiles	0.002	0.002	0.002	0.002
15	Manufacture of leather and related products	0.001	0.002	0.001	0.001
16	Manufacture of wood and products of wood	0.000	0.002	0.000	0.000
17	Manufacture of paper and paper products	0.003	0.004	0.003	0.003
19	Manufacture of coke and refined petroleum products	0.001	0.004	0.001	0.001
20	Manufacture of chemicals and chemical products	0.001	0.001	0.001	0.001
21	Manufacture of basic pharmaceutical products	0.000	0.004	0.000	0.000
22	Manufacture of rubber and plastic products	0.001	0.002	0.001	0.001
23	Manufacture of other non-metallic mineral products	0.001	0.002	0.001	0.001
24	Manufacture of basic metals	0.004	0.012	0.004	0.004
25	Manufacture of fabricated metal products	0.007	0.005	0.007	0.007
26	Manufacture of computer, electronic and optical products	0.003	0.004	0.003	0.003
27	Manufacture of electrical equipment	0.004	0.007	0.004	0.004
28	Manufacture of machinery and equipment	0.001	0.002	0.001	0.001
29	Manufacture of motor vehicles, trailers and semi-trailers	0.002	0.001	0.002	0.002
30	Manufacture of other transport equipment	0.001	0.003	0.001	0.001
31	Manufacture of furniture	0.008	0.012	0.008	0.008

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Table OD2 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-demanding	Non-demanding	AI-demanding	Non-demanding
32	Other manufacturing	0.003	0.008	0.003	0.003
34	Electricity, gas, steam and air conditioning supply	0.006	0.002	0.006	0.006
35	Water collection, treatment and supply	0.002	0.001	0.002	0.002
36	Sewerage	0.000	0.000	0.000	0.000
37	Waste collection, treatment and disposal	0.001	0.004	0.001	0.001
38	Remediation activities and other waste management	0.001	0.001	0.001	0.001
40	Repair and installation of machinery and equipment	0.014	0.022	0.014	0.014
41	Construction of buildings	0.002	0.005	0.002	0.002
42	Civil engineering	0.016	0.045	0.016	0.016
43	Specialised construction activities	0.010	0.019	0.010	0.010
44	Wholesale and retail trade and repair of motor vehicles	0.027	0.045	0.027	0.027
45	Wholesale trade, except of motor vehicles	0.040	0.047	0.041	0.041
46	Retail trade, except of motor vehicles	0.007	0.017	0.007	0.007
47	Land transport and transport via pipelines	0.001	0.001	0.001	0.001
48	Water transport	0.002	0.001	0.002	0.002
49	Air transport	0.004	0.007	0.004	0.004
50	Warehousing and support activities for transportation	0.000	0.003	0.000	0.000
51	Postal and courier activities	0.004	0.020	0.004	0.004
52	Accommodation	0.018	0.060	0.018	0.018
53	Food and beverage service activities	0.022	0.010	0.021	0.021
54	Publishing activities	0.007	0.005	0.007	0.007
55	Motion picture, video and television production	0.001	0.001	0.001	0.001
56	Programming and broadcasting activities	0.019	0.007	0.018	0.018
57	Telecommunications	0.196	0.041	0.194	0.194
58	Computer programming, consultancy and related	0.027	0.011	0.027	0.027

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Table OD2 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-demanding	Non-demanding	AI-demanding	Non-demanding
59	Information service activities	0.055	0.027	0.055	0.055
60	Financial service activities	0.004	0.003	0.004	0.004
61	Insurance, reinsurance and pension funding	0.011	0.007	0.011	0.011
63	Activities auxiliary to financial and insurance	0.038	0.039	0.039	0.039
64	Real estate activities	0.012	0.017	0.012	0.012
65	Legal and accounting activities	0.107	0.045	0.108	0.108
66	Activities of head offices	0.016	0.012	0.016	0.016
67	Architectural and engineering activities	0.017	0.005	0.017	0.017
68	Scientific research and development	0.024	0.012	0.024	0.024
69	Other professional, scientific and technical	0.056	0.036	0.056	0.056
70	Advertising and market research	0.000	0.001	0.000	0.000
71	Veterinary activities	0.004	0.005	0.004	0.004
72	Rental and leasing activities	0.006	0.005	0.006	0.006
73	Employment activities	0.004	0.008	0.004	0.004
74	Travel agency and tour operator activities	0.002	0.019	0.002	0.002
75	Security and investigation activities	0.007	0.006	0.007	0.007
76	Services to buildings and landscape activities	0.002	0.002	0.002	0.002
77	Office administrative and other business support	0.025	0.062	0.025	0.025
78	Public administration and defence	0.018	0.042	0.018	0.018
79	Education	0.004	0.030	0.004	0.004
80	Human health activities	0.007	0.031	0.007	0.007
81	Residential care activities	0.006	0.009	0.006	0.006
82	Social work activities without accommodation	0.001	0.003	0.001	0.001
83	Creative, arts and entertainment activities	0.001	0.001	0.001	0.001
84	Libraries, archives, museums and other cultural	0.007	0.015	0.007	0.007

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Table OD2 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-demanding	Non-demanding	AI-demanding	Non-demanding
85	Gambling and betting activities	0.008	0.014	0.008	0.008
86	Sports activities and amusement and recreation	0.001	0.001	0.001	0.001
87	Activities of membership organisations	0.037	0.039	0.037	0.037
88	Repair of computers and personal household goods	0.000	0.000	0.000	0.000
89	Other personal service activities	0.003	0.006	0.003	0.003
90	Other service activities	0.036	0.021	0.036	0.036
Panel E: Firm Characteristics					
	Publicly Listed	0.013	0.003	0.014	0.014
Panel F: Sample Sizes					
	Observations	17,196	237,255	17,196	237,255

Notes: Table shows mean values of covariates for AI-adopters and non-adopters before and after entropy balancing. After weighting columns show reweighted control group means. Yearly aggregation, mean model. Covariates include all personality traits, skills, location dummies (NUTS1), occupation dummies (SOC 2-digit), industry dummies (SIC 2-digit), and firm characteristics used in entropy balancing.

Table OD3: Covariate Balance - Half-Year model

		Before Weighting		After Weighting	
Code	Variable	AI-demanding	Non-demanding	AI-demanding	Non-demanding
Panel A: Personality Traits and Skills					
	Agreeableness	0.104	0.108	0.104	0.104
	Conscientiousness	0.401	0.380	0.401	0.401
	Emotional Stability	0.132	0.134	0.132	0.132
	Extraversion	0.457	0.460	0.457	0.457
	Openness	0.468	0.307	0.467	0.467
	Soft Skills	0.740	0.634	0.740	0.740
	Hard Skills	0.828	0.572	0.827	0.827
Panel B: Location Distribution (NUTS1)					
UKC	North East	0.015	0.018	0.015	0.015
UKD	North West	0.079	0.093	0.079	0.079
UKE	Yorkshire and The Humber	0.045	0.059	0.045	0.045
UKF	East Midlands	0.036	0.058	0.036	0.036
UKG	West Midlands	0.058	0.072	0.058	0.058
UKH	East of England	0.070	0.098	0.070	0.070
UKI	London	0.396	0.262	0.396	0.396
UKJ	South East	0.157	0.172	0.157	0.157
UKK	South West	0.077	0.084	0.077	0.077
UKL	Wales	0.020	0.024	0.020	0.020
UKM	Scotland	0.042	0.052	0.042	0.042
Panel C: Occupation Distribution (SOC 2-digit)					
11	Corporate managers and directors	0.040	0.049	0.040	0.040
12	Other managers and proprietors	0.009	0.024	0.009	0.009

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Table OD3 – continued from previous page

Code	Variable	Before Weighting		After Weighting	
		AI-demanding	Non-demanding	AI-demanding	Non-demanding
21	Science, research, engineering and technology professionals	0.266	0.057	0.263	0.262
22	Health professionals	0.007	0.016	0.007	0.007
23	Teaching and educational professionals	0.008	0.019	0.008	0.008
24	Business, media and public service professionals	0.134	0.074	0.133	0.133
31	Science, engineering and technology associate professionals	0.049	0.033	0.049	0.049
32	Health and social care associate professionals	0.009	0.024	0.009	0.009
33	Protective service occupations	0.001	0.000	0.001	0.001
34	Culture, media and sports occupations	0.017	0.015	0.017	0.017
35	Business and public service associate professionals	0.217	0.165	0.218	0.218
41	Administrative occupations	0.053	0.090	0.053	0.053
42	Secretarial and related occupations	0.008	0.012	0.008	0.008
51	Skilled agricultural and related trades	0.000	0.004	0.000	0.000
52	Skilled metal, electrical and electronic trades	0.034	0.047	0.034	0.034
53	Skilled construction and building trades	0.005	0.018	0.005	0.005
54	Textiles, printing and other skilled trades	0.007	0.052	0.007	0.007
61	Caring personal service occupations	0.017	0.091	0.017	0.017
62	Leisure, travel and related personal service occupations	0.006	0.016	0.006	0.006
71	Sales occupations	0.016	0.023	0.016	0.016
72	Customer service occupations	0.055	0.043	0.055	0.055
81	Process, plant and machine operatives	0.005	0.012	0.005	0.005
82	Transport and mobile machine drivers and operatives	0.010	0.034	0.010	0.010
91	Elementary trades and related occupations	0.002	0.007	0.002	0.002
92	Elementary administration and service occupations	0.047	0.073	0.047	0.047
Panel D: Industry Distribution (SIC 2-digit)					
01	Crop and animal production	0.003	0.004	0.003	0.003

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Table OD3 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-demanding	Non-demanding	AI-demanding	Non-demanding
02	Forestry and logging	0.000	0.001	0.000	0.000
03	Fishing and aquaculture	0.000	0.001	0.000	0.000
05	Mining of coal and lignite	0.000	0.000	0.000	0.000
07	Extraction of crude petroleum and natural gas	0.000	0.001	0.000	0.000
08	Mining of metal ores	0.001	0.001	0.001	0.001
09	Other mining and quarrying	0.004	0.010	0.004	0.004
10	Manufacture of food products	0.001	0.002	0.001	0.001
12	Manufacture of tobacco products	0.001	0.002	0.001	0.001
13	Manufacture of textiles	0.002	0.002	0.002	0.002
15	Manufacture of leather and related products	0.001	0.002	0.001	0.001
16	Manufacture of wood and products of wood	0.000	0.002	0.000	0.000
17	Manufacture of paper and paper products	0.003	0.003	0.003	0.003
19	Manufacture of coke and refined petroleum products	0.001	0.003	0.001	0.001
20	Manufacture of chemicals and chemical products	0.001	0.001	0.001	0.001
21	Manufacture of basic pharmaceutical products	0.000	0.004	0.000	0.000
22	Manufacture of rubber and plastic products	0.001	0.002	0.001	0.001
23	Manufacture of other non-metallic mineral products	0.001	0.002	0.001	0.001
24	Manufacture of basic metals	0.004	0.012	0.004	0.004
25	Manufacture of fabricated metal products	0.007	0.005	0.007	0.007
26	Manufacture of computer, electronic and optical products	0.003	0.004	0.003	0.003
27	Manufacture of electrical equipment	0.004	0.007	0.004	0.004
28	Manufacture of machinery and equipment	0.001	0.002	0.001	0.001
29	Manufacture of motor vehicles, trailers and semi-trailers	0.002	0.001	0.002	0.002
30	Manufacture of other transport equipment	0.001	0.003	0.001	0.001
31	Manufacture of furniture	0.007	0.012	0.007	0.007

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Table OD3 – continued from previous page

Code	Variable	Before Weighting		After Weighting	
		AI-demanding	Non-demanding	AI-demanding	Non-demanding
32	Other manufacturing	0.003	0.007	0.003	0.003
34	Electricity, gas, steam and air conditioning supply	0.006	0.002	0.006	0.006
35	Water collection, treatment and supply	0.002	0.001	0.002	0.002
36	Sewerage	0.000	0.000	0.000	0.000
37	Waste collection, treatment and disposal	0.001	0.004	0.001	0.001
38	Remediation activities and other waste management	0.001	0.001	0.001	0.001
40	Repair and installation of machinery and equipment	0.014	0.021	0.014	0.014
41	Construction of buildings	0.002	0.005	0.002	0.002
42	Civil engineering	0.016	0.042	0.016	0.016
43	Specialised construction activities	0.011	0.018	0.011	0.011
44	Wholesale and retail trade and repair of motor vehicles	0.026	0.044	0.026	0.026
45	Wholesale trade, except of motor vehicles	0.041	0.047	0.041	0.041
46	Retail trade, except of motor vehicles	0.007	0.017	0.007	0.007
47	Land transport and transport via pipelines	0.000	0.001	0.000	0.000
48	Water transport	0.002	0.001	0.002	0.002
49	Air transport	0.004	0.007	0.004	0.004
50	Warehousing and support activities for transportation	0.000	0.003	0.000	0.000
51	Postal and courier activities	0.004	0.021	0.004	0.004
52	Accommodation	0.018	0.060	0.018	0.018
53	Food and beverage service activities	0.021	0.010	0.020	0.020
54	Publishing activities	0.006	0.005	0.006	0.006
55	Motion picture, video and television production	0.001	0.001	0.001	0.001
56	Programming and broadcasting activities	0.018	0.007	0.018	0.018
57	Telecommunications	0.187	0.040	0.184	0.184
58	Computer programming, consultancy and related	0.026	0.010	0.026	0.026

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Table OD3 – continued from previous page

Code	Variable	Before Weighting		After Weighting	
		AI-demanding	Non-demanding	AI-demanding	Non-demanding
59	Information service activities	0.057	0.028	0.057	0.057
60	Financial service activities	0.004	0.002	0.004	0.004
61	Insurance, reinsurance and pension funding	0.011	0.006	0.011	0.011
63	Activities auxiliary to financial and insurance	0.040	0.039	0.040	0.040
64	Real estate activities	0.012	0.016	0.012	0.012
65	Legal and accounting activities	0.112	0.047	0.113	0.113
66	Activities of head offices	0.016	0.012	0.016	0.016
67	Architectural and engineering activities	0.016	0.005	0.016	0.016
68	Scientific research and development	0.024	0.012	0.024	0.024
69	Other professional, scientific and technical	0.057	0.037	0.057	0.057
70	Advertising and market research	0.000	0.001	0.000	0.000
71	Veterinary activities	0.004	0.005	0.004	0.004
72	Rental and leasing activities	0.006	0.004	0.006	0.006
73	Employment activities	0.004	0.008	0.004	0.004
74	Travel agency and tour operator activities	0.002	0.018	0.002	0.002
75	Security and investigation activities	0.007	0.006	0.007	0.007
76	Services to buildings and landscape activities	0.002	0.003	0.002	0.002
77	Office administrative and other business support	0.026	0.065	0.026	0.026
78	Public administration and defence	0.017	0.045	0.017	0.017
79	Education	0.004	0.031	0.004	0.004
80	Human health activities	0.008	0.033	0.008	0.008
81	Residential care activities	0.006	0.009	0.006	0.006
82	Social work activities without accommodation	0.001	0.003	0.001	0.001
83	Creative, arts and entertainment activities	0.002	0.001	0.002	0.002
84	Libraries, archives, museums and other cultural	0.007	0.015	0.007	0.007

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Table OD3 – continued from previous page

		Before Weighting		After Weighting	
Code	Variable	AI-demanding	Non-demanding	AI-demanding	Non-demanding
85	Gambling and betting activities	0.009	0.014	0.008	0.008
86	Sports activities and amusement and recreation	0.001	0.001	0.001	0.001
87	Activities of membership organisations	0.039	0.038	0.039	0.039
88	Repair of computers and personal household goods	0.000	0.000	0.000	0.000
89	Other personal service activities	0.003	0.006	0.003	0.003
90	Other service activities	0.037	0.022	0.038	0.038
Panel E: Firm Characteristics					
	Publicly Listed	0.014	0.003	0.015	0.015
Panel F: Sample Sizes					
	Observations	25,143	305,899	25,143	305,899

Notes: Table shows mean values of covariates for AI-adopters and non-adopters before and after entropy balancing. After weighting columns show reweighted control group means. Half-yearly aggregation, mean model. Covariates include all personality traits, skills, location dummies (NUTS1), occupation dummies (SOC 2-digit), industry dummies (SIC 2-digit), and firm characteristics used in entropy balancing.

E. Personality traits. Evolution over time

In this section, we examine whether personality traits change over our sample period. In order to do so, we regress personality traits against year dummies in the following way:

$$P(Trait_i = 1) = \beta_0 + \beta_1 X_i + \sum_{t=2018}^{2022} \alpha_t Year_t + \gamma_k + \phi_j + \epsilon_i$$

where X_i is a vector of job posting characteristics including the demand for other personality traits, δ_l are location fixed effects, $Year_t$ are dummy variables (with 2017 as the base year), ϕ_j are firm fixed effects and γ_k are occupation fixed effects. The coefficients α_t capture the evolution of personality trait demands over 2018-2022, after accounting for changes in job requirements, occupational composition, geographical variation, and firm-specific practices.

After accounting for compositional changes in location, occupation, and firm, Table OE4 shows the estimated year effects relative to 2017 for the full period. Conscientiousness shows a gradual decline, becoming significantly negative by 2022. Extraversion shows positive and statistically significant coefficients in 2019–2021, but the 2022 estimate is not statistically significant. Emotional Stability shows temporary positive effects during 2021 (1.1 percentage points), while Openness and Agreeableness show no significant time trends after controlling for composition.

Table OE4: Year effect on each Personality Trait. Baseline: 2017

	(1) P(All PT)	(2) P(Agree.)	(3) P(Consc.)	(4) P(Emot.)	(5) P(Extra.)	(6) P(Open.)
2018	−0.003	0.002	−0.010	0.004	0.009	0.002
	(0.004)	(0.004)	(0.008)	(0.004)	(0.006)	(0.005)
2019	−0.003	0.003	−0.012	0.011	0.017*	0.006
	(0.006)	(0.007)	(0.010)	(0.007)	(0.009)	(0.007)
2020	0.001	0.012	−0.014	0.009	0.028**	−0.000
	(0.008)	(0.013)	(0.011)	(0.007)	(0.012)	(0.007)
2021	−0.005	0.018	−0.016	0.011*	0.022*	−0.001
	(0.008)	(0.016)	(0.010)	(0.007)	(0.012)	(0.008)
2022	−0.014	0.003	−0.021**	0.006	0.017	−0.003
	(0.009)	(0.009)	(0.011)	(0.008)	(0.012)	(0.009)
R^2	0.349	0.364	0.280	0.286	0.341	0.331

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. $N = 9888315$. Regression controls for traits and skills demand, remote work requirements, length of advertisement, contract type, work schedule, bonus offerings, education, and negotiable salary option. Location, occupation and firm fixed effects. SE clustered at the occupation and firm level. Author's creation using Adzuna